

TFS 1.03 Teletext ROM

A reference to the Acorn Teletext Adapter and its Telesoft filing system ROM for the BBC Micro Model B. It covers the register map checked against the redrawn schematic (drawing 104,000/C), the layout of the ROM, every dispatch table, and a fully commented disassembly of the correct image, TELEROMU.ROM. A second part covers its successor, the BBC's Advanced Teletext System 2.50 (ATS250.ROM, 1986), with its own commented disassembly and a summary of what changed.

Use TELEROMU.ROM: TFS103.ROM is corrupt

The two images are the same ROM, but TFS103.ROM is a bad dump. It differs from TELEROMU.ROM, the externally confirmed good copy, in exactly three places:

ADDRESS	TFS103.ROM	TELEROMU.ROM	EFFECT OF THE CORRUPTION
&8000	&00 (BRK)	&4C (JMP &B32F)	*TELETEXT hangs after printing "TFS"
&A000	&00 (BRK)	&91 (STA (&B4),Y)	Breaks the end of a Telesoftware *LOAD / *RUN (&9FEC), when the file length is written back. Inferred from the code, not tested.
&BD2E-&BFFF	left-over code from another build	zeros	None: nothing reaches this area

Why *TELETEXT hung

- *TELETEXT runs &9D1E, which asks the MOS (OSBYTE 142) to start TFS as a language.
- The MOS prints the ROM title TFS, then jumps to &8000, which in the bad image is BRK.
- The MOS BRK handler pages in the current language (now TFS) and jumps through BRKV, which still points at the previous language's handler: &B402 for BASIC II (from memory; the BASIC ROM isn't in the folder). In the TFS ROM that address is a short routine ending in RTS, which "returns" into page &02 (MOS vectors) and the machine hangs.

TFS103P.ROM fixed only the first byte. Use TELEROMU.ROM. To check which image is in an EPROM, this BASIC finds the socket holding TFS and prints its first byte (4C is correct):

```
FOR R%=0 TO 15:IF &F6=9:IF &F7=&80:Y%=R%:IF (USR&FFB9 AND &FF)=84 THEN ?&F6=0:PRINT R%;" ";~
(USR&FFB9 AND &FF)
NEXT
```

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How capture works

The adapter has no SAA5040 acquisition chip. The hardware only slices teletext lines out of the field blanking interval and stores the raw bytes. Everything a TAC would do, the BBC Micro's 6502 does in the interrupt handler.

Tuner & IF

U341 varicap tuner, SL1430 preamp, SW153A SAW filter, TDA2541 IF/demodulator. Four 100K presets (VR2–VR5) set the tuning voltage from a 33 V zener rail.

Slicing

SAA5030 VIP recovers data and a 6.9375 MHz clock. The SAA5020 TIC supplies line and field timing, including the data entry window DEW.

Bytes to RAM

A 74LS164 shift register and 74LS374 latch assemble bytes. After run-in detection, each line is written into 1 KB of 2114 RAM, organised as 16 slots of 64 bytes.

Field interrupt

At the DEW edge IC4 sets INT. With control bit 3 set, IC18 pulls NIRQ low.

Software TAC

[&8D3F](#) turns capture off so the CPU owns the RAM. It reads each slot flagged non-zero, Hamming-decodes magazine and row, and matches row 0 headers against the 25 page request slots. Matching rows go into page buffers, and 8/30 service data is copied. It then checks the row-27 CRC, writes [&FC13](#) to clear the status flip-flops, and turns capture back on.

Signal quality for the [*TUNE](#) bar comes from counters at [&0E52](#) (Hamming bytes decoded) and [&0E53](#) (bytes needing correction or rejected).

What "Bad data" means

The message at the bottom right of the terminal-mode screen (row 24, column 31) reports the page checksum. When a page finishes arriving, [&9105](#) looks for a row 27 packet carrying the page's 16-bit check word (user guide section 9.2). If it finds one, it computes the CRC over the received character bytes ([&91AA](#)–[&9221](#)), counting rows that never arrived as spaces. It records two flags per page, which terminal mode reads with OSWORD [&7A](#) code [&8C](#) and shows at [&B619](#).

CHECK WORD IN ROW 27?	COMPUTED CRC MATCHES?	BOTTOM RIGHT SHOWS
Yes	Yes	Nothing (page verified)
Yes	No	Bad data
No	n/a	No check

Only the address and header bytes are Hamming protected, so the checksum is the only thing that catches corrupted text. "Bad data" usually means at least one character on the page is wrong. It is a warning only; the page is still displayed.

Likely causes

- **Reception errors.** A weak or ghosted signal, or a slightly off-tune channel. Use SHIFT f4 and aim for the longest tuning bar. The page often comes good on a later cycle.
- **Rows mixed from two transmissions.** The page changed while it was being captured, or a row was missed and an older copy kept.
- **A modern teletext source.** UK broadcast teletext ended in 2012, so the adapter is probably fed from an inserter (such as a Raspberry Pi VBIT setup) or a recording. Many modern inserters send row 27 with a check word that isn't calculated the original 1976 way. If every page shows "Bad data" while the text looks perfect, this is the likely reason, and it is harmless. This is inferred from the code, not tested against a particular inserter.

Telesoftware downloads use the same check, but act on it: a failure gives "Bad data. Retrying..." and a retry, controlled by `*OPT2`.

Registers & pins

The adapter decodes &FC10–&FC13 on the 1 MHz bus. IC28, IC27, IC17 and IC33 require A7–A2 = 000100 while NPGFC (page &FC) is active. IC6 (74LS139, first half) splits A1/A0 into four strobes. Chips give the confidence of each entry: **TRACED** means followed wire by wire on the schematic, **PROBABLE** means net names and software agree but not every gate was traced, and **SOFTWARE** means inferred from code only.

&FC10 read: status (IC19 81LS95, enabled by IC6 Y0 pin 12 → IC19 pin 1)

BIT	SIGNAL	IC19 IN → OUT	SOURCE & MEANING	USED BY ROM	
D7	INT	12 → 11	IC4a Q (pin 5). D tied high, clocked by the DEW edge, so it sets once per field.	BIT &FC10 / BPL: no interrupt, pass service 5 on	TRACED
D6	DOR	8 → 9	IC4b Q (pin 9). D = INT, clocked on the next DEW edge, so it sets when an interrupt was not serviced in time (overrun).	Abandons all partly captured pages (&908D)	TRACED
D5	DEW	14 → 13	Data entry window from SAA5020 pin 14 via IC14	Not read	TRACED
D4	FSYN	6 → 7	Field sync (TP9)	Not read	TRACED
D3–D0	LK4–LK7	2/18/16/4 → 5/15/17/3	Option links. Each input can be strapped to 0 V or +5 V, or (lower link rows) to control bits 4 or 5.	AND #&0F = 0 means "no power"; nothing else	TRACED

The "no power" test relies on an unpowered adapter reading zero. RP1/RP2 (2K2 to 0 V and to the adapter's +5 V) would both pull the bus low when the adapter's supply is off. That mechanism is inferred, not measured.

&FC10 write: control latch (IC20 74LS174, clocked by IC27 pin 3 from Y0 + write; cleared by NRST)

BIT	D → Q PINS	GOES TO	FUNCTION	ROM USE	
b0	14 → 15	IC6b A0 (pin 2)	Channel select. IC6b Y0–Y3 → IC32 inverters → Q3–Q6 → presets VR2–VR5 (channels 1–4).	OSWORD &8B, *CH/*BBC/*ITV, f4	TRACED
b1	3 → 2	IC6b A1 (pin 3)			
b2	13 → 12	TTXEN → IC12 pin 13	Capture enable. IC12 makes CPU = NAND(DEW, TTXEN, /INT), which switches the IC22 address multiplexer. The hardware only writes RAM inside the data window, with capture on and no interrupt pending. Otherwise the CPU owns the RAM.	Cleared during the IRQ (&A249), restored after (&A25A)	TRACED
b3	4 → 5	IC18 pin 1	IRQ enable. IC18 (74LS03, open collector) drives NIRQ = NAND(INT, b3) onto bus pin 8.	Same routines as b2	TRACED
b4	11 → 10	TDA2541 pin 6; link row	AFC control on the IF chip. Also strappable back onto a status bit.	Set at init; cleared by *TUNE / SHIFT f4 so hand tuning isn't pulled by AFC, set again on exit	TRACED
b5	6 → 7	Link row only	Spare output, readable only via the option links	Never set	PROBABLE

The ROM keeps a shadow at &0E8E and initialises it to &1C (b2, b3, b4 on; channel 1).

&FC11–&FC13

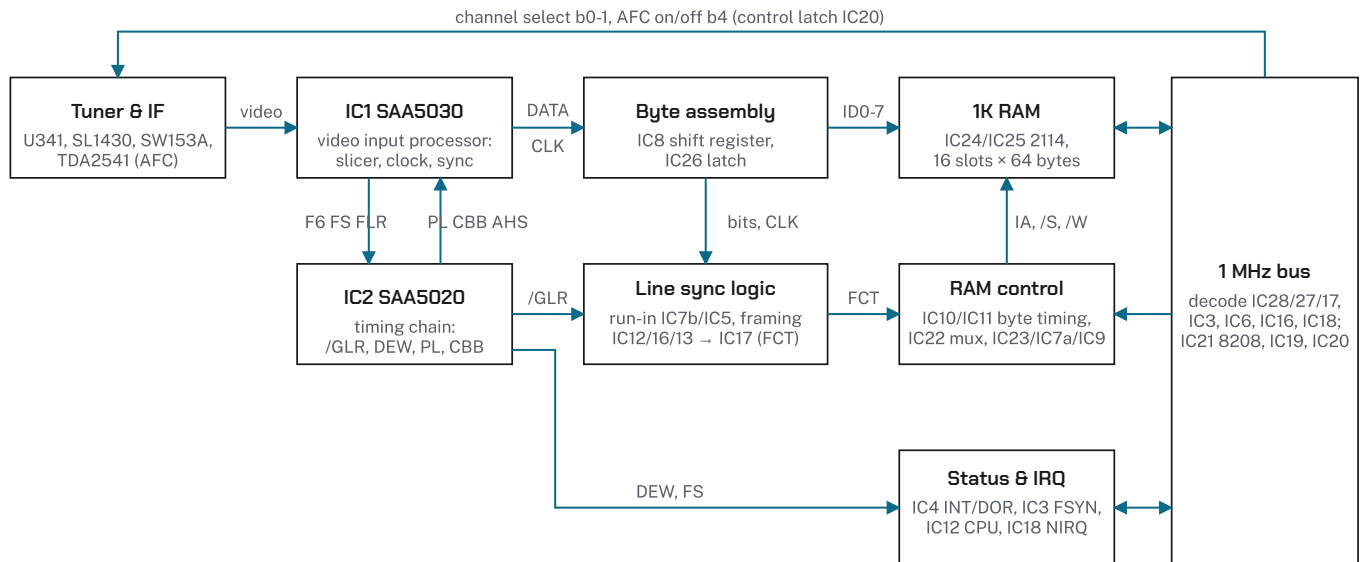
ADDRESS	STROBE	FUNCTION	ROM USE	
&FC11 W	IC6 Y1 (pin 11), ROWREG	Loads the slot number 0–15 into IC9 (RAM address IA6–IA9) and clears the byte counter (IC23, IC7a)	STY &FC11 before each slot	TRACED
&FC12 R/W	IC6 Y2 (pin 10), DATA	RAM data through the IC21 8208 transceiver (enabled for &FC11/&FC12 by IC13). The trailing edge of each access steps the byte counter IA0–IA5 (IC23 + IC7a).	Reads slot bytes. Writes 0 to byte 0 to mark a slot consumed. Also used to stash workspace.	TRACED
&FC13 W	IC6 Y3 (pin 9), STATCLR	Clears IC3, IC4a (INT) and IC4b (DOR). The data value is ignored.	Last act of every interrupt (&9221)	TRACED

Slot format: byte 0 is the framing code &27, written by the hardware as the first byte of every captured line (see the circuit description), so it is non-zero when the slot has been filled. Bytes 1–2 are the Hamming-coded magazine and row address. Bytes 3 onward are the 40 character bytes, or for row 0 the 8 header bytes followed by 32 characters.

Circuit description

This section follows the redrawn schematic (drawing 104,000/C, PCB issue 10) signal by signal. Pin numbers are the drawing's. Statements about the SAA5030 come from its data sheet in this folder. SAA5020 timings come from its data sheet (Mullard, August 1978), also in this folder. Where a conclusion is my deduction from those timings rather than something either data sheet states, the text says so.

Block diagram



The CPU side and the capture side share the RAM: the CPU signal decides which owns it (see 4 below).

1. Bus interface and address decoding (bottom left)

The adapter sits on the BBC Micro's 1 MHz bus: A0–A7, D0–D7, R/W, 1MHzE (the 1 MHz bus clock), NPGFC (low when page &FC, "FRED", is addressed), NRST and NIRQ. The address and data lines are terminated with 2K2 to +5 V and 2K2 to 0 V (RP3/RP4 on the address lines, RP1/RP2 on the data lines), and R/W, 1MHzE and NPGFC with 2K2 pairs (R21/R22, R39/R30, R43/R38). This holds unconnected lines at about 2.5 V and damps reflections on the ribbon cable.

Buffering

IC33 is Schmitt-trigger inverters (the drawing labels it both 74LS14 and 74LS04; the symbols are Schmitt). 1MHzE passes through two of them (IC33 pin 6 = /E, pin 8 = E), R/W through two (pin 4 = R/W), NPGFC through one (pin 10 = PGFC, high when page &FC is addressed).

Page select synchronised to the bus clock

IC3b (74LS74, pins 8–13) clocks PGFC in on the rising edge of 1MHzE. Its /Q output, CNPGFC, goes low at the start of a 1 MHz cycle that addresses page &FC. Because PGFC is also wired to the flip-flop's clear input, CNPGFC goes high again the moment NPGFC is released. The select is therefore clean for exactly one 1 MHz cycle.

Address match &10–&13

IC28 ORs A7|A6 (pin 6) and A5|A3 (pin 3), then those two together (pin 8). IC33 inverts A2, and IC17 (pins 1–3) NANDs A4 with /A2, giving a low output only when A4=1 and A2=0. IC27 ORs that with CNPGFC (pin 6), then with IC28's output (pin 8), then with /E (pin 11). The result, /BUSSEL at IC27 pin 11, is low only when A7=A6=A5=A3=0, A4=1, A2=0, page &FC is selected and 1MHzE is high: the second half of a cycle to &FC10–&FC13.

Register strobes

/BUSSEL enables IC6a (74LS139, pin 15), which decodes A1/A0 into four active-low strobes: Y0 (pin 12) = &FC10, Y1 (pin 11) = &FC11 "ROWREG", Y2 (pin 10) = &FC12 "DATA", Y3 (pin 9) = &FC13 "STATCLR".

Write detection

IC16 (74LS27, pins 9–11, 8) NORs /BUSSEL, R/W and 0 V, giving WR high during a write to the adapter. IC18 (74LS03, open collector, pull-up R10 3K3) NANDs WR with E to give /WRSTB, a write strobe that ends when 1MHzE falls.

ACCESS	LOGIC	EFFECT
read &FC10	IC19 (81LS95) enabled by Y0 (pin 1) and WR low (pin 19)	Status bits driven straight onto the BBC data bus
write &FC10	IC27 pins 1–3 OR Y0 with /WRSTB; the rising edge at the end of the write clocks IC20 (74LS174, pin 9)	Control latch takes D0–D5 from the BBC data bus. NRST clears it at reset: channel 1, capture, IRQ and AFC all off.
&FC11, &FC12	IC13 (pins 9, 10 → 8) ANDs Y1 and Y2: low for either	Enables the IC21 8208 bus transceiver (/OE, pin 9) between the BBC data bus and the internal ID bus. R/W sets its direction (pin 11).
write &FC11	Y1 = /ROWREG through IC22	Loads the slot number into IC9 and clears the byte counter (see 3)
&FC12	Y2 = /DATA through IC22 to the RAM's chip select, /WRSTB to its write enable	Reads or writes one RAM byte; the end of the access steps the byte counter
&FC13	Y3 = /STATCLR, read or write	Clears the INT, DOR and FSYN flip-flops

Only &FC11 and &FC12 go through the 8208; the status buffer and control latch connect to the BBC data bus directly. A read of &FC11 would also produce a ROWREG strobe and load the slot register with whatever is on the internal bus; the software only ever writes it.

2. Timing: the SAA5030 / SAA5020 pair

IC1, the SAA5030 video input processor (VIP), runs from +12 V. It slices the teletext data out of the composite video from the TDA2541, recovers a 6.9375 MHz data clock with the tuned circuit L2, and runs a 6 MHz crystal oscillator (X1, trimmed by C7) that is phase-locked to the line sync. IC2, the SAA5020 timing chain (TIC), divides that 6 MHz down into line and field timing. The two chips exchange the signals below.

SIGNAL	FROM → TO	WHAT IT IS	WHAT IT DOES ON THIS BOARD
F6	IC1 pin 6 → IC2 pin 2	6 MHz clock from the crystal oscillator	Master clock for the timing chain (diode-clamped input). Inside the SAA5020 it is divided by 6 to 1 MHz (the F1 character clock), by 64 to the 15.625 kHz line rate, then by 312/313 alternately to the 50 Hz field rate: 625 lines per frame. Every time below is measured from the counter's "time zero" of the line, which the phase lock (PL) holds on the broadcast's line sync.
FS	IC1 pin 13 → IC2 pin 10; TP9 "FSYN"	Field sync separated from the video (data sheet)	Resets the SAA5020's field-rate (line-number) counter once a field, so the DEW lines below are real broadcast line numbers. Also clocks IC3a, the FSYN status latch (see 4).
FLR	IC1 pin 3 → IC2 pin 6	Fast line reset: a positive pulse of about 4.6 μs, issued while the phase-locked loop is first acquiring the signal	Resets the SAA5020's line-rate (μs) counter, for rapid lock when a signal first appears or the channel changes

SIGNAL	FROM → TO	WHAT IT IS	WHAT IT DOES ON THIS BOARD
PL	IC2 pin 8 → IC1 pin 5	/PL, phase lock: a 4 μ s negative pulse from 62 μ s, so it straddles the line boundary (62 μ s to 2 μ s into the next line). Open-drain output.	Compared with the separated sync in the VIP's phase detector, closing the loop that locks the 6 MHz oscillator to the broadcast, so time zero sits on line sync
CBB	IC2 pin 9 → R20 6K8 → IC1 pin 5 (TP8)	/CBB, colour-burst blanking: an 8 μ s negative pulse starting at time zero. Open-drain output. The data sheet says it resets the VIP's internal data-processing and sync circuits each line.	Because /PL and /CBB are both open drain they can share one pin: /PL pulls IC1 pin 5 hard low, /CBB pulls it only part way through R20, giving a two-level "sandcastle" on one pin: the VIP slices PL at about 2.5 V and CBB at about 5 V. CBB stops the colour burst from offsetting the data slicer.
AHS	IC2 pin 5 → IC1 pin 11	"After hours sync": composite sync generated inside the timing chain (data sheet Fig. 2: 4.66 μ s line syncs with the usual equalising and broad pulses)	For driving a TV's sync when there is no programme; not used further here
/GLR	IC2 pin 7 → IC14 (CD40106 Schmitt inverters) → GLR (pin 2, TP5) and /GLR (pin 4)	General line reset (the data sheet's pin name is GLR with a bar, i.e. active low): a 1 μ s negative pulse from 5 μ s to 6 μ s into every line	(a) Resets the IC17 "framing found" latch, so each line is captured afresh. (b) Through IC22, resets the byte counter and steps IC9 to the next RAM slot. (c) Clears IC5a. (d) Stretched by D1/R23/C20 (to about 21 μ s into the line) into the window in which a framing code may be recognised.
DEW	IC2 pin 14 → IC14 pin 13 → /DEW (pin 12); IC15d re-inverts to DEW; TP6	Data entry window: a positive pulse from the end of line 5 to the end of line 22 (and 318 to 335 in the other field). Each edge comes 7.5–8.5 μ s after /CBB falls, i.e. about 8 μ s into lines 6 and 23 (319 and 336).	(a) Hands the RAM to the capture logic (IC12 pin 2). (b) Its falling edge, at the end of the window, sets INT (IC4a). (c) Its rising edge, at the start of the next window, sets DOR (IC4b) if INT is still set.

One line, in time order (times from the SAA5020 data sheet; teletext positions from the broadcast specification):

TIME IN LINE	EVENT
62 → 2 μ s	/PL low, centred on time zero (line sync)
0 → 8 μ s	/CBB low: the VIP's slicer and sync circuits are reset; colour burst ignored
5 → 6 μ s	/GLR low: FCT reset; in capture mode, next slot and byte 0; window for the framing code opens
≈8 μ s	DEW changes, but only in lines 6 and 23 (319 and 336)
≈10.3 μ s	teletext clock run-in begins (16 bits of 1010...)
≈13.8 μ s	framing code &27 complete; FCT sets and the first byte is written
≈21 μ s	stretched-GLR window closes: no framing code is accepted after this
≈62 μ s	last of the 45 bytes (360 bits at 6.9375 MHz ≈ 52 μ s) has gone by

Which lines land in which slot (my deduction from the timings above). DEW rises about 8 μ s into line 6, after that line's /GLR, and falls about 8 μ s into line 23, after that line's /GLR. So the 17 /GLR pulses that fall inside the window are those of lines 7 to 23, and each steps IC9 once. TFS's interrupt routine scans slots 0 to 15 in order, so it leaves slot 15 selected. The next window's line 7 therefore goes into slot 0, and lines 7–22 fill slots 0–15 in order (lines 320–335 in the other field). Line 23's /GLR steps on once more, but DEW ends before any data could arrive on it. Sixteen slots match the sixteen lines 7–22 that UK teletext used when the adapter was designed.

Line 6 is the exception. DEW rises in time for its data, but no /GLR has reset the counters, so a packet on line 6 would be written into slot 15 starting from whatever byte the interrupt routine last read, wrapping within the slot. If line 22 also carries teletext it overwrites the slot completely; if not, the software could see a misaligned packet in slot 15. Later broadcasts did use line 6 (and 318), so this matters only for those signals, and I haven't tested it.

SAA5020 pins this board doesn't use. The chip was designed for a TV decoder with an SAA5040 and SAA5050, and most of it serves the display: TR6 (pin 3, 6 MHz dot clock), F1 (pin 4, 1 MHz character clock), CRS (pin 11, character rounding), LOSE (pin 13, a 40 μ s display-line enable on lines 49–288/362–601), A0–A4 (pins 19–23, display row addresses) and RACK (pin 24, display address clock) are all outputs left unconnected. The inputs TLC (15), HIE (16), BCS (17) and T/B (18) control only the display row addressing. The redrawn schematic doesn't show pins 3, 4, 11, 13 or 15–24 at all, so I can't tell from it how the four inputs are tied. Since they affect nothing but those unused outputs, it makes no difference to capture.

Data and clock leave the VIP at 12 V logic levels (about 5.5 V peak to peak). Each is AC-coupled through 1 nF (C15, C18), clamped by a diode (D8, D7) with a 15K bias resistor, and buffered by an emitter follower (Q1, Q2 BC239C, 470R emitter loads) to TTL levels: TP1 DATA and TP2 CLK. IC13a (pins 1–3) gates the clock with TTXEN (control bit 2), so no clock reaches the capture logic when capture is off. That gated clock is CLK; IC15a inverts it to /CLK.

3. Byte timing and RAM addressing: IC10, IC11, IC15, IC22, IC23, IC7a, IC9

Assembling bytes. IC8 (74LS164) shifts DATA in on each rising CLK edge; its two serial inputs (pins 1–2, drawn as an AND gate) have pin 1 tied high. Teletext is sent least significant bit first, so after eight clocks the first bit of a byte is in QH (pin 13) and the last in QA (pin 3). The internal bus numbering follows this: QH is ID0, QA is ID7, and bytes land in RAM in their normal bit order, as the software's Hamming table expects. The drawing labels IC8's pins Q7 (pin 3) to Q0 (pin 13), the reverse of the 74LS164's QA–QH naming. IC26 (74LS374) copies all eight bits on a rising edge of LOAD and drives them onto the internal ID bus while CPU is low.

Counting bits after the framing code. IC10 (74LS161) has its load, enable and data inputs all tied high, so it is a plain 4-bit counter. It counts falling CLK edges (IC15a pin 6 into pin 2) and is held clear while FCT is low (pin 1). Once the framing code has been found, its QC output (pin 12) is low for four bit-times and high for four: a square wave at the byte rate.

IC11 (74LS51, AND-OR-INVERT) turns that into the two strobes:

OUTPUT	EQUATION [DRAWING PINS]	BEHAVIOUR
LOAD (pin 8)	$\text{NOT}(\text{QC} \cdot 1 \cdot 1 [10,9,11] + 1 \cdot \text{CLK} \cdot \text{FCT} [13,1,12])$	Before the framing code: $\text{LOAD} = \text{/CLK}$, so IC26 copies the shift register on every falling clock edge, half a bit after each shift. When FCT goes high, LOAD rises and IC26 is left holding the framing code itself. From then on, $\text{LOAD} = \text{/QC}$: it rises every eighth falling clock edge, just after a complete byte has been shifted in, and IC26 latches that byte.
/WRITE (pin 6)	$\text{NOT}(\text{QC} \cdot \text{FCT} [2,3] + \text{FCT} \cdot 0 [4,5])$	Low for bit-times 4–7 of each byte period while FCT is high: one write pulse of about 577 ns per byte, writing the byte latched at the start of that period.

The first /WRITE after the framing code writes the latched framing code &27 into byte 0 of the slot, followed by the two Hamming-coded magazine/row bytes and the 40 data bytes. **So the "slot full" byte the software tests is the framing code**, and an empty slot is one the CPU has set to zero. Writing stops when /GLR resets FCT at the next line, or when the VIP's regenerated clock dies away after the data.

IC22 (74LS157) chooses who drives the RAM controls. Its select input (pin 1) is the CPU signal from section 4. The drawing's A/B labels are the reverse of the 74LS157's, so the table goes by pin number.

SECTION	CPU LOW: CAPTURE (PINS 2, 5, 11, 14)	CPU HIGH: COMPUTER (PINS 3, 6, 10, 13)	OUTPUT GOES TO
1Y pin 4 = /SEL	/WRITE	/DATA (&FC12 strobe)	2114 chip selects /S (pin 8) and clock of IC23
4Y pin 12	/WRITE	/WRSTB (bus write)	2114 write enables /W (pin 10)
2Y pin 7 = row clock	/GLR	/ROWREG (&FC11 strobe)	IC9 clock, IC23 clears, IC15b → IC7a clear, and IC22 pin 10
3Y pin 9	+5 V	row clock (2Y)	IC9 /LOAD

The RAM address (IA0–IA9, 1K × 4 bits in each of IC24 and IC25, which hold ID0–3 and ID4–7):

- **IA0–IA1: IC23**, two 74LS74 halves wired as toggle flip-flops (/Q to D). The first is clocked by /SEL, the second by the first one's /Q. Each RAM access, capture or CPU, steps the address on its trailing edge.
- **IA2–IA5: IC7a** (74LS393, pins 1–6), clocked by the falling edge of IA1. With IC23 this is a 6-bit byte counter, 0–63.
- **IA6–IA9: IC9** (74LS161, the slot number, 0–15). In capture mode its /LOAD input is held high, so each row clock (/GLR) counts it on to the next slot: one teletext line per 64-byte slot. In CPU mode a write to &FC11 pulls /LOAD low through the 157 and clocks it on the same pulse's rising edge, loading ID0–3. The 157's second gate delay on /LOAD makes it rise just after the clock, which is what makes this single-strobe load work.
- While the row clock is low it clears the byte counter: IC23 directly (pins 1, 13) and IC7a through IC15b (pin 3), an inverter because the 393's clear is active high.

The software relies on all this. Writing a slot number to &FC11 starts that slot at byte 0, and each &FC12 access moves to the next byte. During capture, broadcast lines 7–22 of each field fill slots 0–15 in order. The slot counter is stepped 17 times per window and simply wraps round; see the line-to-slot note in section 2.

4. Line synchronisation, RAM ownership and the IC17 flip-flop (centre / right)

Run-in detector. Each teletext line starts with a clock run-in of alternating bits (1010...) and then the framing code &27. IC5b (74LS74, pins 8–13) copies IC8's oldest bit (QH) on each clock, so IC15c (pins 9, 10 → 8) compares the two bits leaving the shift register: its output is 1 when they are equal. IC5a (pins 1–6) samples that, and its Q clears IC7b (74LS393 pins 8–13) whenever two equal bits pass. IC7b counts clock pulses through IC17a (pins 11–13, a NAND of CLK and /RUN IN DETECT). After a run of alternating bits, 4 or 8 depending on link LK2 (IC7b QC or QD), IC32 (pins 5 → 6) produces /RUN IN DETECT. That also stops IC7b's clock, so the indication holds. /GLR clears IC5a at each line.

Framing code detector. A combinational decoder watches the shift register's eight outputs (bus 0 = oldest bit ... 7 = newest):

GATE	INPUTS	TESTS
IC12a 74LS10 (3, 4, 5 → 6)	bits 0, 1, 2	all 1 (output low)
IC16a 74LS27 (13, 1, 2 → 12)	IC12a, bits 3, 4	bits 0–2 = 1 and bits 3, 4 = 0
IC13a 74LS08 (4, 5 → 6)	bit 5, IC5b /Q	bit 5 = 1, and the bit before the byte (the last run-in bit) was 0
IC13b 74LS08 (12, 13 → 11)	IC13a, /CLK	only during the second half of a bit, when the shift register is stable
IC28d 74LS32 (12, 13 → 11)	/RUN IN DETECT, stretched-GLR window (IC14 pin 6)	low only if the run-in has been seen and we are within about 15 µs of the line start

GATE	INPUTS	TESTS
IC16b 74LS27 (3, 4, 5 → 6)	bits 6, 7, IC28d	bits 6, 7 = 0 and the window/run-in condition
IC12c 74LS10 (9, 10, 11 → 8)	IC16a, IC16b, IC13b	low = "framing code here"

Bits 0–7 = 1,1,1,0,0,1,0,0 is &27, the teletext framing code (sent LSB first), and the check also requires the preceding run-in bit to be 0. The window comes from D1, R23 (150K) and C20 (100 pF): GLR (5–6 μ s into the line) charges C20 through the diode at once and R23 discharges it over roughly 15 μ s, so the window runs from about 5 to 21 μ s. The run-in starts at about 10.3 μ s and the framing code ends at about 13.8 μ s, comfortably inside it. IC14 (pins 5 → 6) squares this up, so a framing code is only accepted near the start of a line, not in the middle of picture or data.

The IC17 flip-flop. IC17b (pins 4–6) and IC17c (pins 8–10) are cross-coupled NANDs forming a set/reset latch. A low from IC12c on pin 10 sets it: pin 8, FCT, goes high and pin 6, /FCT, goes low. /GLR on pin 4 resets it at the start of every line. FCT therefore means "this line's framing code has been found", and it:

- releases IC10's clear, starting the byte counter
- enables /WRITE in IC11 and switches LOAD from shadowing the register to latching whole bytes (/FCT)

All four gates of IC17 (74LS00) are used. The fourth, pins 1–3, is the NAND in the address decoder.

Who owns the RAM: the CPU signal. IC12b (74LS10, pins 1, 2, 13 → 12) computes CPU = NAND(DEW, TTXEN, /INT). CPU is low (capture owns the RAM) only inside the data entry window, with capture enabled (control bit 2), and with no unserved interrupt. At all other times CPU is high:

- IC26's outputs are switched off (its /OE is CPU)
- IC22 hands the RAM controls to the bus strobes

So the computer can read the slots at any time outside the window. While an interrupt is pending, the capture logic is locked out, so unread lines are never overwritten. The lines it misses are flagged by DOR instead. The software still turns TTXEN off before reading, which also stops the data clock.

5. Status and interrupt

- **INT:** IC4a (74LS74, pins 1–6), D tied high, clocked by /DEW, so it sets at the *end* of each data window, once every teletext line of the field is in RAM. Q (pin 5) goes to status bit 7 and to IC18 (pins 1, 2 → 3, open collector), which pulls NIRQ low when control bit 3 is set. /Q (pin 6) is /INT, used in the CPU equation.
- **DOR:** IC4b (pins 8–13), D = INT, clocked by DEW rising: it sets at the start of the next window if the last interrupt is still pending. Status bit 6.
- **FSYN:** IC3a (pins 1–6), D tied high, clocked by the field sync through IC14 (pins 11 → 10 → 9 → 8). Status bit 4 means "a field sync has arrived since the last STATCLR". ATS 2.50 uses this to tell "No Signal" from a working adapter.
- The status buffer IC19 also carries DEW (bit 5) and the four option links LK4–LK7 (bits 3–0). Each link can be strapped to 0 V, +5 V, or control bit 4 or 5.
- Any access to &FC13 (Y3, /STATCLR) clears IC3a, IC4a and IC4b together.

6. Tuner, IF and tuning voltage

- The aerial feeds U341, a varicap UHF tuner. Its IF output goes through IC30 (SL1430 IF preamplifier) and IC31 (SW153A SAW filter, which sets the IF passband) into IC29 (TDA2541 IF amplifier and video demodulator).

- Composite video comes from TDA2541 pin 12 through L3, R29 (75R) and the ceramic trap ST1/L6. LK3 selects where the VIDEO net is taken from.
- The tuning voltage comes from +40 V through R24 (820R) and ZD1 (HZT33 / TAA550), a 33 V regulator. Each channel has a 100K preset (VR2–VR5) across that rail. The selected channel's transistor (Q3–Q6, driven through IC32 inverters from IC6b's decode of control bits 0–1) grounds the bottom of its preset.
- Diodes D2–D5 pass the lowest wiper voltage to the common tuning line, which R26 (220K) pulls up from 33 V. The unselected presets float at 33 V, so only the selected one sets the voltage. R25 (47K) feeds it on to the tuner.
- The TDA2541's AFC output (pin 5) is summed into the tuning voltage through R31, R33 and R36. Its AFC switch input (pin 6) comes from control bit 4, which the software clears while you tune by hand.

Supplies: +5 V for the logic, +12 V for the SAA5030 and IF stages, and +40 V for the tuning regulator. Decoupling is eight 47 nF capacitors plus C37/C38 (33 μ F).

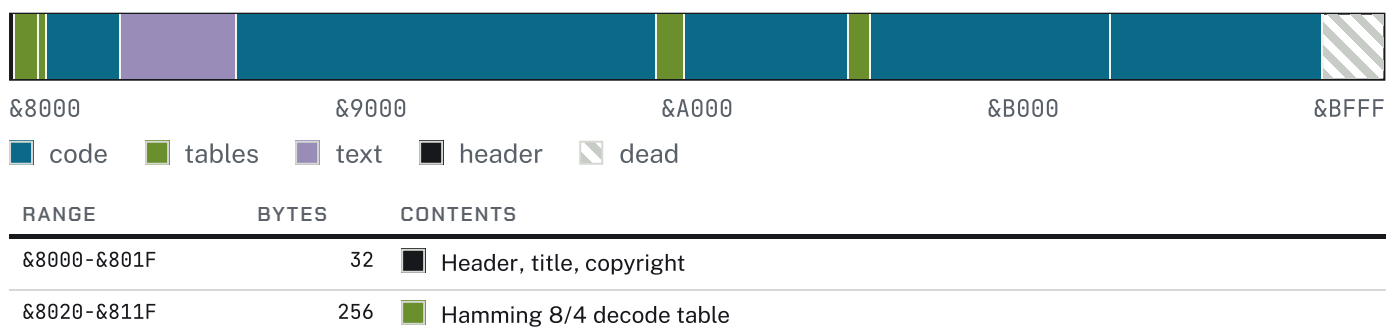
7. One field, start to finish

1. The VIP locks its 6 MHz oscillator to the broadcast through /PL, and FS keeps the SAA5020's line count in step with the field. The timing chain produces /GLR 5 μ s into every line, and DEW from about 8 μ s into line 6 to about 8 μ s into line 23.
2. During DEW (with capture on and no interrupt pending), each line from 7 to 23 begins with /GLR. That resets FCT, moves to the next RAM slot and zeroes the byte counter, so lines 7–22 land in slots 0–15.
3. If the line carries teletext, the run-in and framing code are recognised by about 14 μ s, inside the ~5–21 μ s window, and FCT sets. The framing code and the following 42 bytes are written to the slot, one byte every 8 bit-times.
4. DEW ends about 8 μ s into line 23: INT sets, and NIRQ goes low if enabled. The 6502 then has the rest of the field, about 19 ms, before the next window.
5. The 6502 turns capture off and reads the 16 slots through &FC11/&FC12. It zeroes byte 0 of each one it has used, writes &FC13 to clear INT, and turns capture back on before the next window.

Notes on the drawing

- IC8's outputs are labelled Q7 (pin 3) to Q0 (pin 13). On a 74LS164, pin 3 is QA, the newest bit, and pin 13 is QH, the oldest. The pin numbers and bus numbering are consistent with the software; only the Q labels are reversed.
- IC22's A/B input labels are swapped relative to the 74LS157 data sheet (pins 2, 5, 11, 14 are its A inputs, selected when pin 1 is low).
- IC33 is labelled 74LS14 in some places and 74LS04 in others; all its symbols are Schmitt triggers.
- IC17 (74LS00), IC13 (74LS08), IC12 (74LS10), IC16 (74LS27) and IC28/IC27 (74LS32) are each shared between unrelated parts of the circuit, so one package appears in several places on the sheet.

ROM map



RANGE	BYTES	CONTENTS
&8120-&816B	76	 OSWORD &7A dispatch table
&816C-&84D8	877	 Service entry, OSWORD glue, workspace stash
&84D9-&8A45	1,389	 *HELP text
&8A46-&9E14	5,071	 OSWORD handlers, IRQ capture engine, page store, *TUNE
&9E15-&9F4B	311	 FS vectors, * command table
&9F4C-&A6EE	1,955	 * commands, FSC, date/time, messages
&A6EF-&A7EB	253	 Telesoftware command table
&A7EC-&B32E	2,883	 Telesoftware decoder
&B32F-&BD2B	2,557	 Terminal mode (language)
&BD2C-&BFFF	724	 Unused (zeros)

Code regions are approximate at their edges. The tracer marks 11,963 bytes as reachable code.

Entry points & dispatch tables

Service calls (table at &88393)

CALL	HANDLER	PURPOSE
&00	&820B	none
&01	&8250	Absolute workspace: raises the top to &1A00 (claims &0E00-&19FF)
&02	&825C	Private workspace: claims 8 pages and initialises TFS
&03	&82B4	Auto-boot: "Acorn TFS", "no power", T+BREAK
&04	&81FC	Unrecognised * command
&05	&8D3F	Interrupt: the capture engine
&06	&820B	BRK (not handled)
&07	&820B	OSBYTE (not handled)
&08	&81B2	OSWORD &7A
&09	&8330	*HELP TELETEX / TELESOFT / OPT
&0A	&8336	Static workspace claimed by another ROM: stash state
&12	&8246	Select filing system (Y=6 selects TFS)

Filing system vectors (&89E15)

VECTOR	HANDLER
FILEV	&819D
ARGSV	&9E32
BGETV	&83A9
BPUTV	&81E6
GBPBV	&9D1B
FINDV	&AC84
FSCV	&A1AF

* commands (&89EE9)

COMMAND	HANDLER	NOTE
*E	&9B38	
*DISPLAY	&A15C	
*TRANSFER	&A14E	
*DATE	&A4D2	
*TIME	&A34B	
*BBC	&A055	*BBC1, *BBC2 (channel 1/2)
*ITV	&A04D	*ITV1, *ITV2 (channel 3/4)
*CH	&A055	Also used by *CH1-*CH4
*PAGE	&A06C	
*HON	&9ECC	Undocumented. Headers on.
*HOFF	&9ED2	Undocumented. Headers off.
*TELETEX	&9D1E	

COMMAND	HANDLER	NOTE
*TELES0FT	&9D34	

OSWORD &7A sub-commands (table at &8120)

The user guide documents 12 of the 38 codes. Rows with a cyan edge are undocumented; their names are inferred from the code.

CODE	NAME	HANDLER	NOTES
&80	RELOAD PAGE	&8BF7	Documented. In this version &80, &86 and &88 share one routine: follow link 0 and wait.
&81	read position: start := mark	&8A46	Undocumented. Used by the Telesoftware decoder.
&82	read position: rewind to start	&8A55	Undocumented.
&83	read position: limit := mark	&8A64	Undocumented. BGET #15 stops at the limit.
&84	read position: mark := current	&8A73	Undocumented.
&85	FETCH PAGE	&9800	Documented. Seven page digits, &FF or "#" = wild.
&86	CHAINED PAGE	&8BF7	Documented. Same routine as &80.
&87	reserved	&A036	Raises error &CF "Data not available".
&88	HEAD OF CHAIN	&8BF7	Documented. Same routine as &80.
&89	TEST ARRIVED	&8B75	Documented. Locks the page on success.
&8A	UNLOCK PAGE	&8BC8	Documented.
&8B	SELECT CHANNEL	&8A91	Documented. Writes bits 0-1 of &FC10 and abandons all captures.
&8C	read page flags	&8184	Undocumented. b7 check word present, b6 checksum OK.
&8D	READ TSDP	&9297	Documented. The 20-character status message.
&8E	HAMMING DECODE	&9286	Documented. Uses the table at &8020.
&8F	TRANSFER	&8AF0	Documented. 32-bit address, Tube aware; handles double height.
&90	reserved	&A036	Raises error &CF "Data not available".
&91	read previous language and FS	&8AD7	Undocumented. Used by terminal mode for f9.
&92	HEADERS	&8A82	Documented. Rolling-header destination (0 = off).
&93	WHAT CHANNEL	&8ACB	Documented. Reads bits 0-1 of the &FC10 shadow.
&94	set up request slot	&95D6	Undocumented. Gives slot n a buffer and clears it.
&95	wait for page	&8C05	Undocumented. Waits, applying the *OPT2 checksum rule.
&96	select link	&9405	Undocumented. Link 0-23 from row 27; result 0-3.
&97	unused	&81D7	"Bad OSWORD".
&98	unused	&81D7	"Bad OSWORD".
&99	unused	&81D7	"Bad OSWORD".
&9A	unlock and invalidate	&8BBA	Undocumented. Used by the decoder after each page.
&9B	allocate page buffers	&9357	Undocumented. &602-byte buffers from PAGE up to &7100.
&9C	keep page	&92B4	Undocumented. f3; result 1 when every buffer is kept.

CODE	NAME	HANDLER	NOTES
&9D	release page	&92E7	Undocumented. SHIFT f3; result 1 if it was not kept.
&9E	swap current slot into slot 0	&9305	Undocumented. No effect here: the current slot is always 0.
&9F	show slot 0 again	&9331	Undocumented. Flags the page as arrived again.
&A0	request initial page	&9693	Undocumented. Page named in the 8/30 packet, else x00.
&A1	deactivate slots 1-24	&960A	Undocumented.
&A2	slot 0 back to private buffer	&9626	Undocumented. Used when terminal mode exits.
&A3	read page number	&816C	Undocumented. Returns seven ASCII digits.
&A4	suspend all requests	&9671	Undocumented. Used around f5 * commands.
&A5	resume all requests	&9682	Undocumented.

Telesoftware commands (&A6EF)

Grey rows share the generic handler &AA0D.

NAME	FLAGS	HANDLER
<CEB>	&11	&AA0D
<CET>	&90	&AA0D
<CSB>	&10	&AA0D
<CE0>	&12	&AA0D
<CE1>	&12	&AA0D
<CE2>	&12	&AA0D
<CE3>	&12	&AA0D
<CE4>	&12	&AA0D
<CE5>	&12	&AA0D
<DAC>	&92	&AFB7
<DC0>	&92	&B0AA
<DDT>	&92	&AFB7
<DEB>	&91	&B26A
<DEC>	&92	&B1EC
<DEF>	&92	&AFA3
<DES>	&92	&AFD7
<DET>	&92	&AFAD
<DIG>	&92	&AFB7
<DIR>	&92	&B1AF
<DLA>	&92	&AEED
<DLC>	&92	&B1DB
<DLR>	&92	&AF20
<DLS>	&92	&B034
<DNC>	&92	&AFB7
<DND>	&92	&AFB2
<DSA>	&90	&AA0D

Terminal-mode keys (&B871)

Function keys are set to &90+ (OSBYTE 225) and SHIFT function keys to &9A+ (OSBYTE 226). Cursor keys return codes (OSBYTE 4,1).

CODE	KEY	HANDLER
&87	COPY	&B8AD
&88	Cursor left	&B8C7
&89	Cursor right	&B8EA
&8A	Cursor down	&B90E
&8B	Cursor up	&B91C
&90	f0 select page	&B9D9
&91	f1 last explicit page	&B9EB
&92	f2 reveal	&BB78
&93	f3 keep page	&B97D
&94	f4 channel	&B9FA
&95	f5 * command	&BB22
&96	f6 load page	&BA57
&97	f7 save page	&BA6C
&98	f8 hold	&BB7F
&99	f9 exit to language	&BBA0
&9B	SHIFT f1 index page	&BB51
&9C	SHIFT f2 conceal	&BB99
&9D	SHIFT f3 release	&B946
&9E	SHIFT f4 tune	&BB57
&A3	SHIFT f9 exit to Telesoft	&BBB7

NAME	FLAGS	HANDLER
<DSB>	&90	&AE2F
<DSL>	&92	&AA0D
<DSN>	&90	&AA0D
<DST>	&92	&AA0D
<DTL>	&92	&B10C
<DXA>	&92	&AF0E
<DXR>	&92	&AF44
<ESC>	&00	none
<EMP>	&90	&AA0D
<TXT>	&90	&AD99
<TX8>	&90	&AA0D
<UER>	&1F	none
<ULB>	&12	&B1B4
<URB>	&12	&B1BD
<VDX>	&90	&AA0D
<TX0>	&90	&AD9C

Inline-data calls

JSR &A2AF prints the zero-terminated string after it. **JSR &A198** copies the error number and message that follow to &0100 and executes BRK there. **JSR &B73A** and **&B737** print to row 24 until an &EA byte, which then runs as a NOP.

Workspace & page requests

- Service 1 raises the absolute workspace top to &1A00, so TFS uses &0E00-&19FF. This is skipped when the adapter is off, which is why "no power" frees the memory.
- &0E00-&0E14: interrupt scratch (decoded header, current row and magazine). &0E15-&0E1C: the request slot being captured in each magazine (&FF = none).
- &0E5C: the 8/30 broadcast service data packet (initial page, time offset, date, UTC, status). &0E8E: &FC10 shadow. &0E99/&0E9A: language and filing system to return to.
- &0F9A-&1999: the Telesoftware decoder: six 256-byte name tables, a frame stack, its state, and a heap and output queue that grow towards each other.
- When another ROM claims static workspace (service &0A), **&8346** writes 1 KB of state into the **adapter's own RAM** and 2 KB to private workspace. ***TELESOFT** restores both.

Request slots (&0EAO, 25 × 10 bytes)

Slot 0 is the page being shown or fetched. The other 24 hold kept pages (f3) and pages prefetched through the links in a page's row 27, so a linked page appears at once when selected. FETCH PAGE swaps a slot that already holds the wanted page into slot 0.

BYTE	CONTENTS	BIT 6 OF THIS BYTE MEANS
+0, +1	Page buffer address (2 header bytes + 48-byte row records)	n/a
+2	Magazine digit	LOCKED (being read)

BYTE	CONTENTS	BIT 6 OF THIS BYTE MEANS
+3	Page tens	ARRIVED (new copy)
+4	Page units	CAPTURING
+5	Sub-code thousands	VALID (complete copy in buffer)
+6	Sub-code hundreds	CHECK WORD present
+7	Sub-code tens	CHECKSUM OK
+8	Sub-code units	KEPT
+9	State: &40 = active	n/a

Digit bytes hold the value in bits 0–3, and bit 7 marks a wild digit.

Other findings

- **Selective re-capture.** A page already held in full is only captured again if the broadcaster sets the "update" control bit (C8) or the stored copy failed its checksum.
- **Shared OSWORD code.** RELOAD (&80), CHAINED PAGE (&86) and HEAD OF CHAIN (&88) all run the same routine: follow link 0 of the current page and wait for it.
- **The "current slot" is always 0.** OSWORD calls act on &0E9B's slot, but nothing ever sets it to anything else.
- **<ULB> and <URB> are swapped** relative to the user guide: <TEXT> makes lone &FE subtract &58 and lone &E0 add it.
- **Undocumented commands.** *H0N sends rolling headers to &7C08 (the top row of the MODE 7 screen); *H0FF turns them off.
- ***DATE** counts forward from 1 January 1982, so it can't show earlier dates.
- **Apparent bugs (not tested):**
 - The workspace stash covers &0E8E–&1A8D, so reselecting TFS writes back 142 bytes above &1A00, which is normally the start of user memory (&9D34).
 - The <DEB> check value is compared with a counter that is never updated, so any non-zero value forces a page reload (&B26A).
 - A branch-order slip makes most of the decoder-information OSWORD codes below &80 unreachable (&AD5A).
- **Hamming table.** Values &0x are clean nibbles, &4x single-bit corrected, &FF uncorrectable. Both corrected and rejected bytes count towards the tuning-bar error figure.

ATS 2.50: what changed from TFS 1.03

ATS250.ROM is the BBC's Advanced Teletext System, version 2.50, "(C)BBC 1986". It drives the same Acorn adapter through the same four registers, but it is a complete rewrite: no routine is shared with TFS 1.03. It needs MOS 1.20 or later and also runs on the B+ and Master 128. Sources: the code itself and the *Advanced Teletext System User Guide* (BBC, 1986).

AREA	TFS 1.03 [ACORN, 1982]	ATS 2.50 [BBC, 1986]
What the ROM is	A filing system (Telesoft, number 6) plus a language (terminal mode)	A service ROM with OS commands and OSWORD calls, plus a terminal. Not a filing system. The terminal runs as a language without a Tube, or as an OS command with one (service &FF hides the language flag).
Memory	Static workspace &0E00–&19FF (shared with other filing systems, stashed in the adapter RAM) plus 8 private pages	One private page (layout), in hidden RAM on a Master. If the adapter is off or absent at BREAK, nothing is claimed at all.

AREA	TFS 1.03 (ACORN, 1982)	ATS 2.50 (BBC, 1986)
Page capture	25 request slots; every magazine captured in parallel; rows stored by row number in 48-byte records	A "page grabber" with 20 records (page + buffer address); one page captured at a time; rows appended as 42-byte records (magazine, row, 40 bytes); rows per page settable (24 or 30) (interrupt)
Programming interface	OSWORD &7A codes &80–&A5 (12 documented)	OSWORD &7A codes 0–27, 26 documented: grabber records, flags, headers, subtitles, adaptor status, time, date, CRC, download
Data format	*TRANSFER gives 960 bytes of MODE 7 screen codes	*TRANSFER gives 24 rows and *DATA all rows in the 42-byte format; *TELESOFT switches to TFS's 40-byte screen format for old programs
Telesoftware	Read through the filing system (*EXEC, *LOAD, *RUN, BGET), decoded as it is read	A download terminal (f5): reads the broadcast catalogue, lists the BBC files, shows a block map, and saves each block to disc or network as it arrives (tape: all in memory, then saved). Also OSWORD 24 (downloader).
Terminal keys	f0 page, f1 last page, f2 reveal, f3 keep, f4 channel, f5 *, f6/f7 load/save, f8 hold, f9 exit; links by number	Digits enter a page at once; f0 index and f1–f4 coloured link keys (Fastext style); SHIFT f1–f4 channels; f6 reveal, f7 hold, f8 sub-page; COPY keep, TAB previous page, RETURN status, * OS command
Checksum shown as	"Bad data" / "No check" on the bottom row	The page number turns white (CRC good), red (failed) or yellow (no check word)
Adaptor detection	Option links read as zero: "no power"	Also watches field-sync pulses: "No Power", "No Signal" or "No TTX Unit" (adaptor_status)
* commands	TELETEXT TELESOFT PAGE TRANSFER DISPLAY TIME DATE TUNE BBC ITV CH HON HOFF, plus filing system *OPT, *EXEC, *RUN	Adds TTXON / TTXOFF, MAGAZINE, S4C, NOTTX, DATA; HON takes a clock option; any command can take extra leading T's (*TTIME)

ATS 2.50: new features and notable details

- **Live subtitles.** OSWORD 18 names a subtitle page (normally 888). Its rows are written straight onto the MODE 7 screen as they arrive, and the screen is cleared on "erase page" ([&8ED4](#)).
- **Fastext-style links.** When a page arrives, the six links in its row 27 are loaded into grabber records 1–6, each with its own buffer, so linked pages are captured in the background. f1–f4 and f0 select them, and row 24 shows coloured page numbers or keywords ([fetch_links](#)).
- **Kept pages** fill 5-page buffers from OSHWM up to &4A00 (7 to 11 pages). **TAB** swaps with the previous screen, kept at &7200.
- **Master 128 support:** workspace in hidden RAM via services &22/&24, reset service &27.
- ***NOTTX** switches the ROM off until power-off: "Press CTRL-BREAK", then it ignores every service call.
- **The interrupt handler saves and restores zero page &70–&83**, so the grabber can use fast zero-page code without disturbing the foreground program.
- **Telesoftware catalogue:** the catalogue page is found through link 5 of the initial (index) page. Entries are filtered by computer type, recognised by the *sum* of its three letters: &C7 "BBC", &E8 "ANY", &7E "****" (end) ([&BEC2](#)).
- "Finding catalogue..." becomes "Loading catalogue..." by overprinting just the first three letters.
- **Quirks:**
 - Reveal (f6) swaps the conceal code &98 with &80 on the screen, so any &80 already on the page turns into a conceal code when concealing again.
 - The CRC check (OSWORD 23) covers rows 1–24, where TFS 1.03 also included row 25.
 - Unused function keys issue OSWORD &7A with code &A0+, which ATS itself ignores.

ATS 2.50: tables & ROM map



RANGE	BYTES	CONTENTS
&8000-&8023	36	 Header, title, copyright
&8024-&82E9	710	 Service entry, *HELP, OSWORD dispatch
&82EA-&8321	56	 OSWORD &7A table (codes 0-27)
&8322-&84BC	411	 OSWORD 0-19, 26, 27
&84BD-&8541	133	 * command table
&8542-&8B8E	1,613	 * commands, Tube transfers
&8B8F-&91AE	1,568	 Page grabber (interrupt), header decoding, adaptor status
&91AF-&92AE	256	 Hamming 8/4 decode table
&92AF-&980F	1,377	 OSWORD 20-25: tune, time, date, CRC, download, Hamming
&9810-&ACA2	5,267	 Terminal: page entry, links, keep, hold, reveal, status
&ACA3-&B14F	1,197	 Telesoftware terminal and RTF decoder
&B150-&B1F2	163	 RTF command table
&B1F3-&BB5F	2,413	 RTF commands, block display, page reading
&BB60-&BFFF	1,184	 Catalogue browser

The tracer marks 14,707 bytes as reachable code.

OSWORD 87A codes (table at 882EA)

Cyan edge: undocumented.

CODE	FUNCTION	HANDLER
0	Read channel	⌘8338
1	Set channel	⌘8322
2	Read record flag	⌘8345
3	Set record flag	⌘8355
4	Read grabber record	⌘8368
5	Set grabber record	⌘8379
6	Read rows per page	⌘83B1
7	Set rows per page	⌘83B6
8	Read header status	⌘83C0
9	Set header status	⌘83C5
10	Read header magazine	⌘83DA
11	Set header magazine	⌘83E5
12	Interrupts off (*TTXOFF)	⌘83EF

* commands [684BD]

Stored without the leading T, which the matcher skips.

COMMAND	STORED AS	HANDLER
*TELETEXT	ELETEXT	&8542
*HON	HON	&8592
*HOFF	HOFF	&85F4
*BBC	BBC	&8645
*ITV	ITV	&8673
*CH	CH	&8645
*S4C	S\x14C	&868B
*NOTTX	NOTTX	&8568
*TTX (ON / OFF)	X	&8698
*TUNE	UNE	&86F4
*TIME	IME	&8714
*DATE	DATE	&875B
*MAGAZINE	MAGAZINE	&860B

CODE	FUNCTION	HANDLER
13	Interrupts on (*TTXON)	&8403
14	Clear all grabber records	&8419
15	Read TSDP	&844E
16	Stop the grabber	&846B
17	Read adaptor status	&8479
18	Subtitles on/off	&848B
19	Interrupt handling flag	&84A2
20	Tuning screen	&92AF
21	Time from TSDP	&944D
22	Date from TSDP	&9546
23	Check CRC	&96DF
24	Download a Telesoftware file	&97B6
25	Hamming decode	&97F7
26	Set row length (42/40)	&84AC
27	Read row length	&84B3

COMMAND	STORED AS	HANDLER
*PAGE	PAGE	&8795
*TRANSFER	RANSFER	&883E
*DATA	DATA	&8835
*DISPLAY	DISPLAY	&8996
*TELESOFT	ELESOFT	&86F0

Terminal keys ([&9D08](#) and others)

KEY	ACTION	HANDLER
&A0	f0 index page	&A527
&A1	f1 red link	&A4D0
&A2	f2 green link	&A4D9
&A3	f3 yellow link	&A4F3
&A4	f4 cyan link	&A50D
&A5	f5 Telesoftware	&ACA3
&A6	f6 reveal / conceal	&9F2F
&A7	f7 hold / release	&9E07
&A8	f8 sub-page entry	&9E70
&A9	f9 exit	&A2B6
&B0	SHIFT f0 tune	&AA8E
&B1	SHIFT f1 channel 1	&9D89
&B2	SHIFT f2 channel 2	&9D8D
&B3	SHIFT f3 channel 3	&9D91
&B4	SHIFT f4 channel 4	&9D95
&B6	SHIFT f6 load page	&9F96
&B7	SHIFT f7 save page	&9F6B
0-9	page / sub-page digits	&9A27
TAB	previous page	&A738
COPY	keep / release page	&A8EC
SHIFT COPY	release all kept pages	&9BF1
RETURN	status page	&AB28
*	OS command	&A776

Full listings

The commented disassemblies of TELEROMU.ROM and ATS250.ROM follow as Appendix A and Appendix B. Each has an overview and symbol table at the top, a header for every routine, and notes on the important lines. Labels in the listings and in the tables above are links. Names and comments are the analyst's reconstruction, not Acorn's source.

Appendix A: TELEROMU.ROM (TFS 1.03) listing

```
; =====
; TFS 1.03 - Acorn Teletext System ROM ("TELEROM"), for the BBC Microcomputer
; Commented disassembly of TELEROMU.ROM (16K, &8000-&BFFF)
; =====
;
; Generated by dis.py from teleromu.json + teleromu.ann. Labels and comments
; are an analyst's reconstruction, not Acorn's original source.
;
; WHAT THE ROM DOES
; The ROM drives the Acorn Teletext Adapter, a box on the 1MHz bus with its
; own tuner, an SAA5030 video input processor and SAA5020 timing chain, and
; 1K of RAM. The adapter has no acquisition chip (SAA5040 TAC): it just
; stores every teletext line it finds in the field blanking interval. This
; ROM does the rest in software, in the interrupt handler, and provides:
; - OSWORD &7A: a teletext system call (fetch page, test arrived, transfer,
;   channel select, links, headers, ...)
; - the Telesoft filing system (filing system 6): read-only, with
;   *PAGE *TRANSFER *DISPLAY *TIME *DATE *TUNE *CH1-4 *BBC1/2 *ITV1/2,
;   and Telesoftware download (*EXEC, *LOAD, *RUN) through a decoder for the
;   Telesoftware name/command format
; - terminal mode (*TELETEXT): a language ROM that works as a teletext set
;   with function-key commands, kept pages, links and rolling headers
;
; ADAPTER REGISTERS (page &FC)
; &FC10 read  status: b7 INT, b6 DOR (overrun), b5 DEW, b4 FSYN,
;             b3-b0 option links (all 0 when the adapter has no power)
; &FC10 write control latch: b0-b1 channel, b2 TTXEN (capture enable),
;             b3 IRQ enable, b4 AFC on, b5 spare (shadow copy in &0E8E)
; &FC11 write select one of the 16 64-byte line slots in the adapter RAM
; &FC12 r/w   read/write the selected slot; the address steps on each access
; &FC13 write STATCLR: cclear the INT and DOR flip-flops
;
; MEMORY USE
; &0E00-&19FF static workspace (shared by filing systems; stashed into the
;             adapter RAM and private workspace when another FS takes over)
; &0E00-&0E14 interrupt scratch: decoded header, current row/magazine
; &0E15-&0E1C page being captured in each magazine (&FF = none)
; &0E5C-&0E83 broadcast service data packet (magazine 8 row 30)
; &0E8E      control latch shadow
; &0EA0-&0F99 25 page request slots of 10 bytes
; &0F9A-&1599 Telesoftware name tables (3 lone + 3 escaped, 256 bytes each)
; &159A-&1619 Telesoftware decoder frame stack
; &161A-&1699 Telesoftware decoder state
; &169A-&1999 Telesoftware heap (grows up) and output queue (grows down)
; private workspace: 8 pages; slot 0's page buffer while TFS is selected,
;                   the stashed workspace while it isn't
; terminal mode: page buffers from PAGE upwards (&602 bytes each),
;               page copy at &7800, MODE 7 screen at &7C00
;
; DIFFERENCES FROM TFS103.ROM (a corrupt dump of the same ROM)
; &8000 &4C here (JMP terminal_entry); &00 in TFS103.ROM, which made
;       *TELETEXT crash into BRK
; &A000 &91 here (STA (&B4),Y); &00 in TFS103.ROM, which broke the end
;       of OSFILE load
; &BD2E- zeros here; left-over code from another build in TFS103.ROM
;
; CONVENTIONS
; &xx      hexadecimal, as in BBC BASIC
; Lxxxx    an address with no better name
; "slot"   either a page request slot (&0EA0) or an adapter RAM line
;          slot (&FC11); the comments say which
;
; =====
; SYMBOLS (RAM, zero page and I/O addresses named in this listing)
; =====
; &0000 t_ptr
; &0001 t_ptr+1
; &0009 t_reveal
; &000A t_link
; &000B t_hold
; &000C t_osw
; &000D t_osw+1
; &000E t_osw+2
; &000F t_osw+3
; &0010 t_osw+4
; &0020 t_last_page
; &0029 t_shown_page
; &0031 t_prev_lang
; &0032 t_fs
; &0034 t_flags
; &0035 t_delay
; &0037 t_line
; &00B0 zp_b0
; &00B1 zp_b1
; &00B4 fblk
; &00B5 fblk+1
```



```
; &00B6 count
; &00B7 count+1
; &00B8 xfer_blk
; &00B9 xfer_blk+1
; &00BA xfer_byte
; &00BC ptr
; &00BD ptr+1
; &00C0 ptr2
; &00C1 ptr2+1
; &00C4 irq_ptr
; &00C5 irq_ptr+1
; &00C6 nm_kind
; &00C7 nm_kind+1
; &00C8 nm_lo
; &00C9 nm_lo+1
; &00CA nm_hi
; &00CB nm_hi+1
; &00CC ts_ptr
; &00CD ts_ptr+1
; &00CE ts_tmp
; &00EF osw_a
; &00F0 osw_x
; &00F1 osw_y
; &00F2 text_ptr
; &00F3 text_ptr+1
; &00F4 romsel_copy
; &00FD brk_ptr
; &00FE brk_ptr+1
; &00FF escape_flag
; &0100 stack_page
; &0200 USERV
; &0202 BRKV
; &0203 BRKV+1
; &0212 FILEV
; &0213 FILEV+1
; &021E FSCV
; &0406 TUBE_CODE
; &0DF0 rom_wksp_table
; &0E00 hdr_bytes
; &0E08 row_tmp
; &0E09 row_tmp_hi
; &0E0A slot_scan
; &0E0B cur_row
; &0E0C cur_mag
; &0E0D hdr_tens
; &0E0E hdr_units
; &0E0F hdr_s1
; &0E10 hdr_s2
; &0E11 hdr_s3
; &0E12 hdr_s4
; &0E13 hdr_ctrl
; &0E14 cur_slot_no
; &0E15 mag_capture
; &0E1E req_scan
; &0E1F tsdp_timeout
; &0E21 time_buf
; &0E3A osw_blk
; &0E3B osw_blk+1
; &0E3C osw_blk+2
; &0E3D osw_blk+3
; &0E4A want_mag
; &0E4B want_tens
; &0E4C want_units
; &0E4D want_s1
; &0E4E want_s2
; &0E4F want_s3
; &0E50 want_s4
; &0E52 ham_count
; &0E53 ham_errors
; &0E54 tune_err_sum
; &0E55 tune_err_sum+1
; &0E56 tune_byte_sum
; &0E57 tune_byte_sum+1
; &0E5A tune_frames
; &0E5C tsdp
; &0E5D tsdp+1
; &0E5E tsdp+2
; &0E5F tsdp+3
; &0E60 tsdp+4
; &0E61 tsdp+5
; &0E62 tsdp+6
; &0E70 tsdp_status
; &0E84 crc_col
; &0E85 crc_row
; &0E86 crc_tmp0
; &0E87 crc_tmp1
; &0E88 crc_tmp2
; &0E89 crc_tmp3
; &0E8A crc_lo
; &0E8B crc_hi
```

```
; &0E8C crc_rx_hi
; &0E8D crc_rx_lo
; &0E8E ctrl_shadow
; &0E8F rd_row
; &0E90 rd_col
; &0E91 rd_mark_row
; &0E92 rd_mark_col
; &0E93 rd_start_row
; &0E94 rd_start_col
; &0E95 rd_limit_row
; &0E96 rd_limit_col
; &0E98 opt2_errors
; &0E99 prev_language
; &0E9A prev_fs
; &0E9B cur_req
; &0E9C hdr_dest
; &0E9D hdr_dest+1
; &0EA0 req_buf
; &0EA1 req_buf+1
; &0EA2 req_mag
; &0EA3 req_tens
; &0EA4 req_units
; &0EA5 req_s1
; &0EA6 req_s2
; &0EA7 req_s3
; &0EA8 req_s4
; &0EA9 req_state
; &0F9A lone_kind
; &109A lone_lo
; &119A lone_hi
; &129A esc_kind
; &139A esc_lo
; &13B5 esc_kind+&1B
; &1416 esc_kind+&7C
; &1435 esc_kind+&9B
; &1496 esc_kind+&FC
; &149A esc_hi
; &161A frame_sp
; &161B frame_sp+1
; &161C outq_wr
; &161D outq_wr+1
; &161E outq_rd
; &161F outq_rd+1
; &1620 ts_state
; &1621 ts_hex
; &1622 ts_blocks_done
; &1623 ts_blocks_total
; &1624 ts_eof
; &1625 ts_block_map
; &1649 ts_comment
; &164A ts_comment+1
; &164B ts_operand
; &164C ts_load_base
; &1650 ts_offset
; &1651 ts_offset+1
; &1652 ts_offset+2
; &1653 ts_offset+3
; &1654 ts_user_load
; &1655 ts_name
; &1656 ts_name+1
; &1657 ts_load
; &165B ts_exec
; &165C ts_exec+1
; &165D ts_exec+2
; &165E ts_exec+3
; &165F ts_attr
; &1660 opt3_keep
; &1661 ts_newaddr
; &1662 ts_pending
; &1663 ts_to_memory
; &1664 ts_title
; &1665 ts_title+1
; &1666 ts_savey
; &1667 ts_block_no
; &1668 ts_random
; &1669 ts_blk_check
; &166B ts_deb_check
; &166D ts_no_run
; &166E ts_total_save
; &166F ts_osw
; &167E ts_hexmode
; &1683 ts_cmdtab2
; &1692 heap
; &1693 heap+1
; &1694 opt1_msgs
; &199A run_flag
; &FC10 TT_STATUS_CTRL
; &FC11 TT_SLOT
; &FC12 TT_DATA
; &FC13 TT_STATCLR
```

```

; &FEE4 TUBE_R3_STATUS
; &FEE5 TUBE_R3_DATA
; &FFB9 OSRDRM
; &FFCE OSFIND
; &FFD1 OSGBPB
; &FFD4 OSBPUT
; &FFD7 OSBGET
; &FFDA OSARGS
; &FFDD OSFILE
; &FFE0 OSRDCH
; &FFE3 OSASCI
; &FFE7 OSNEWL
; &FFEE OSWRCH
; &FFF1 OSWORD
; &FFF4 OSBYTE
; &FFF7 OSCLI
;
;-----
; SIDEWAYS ROM HEADER
;
; &8000 JMP terminal_entry    language entry. (TFS103.ROM has &00 here: BRK, which hangs *TELETEXT.)
; &8003 JMP service_entry    service entry
; &8006 &C2                ROM type: service + language, 6502 code
; &8007 &11                offset of the copyright string
; &8008 &01                binary version number
; &8009 "TFS",0,"1.03",0    title (printed by the MOS when the language starts) and version
; &8011 0,"(C)1982 Acorn",0 copyright string the MOS checks for
;-----
.L8000
8000 4C 2F B3 JMP terminal_entry    ; language entry: start terminal mode
;-----
; Service entry vector, ROM type, copyright offset, version, title and copyright strings.
;-----
8003 .db 4C 19 82 C2 11 01 54 46 53 00 31 2E 30 33 00 28 ; L....TFS.1.03.(
8013 .db 43 29 31 39 38 32 20 41 63 6F 72 6E 00          ; C)1982 Acorn.
;-----
; HAMMING 8/4 DECODE TABLE (256 bytes), indexed by the received byte.
;
; Teletext protects addresses, page numbers and control bits with Hamming 8/4 coding:
; each byte carries 4 data bits plus 4 protection bits.
; &0x the byte was received cleanly; x is the decoded nibble
; &4x one bit was wrong and has been corrected; x is the nibble
; &FF two or more bits wrong: cannot be decoded
; hamming_decode_count (&90C1) also counts decodes and errors for the *TUNE meter.
;-----
.hamming_table
8020 .db 41 FF 01 41 FF 40 41 FF FF 42 41 FF 4A FF FF 47 ; A..A.@A..BA.J..G
8030 .db FF 40 41 FF 40 00 FF 40 46 FF FF 4B FF 40 43 FF ; .@A.@..@F..K.@C.
8040 .db FF 4C 41 FF 44 FF FF 47 46 FF FF 47 FF 47 47 07 ; .LA.D..GF..G.GG.
8050 .db 46 FF FF 45 FF 40 4D FF 06 46 46 FF 46 FF FF 47 ; F..E.@M..FF.F..G
8060 .db FF 42 41 FF 44 FF FF 49 42 02 FF 42 FF 42 43 FF ; .BA.D..IB..B.BC.
8070 .db 48 FF FF 45 FF 40 43 FF FF 42 43 FF 43 FF 03 43 ; H..E.@C..BC.C..C
8080 .db 44 FF FF 45 04 44 44 FF FF 42 4F FF 44 FF FF 47 ; D..E.DD..BO.D..G
8090 .db FF 45 45 05 44 FF FF 45 46 FF FF 45 FF 4E 43 FF ; .EE.D..EF..E.NC.
80A0 .db FF 4C 41 FF 4A FF FF 49 4A FF FF 4B 0A 4A 4A FF ; .LA.J..IJ..K.JJ.
80B0 .db 48 FF FF 4B FF 40 4D FF FF 4B 4B 0B 4A FF FF 4B ; H..K.@M..KK.J..K
80C0 .db 4C 0C FF 4C FF 4C 4D FF FF 4C 4F FF 4A FF FF 47 ; L..L.LM..LO.J..G
80D0 .db FF 4C 4D FF 4D FF 0D 4D 46 FF FF 4B FF 4E 4D FF ; .LM.M..MF..K.NM.
80E0 .db 48 FF FF 49 FF 49 49 09 FF 42 4F FF 4A FF FF 49 ; H..I.II..BO.J..I
80F0 .db 08 48 48 FF 48 FF FF 49 48 FF FF 4B FF 4E 43 FF ; .HH.H..IH..K.NC.
8100 .db FF 4C 4F FF 44 FF FF 49 4F FF 0F 4F FF 4E 4F FF ; .LO.D..IO..O.NO.
8110 .db 48 FF FF 45 FF 4E 4D FF FF 4E 4F FF 4E 0E FF 4E ; H..E.NM..NO.N..N
;-----
; OSWORD &7A DISPATCH TABLE: 38 handler addresses for sub-commands &80-&A5.
; The first byte of the OSWORD parameter block selects the sub-command; see svc08_osword.
;-----
.osword_table
8120 .db F7 8B 46 8A 55 8A 64 8A 73 8A 00 98 F7 8B 36 A0 ; ..F.U.d.s....6.
8130 .db F7 8B 75 8B C8 8B 91 8A 84 81 97 92 86 92 F0 8A ; ..U.....
8140 .db 36 A0 D7 8A 82 8A CB 8A D6 95 05 8C 05 94 D7 81 ; 6.....
8150 .db D7 81 D7 81 BA 8B 57 93 B4 92 E7 92 05 93 31 93 ; .....W.....1.
8160 .db 93 96 0A 96 26 96 6C 81 71 96 82 96          ; ....&.l.q...
;-----
; OSWORD &7A &A3: READ PAGE NUMBER
; Returns the seven digits of the current request slot (mag, tens, units, sub-code x4)
; as ASCII characters in bytes 1-7 of the parameter block.
;-----
.osw_A3_read_page_number
816C A0 01 LDY #&01
816E AE 9B 0E LDX cur_req          ; X = offset of the current request slot
.L8171
8171 BD A2 0E LDA req_mag,X
8174 29 0F AND #&0F                ; digit value is in the low nibble
8176 18 CLC
8177 69 30 ADC #&30                ; convert to ASCII
8179 91 F0 STA (osw_x),Y
817B E8 INX
817C C8 INY
817D C0 08 CPY #&08
817F D0 F0 BNE L8171

```

```

8181 4C 11 82 JMP svc_exit_claimed
;-----
; OSWORD &7A &8C: READ PAGE FLAGS
; Returns in byte 1: bit 7 = page carried a row 27 check word (req_s2 bit 6)
;                   bit 6 = computed checksum matched it (req_s3 bit 6)
; Terminal mode uses this to show "Bad data" / "No check" on the status line.
;-----
.osw_8C_read_page_flags
8184 AE 9B 0E LDX cur_req
8187 BD A6 0E LDA req_s2,X
818A 29 40 AND #&40
818C 0A ASL A
818D 85 BC STA ptr
818F BD A7 0E LDA req_s3,X
8192 29 40 AND #&40
8194 05 BC ORA ptr
8196 A0 01 LDY #&01
8198 91 F0 STA (osw_x),Y
819A 4C 11 82 JMP svc_exit_claimed
;-----
; FILEV (OSFILE) for the Telesoft filing system.
; A=5 read catalogue information → LAD1A
; A=7 (create) → LAC94
; A=&FF load a file → L9F4C
; anything else: "Illegal operation" (TFS is read-only)
;-----
.filev
819D C9 05 CMP #&05
819F F0 0B BEQ L81AC
81A1 C9 07 CMP #&07
81A3 F0 0A BEQ L81AF
81A5 C9 FF CMP #&FF
81A7 D0 3D BNE bputv_illegal
81A9 4C 4C 9F JMP L9F4C
.L81AC
81AC 4C 1A AD JMP LAD1A
.L81AF
81AF 4C 94 AC JMP LAC94
;-----
; SERVICE CALL 8: UNRECOGNISED OSWORD
; Claims OSWORD &7A (the Teletext system call). The parameter block is at (osw_x).
; Sub-commands &80-&A5 dispatch through osword_table; codes below &80 go to LAD5A.
;-----
.svc08_osword
81B2 A5 EF LDA osw_a ; OSWORD number the MOS was given
81B4 C9 7A CMP #&7A ; 'z'
81B6 F0 03 BEQ L81BB
81B8 4C 0B 82 JMP svc_exit_unclaimed
.L81BB
81BB A0 00 LDY #&00
81BD B1 F0 LDA (osw_x),Y ; sub-command byte from the parameter block
81BF 30 03 BMI L81C4
81C1 4C 5A AD JMP LAD5A
.L81C4
81C4 0A ASL A ; two bytes per table entry
81C5 C9 4C CMP #&4C ; past &A5?
81C7 B0 0E BCS osw_bad
81C9 AA TAX
81CA BD 20 81 LDA osword_table,X
81CD 85 BC STA ptr
81CF BD 21 81 LDA &8121,X
81D2 85 BD STA ptr+1
81D4 6C BC 00 JMP (ptr) ; jump to the handler; handlers finish with JMP svc_exit_claimed
;-----
; OSWORD &7A sub-commands &97-&99 and anything above &A5: "Bad OSWORD" (error &FF).
;-----
.osw_bad
81D7 20 98 A1 JSR error_inline
81DA .str '\xffBad OSWORD\x00'
;-----
; BPUTV: the Telesoft filing system is read-only, so any write gives "Illegal operation".
;-----
.bputv_illegal
81E6 20 98 A1 JSR error_inline
81E9 .str '\x9dIllegal operation\x00'
;-----
; SERVICE CALL 4: UNRECOGNISED * COMMAND
; Looks the command up in the * command table (&9EE9). X=&4F is the table offset bias
; used by match_star_command. Carry set on return means the command was found and run.
;-----
.svc04_star_command
81FC A5 F2 LDA text_ptr
81FE 85 BC STA ptr
8200 A5 F3 LDA text_ptr+1
8202 85 BD STA ptr+1
8204 A2 4F LDX #&4F
8206 20 87 9E JSR match_star_command
8209 B0 06 BCS svc_exit_claimed
;-----
; Return from a service call WITHOUT claiming it: restore X, Y and A (the service number)

```

```

; from the frame pushed by service_entry.
;-----
.svc_exit_unclaimed
820B 68      PLA
820C AA      TAX
820D 68      PLA
820E A8      TAY
820F 68      PLA
8210 60      RTS
;-----
; Return from a service call CLAIMING it: restore X and Y, and return A=0.
;-----
.svc_exit_claimed
8211 68      PLA
8212 AA      TAX
8213 68      PLA
8214 A8      TAY
8215 68      PLA
8216 A9 00    LDA #000
8218 60      RTS
;-----
; PAGED ROM SERVICE ENTRY
;
; On entry A = service call number, X = this ROM's number, Y = parameter.
; A, Y and X are pushed so that every handler can reach them on the stack
; (&0103,X after TSX = the original A) and exit through svc_exit_claimed or
; svc_exit_unclaimed.
;
; Service 3 (auto-boot) is always handled so that "Acorn TFS / no power" can be
; printed. Every other service is ignored when the adapter is switched off: an
; unpowered adapter reads as zero in the link bits of TT_STATUS_CTRL, and then TFS
; claims no workspace and does nothing at all.
;-----
.service_entry
8219 48      PHA
821A 98      TYA
821B 48      PHA
821C 8A      TXA
821D 48      PHA
821E BA      TSX
821F BD 03 01 LDA &0103,X          ; A = service call number
8222 C9 03    CMP #003
8224 D0 03    BNE L8229
8226 4C B4 82 JMP svc03_autoboot
.L8229
8229 AD 10 FC LDA TT_STATUS_CTRL    ; link bits D3-D0 read 0 when the adapter has no power
822C 29 0F    AND #00F
822E F0 DB    BEQ svc_exit_unclaimed
8230 BD 03 01 LDA &0103,X
8233 C9 12    CMP #12              ; service &12: select filing system
8235 F0 0F    BEQ svc12_select_fs
8237 C9 0B    CMP #00B            ; only services 0-&0A are dispatched by table
8239 B0 D0    BCS svc_exit_unclaimed
823B 0A      ASL A
823C AA      TAX
823D BD 94 83 LDA &8394,X          ; RTS-dispatch: push handler address-1, then RTS
8240 48      PHA
8241 BD 93 83 LDA service_table,X
8244 48      PHA
8245 60      RTS
;-----
; SERVICE CALL &12: SELECT FILING SYSTEM Y. TFS is filing system number 6.
;-----
.svc12_select_fs
8246 C0 06    CPY #006
8248 D0 C1    BNE svc_exit_unclaimed
824A 20 34 9D JSR cmd_TELESOFT
824D 4C 11 82 JMP svc_exit_claimed
;-----
; SERVICE CALL 1: ABSOLUTE WORKSPACE CLAIM
; Raise the claim so that pages &0E-&19 belong to the filing systems.
; Y is the highest page wanted so far; TFS needs it to be at least &1A.
; A is left as 1 so the call passes on to other ROMs.
;-----
.svc01_abs_workspace
8250 68      PLA
8251 AA      TAX
8252 68      PLA
8253 A8      TAY
8254 C0 1A    CPY #1A
8256 B0 02    BCS L825A
8258 A0 1A    LDY #1A
.L825A
825A 68      PLA
825B 60      RTS
;-----
; SERVICE CALL 2: PRIVATE WORKSPACE CLAIM
; Y = first free private page. TFS takes 8 pages (2K) there and initialises its state:
; - remember the private page in the MOS table at &0DF0+ROM number
; - initialise the Telesoftware decoder tables (LA80A)

```

```

; - clear all 25 request slots; slot 0 becomes the current page, using the private
;   workspace as its page buffer, and is marked active
; - defaults: *OPT1,128 *OPT2,1 *OPT3,0, control latch &1C (capture on, IRQ on, AFC on, channel 1)
; - immediately stash the static workspace (TFS is not the selected filing system yet)
; Returns Y = Y+8.
;-----
.svc02_private_workspace
825C A6 F4 LDX romsel_copy
825E 98 TYA
825F 9D F0 0D STA rom_wksp_table,X ; private workspace page for this ROM
8262 48 PHA
8263 20 0A A8 JSR ts_init ; initialise the Telesoftware name tables
8266 A6 F4 LDX romsel_copy
8268 8E 99 0E STX prev_language ; "previous language" defaults to this ROM
826B A2 F9 LDX #&F9
826D A9 00 LDA #&00
.L826F
826F 9D A0 0E STA req_buf,X ; clear &0EA1-&0F99 (25 request slots of 10 bytes)
8272 CA DEX
8273 D0 FA BNE L826F
8275 8D 9D 0E STA hdr_dest+1
8278 8D 9B 0E STA cur_req
827B 8D 9E 0E STA &0E9E
827E 8D 9F 0E STA &0E9F
8281 8D A0 0E STA req_buf ; slot 0 buffer = private workspace page, offset 0
8284 68 PLA
8285 48 PHA
8286 8D A1 0E STA req_buf+1
8289 A9 40 LDA #&40 ; slot 0 is the active current-page request
828B 8D A9 0E STA req_state
828E A9 01 LDA #&01
8290 8D 98 0E STA opt2_errors ; *OPT2,1: retry on checksum errors
8293 A9 80 LDA #&80
8295 20 7C AC JSR LAC7C ; *OPT1,128
8298 A9 00 LDA #&00
829A 20 80 AC JSR LAC80 ; *OPT3,0
829D A9 01 LDA #&01
829F 8D 5D 0E STA tsdp+1
82A2 A9 1C LDA #&1C
82A4 8D 8E 0E STA ctrl_shadow ; TT_STATUS_CTRL shadow: b4 AFC on, b3 IRQ enable, b2 capture enable, channel 0
82A7 68 PLA
82A8 20 46 83 JSR stash_workspace ; save the static workspace in the adapter RAM and private workspace
82AB 68 PLA
82AC AA TAX
82AD 68 PLA
82AE 18 CLC
82AF 69 08 ADC #&08 ; claim 8 pages
82B1 A8 TAY
82B2 68 PLA
82B3 60 RTS
;-----
; SERVICE CALL 3: AUTO-BOOT
; Runs after BREAK. If no key is held, or T is held (internal key number &23),
; print "Acorn TFS" and make Telesoft the filing system.
; If the adapter has no power, print "no power" instead and claim nothing else.
; Any other key held: pass the call on so that another filing system can boot.
; The keyboard buffer is flushed and &CA re-inserted: &CA is the code that makes the
; MOS run the *KEY10 string after BREAK. OSMODE &78 records T as the key being held so
; the T used for T+BREAK does not also appear as a typed character.
;-----
.svc03_autoboot
82B4 A9 7A LDA #&7A
82B6 20 F4 FF JSR OSMODE ; OSMODE &7A: keyboard scan from key 16
82B9 8A TXA
82BA C9 FF CMP #&FF
82BC F0 04 BEQ L82C2
82BE C9 23 CMP #&23 ; internal key number for T
82C0 D0 6B BNE L832D
.L82C2
82C2 20 AF A2 JSR print_inline
82C5 .str 'Acorn TFS\x00'
82CF AD 10 FC LDA TT_STATUS_CTRL ; adapter powered?
82D2 29 0F AND #&0F
82D4 D0 2F BNE L8305
82D6 20 AF A2 JSR print_inline
82D9 .str ' no power\r\n\n\x00'
82E7 A9 15 LDA #&15
82E9 A2 00 LDX #&00
82EB A0 00 LDY #&00
82ED 20 F4 FF JSR OSMODE ; OSMODE &15: flush buffer X
82F0 A9 8A LDA #&8A
82F2 A2 00 LDX #&00
82F4 A0 CA LDY #&CA
82F6 20 F4 FF JSR OSMODE ; OSMODE &8A: insert character into buffer
82F9 A9 78 LDA #&78
82FB A2 23 LDX #&23
82FD A0 00 LDY #&00
82FF 20 F4 FF JSR OSMODE ; OSMODE &78: write keys-pressed information
8302 4C 11 82 JMP svc_exit_claimed ; claim the call: no other ROM boots
.L8305

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8305 20 AF A2 JSR print_inline
8308 .str '\r\n\r\n\x00'
830C 20 34 9D JSR cmd_TELESOFT ; select Telesoft (TFS) as the filing system
830F A9 15 LDA #&15
8311 A2 00 LDX #&00
8313 A0 00 LDY #&00
8315 20 F4 FF JSR OSBYTE ; OSBYTE &15: flush buffer X
8318 A9 8A LDA #&8A
831A A2 00 LDX #&00
831C A0 CA LDY #&CA
831E 20 F4 FF JSR OSBYTE ; OSBYTE &8A: insert character into buffer
8321 A9 78 LDA #&78
8323 A2 23 LDX #&23
8325 A0 00 LDY #&00
8327 20 F4 FF JSR OSBYTE ; OSBYTE &78: write keys-pressed information
832A 4C 11 82 JMP svc_exit_claimed
.L832D
832D 4C 0B 82 JMP svc_exit_unclaimed
;-----
; SERVICE CALL 9: *HELP. Handled by help_command; the call is always passed on.
;-----
.svc09_help
8330 20 C3 83 JSR help_command
8333 4C 0B 82 JMP svc_exit_unclaimed
;-----
; SERVICE CALL &0A: ANOTHER ROM WANTS THE STATIC WORKSPACE
; If TFS's state isn't already stashed (bit 7 of its &0DF0 entry clear), stash it now.
;-----
.svc0A_static_workspace
8336 A6 F4 LDX romsel_copy
8338 BD F0 0D LDA rom_wksp_table,X
833B 10 03 BPL L8340
833D 4C 0B 82 JMP svc_exit_unclaimed
.L8340
8340 20 46 83 JSR stash_workspace
8343 4C 11 82 JMP svc_exit_claimed
;-----
; STASH TFS STATE (A = private workspace page)
;
; The filing systems share &0E00-&19FF, so TFS must save its state when another
; filing system takes over. It saves 3K starting at &0E8E:
; - the first 1K (&0E8E-&128D) goes into the ADAPTER'S OWN 1K RAM, 64 bytes into each
; of the 16 line slots (capture is switched off first so the hardware can't overwrite it)
; - the next 2K (&128E-&1A8D) goes to the 8 pages of private workspace
; Bit 7 of the &0DF0 entry marks the state as stashed; cmd_TELESOFT reverses this.
; NOTE: 3K from &0E8E ends at &1A8D, so the last 142 bytes come from &1A00-&1A8D,
; above the claimed workspace (normally the start of user memory). See cmd_TELESOFT.
;-----
.stash_workspace
8346 A6 F4 LDX romsel_copy
8348 85 C1 STA ptr2+1
834A 09 80 ORA #&80 ; mark "state stashed"
834C 9D F0 0D STA rom_wksp_table,X
834F 29 7F AND #&7F
8351 20 49 A2 JSR capture_off ; stop capture and IRQs so the adapter RAM is free
8354 08 PHP
8355 78 SEI
8356 A9 8E LDA #&8E
8358 85 BC STA ptr
835A A9 0E LDA #&0E
835C 85 BD STA ptr+1
835E A2 0F LDX #&0F ; 16 slots
.L8360
8360 A0 3F LDY #&3F ; 64 bytes per slot
8362 8E 11 FC STX TT_SLOT ; select adapter RAM slot X (resets its byte counter)
.L8365
8365 B1 BC LDA (ptr),Y
8367 8D 12 FC STA TT_DATA ; write one byte; the address counter steps on
836A 88 DEY
836B 10 F8 BPL L8365
836D A5 BC LDA ptr
836F 18 CLC
8370 69 40 ADC #&40
8372 85 BC STA ptr
8374 90 02 BCC L8378
8376 E6 BD INC ptr+1
.L8378
8378 CA DEX
8379 10 E5 BPL L8360
837B A9 00 LDA #&00
837D 85 C0 STA ptr2
837F A2 07 LDX #&07 ; 8 pages to private workspace
.L8381
8381 A0 00 LDY #&00
.L8383
8383 B1 BC LDA (ptr),Y
8385 91 C0 STA (ptr2),Y
8387 C8 INY
8388 D0 F9 BNE L8383
838A E6 BD INC ptr+1

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838C E6 C1    INC ptr2+1
838E CA       DEX
838F 10 F0    BPL L8381
8391 28       PLP
8392 60       RTS
;-----
; SERVICE CALL DISPATCH TABLE (address-1 for RTS dispatch), services 0-&0A:
; 0 none, 1 abs workspace, 2 private workspace, 3 auto-boot, 4 * command,
; 5 interrupt, 6 BRK (ignored), 7 OSBYTE (ignored), 8 OSWORD, 9 *HELP, &0A static workspace.
;-----
.service_table
8393 .db 0A 82 4F 82 5B 82 B3 82 FB 81 3E 8D 0A 82 0A 82 ; ..0.[.....>.....
83A3 .db B1 81 2F 83 35 83 ; ../.5.
;-----
; BGETV (OSBGET). Y = channel.
; channel 15: read the next character of the current Teletext page (L9510)
; channel 14: read the next byte of a decoded Telesoftware file (LABD1)
; anything else: "Channel" error
;-----
.bgetv
83A9 C0 0F     CPY #&0F
83AB F0 07     BEQ L83B4
83AD C0 0E     CPY #&0E
83AF D0 06     BNE L83B7
83B1 4C D1 AB  JMP LABD1
.L83B4
83B4 4C 10 95  JMP L9510
.L83B7
83B7 20 98 A1  JSR error_inline
83BA .str '\xdeChannel\x00'
;-----
; *HELP handler. With no keyword, prints the ROM title and version and the list of
; keywords (TELETEXT, TELESOFT, OPT). With a keyword (abbreviations with "." allowed),
; prints the matching help text from &84FF / &872C / &895A.
; text_ptr is preserved for the other ROMs that also see the *HELP.
;-----
.help_command
83C3 98       TYA
83C4 48       PHA
83C5 8A       TXA
83C6 48       PHA
83C7 A5 F2    LDA text_ptr
83C9 48       PHA
83CA A5 F3    LDA text_ptr+1
83CC 48       PHA
83CD 20 5B 84 JSR L845B ; skip "HELP" (Y = offset past the command name)
.L83D0
83D0 A0 00     LDY #&00
83D2 B1 F2    LDA (text_ptr),Y
83D4 C9 0D     CMP #&0D
83D6 F0 2F     BEQ L8407
83D8 C9 20     CMP #&20
83DA D0 08     BNE L83E4
83DC E6 F2     INC text_ptr
83DE D0 F0     BNE L83D0
83E0 E6 F3     INC text_ptr+1
83E2 D0 EC     BNE L83D0
.L83E4
83E4 20 66 84 JSR L8466 ; look the word up in the keyword table
83E7 B0 05     BCS L83EE
83E9 20 C8 84 JSR L84C8
83EC A0 00     LDY #&00
.L83EE
83EE B1 F2     LDA (text_ptr),Y
83F0 C9 0D     CMP #&0D
83F2 F0 5C     BEQ L8450
83F4 C9 20     CMP #&20
83F6 F0 08     BEQ L8400
83F8 C9 2E     CMP #&2E ; '.'
83FA F0 04     BEQ L8400
83FC C8        INY
83FD 4C EE 83 JMP L83EE
.L8400
8400 C8        INY
8401 20 5B 84 JSR L845B
8404 4C E4 83 JMP L83E4
.L8407
8407 20 E7 FF JSR OSNEWL ; plain *HELP: print the title and version from the ROM header
840A A9 09     LDA #&09
840C 85 BC     STA ptr
840E A9 80     LDA #&80
8410 85 BD     STA ptr+1
8412 20 C8 84 JSR L84C8
8415 A9 20     LDA #&20
8417 20 EE FF JSR OSWRCH ; print ' '
841A C8        INY
841B 20 C8 84 JSR L84C8
841E 20 E7 FF JSR OSNEWL
8421 A9 D9     LDA #&D9 ; then the keyword list at &84D9
8423 85 BC     STA ptr

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8425 A9 84 LDA #84
8427 85 BD STA ptr+1
8429 A0 00 LDY #00
842B 20 C8 84 JSR L84C8
842E A2 01 LDX #01
.L8430
8430 E0 03 CPX #03
8432 B0 17 BCS L844B
8434 A9 0A LDA #0A
8436 20 EE FF JSR OSWRCH ; VDU 10: line feed
8439 A9 20 LDA #20
843B 20 EE FF JSR OSWRCH ; print ' '
843E 20 EE FF JSR OSWRCH
8441 C8 INY
8442 C8 INY
8443 C8 INY
8444 20 C8 84 JSR L84C8
8447 E8 INX
8448 4C 30 84 JMP L8430
.L844B
844B A9 0A LDA #0A
844D 20 EE FF JSR OSWRCH ; VDU 10: line feed
.L8450
8450 68 PLA
8451 85 F3 STA text_ptr+1
8453 68 PLA
8454 85 F2 STA text_ptr
8456 68 PLA
8457 AA TAX
8458 68 PLA
8459 A8 TAY
845A 60 RTS
; text_ptr += Y
.L845B
845B 98 TYA
845C 18 CLC
845D 65 F2 ADC text_ptr
845F 85 F2 STA text_ptr
8461 90 02 BCC L8465
8463 E6 F3 INC text_ptr+1
.L8465
8465 60 RTS
;
; -----
; Match the word at (text_ptr) against the *HELP keyword table at &84DB (each entry is
; a word, CR, 0, then a 2-byte text address). Returns C clear and (ptr) = the help text
; on a match, C set (and text_ptr past the word) if nothing matches.
; -----
.L8466
8466 A9 DB LDA #DB
8468 85 BC STA ptr
846A A9 84 LDA #84
846C 85 BD STA ptr+1
846E A2 01 LDX #01
.L8470
8470 A0 FF LDY #FF
.L8472
8472 C8 INY
8473 B1 F2 LDA (text_ptr),Y
8475 C9 2E CMP #2E ; '.'
8477 F0 26 BEQ L849F
8479 D1 BC CMP (ptr),Y
847B F0 1E BEQ L849B
847D 38 SEC
847E E9 20 SBC #20
8480 D1 BC CMP (ptr),Y
8482 F0 17 BEQ L849B
8484 C9 00 CMP #00
8486 D0 06 BNE L848E
8488 B1 BC LDA (ptr),Y
848A C9 00 CMP #00
848C F0 11 BEQ L849F
.L848E
848E E0 03 CPX #03
8490 B0 25 BCS L84B7
8492 E8 INX
8493 A9 0C LDA #0C
8495 20 BE 84 JSR L84BE
8498 4C 70 84 JMP L8470
.L849B
849B C9 00 CMP #00
849D D0 D3 BNE L8472
.L849F
849F 20 5B 84 JSR L845B
84A2 A9 0A LDA #0A
84A4 20 BE 84 JSR L84BE
84A7 A0 00 LDY #00
84A9 B1 BC LDA (ptr),Y
84AB 48 PHA
84AC C8 INY
84AD B1 BC LDA (ptr),Y

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84AF 85 BD STA ptr+1
84B1 68 PLA
84B2 85 BC STA ptr
84B4 18 CLC
84B5 90 04 BCC L84BB
.L84B7
84B7 20 5B 84 JSR L845B
84BA 38 SEC
.L84BB
84BB A0 00 LDY #000
84BD 60 RTS
.L84BE
84BE 18 CLC
84BF 65 BC ADC ptr
84C1 85 BC STA ptr
84C3 90 02 BCC L84C7
84C5 E6 BD INC ptr+1
.L84C7
84C7 60 RTS
;-----
; Print the zero-terminated string at (ptr)+Y. Y and ptr are left pointing past it.
;-----
.L84C8
84C8 B1 BC LDA (ptr),Y
84CA C9 00 CMP #000
84CC F0 0A BEQ L84D8
84CE 20 EE FF JSR OSWRCH
84D1 C8 INY
84D2 D0 F4 BNE L84C8
84D4 E6 BD INC ptr+1
84D6 D0 F0 BNE L84C8
.L84D8
84D8 60 RTS
; *HELP keyword table and help texts.
.help_text
84D9 .db 20 20 54 45 4C 45 54 45 58 54 0D 00 FF 84 54 45 ; TELETTEXT...TE
84E9 .db 4C 4C 45 53 4F 46 54 0D 00 2C 87 4F 50 54 0D 0D 0D ; LESOFT...OPT...
84F9 .db 0D 0D 0D 0D 0D 5A 89 20 20 20 20 20 54 45 4C 45 54 ; ....Z. TELET
8509 .db 45 58 54 20 54 45 52 4D 49 4E 41 4C 20 4D 4F 44 ; EXT TERMINAL MOD
8519 .db 45 20 43 4F 4D 4D 41 4E 44 53 0D 0A 20 20 20 66 ; E COMMANDS.. f
8529 .db 30 20 53 65 6C 65 63 74 20 61 20 70 61 67 65 20 ; 0 Select a page
8539 .db 2F 20 77 69 6C 64 20 63 61 72 64 0D 0A 0A 20 20 ; / wild card...
8549 .db 20 66 31 20 53 65 6C 65 63 74 20 6C 61 73 74 20 ; f1 Select last
8559 .db 65 78 70 6C 69 63 69 74 20 70 61 67 65 0D 0A 53 ; explicit page..S
8569 .db 48 20 66 31 20 53 65 6C 65 63 74 20 69 6E 64 65 ; H f1 Select inde
8579 .db 78 20 70 61 67 65 0D 0A 0A 20 20 20 66 32 20 52 ; x page... f2 R
8589 .db 65 76 65 61 6C 20 68 69 64 64 65 6E 20 74 65 78 ; eveal hidden tex
8599 .db 74 20 61 6E 64 20 68 6F 6C 64 20 70 61 67 65 0D ; t and hold page.
85A9 .db 0A 53 48 20 66 32 20 43 6F 6E 63 65 61 6C 20 74 ; .SH f2 Conceal t
85B9 .db 65 78 74 20 26 20 72 65 74 61 69 6E 20 68 6F 6C ; ext & retain hol
85C9 .db 64 20 73 74 61 74 65 0D 0A 0A 20 20 20 66 33 20 ; d state... f3
85D9 .db 4B 65 65 70 20 70 61 67 65 20 75 70 20 74 6F 20 ; Keep page up to
85E9 .db 64 61 74 65 20 69 6E 20 6D 65 6D 6F 72 79 0D 0A ; date in memory..
85F9 .db 53 48 20 66 33 20 52 65 6C 65 61 73 65 20 61 20 ; SH f3 Release a
8609 .db 6B 65 70 74 20 70 61 67 65 0D 0A 0A 20 20 20 66 ; kept page... f
8619 .db 34 20 53 65 6C 65 63 74 20 63 68 61 6E 6E 65 6C ; 4 Select channel
8629 .db 20 28 31 2C 32 2C 33 20 6F 72 20 34 29 0D 0A 53 ; (1,2,3 or 4)..S
8639 .db 48 20 66 34 20 54 75 6E 65 20 63 68 61 6E 6E 65 ; H f4 Tune channe
8649 .db 6C 20 75 73 69 6E 67 20 73 63 61 6C 65 0D 0A 0A ; l using scale...
8659 .db 20 20 20 66 35 20 45 6E 74 65 72 20 6F 70 65 72 ; f5 Enter oper
8669 .db 61 74 69 6E 67 20 73 79 73 74 65 6D 20 2A 20 63 ; ating system * c
8679 .db 6F 6D 6D 61 6E 64 0D 0A 20 20 20 66 36 20 4C 6F ; ommand.. f6 Lo
8689 .db 61 64 20 6D 20 70 61 67 65 20 66 72 6F 6D 20 66 ; ad a page from f
8699 .db 69 6C 65 0D 0A 20 20 20 66 37 20 53 61 76 65 20 ; ile.. f7 Save
86A9 .db 64 69 73 70 6C 61 79 65 64 20 70 61 67 65 20 74 ; displayed page t
86B9 .db 6F 20 66 69 6C 65 0D 0A 20 20 20 66 38 20 48 6F ; o file.. f8 Ho
86C9 .db 6C 64 20 74 68 65 20 64 69 73 70 6C 61 79 65 64 ; ld the displayed
86D9 .db 20 70 61 67 65 0D 0A 0A 20 20 20 66 39 20 45 78 ; page... f9 Ex
86E9 .db 69 74 20 74 6F 20 70 72 65 76 69 6F 75 73 20 6C ; it to previous l
86F9 .db 61 6E 67 75 61 67 65 0D 0A 53 48 20 66 39 20 45 ; anguage..SH f9 E
8709 .db 78 69 74 20 74 6F 20 54 65 6C 65 73 6F 66 74 20 ; xit to Telesoft
8719 .db 28 26 20 65 78 65 63 20 70 72 6F 67 72 61 6D 29 ; (& exec program)
8729 .db 0D 0A 0D 20 20 20 20 20 20 20 54 45 4C 45 53 4F ; ... TELESO
8739 .db 46 54 20 4D 4F 44 45 20 43 4F 4D 4D 41 4E 44 53 ; FT MODE COMMANDS
8749 .db 0D 0A 0A 42 42 43 31 2C 42 42 43 32 20 20 20 20 ; ...BBC1,BBC2
8759 .db 20 20 20 20 53 65 6C 65 63 74 20 63 68 61 6E ; Select chan
8769 .db 6E 65 6C 20 31 20 6F 72 20 32 0D 0A 43 48 31 20 ; nel 1 or 2..CH1
8779 .db 20 65 74 63 2E 20 20 20 20 20 20 20 20 53 65 ; etc. Se
8789 .db 6C 65 63 74 20 61 6E 79 20 63 68 61 6E 6E 65 6C ; lect any channel
8799 .db 0D 0A 44 41 54 45 20 20 20 20 20 20 20 20 20 ; ..DATE
87A9 .db 20 20 20 20 53 68 6F 77 20 74 6F 64 61 79 73 20 ; Show todays
87B9 .db 64 61 74 65 0D 0A 44 49 53 50 4C 41 59 20 3C 61 ; date..DISPLAY <a
87C9 .db 64 64 72 3E 20 20 20 20 44 69 73 70 6C 61 79 20 ; ddr> Display
87D9 .db 73 74 6F 72 65 64 20 70 61 67 65 0D 0A 45 58 45 ; stored page..EXE
87E9 .db 43 20 3C 66 73 70 3E 20 20 20 20 20 20 20 44 ; C <fsp> D
87F9 .db 6F 77 6E 6C 6F 61 64 20 41 53 43 49 49 20 64 61 ; ownload ASCII da
8809 .db 74 61 0D 0A 49 54 56 31 2C 49 54 56 32 20 20 20 ; ta..ITV1,ITV2
8819 .db 20 20 20 20 20 53 65 6C 65 63 74 20 63 68 61 ; Select cha
8829 .db 6E 6E 65 6C 20 33 20 6F 72 20 34 0D 0A 50 41 47 ; nnel 3 or 4..PAG
8839 .db 45 20 3C 6E 75 6D 62 65 72 3E 20 20 20 20 53 ; E <number> S

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8849 .db 65 6C 65 63 74 20 63 75 72 72 65 6E 74 20 70 61 ; elect current pa
8859 .db 67 65 0D 0A 54 45 4C 45 54 45 58 54 20 20 20 20 ; ge..TELETEXT
8869 .db 20 20 20 20 20 20 45 6E 74 65 72 20 74 65 72 6D ; Enter term
8879 .db 69 6E 61 6C 20 6D 6F 64 65 0D 0A 20 20 20 20 20 ; inal mode..
8889 .db 20 20 20 20 20 20 20 20 20 20 20 20 20 28 53 65 ; (Se
8899 .db 65 20 2A 48 45 4C 50 20 54 45 4C 45 54 45 58 54 ; e *HELP TELETEXT
88A9 .db 29 0D 0A 54 49 4D 45 20 20 20 20 20 20 20 20 20 ; )..TIME
88B9 .db 20 20 20 20 20 53 68 6F 77 20 63 6C 6F 63 6B 20 ; Show clock
88C9 .db 74 69 6D 65 0D 0A 54 52 41 4E 53 46 45 52 20 3C ; time..TRANSFER <
88D9 .db 61 64 64 72 3E 20 20 20 53 74 6F 72 65 20 63 75 ; addr> Store cu
88E9 .db 72 72 65 6E 74 20 70 61 67 65 0D 0A 54 55 4E 45 ; rrent page..TUNE
88F9 .db 20 20 20 20 20 20 20 20 20 20 20 20 20 54 75 ; Tu
8909 .db 6E 65 20 63 75 72 72 65 6E 74 20 63 68 61 6E 6E ; ne current chann
8919 .db 65 6C 0D 0A 0A 3C 66 73 70 3E 20 20 3D 20 66 69 ; el...<fsp> = fi
8929 .db 6C 65 20 73 70 65 63 69 66 69 63 61 74 69 6F 6E ; le specification
8939 .db 0D 0A 3C 61 64 64 72 3E 20 3D 20 68 65 78 61 64 ; ..<addr> = hexad
8949 .db 65 63 69 6D 61 6C 20 61 64 64 72 65 73 73 0D 0A ; ecimal address..
8959 .db 00 4F 50 54 20 30 20 20 20 20 20 20 20 52 65 73 ; .OPT 0 Res
8969 .db 65 74 20 64 65 66 61 75 6C 74 20 6F 70 74 69 6F ; et default optio
8979 .db 6E 73 0D 0A 0A 4F 50 54 20 31 2C 30 20 20 20 20 ; ns...OPT 1,0
8989 .db 20 53 75 70 70 72 65 73 73 20 61 6C 6C 20 6D 65 ; Suppress all me
8999 .db 73 73 61 67 65 73 0D 0A 4F 50 54 20 31 2C 31 32 ; ssages..OPT 1,12
89A9 .db 38 20 20 20 45 6E 61 62 6C 65 20 27 53 65 61 72 ; 8 Enable 'Sear
89B9 .db 63 68 69 6E 67 27 20 6D 65 73 73 61 67 65 73 0D ; ching' messages.
89C9 .db 0A 20 20 20 20 20 20 20 20 20 20 20 20 28 64 65 ; . (de
89D9 .db 66 61 75 6C 74 29 0D 0A 0A 4F 50 54 20 32 2C 30 ; fault)...OPT 2,0
89E9 .db 20 20 20 20 20 49 67 6E 6F 72 65 20 64 61 74 61 ; Ignore data
89F9 .db 20 65 72 72 6F 72 73 0D 0A 4F 50 54 20 32 2C 31 ; errors..OPT 2,1
8A09 .db 20 20 20 20 20 52 65 74 72 79 20 6F 6E 20 65 72 ; Retry on er
8A19 .db 72 6F 72 73 20 28 64 65 66 61 75 6C 74 29 0D 0A ; rors (default)..
8A29 .db 4F 50 54 20 32 2C 32 20 20 20 20 20 41 62 6F 72 ; OPT 2,2 Abor
8A39 .db 74 20 6F 6E 20 65 72 72 6F 72 0D 0A 00 ; t on error...
;
;-----
; OSWORD &7A &81-&84: PAGE READ-POSITION BOOKKEEPING
; BGET on channel 15 reads the current page character by character, from position
; (rd_row, rd_col). The Telesoftware decoder uses these calls to remember and return to
; positions while it parses a block:
; &84 mark := current position
; &81 start := mark
; &82 current position := start (rewind)
; &83 limit := mark (BGET reports end of data at the limit)
;
;-----
.osw_81_start_from_mark
8A46 AD 91 0E LDA rd_mark_row
8A49 8D 93 0E STA rd_start_row
8A4C AD 92 0E LDA rd_mark_col
8A4F 8D 94 0E STA rd_start_col
8A52 4C 11 82 JMP svc_exit_claimed
.osw_82_rewind_to_start
8A55 AD 93 0E LDA rd_start_row
8A58 8D 8F 0E STA rd_row
8A5B AD 94 0E LDA rd_start_col
8A5E 8D 90 0E STA rd_col
8A61 4C 11 82 JMP svc_exit_claimed
.osw_83_limit_at_mark
8A64 AD 91 0E LDA rd_mark_row
8A67 8D 95 0E STA rd_limit_row
8A6A AD 92 0E LDA rd_mark_col
8A6D 8D 96 0E STA rd_limit_col
8A70 4C 11 82 JMP svc_exit_claimed
.osw_84_mark_position
8A73 AD 8F 0E LDA rd_row
8A76 8D 91 0E STA rd_mark_row
8A79 AD 90 0E LDA rd_col
8A7C 8D 92 0E STA rd_mark_col
8A7F 4C 11 82 JMP svc_exit_claimed
;
;-----
; OSWORD &7A &92: HEADERS [&92, addr lo, addr hi]
; Store the address where the interrupt handler writes the 32 rolling header characters
; of every row 0 received (hdr_dest; 0 = off). Note bytes 1-2 land in hdr_dest via
; &0E9B+Y, Y=1..2.
;
;-----
.osw_92_headers
8A82 A0 01 LDY #&01
.L8A84
8A84 B1 F0 LDA (osw_x),Y
8A86 99 9B 0E STA cur_req,Y
8A89 C8 INY
8A8A C0 03 CPY #&03
8A8C D0 F6 BNE L8A84
8A8E 4C 11 82 JMP svc_exit_claimed
;
;-----
; OSWORD &7A &8B: SELECT CHANNEL [&8B, channel 0-3]
; Writes the channel number into bits 0-1 of the control latch: IC6b decodes them and
; one of Q3-Q6 connects tuning preset VR2-VR5. All captures in progress are abandoned
; and every magazine is marked idle, because rows from the old channel are useless.
;
;-----
.osw_8B_select_channel
8A91 A0 01 LDY #&01
8A93 B1 F0 LDA (osw_x),Y

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.L8A95
8A95 4D 8E 0E EOR ctrl_shadow      ; merge the new channel into bits 0-1 of the shadow
8A98 29 03    AND #003
8A9A 4D 8E 0E EOR ctrl_shadow
8A9D 8D 8E 0E STA ctrl_shadow
8AA0 8D 10 FC STA TT_STATUS_CTRL    ; select the tuning preset
8AA3 A9 01    LDA #001              ; reset the broadcast-data packet (no time/date yet)
8AA5 8D 5D 0E STA tsdp+1
8AA8 A2 00    LDX #000

.L8AAA
8AAA 8A      TXA
8AAB 48      PHA
8AAC BD A9 0E LDA req_state,X      ; active request slot?
8AAF 2A      ROL A
8AB0 10 03    BPL L8AB5
8AB2 20 A2 90 JSR clear_capture_flags ; abandon any capture in progress for it

.L8AB5
8AB5 68      PLA
8AB6 18      CLC
8AB7 69 0A    ADC #00A
8AB9 AA      TAX
8ABA C9 FA    CMP #0FA
8ABC D0 EC    BNE L8AAA
8ABE A9 FF    LDA #0FF
8AC0 A2 07    LDX #007

.L8AC2
8AC2 9D 15 0E STA mag_capture,X    ; no page being captured in any magazine
8AC5 CA      DEX
8AC6 10 FA    BPL L8AC2
8AC8 4C 11 82 JMP svc_exit_claimed

;-----
; OSWORD &7A &93: WHAT CHANNEL. Returns bits 0-1 of the control latch in byte 1.
;-----
.osw_93_what_channel
8ACB A0 01    LDY #001
8ACD AD 8E 0E LDA ctrl_shadow
8AD0 29 03    AND #003
8AD2 91 F0    STA (osw_x),Y
8AD4 4C 11 82 JMP svc_exit_claimed

;-----
; OSWORD &7A &91: READ PREVIOUS LANGUAGE AND FILING SYSTEM
; Byte 1 = language ROM that was running before *TELETEXT (for f9),
; byte 2 = filing system number that was selected before *TELESOFT.
;-----
.osw_91_read_prev_lang
8AD7 A0 01    LDY #001
8AD9 AD 99 0E LDA prev_language
8ADC 91 F0    STA (osw_x),Y
8ADE C8      INY
8ADF AD 9A 0E LDA prev_fs
8AE2 91 F0    STA (osw_x),Y
8AE4 4C 11 82 JMP svc_exit_claimed

;-----
; Return A to the caller of a BGET-style routine: restore X and Y from the stack.
;-----
.L8AE7
8AE7 85 BC    STA ptr
8AE9 68      PLA
8AEA AA      TAX
8AEB 68      PLA
8AEC A8      TAY
8AED A5 BC    LDA ptr
8AEF 60      RTS

;-----
; OSWORD &7A &8F: TRANSFER [&8F, 32-bit address]
; Wait for the current page to arrive (or ESCAPE), then copy it to the address given,
; as 24 rows x 40 characters (960 bytes) of MODE 7 screen codes:
; - every character has bit 7 set, so control codes &00-&1F become the MODE 7 teletext
;   control codes &80-&9F
; - rows that never arrived are filled with spaces
; - a row containing "double height" (&8D) is written twice and the next transmitted
;   row is skipped, which is how double height works on a teletext screen
; The address may be in a second processor: LA28C writes through the Tube when needed.
; The page is unlocked afterwards.
;-----
.osw_8F_transfer
8AF0 A5 F0    LDA osw_x            ; point ptr at the 4-byte address in the parameter block
8AF2 18      CLC
8AF3 69 01    ADC #001
8AF5 85 BC    STA ptr
8AF7 A5 F1    LDA osw_y
8AF9 69 00    ADC #000
8AFB 85 BD    STA ptr+1
8AFD 20 76 A2 JSR LA276            ; copy the destination address to ptr2..&C3 and set up Tube transfer if needed
8B00 A5 EF    LDA osw_a
8B02 48      PHA
8B03 A5 F0    LDA osw_x
8B05 48      PHA
8B06 A5 F1    LDA osw_y
8B08 48      PHA

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8B09 08      PHP
8B0A 58      CLI
8B0B AE 9B 0E LDX cur_req          ; X = current request slot
8B0E BD A0 0E LDA req_buf,X
8B11 85 BC    STA ptr
8B13 BD A1 0E LDA req_buf+1,X
8B16 85 BD    STA ptr+1
8B18 20 2B 8C JSR L8C2B          ; wait until the page has arrived, then lock it
8B1B 24 FF    BIT escape_flag
8B1D 30 2F    BMI L8B4E
8B1F A2 18    LDX #&18          ; 24 rows
8B21 A9 00    LDA #&00
8B23 85 B0    STA zp_b0        ; zp_b0 = double-height state
.L8B25
8B25 A0 02    LDY #&02          ; row tag: &FF means the row was never received
8B27 B1 BC    LDA (ptr),Y
8B29 10 0C    BPL L8B37
8B2B A0 28    LDY #&28          ; write 40 spaces instead
8B2D A9 20    LDA #&20
.L8B2F
8B2F 20 8C A2 JSR LA28C
8B32 88      DEY
8B33 D0 FA    BNE L8B2F
8B35 F0 14    BEQ L8B4B
.L8B37
8B37 A0 0A    LDY #&0A          ; characters are at offsets 10-49 of a row record
.L8B39
8B39 B1 BC    LDA (ptr),Y
8B3B 09 80    ORA #&80          ; make it a MODE 7 code
8B3D 20 8C A2 JSR LA28C
8B40 C9 8D    CMP #&8D          ; double height in this row?
8B42 D0 02    BNE L8B46
8B44 E6 B0    INC zp_b0
.L8B46
8B46 C8      INY
8B47 C0 32    CPY #&32          ; '2'
8B49 D0 EE    BNE L8B39
.L8B4B
8B4B CA      DEX
8B4C D0 0D    BNE L8B5B
.L8B4E
8B4E 28      PLP
8B4F 68      PLA
8B50 85 F1    STA osw_y
8B52 68      PLA
8B53 85 F0    STA osw_x
8B55 68      PLA
8B56 85 EF    STA osw_a
8B58 4C C8 8B JMP osw_8A_unlock
.L8B5B
8B5B A5 B0    LDA zp_b0        ; double-height state machine
8B5D F0 0E    BEQ L8B6D
8B5F 49 80    EOR #&80
8B61 85 B0    STA zp_b0
8B63 30 C0    BMI L8B25        ; first pass: send the same row again
8B65 A9 00    LDA #&00
8B67 85 B0    STA zp_b0
8B69 A9 02    LDA #&02          ; second pass done: skip the transmitted copy of the next row
8B6B D0 02    BNE L8B6F
.L8B6D
8B6D A9 01    LDA #&01          ; normal: advance one row record
.L8B6F
8B6F 20 1C 8D JSR row_offset    ; ptr += A * 48 (row records are 48 bytes apart)
8B72 4C 25 8B JMP L8B25

;-----
; OSWORD &7A &89: TEST ARRIVED [&89, result]
; Result &FF if a new copy of the current page has arrived (and is complete), else 0.
; On success the page is LOCKED so that the interrupt handler won't overwrite it while
; it is read; UNLOCK (&8A) or TRANSFER releases it.
;
; REQUEST SLOTS: 25 slots of 10 bytes at &0EA0. Slot 0 is the current page; the others
; hold kept pages and pages prefetched through links. cur_req selects the slot OSWORD
; calls act on, but this version only ever sets it to 0.
; +0,+1 page buffer address +2..+8 page digits: mag, tens, units, sub-code x4
; +9 state: &40 = active (digits: bits 0-3 value, bit 7 = wild)
; Status bits are kept in bit 6 of the digit bytes:
; req_mag b6 LOCKED req_s1 b6 VALID (a complete copy is in the buffer)
; req_tens b6 ARRIVED req_s2 b6 CHECK WORD present
; req_units b6 CAPTURING req_s3 b6 CHECKSUM OK
; req_s4 b6 KEPT (f3)
;-----
.osw_89_test_arrived
8B75 AE 9B 0E LDX cur_req
8B78 BD A3 0E LDA req_tens,X    ; new arrival?
8B7B 2A      ROL A
8B7C 10 33    BPL L8BB1
8B7E BD A2 0E LDA req_mag,X
8B81 09 40    ORA #&40          ; lock it
8B83 9D A2 0E STA req_mag,X
8B86 BD A4 0E LDA req_units,X   ; capture of a newer copy already started?

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8B89 2A      ROL A
8B8A 30 1D   BMI L8BA9
8B8C BD A3 0E LDA req_tens,X
8B8F 2A      ROL A
8B90 10 17   BPL L8BA9 ; (re-test after locking)
8B92 BD A3 0E LDA req_tens,X ; consume the "arrived" event
8B95 29 BF   AND #&BF
8B97 9D A3 0E STA req_tens,X
8B9A BD A5 0E LDA req_s1,X ; complete copy available?
8B9D 2A      ROL A
8B9E 10 11   BPL L8BB1
8BA0 A0 01   LDY #&01
8BA2 A9 FF   LDA #&FF
8BA4 91 F0   STA (osw_x),Y
8BA6 4C 11 82 JMP svc_exit_claimed
.L8BA9
8BA9 BD A2 0E LDA req_mag,X
8BAC 29 BF   AND #&BF
8BAE 9D A2 0E STA req_mag,X
.L8BB1
8BB1 A0 01   LDY #&01
8BB3 A9 00   LDA #&00
8BB5 91 F0   STA (osw_x),Y
8BB7 4C 11 82 JMP svc_exit_claimed
;-----
; OSWORD &7A &9A: UNLOCK and also forget that the buffer holds a valid page.
;-----
.osw_9A_unlock_variant
8BBA 20 CE 8B JSR unlock_current
8BBD BD A5 0E LDA req_s1,X
8BC0 29 BF   AND #&BF
8BC2 9D A5 0E STA req_s1,X
8BC5 4C 11 82 JMP svc_exit_claimed
; OSWORD &7A &8A: UNLOCK PAGE
.osw_8A_unlock
8BC8 20 CE 8B JSR unlock_current
8BCB 4C 11 82 JMP svc_exit_claimed
;-----
; Clear LOCKED and ARRIVED on the current slot. If a capture was in progress the copy in
; the buffer is incomplete, so VALID and CAPTURING are cleared too.
;-----
.unlock_current
8BCE AE 9B 0E LDX cur_req
8BD1 BD A3 0E LDA req_tens,X
8BD4 29 BF   AND #&BF
8BD6 9D A3 0E STA req_tens,X
8BD9 BD A4 0E LDA req_units,X
8BDC 48      PHA
8BDD 29 BF   AND #&BF
8BDF 9D A4 0E STA req_units,X
8BE2 68      PLA
8BE3 2A      ROL A
8BE4 10 08   BPL L8BEE
8BE6 BD A5 0E LDA req_s1,X
8BE9 29 BF   AND #&BF
8BEB 9D A5 0E STA req_s1,X
.L8BEE
8BEE BD A2 0E LDA req_mag,X
8BF1 29 BF   AND #&BF
8BF3 9D A2 0E STA req_mag,X
8BF6 60      RTS
;-----
; OSWORD &7A &80 RELOAD, &86 CHAINED PAGE, &88 HEAD OF CHAIN
; In this version all three share one implementation: follow link 0 of the current
; page's row 27 (L9420 with A=0), request that page, and wait for it.
;
; Stack trick: &8C04 is pushed as a return address and three zeros as a dummy
; service-call frame. L9420 eventually leaves through svc_exit_claimed, which pops the
; dummy frame and "returns" to &8C05 (osw_95_internal) to wait for the page; that then
; exits through the real frame.
;-----
.osw_80_86_88_reload_chain
8BF7 A9 8C   LDA #&8C
8BF9 48      PHA
8BFA A9 04   LDA #&04
8BFC 48      PHA
8BFD A9 00   LDA #&00
8BFF 48      PHA
8C00 48      PHA
8C01 48      PHA
8C02 4C 20 94 JMP L9420
;-----
; OSWORD &7A &95: WAIT FOR PAGE (with checksum handling)
; Reset the read position, then wait for the current page to arrive, honouring *OPT2.
; Interrupts are enabled while waiting.
;-----
.osw_95_wait_page
8C05 20 E8 97 JSR reset_read_position
8C08 A5 EF   LDA osw_a
8C0A 48      PHA

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8C0B A5 F0 LDA osw_x
8C0D 48 PHA
8C0E A5 F1 LDA osw_y
8C10 48 PHA
8C11 08 PHP
8C12 58 CLI
8C13 20 C7 8C JSR L8CC7
8C16 28 PLP
8C17 68 PLA
8C18 85 F1 STA osw_y
8C1A 68 PLA
8C1B 85 F0 STA osw_x
8C1D 68 PLA
8C1E 85 EF STA osw_a
8C20 4C 11 82 JMP svc_exit_claimed
;-----
; Wait for the current page: loop until VALID, lock it, wait for any capture in progress
; to finish, and retry if the copy was invalidated meanwhile. ESCAPE aborts.
;-----
.L8C23
8C23 BD A2 0E LDA req_mag,X
8C26 29 BF AND #&BF
8C28 9D A2 0E STA req_mag,X
.L8C2B
8C2B 24 FF BIT escape_flag
8C2D 30 21 BMI L8C50
8C2F AE 9B 0E LDX cur_req
8C32 BD A5 0E LDA req_s1,X
8C35 2A ROL A
8C36 10 F3 BPL L8C2B
8C38 BD A2 0E LDA req_mag,X
8C3B 09 40 ORA #&40
8C3D 9D A2 0E STA req_mag,X
.L8C40
8C40 24 FF BIT escape_flag
8C42 30 0C BMI L8C50
8C44 BD A4 0E LDA req_units,X
8C47 2A ROL A
8C48 30 F6 BMI L8C40
8C4A BD A5 0E LDA req_s1,X
8C4D 2A ROL A
8C4E 10 D3 BPL L8C23
.L8C50
8C50 60 RTS
;-----
; CHECKSUM FAILED: act on *OPT2 (opt2_errors)
; 0 ignore the error and accept the page
; 1 retry: wait for the page to be transmitted again (default)
; 2 give error &CA "Bad checksum on page"
; While retrying, the four sub-code digits are made wild (bit 7) so that any version of
; the page will do. "Bad data. Retrying..." is printed when Telesoftware messages are on.
;-----
.L8C51
8C51 AD 98 0E LDA opt2_errors
8C54 D0 03 BNE L8C59
8C56 4C FB 8C JMP L8CFB
.L8C59
8C59 C9 02 CMP #&02
8C5B D0 19 BNE L8C76
8C5D 20 98 A1 JSR error_inline
8C60 .str '\xcaBad checksum on page\x00'
.L8C76
8C76 20 A2 90 JSR clear_capture_flags ; abandon the bad copy
8C79 BD A5 0E LDA req_s1,X
8C7C 09 80 ORA #&80
8C7E 9D A5 0E STA req_s1,X
8C81 BD A6 0E LDA req_s2,X
8C84 09 80 ORA #&80
8C86 9D A6 0E STA req_s2,X
8C89 BD A7 0E LDA req_s3,X
8C8C 09 80 ORA #&80
8C8E 9D A7 0E STA req_s3,X
8C91 BD A8 0E LDA req_s4,X
8C94 09 80 ORA #&80
8C96 9D A8 0E STA req_s4,X
8C99 AD 94 16 LDA opt1_msgs ; Telesoftware *OPT1 'Searching' messages enabled?
8C9C 10 21 BPL L8CBF
8C9E 8A TXA
8C9F 48 PHA
8CA0 20 AF A2 JSR print_inline
8CA3 .str '\r\nBad data. Retrying...\r\n\x00'
8CBD 68 PLA
8CBE AA TAX
.L8CBF
8CBF BD A2 0E LDA req_mag,X
8CC2 29 BF AND #&BF
8CC4 9D A2 0E STA req_mag,X
;-----
; Wait for the current page, then check its checksum (if it had one) and apply *OPT2.
;-----

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.L8CC7
8CC7 24 FF BIT escape_flag
8CC9 30 50 BMI L8D1B
8CCB AE 9B 0E LDX cur_req
8CCE BD A5 0E LDA req_s1,X
8CD1 2A ROL A
8CD2 10 F3 BPL L8CC7
8CD4 BD A2 0E LDA req_mag,X
8CD7 09 40 ORA #&40
8CD9 9D A2 0E STA req_mag,X

.L8CDC
8CDC 24 FF BIT escape_flag
8CDE 30 3B BMI L8D1B
8CE0 BD A4 0E LDA req_units,X
8CE3 2A ROL A
8CE4 30 F6 BMI L8CDC
8CE6 BD A5 0E LDA req_s1,X
8CE9 2A ROL A
8CEA 10 D3 BPL L8CBF
8CEC BD A6 0E LDA req_s2,X ; page carried a check word?
8CEF 2A ROL A
8CF0 10 06 BPL L8CF8
8CF2 BD A7 0E LDA req_s3,X ; and it matched?
8CF5 2A ROL A
8CF6 30 03 BMI L8CFB

.L8CF8
8CF8 4C 51 8C JMP L8C51
;
; Page accepted: clear the wild bits on the sub-code digits, so the request is pinned to
; the version just received.
;
.L8CFB
8CFB BD A5 0E LDA req_s1,X
8CFE 29 7F AND #&7F
8D00 9D A5 0E STA req_s1,X
8D03 BD A6 0E LDA req_s2,X
8D06 29 7F AND #&7F
8D08 9D A6 0E STA req_s2,X
8D0B BD A7 0E LDA req_s3,X
8D0E 29 7F AND #&7F
8D10 9D A7 0E STA req_s3,X
8D13 BD A8 0E LDA req_s4,X
8D16 29 7F AND #&7F
8D18 9D A8 0E STA req_s4,X

.L8D1B
8D1B 60 RTS
;
; ptr += A * 48. Page buffers hold one 48-byte record per row.
; (A*16 is formed first; a carry out adds the 512 that A*16 overflowed by.)
;
.row_offset
8D1C A0 04 LDY #&04

.L8D1E
8D1E 0A ASL A
8D1F 88 DEY
8D20 D0 FC BNE L8D1E
8D22 48 PHA
8D23 90 06 BCC L8D2B
8D25 A9 02 LDA #&02
8D27 65 BD ADC ptr+1
8D29 85 BD STA ptr+1

.L8D2B
8D2B A0 03 LDY #&03

.L8D2D
8D2D 18 CLC
8D2E 68 PLA
8D2F 48 PHA
8D30 65 BC ADC ptr
8D32 85 BC STA ptr
8D34 A5 BD LDA ptr+1
8D36 69 00 ADC #&00
8D38 85 BD STA ptr+1
8D3A 88 DEY
8D3B D0 F0 BNE L8D2D
8D3D 68 PLA
8D3E 60 RTS
;
; SERVICE CALL 5: INTERRUPT (the heart of the system)
;
; The adapter interrupts once per field, after the vertical-blanking lines that can carry
; teletext. By then the hardware has written every teletext line it found into its 1K
; RAM: 16 slots of 64 bytes, one per line. A slot whose first byte is non-zero holds:
; byte 1-2 magazine and row address (Hamming 8/4)
; byte 3-42 40 data bytes; for row 0, 8 Hamming header bytes then 32 characters
; This handler does what a dedicated SAA5040 "TAC" chip would do in a normal decoder:
; 1. switch capture off so the CPU owns the RAM
; 2. for each full slot: Hamming-decode magazine and row, then
; row 0 → a new page header in that magazine (L8E58)
; mag 8 row 30 → broadcast service data (time, date, status) (8DC3)
; other rows → store in the buffer of the page being captured in that magazine

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; and mark the slot empty
; 3. finish with the checksum of the current page, clear the status latch (TT_STATCLR)
; and switch capture back on
; If DOR (data overrun) is set, a field was missed: every capture in progress is abandoned
; because rows may be missing.
;-----
.svc05_irq_capture
8D3F 2C 10 FC BIT TT_STATUS_CTRL ; b7 = INT: did the adapter interrupt?
8D42 10 07 BPL L8D4B
8D44 A9 08 LDA #00
8D46 2C 8E 0E BIT ctrl_shadow ; and are TFS interrupts enabled (b3 of the shadow)?
8D49 D0 03 BNE L8D4E
.L8D4B
8D4B 4C 0B 82 JMP svc_exit_unclaimed
.L8D4E
8D4E 20 49 A2 JSR capture_off ; capture off, IRQ off: the CPU owns the adapter RAM from here
8D51 58 CLI ; allow other interrupts while we work
8D52 A0 00 LDY #000
.L8D54
8D54 8C 14 0E STY cur_slot_no
8D57 8C 11 FC STY TT_SLOT ; select line slot Y
8D5A AD 12 FC LDA TT_DATA ; first byte: non-zero = the hardware wrote a line here
8D5D D0 28 BNE slot_has_data
.next_slot
8D5F AC 14 0E LDY cur_slot_no
8D62 C8 INY
8D63 C0 10 CPY #010 ; all 16 slots done?
8D65 D0 ED BNE L8D54
8D67 AD 10 FC LDA TT_STATUS_CTRL ; read status again: b6 = DOR (data overrun)
8D6A 85 C4 STA irq_ptr
8D6C 24 C4 BIT irq_ptr
8D6E 50 03 BVC L8D73
8D70 20 8D 90 JSR abort_all_captures ; overrun: abandon every capture in progress
.L8D73
8D73 4C 05 91 JMP irq_finish_crc
.L8D76
8D76 20 8D 90 JSR abort_all_captures ; address byte uncorrectable: rows may be missing, so abandon every capture
.L8D79
8D79 AC 14 0E LDY cur_slot_no
8D7C 8C 11 FC STY TT_SLOT ; mark the slot empty
8D7F A9 00 LDA #000
8D81 8D 12 FC STA TT_DATA
8D84 4C 5F 8D JMP next_slot
;-----
; A line is waiting in slot Y. Mark it empty (write 0 to byte 0), then decode the
; magazine/row address: 3 bits of magazine and 5 bits of row spread over two
; Hamming bytes.
;-----
.slot_has_data
8D87 8C 11 FC STY TT_SLOT
8D8A A9 00 LDA #000
8D8C 8D 12 FC STA TT_DATA ; byte 0 := 0 (slot consumed); the counter now points at byte 1
8D8F AE 12 FC LDX TT_DATA ; first address byte
8D92 20 C1 90 JSR hamming_decode_count; decode (and count for the tuning meter)
8D95 30 DF BMI L8D76
8D97 48 PHA
8D98 29 07 AND #007 ; magazine 1-8 is sent as 1-7, 0 (= 8)
8D9A 8D 0C 0E STA cur_mag
8D9D A8 TAY
8D9E 68 PLA
8D9F C9 08 CMP #008 ; bit 3 is the low bit of the row number
8DA1 08 PHP
8DA2 AE 12 FC LDX TT_DATA ; second address byte
8DA5 20 C1 90 JSR hamming_decode_count
8DA8 10 04 BPL L8DAE
8DAA 28 PLP
8DAB 4C 76 8D JMP L8D76
.L8DAE
8DAE 28 PLP
8DAF 2A ROL A ; row = (byte2 << 1) | bit
8DB0 8D 0B 0E STA cur_row
8DB3 D0 03 BNE L8DB8
8DB5 4C 58 8E JMP row0_page_header ; row 0: page header
.L8DB8
8DB8 C0 00 CPY #000 ; magazine 8 (sent as 0) row 30?
8DBA D0 16 BNE L8DD2
8DBC AD 0B 0E LDA cur_row
8DBF C9 1E CMP #01E
8DC1 D0 0F BNE L8DD2
;-----
; MAGAZINE 8 ROW 30: BROADCAST SERVICE DATA PACKET → tsdp (&0E5C), 40 bytes:
; initial page, network id, time offset, Modified Julian Date, UTC time, status message.
;-----
.packet_8_30
8DC3 A9 5C LDA #05C
8DC5 85 C4 STA irq_ptr
8DC7 A9 0E LDA #00E
8DC9 85 C5 STA irq_ptr+1
8DCB A2 28 LDX #028
8DCD A0 00 LDY #000

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8DCF 4C 3C 8E JMP L8E3C
;-----
; Ordinary row: store it in the buffer of the page currently being captured in this
; magazine (mag_capture, &FF = none). Each buffer starts with two bytes:
; +0 capacity in 48-byte row records
; +1 next free record for rows 26-31 (starts at 26)
; Rows 0-25 go to the record with their own number; rows 26-31 (Links, enhancements)
; are added at the next free record. Each record holds the row number at +2 and the
; 40 data bytes at +10.
;-----
.L8DD2
8DD2 BE 15 0E LDX mag_capture,Y
8DD5 E8 INX ; no capture in progress in this magazine: ignore the row
8DD6 F0 87 BEQ next_slot
8DD8 BD 9F 0E LDA &0E9F,X ; X = slot + 1, so &0E9F,X / req_buf,X are the slot's buffer address
8DDB 85 C4 STA irq_ptr
8DDD BD A0 0E LDA req_buf,X
8DE0 85 C5 STA irq_ptr+1
8DE2 AD 0B 0E LDA cur_row
8DE5 C9 1A CMP #&1A ; row 26 or above?
8DE7 A0 01 LDY #&01
8DE9 90 08 BCC L8DF3
8DEB B1 C4 LDA (irq_ptr),Y ; allocate the next extra-row record
8DED AA TAX
8DEE 69 00 ADC #&00
8DF0 91 C4 STA (irq_ptr),Y
8DF2 8A TXA
.L8DF3
8DF3 88 DEY
8DF4 D1 C4 CMP (irq_ptr),Y ; beyond the buffer's capacity: drop the row
8DF6 B0 4D BCS L8E45
8DF8 8D 08 0E STA row_tmp ; row_tmp = record number; multiply by 48
8DFB A9 00 LDA #&00
8DFD 8D 09 0E STA row_tmp_hi
8E00 A0 04 LDY #&04
.L8E02
8E02 18 CLC
8E03 2E 08 0E ROL row_tmp
8E06 2E 09 0E ROL row_tmp_hi
8E09 88 DEY
8E0A D0 F6 BNE L8E02
8E0C 18 CLC
8E0D A5 C4 LDA irq_ptr
8E0F 6D 08 0E ADC row_tmp
8E12 85 C4 STA irq_ptr
8E14 A5 C5 LDA irq_ptr+1
8E16 6D 09 0E ADC row_tmp_hi
8E19 85 C5 STA irq_ptr+1
8E1B 18 CLC
8E1C 2E 08 0E ROL row_tmp
8E1F 2E 09 0E ROL row_tmp_hi
8E22 18 CLC
8E23 A5 C4 LDA irq_ptr
8E25 6D 08 0E ADC row_tmp
8E28 85 C4 STA irq_ptr
8E2A A5 C5 LDA irq_ptr+1
8E2C 6D 09 0E ADC row_tmp_hi
8E2F 85 C5 STA irq_ptr+1
8E31 A0 02 LDY #&02 ; record +2 = row number (marks the row as received)
8E33 AD 0B 0E LDA cur_row
8E36 91 C4 STA (irq_ptr),Y
8E38 A0 0A LDY #&0A ; copy 40 data bytes
8E3A A2 28 LDX #&28
.L8E3C
8E3C AD 12 FC LDA TT_DATA ; read the next byte from the adapter (address auto-increments)
8E3F 91 C4 STA (irq_ptr),Y
8E41 C8 INY
8E42 CA DEX
8E43 D0 F7 BNE L8E3C
.L8E45
8E45 4C 5F 8D JMP next_slot
;-----
; Bad Hamming code in a header: abandon the capture in this magazine.
;-----
.L8E48
8E48 AC 0C 0E LDY cur_mag
8E4B BE 15 0E LDX mag_capture,Y
8E4E E8 INX
8E4F F0 04 BEQ L8E55
8E51 CA DEX
8E52 20 9D 90 JSR abort_capture
.L8E55
8E55 4C 79 8D JMP L8D79
;-----
; ROW 0: PAGE HEADER
; Decode the 8 Hamming header bytes into hdr_bytes: page units, tens, sub-code S1-S4
; (the sub-code bytes also carry control bits C4-C6), C7-C10 and C11-C14. The control
; bits the decoder acts on are gathered into hdr_ctrl:
; b7 C4 erase page b6 C8 update b3 C10 inhibit display
; b2 C9 interrupted seq b1 C7 suppress header b0 C11 magazine serial

```

```
; Then:
; - copy the 32 header characters to hdr_dest (rolling headers) unless C7, C9 or C10 is
;   set; in parallel-magazine mode only headers from the requested page's magazine are shown
; - the header ends the previous page in this magazine: if one was being captured it
;   is now complete (VALID + ARRIVED)
; - compare the new page number with every active request; on a match start capturing
;   it (CAPTURING) into that request's buffer
;-----
```

.row0_page_header

```
8E58 A0 00 LDY #000
.L8E5A
8E5A AE 12 FC LDX TT_DATA ; 8 header bytes
8E5D 20 C1 90 JSR hamming_decode_count
8E60 30 E6 BMI L8E48
8E62 99 00 0E STA hdr_bytes,Y
8E65 C8 INY
8E66 C0 08 CPY #008
8E68 D0 F0 BNE L8E5A
8E6A A9 00 LDA #000
8E6C 8D 13 0E STA hdr_ctrl
8E6F AD 05 0E LDA &0E05 ; sub-code thousands (2 bits)
8E72 29 03 AND #003
8E74 8D 0F 0E STA hdr_s1
8E77 AD 04 0E LDA &0E04 ; sub-code hundreds (4 bits)
8E7A 29 0F AND #00F
8E7C 8D 10 0E STA hdr_s2
8E7F AD 03 0E LDA &0E03 ; sub-code tens (3 bits)
8E82 29 07 AND #007
8E84 8D 11 0E STA hdr_s3
8E87 AD 02 0E LDA &0E02 ; sub-code units (4 bits)
8E8A 29 0F AND #00F
8E8C 8D 12 0E STA hdr_s4
8E8F AD 01 0E LDA &0E01 ; page tens
8E92 29 0F AND #00F
8E94 8D 0D 0E STA hdr_tens
8E97 AD 00 0E LDA hdr_bytes ; page units
8E9A 29 0F AND #00F
8E9C 8D 0E 0E STA hdr_units
8E9F A9 04 LDA #004 ; C9 interrupted sequence
8EA1 2C 06 0E BIT &0E06
8EA4 F0 08 BEQ L8EAE
8EA6 AD 13 0E LDA hdr_ctrl
8EA9 09 04 ORA #004
8EAB 8D 13 0E STA hdr_ctrl
.L8EAE
8EAE A9 08 LDA #008 ; C10 inhibit display
8EB0 2C 06 0E BIT &0E06
8EB3 F0 08 BEQ L8EBD
8EB5 AD 13 0E LDA hdr_ctrl
8EB8 09 08 ORA #008
8EBA 8D 13 0E STA hdr_ctrl
.L8EBD
8EBD A9 01 LDA #001 ; C11 magazine serial
8EBF 2C 07 0E BIT &0E07
8EC2 F0 08 BEQ L8ECC
8EC4 AD 13 0E LDA hdr_ctrl
8EC7 09 01 ORA #001
8EC9 8D 13 0E STA hdr_ctrl
.L8ECC
8ECC A9 08 LDA #008 ; C4 erase page
8ECE 2C 03 0E BIT &0E03
8ED1 F0 08 BEQ L8EDB
8ED3 AD 13 0E LDA hdr_ctrl
8ED6 09 80 ORA #80
8ED8 8D 13 0E STA hdr_ctrl
.L8EDB
8EDB A9 02 LDA #002 ; C8 update
8EDD 2C 06 0E BIT &0E06
8EE0 F0 08 BEQ L8EEA
8EE2 AD 13 0E LDA hdr_ctrl
8EE5 09 40 ORA #40
8EE7 8D 13 0E STA hdr_ctrl
.L8EEA
8EEA A9 01 LDA #001 ; C7 suppress header
8EEC 2C 06 0E BIT &0E06
8EEF F0 08 BEQ L8EF9
8EF1 AD 13 0E LDA hdr_ctrl
8EF4 09 02 ORA #002
8EF6 8D 13 0E STA hdr_ctrl
.L8EF9
8EF9 AD 9D 0E LDA hdr_dest+1 ; rolling headers wanted?
8EFC F0 4B BEQ L8F49
8EFE A9 04 LDA #004
8F00 2C 13 0E BIT hdr_ctrl
8F03 D0 44 BNE L8F49
8F05 A9 02 LDA #002
8F07 2C 13 0E BIT hdr_ctrl
8F0A D0 3D BNE L8F49
8F0C A9 08 LDA #008
8F0E 2C 13 0E BIT hdr_ctrl
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8F11 D0 36 BNE L8F49
8F13 A9 01 LDA #01
8F15 2C 13 0E BIT hdr_ctrl
8F18 D0 16 BNE L8F30 ; serial mode: every header is shown
8F1A AE 9B 0E LDX cur_req ; parallel mode: skip headers from other magazines when the current page names a magazine
8F1D BD A9 0E LDA req_state,X
8F20 C9 40 CMP #040 ; '@'
8F22 D0 0C BNE L8F30
8F24 BD A2 0E LDA req_mag,X
8F27 30 07 BMI L8F30
8F29 29 07 AND #07
8F2B CD 0C 0E CMP cur_mag
8F2E D0 19 BNE L8F49
.L8F30
8F30 AD 9C 0E LDA hdr_dest ; write the 32 header characters for display
8F33 85 C4 STA irq_ptr
8F35 AD 9D 0E LDA hdr_dest+1
8F38 85 C5 STA irq_ptr+1
8F3A A0 00 LDY #00
8F3C A2 20 LDX #020
.L8F3E
8F3E AD 12 FC LDA TT_DATA
8F41 09 80 ORA #080 ; MODE 7 code
8F43 91 C4 STA (irq_ptr),Y
8F45 C8 INY
8F46 CA DEX
8F47 D0 F5 BNE L8F3E
.L8F49
8F49 AC 0C 0E LDY cur_mag ; was a page being captured in this magazine?
8F4C BE 15 0E LDX mag_capture,Y
8F4F E8 INX
8F50 F0 60 BEQ L8FB2
8F52 E8 INX ; X = slot + 3, so req_buf,X walks req_tens..req_s4 of that slot
8F53 E8 INX
8F54 A0 01 LDY #01
.L8F56
8F56 BD A0 0E LDA req_buf,X ; compare the new header's page number with it (magazine already matches)
8F59 29 0F AND #0F
8F5B D9 0C 0E CMP cur_mag,Y
8F5E D0 0F BNE L8F6F
8F60 E8 INX
8F61 C8 INY
8F62 C0 07 CPY #07
8F64 D0 F0 BNE L8F56
8F66 BD 9B 0E LDA cur_req,X ; (&0E9B+X = req_units of the slot) same page, still capturing: keep going
8F69 2A ROL A
8F6A 10 03 BPL L8F6F
8F6C 4C 5F 8D JMP next_slot
.L8F6F
8F6F AC 0C 0E LDY cur_mag ; different page: the old capture is finished
8F72 BE 15 0E LDX mag_capture,Y
8F75 A9 FF LDA #FF
8F77 99 15 0E STA mag_capture,Y
8F7A E8 INX
8F7B F0 35 BEQ L8FB2
8F7D CA DEX
8F7E BD A9 0E LDA req_state,X
8F81 2A ROL A
8F82 10 2E BPL L8FB2
8F84 BD A4 0E LDA req_units,X ; was it CAPTURING?
8F87 2A ROL A
8F88 10 28 BPL L8FB2
8F8A BD A4 0E LDA req_units,X ; CAPTURING off, VALID on, ARRIVED on; checksum flags cleared
8F8D 29 BF AND #0BF
8F8F 9D A4 0E STA req_units,X
8F92 BD A5 0E LDA req_s1,X
8F95 09 40 ORA #040
8F97 9D A5 0E STA req_s1,X
8F9A BD A3 0E LDA req_tens,X
8F9D 09 40 ORA #040
8F9F 9D A3 0E STA req_tens,X
8FA2 BD A7 0E LDA req_s3,X
8FA5 29 BF AND #0BF
8FA7 9D A7 0E STA req_s3,X
8FAA BD A6 0E LDA req_s2,X
8FAD 29 BF AND #0BF
8FAF 9D A6 0E STA req_s2,X
;-----
; Look for a request (slots 0-24, 10 bytes apart) that wants this page.
;-----
.L8FB2
8FB2 A2 F6 LDX #0F6
8FB4 8E 0A 0E STX slot_scan
.L8FB7
8FB7 AD 0A 0E LDA slot_scan
8FBA C9 F0 CMP #0F0
8FBC D0 03 BNE L8FC1
8FBE 4C 5F 8D JMP next_slot
.L8FC1
8FC1 18 CLC

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8FC2 69 0A    ADC #&0A
8FC4 AA      TAX
8FC5 8D 0A 0E STA slot_scan
8FC8 BD A9 0E LDA req_state,X      ; must be an active slot (&40)
8FCB C9 40    CMP #&40             ; '@'
8FCD D0 E8    BNE L8FB7
8FCF BD A4 0E LDA req_units,X      ; marked CAPTURING? Only skip it if that capture is really still running
8FD2 2A      ROL A
8FD3 10 13    BPL L8FE8
8FD5 BD A2 0E LDA req_mag,X
8FD8 29 07    AND #&07
8FDA A8      TAY
8FDB B9 15 0E LDA mag_capture,Y
8FDE C9 FF    CMP #&FF
8FE0 D0 D5    BNE L8FB7
8FE2 20 A2 90 JSR clear_capture_flags
8FE5 4C B7 8F JMP L8FB7

.L8FE8
8FE8 BD A2 0E LDA req_mag,X      ; locked slots are not overwritten
8FEB 2A      ROL A
8FEC 30 C9    BMI L8FB7
8FEE BD A5 0E LDA req_s1,X      ; already hold a valid copy? Then only re-capture if C8 (update) is set, or the copy
8FF1 2A      ROL A
8FF2 10 13    BPL L9007
8FF4 2C 13 0E BIT hdr_ctrl
8FF7 70 0E    BVS L9007
8FF9 BD A7 0E LDA req_s3,X      ; failed its checksum (CHECK WORD present but CHECKSUM OK clear)
8FFC 2A      ROL A
8FFD 85 C4    STA irq_ptr
8FFF BD A6 0E LDA req_s2,X
9002 2A      ROL A
9003 45 C4    EOR irq_ptr
9005 10 B0    BPL L8FB7

.L9007
9007 A0 00    LDY #&00             ; compare the 7 digits; wild digits (bit 7) match any decimal digit
9009 E8      INX                   ; X = slot + 2; req_buf,X then walks the 7 digits
900A E8      INX

.L900B
900B BD A0 0E LDA req_buf,X
900E 30 09    BMI L9019
9010 29 0F    AND #&0F
9012 D9 0C 0E CMP cur_mag,Y
9015 D0 A0    BNE L8FB7
9017 F0 07    BEQ L9020

.L9019
9019 B9 0C 0E LDA cur_mag,Y
901C C9 0A    CMP #&0A
901E B0 97    BCS L8FB7

.L9020
9020 E8      INX
9021 C8      INY
9022 C0 07    CPY #&07
9024 D0 E5    BNE L900B
9026 AE 0A 0E LDX slot_scan      ; match: record the digits actually received in place of the wild ones
9029 A0 00    LDY #&00

.L902B
902B BD A2 0E LDA req_mag,X
902E 10 08    BPL L9038
9030 29 F0    AND #&F0
9032 19 0C 0E ORA cur_mag,Y
9035 9D A2 0E STA req_mag,X

.L9038
9038 E8      INX
9039 C8      INY
903A C0 07    CPY #&07
903C D0 ED    BNE L902B
903E AD 0A 0E LDA slot_scan      ; this magazine is now capturing into this slot
9041 AC 0C 0E LDY cur_mag
9044 99 15 0E STA mag_capture,Y
9047 AE 0A 0E LDX slot_scan
904A BD A4 0E LDA req_units,X      ; CAPTURING
904D 09 40    ORA #&40
904F 9D A4 0E STA req_units,X
9052 AE 0A 0E LDX slot_scan
9055 BD A5 0E LDA req_s1,X      ; valid copy + C8 update + no C4 erase: keep the old rows and overlay the new ones
9058 2A      ROL A
9059 10 07    BPL L9062
905B 2C 13 0E BIT hdr_ctrl
905E 30 02    BMI L9062
9060 70 06    BVS L9068

.L9062
9062 AE 0A 0E LDX slot_scan
9065 20 D6 90 JSR clear_page_buffer ; otherwise clear the page buffer (all rows "not received")

.L9068
9068 AD 14 0E LDA cur_slot_no      ; re-select the adapter slot and skip its flag and two address bytes, then store
906B 8D 11 FC STA TT_SLOT
906E A0 03    LDY #&03             ; the whole of row 0 (8 Hamming header bytes + 32 characters) as record 0

.L9070
9070 AD 12 FC LDA TT_DATA
9073 88      DEY

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9074 D0 FA BNE L9070
9076 AE 0A 0E LDX slot_scan
9079 BD A0 0E LDA req_buf,X
907C 85 C4 STA irq_ptr
907E BD A1 0E LDA req_buf+1,X
9081 85 C5 STA irq_ptr+1
9083 A9 1A LDA #&1A ; extra rows start at record 26
9085 C8 INY
9086 91 C4 STA (irq_ptr),Y
9088 A9 00 LDA #&00
908A 4C F3 8D JMP L8DF3
;-----
; Abandon the capture in every magazine (after DOR or a channel change).
; Called with each magazine's slot in turn.
;-----
.abort_all_captures
908D A0 07 LDY #&07
.L908F
908F BE 15 0E LDX mag_capture,Y
9092 E8 INX
9093 F0 04 BEQ L9099
9095 CA DEX
9096 20 9D 90 JSR abort_capture
.L9099
9099 88 DEY
909A 10 F3 BPL L908F
909C 60 RTS
;-----
; Abandon the capture in magazine Y (slot offset X): magazine idle, and the slot's
; CAPTURING, VALID and ARRIVED bits cleared.
;-----
.abort_capture
909D A9 FF LDA #&FF
909F 99 15 0E STA mag_capture,Y
;-----
; Clear CAPTURING, VALID and ARRIVED on slot X (the magazine is not touched).
;-----
.clear_capture_flags
90A2 BD A4 0E LDA req_units,X
90A5 29 BF AND #&BF
90A7 9D A4 0E STA req_units,X
90AA BD A5 0E LDA req_s1,X
90AD 29 BF AND #&BF
90AF 9D A5 0E STA req_s1,X
90B2 BD A3 0E LDA req_tens,X
90B5 29 BF AND #&BF
90B7 9D A3 0E STA req_tens,X
90BA 60 RTS
;-----
; X = received byte. Returns A = 4-bit value, N set if uncorrectable.
;-----
.hamming_decode
90BB BD 20 80 LDA hamming_table,X
90BE 29 8F AND #&8F
90C0 60 RTS
;-----
; As hamming_decode, also counting decoded bytes (ham_count) and errors (ham_errors,
; corrected or not) for the *TUNE quality meter.
;-----
.hamming_decode_count
90C1 EE 52 0E INC ham_count
90C4 BD 20 80 LDA hamming_table,X
90C7 85 C4 STA irq_ptr
90C9 24 C4 BIT irq_ptr
90CB 30 04 BMI L90D1 ; N: uncorrectable
90CD 50 05 BVC L90D4 ; V: corrected single-bit error
90CF 29 0F AND #&0F
.L90D1
90D1 EE 53 0E INC ham_errors
.L90D4
90D4 AA TAX
90D5 60 RTS
;-----
; Clear a page buffer: mark every row record as "not received" (row tag &FF).
;-----
.clear_page_buffer
90D6 BD A0 0E LDA req_buf,X
90D9 85 C4 STA irq_ptr
90DB BD A1 0E LDA req_buf+1,X
90DE 85 C5 STA irq_ptr+1
90E0 8A TXA
90E1 48 PHA
90E2 A2 1A LDX #&1A
90E4 A0 00 LDY #&00
90E6 B1 C4 LDA (irq_ptr),Y
90E8 C9 1A CMP #&1A
90EA B0 01 BCS L90ED
90EC AA TAX
.L90ED
90ED A0 02 LDY #&02

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.L90EF
90EF A5 C4 LDA irq_ptr
90F1 18 CLC
90F2 69 30 ADC #&30
90F4 85 C4 STA irq_ptr
90F6 90 02 BCC L90FA
90F8 E6 C5 INC irq_ptr+1

.L90FA
90FA A9 FF LDA #&FF
90FC 91 C4 STA (irq_ptr),Y
90FE 98 TYA
90FF CA DEX
9100 D0 ED BNE L90EF
9102 68 PLA
9103 AA TAX
9104 60 RTS
;-----
; END OF INTERRUPT: PAGE CHECKSUM (current page, slot 0 only)
; When slot 0 holds a newly completed page that hasn't been checked yet:
; - look through its extra rows for row 27 with designation code 0, which carries the
;   16-bit page check word (bytes &2E-&2F of the record)
; - none: mark checksum OK so that "No check" is shown and the page is accepted
; - found: set CHECK WORD present, compute the CRC over the page and compare
; The CRC covers row 0 characters 8-31 (the clock at the end of row 0 is excluded) and
; rows 1-25, 40 characters each, with missing rows counted as spaces: 1024 bytes in all.
;-----
 irq_finish_crc
9105 2C A9 0E BIT req_state ; slot 0 active?
9108 50 14 BVC L911E
910A 2C A5 0E BIT req_s1 ; complete?
910D 50 0F BVC L911E
910F 2C A7 0E BIT req_s3 ; checksum already done?
9112 70 0A BVS L911E
9114 2C A6 0E BIT req_s2
9117 70 05 BVS L911E
9119 2C A2 0E BIT req_mag ; not locked by a reader
911C 50 03 BVC L9121

.L911E
911E 4C 21 92 JMP irq_statclr_exit

.L9121
9121 A9 11 LDA #&11 ; start at column &11 of row 0
9123 8D 84 0E STA crc_col
9126 A9 00 LDA #&00
9128 8D 85 0E STA crc_row
912B AD A0 0E LDA req_buf
912E 85 C4 STA irq_ptr
9130 AD A1 0E LDA req_buf+1
9133 85 C5 STA irq_ptr+1
9135 A0 00 LDY #&00
9137 B1 C4 LDA (irq_ptr),Y ; any extra rows (count > 26)?
9139 C9 1B CMP #&1B
913B B0 0B BCS L9148

.L913D
913D AD A7 0E LDA req_s3 ; no row 27: treat the checksum as OK
9140 09 40 ORA #&40
9142 8D A7 0E STA req_s3
9145 4C 21 92 JMP irq_statclr_exit

.L9148
9148 C8 INY ; number of extra rows
9149 B1 C4 LDA (irq_ptr),Y
914B 38 SEC
914C E9 1B SBC #&1B
914E 48 PHA
914F 18 CLC ; ptr += &4E2 = 26 records * 48 + 2: first extra-row record
9150 A5 C4 LDA irq_ptr
9152 69 E2 ADC #&E2
9154 85 C4 STA irq_ptr
9156 A5 C5 LDA irq_ptr+1
9158 69 04 ADC #&04
915A 85 C5 STA irq_ptr+1
915C 68 PLA

.L915D
915D C9 00 CMP #&00
915F 30 DC BMI L913D
9161 48 PHA
9162 A0 00 LDY #&00
9164 B1 C4 LDA (irq_ptr),Y
9166 C9 1B CMP #&1B ; row 27?
9168 D0 1F BNE L9189
916A A0 08 LDY #&08 ; designation code (Hamming) must be 0
916C B1 C4 LDA (irq_ptr),Y
916E A8 TAY
916F B9 20 80 LDA hamming_table,Y
9172 30 15 BMI L9189
9174 29 0F AND #&0F
9176 D0 11 BNE L9189
9178 A0 2F LDY #&2F ; received check word
917A B1 C4 LDA (irq_ptr),Y
917C 8D 8C 0E STA crc_rx_hi
917F 88 DEY

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9180 B1 C4 LDA (irq_ptr),Y
9182 8D 8D 0E STA crc_rx_lo
9185 68 PLA
9186 4C 9D 91 JMP L919D
.L9189
9189 68 PLA
918A A8 TAY
918B 88 DEY
918C A5 C4 LDA irq_ptr
918E 18 CLC
918F 69 30 ADC #&30
9191 85 C4 STA irq_ptr
9193 A5 C5 LDA irq_ptr+1
9195 69 00 ADC #&00
9197 85 C5 STA irq_ptr+1
9199 98 TYA
919A 4C 5D 91 JMP L915D
.L919D
919D 20 2F 99 JSR prefetch_links ; prefetch linked pages named in row 27 (L992F)
91A0 AD A0 0E LDA req_buf
91A3 85 C4 STA irq_ptr
91A5 AD A1 0E LDA req_buf+1
91A8 85 C5 STA irq_ptr+1
91AA AD A6 0E LDA req_s2 ; CHECK WORD present
91AD 09 40 ORA #&40
91AF 8D A6 0E STA req_s2
91B2 A9 00 LDA #&00
91B4 8D 8A 0E STA crc_lo
91B7 8D 8B 0E STA crc_hi
.L91BA
91BA AC 84 0E LDY crc_col ; CRC loop: Y = column, crc_row = row
91BD AD 85 0E LDA crc_row
91C0 D0 09 BNE L91CB
91C2 C0 29 CPY #&29 ; in row 0 skip from column &29 to &31 (the clock)
91C4 D0 05 BNE L91CB
91C6 A0 31 LDY #&31
91C8 8C 84 0E STY crc_col
.L91CB
91CB AC 84 0E LDY crc_col
91CE C8 INY
91CF C0 32 CPY #&32 ; '2'
91D1 D0 19 BNE L91EC
91D3 18 CLC
91D4 69 01 ADC #&01
91D6 8D 85 0E STA crc_row
91D9 C9 1A CMP #&1A ; 26 rows done?
91DB F0 2A BEQ L9207
91DD A5 C4 LDA irq_ptr
91DF 18 CLC
91E0 69 30 ADC #&30
91E2 85 C4 STA irq_ptr
91E4 A5 C5 LDA irq_ptr+1
91E6 69 00 ADC #&00
91E8 85 C5 STA irq_ptr+1
91EA A0 0A LDY #&0A
.L91EC
91EC 8C 84 0E STY crc_col
91EF A0 02 LDY #&02
91F1 B1 C4 LDA (irq_ptr),Y
91F3 C9 FF CMP #&FF ; missing row?
91F5 D0 05 BNE L91FC
91F7 A9 20 LDA #&20 ; counts as spaces
91F9 4C 01 92 JMP L9201
.L91FC
91FC AC 84 0E LDY crc_col
91FF B1 C4 LDA (irq_ptr),Y
.L9201
9201 20 2A 92 JSR crc_byte ; add the byte to the CRC
9204 4C BA 91 JMP L91BA
.L9207
9207 A9 00 LDA #&00 ; compare the computed CRC with the received check word
9209 AC 8C 0E LDY crc_rx_hi
920C CC 8A 0E CPY crc_lo
920F D0 0A BNE L921B
9211 AC 8D 0E LDY crc_rx_lo
9214 CC 8B 0E CPY crc_hi
9217 D0 02 BNE L921B
9219 A9 40 LDA #&40 ; CHECKSUM OK
.L921B
921B 0D A7 0E ORA req_s3
921E 8D A7 0E STA req_s3
;-----
; Clear the INT and DOR flip-flops (any write to TT_STATCLR), re-enable capture and the
; adapter interrupt, and claim the interrupt.
;-----
.L921B_exit
9221 8D 13 FC STA TT_STATCLR
9224 20 5A A2 JSR capture_on
9227 4C 11 82 JMP svc_exit_claimed
;-----

```

; Add the byte in A to the 16-bit page CRC (crc_lo/crc_hi). This is the teletext page
; check word polynomial $x^{16} + x^{12} + x^9 + x^7 + 1$, worked a byte at a time with shifts.

;crc_byte

```
922A A8      TAY
922B AD 8A 0E LDA crc_lo
922E 8D 86 0E STA crc_tmp0
9231 AD 8B 0E LDA crc_hi
9234 8D 87 0E STA crc_tmp1
9237 A2 04      LDX #04
```

.L9239

```
9239 0E 8A 0E ASL crc_lo
923C 2E 8B 0E ROL crc_hi
923F CA      DEX
9240 D0 F7      BNE L9239
9242 AD 8B 0E LDA crc_hi
9245 8D 88 0E STA crc_tmp2
9248 A2 03      LDX #03
```

.L924A

```
924A 0E 8A 0E ASL crc_lo
924D 2E 8B 0E ROL crc_hi
9250 CA      DEX
9251 D0 F7      BNE L924A
9253 AD 8B 0E LDA crc_hi
9256 8D 89 0E STA crc_tmp3
9259 A2 02      LDX #02
```

.L925B

```
925B 0E 8A 0E ASL crc_lo
925E 2E 8B 0E ROL crc_hi
9261 CA      DEX
9262 D0 F7      BNE L925B
9264 98      TYA
9265 4D 87 0E EOR crc_tmp1
9268 4D 88 0E EOR crc_tmp2
926B 4D 89 0E EOR crc_tmp3
926E 4D 8B 0E EOR crc_hi
9271 8D 87 0E STA crc_tmp1
9274 2A      ROL A
9275 A9 00      LDA #00
9277 2A      ROL A
9278 4D 87 0E EOR crc_tmp1
927B 8D 8A 0E STA crc_lo
927E AD 86 0E LDA crc_tmp0
9281 8D 8B 0E STA crc_hi
9284 98      TYA
9285 60      RTS
```

; OSWORD &7A &8E: HAMMING DECODE [&8E, byte]. Result replaces the byte; &FF = error.

;osw_8E_hamming

```
9286 A0 01      LDY #01
9288 B1 F0      LDA (osw_x),Y
928A AA      TAX
928B BD 20 80 LDA hamming_table,X
928E 30 02      BMI L9292
9290 29 0F      AND #0F
```

.L9292

```
9292 91 F0      STA (osw_x),Y
9294 4C 11 82 JMP svc_exit_claimed
```

; OSWORD &7A &8D: READ TSDP [&8D, addr lo, addr hi]
; Copy the 20-character status message of the broadcast service data packet to addr.

;osw_8D_read_tsdp

```
9297 A0 01      LDY #01
9299 B1 F0      LDA (osw_x),Y
929B 85 BC      STA ptr
929D C8      INY
929E B1 F0      LDA (osw_x),Y
92A0 85 BD      STA ptr+1
92A2 A2 14      LDX #14
92A4 A0 00      LDY #00
```

.L92A6

```
92A6 BD 5C 0E LDA tsdp,X
92A9 91 BC      STA (ptr),Y
92AB E8      INX
92AC C8      INY
92AD C0 14      CPY #14
92AF D0 F5      BNE L92A6
92B1 4C 11 82 JMP svc_exit_claimed
```

; OSWORD &7A &9C: KEEP PAGE (f3)
; Result 0 and the current page is marked KEPT, if at least one active slot is not kept.
; Result 1 if every buffer is already kept ("Buffers full - Please release a page").

;osw_9C_keep_page

```
92B4 A0 00      LDY #00
```

.L92B6

```
92B6 B9 A9 0E LDA req_state,Y
92B9 2A      ROL A
```

```

92BA 10 06 BPL L92C2
92BC B9 A8 0E LDA req_s4,Y
92BF 2A ROL A
92C0 10 11 BPL L92D3
.L92C2
92C2 98 TYA
92C3 18 CLC
92C4 69 0A ADC #&0A
92C6 A8 TAY
92C7 C9 FA CMP #&FA
92C9 D0 EB BNE L92B6
92CB A9 01 LDA #&01
92CD A8 TAY
92CE 91 F0 STA (osw_x),Y
92D0 4C 11 82 JMP svc_exit_claimed
.L92D3
92D3 AE 9B 0E LDX cur_req
92D6 BD A8 0E LDA req_s4,X
92D9 09 40 ORA #&40
92DB 9D A8 0E STA req_s4,X
92DE A9 00 LDA #&00
92E0 A0 01 LDY #&01
92E2 91 F0 STA (osw_x),Y
92E4 4C 11 82 JMP svc_exit_claimed
;-----
; OSWORD &7A &9D: RELEASE PAGE (SHIFT f3)
; Clears KEPT on the current slot. Result 0 if it was kept, 1 if not ("Page was not kept").
;-----
.osw_9D_release_page
92E7 AE 9B 0E LDX cur_req
92EA BD A8 0E LDA req_s4,X
92ED 48 PHA
92EE 29 BF AND #&BF
92F0 9D A8 0E STA req_s4,X
92F3 68 PLA
92F4 0A ASL A
92F5 10 09 BPL L9300
92F7 A9 00 LDA #&00
.L92F9
92F9 A0 01 LDY #&01
92FB 91 F0 STA (osw_x),Y
92FD 4C 11 82 JMP svc_exit_claimed
.L9300
9300 A9 01 LDA #&01
9302 4C F9 92 JMP L92F9
;-----
; OSWORD &7A &9E: SWAP THE CURRENT SLOT INTO SLOT 0
; If the current request is not slot 0, swap it into slot 0 (L9714) with a page number of
; 000 and wild sub-code, then make slot 0 current.
;-----
.osw_9E_swap_into_slot0
9305 AD 9B 0E LDA cur_req
9308 F0 1F BEQ L9329
930A 8D 1E 0E STA req_scan
930D A9 00 LDA #&00
930F 8D 4A 0E STA want_mag
9312 8D 4B 0E STA want_tens
9315 8D 4C 0E STA want_units
9318 A9 FF LDA #&FF
931A 8D 4D 0E STA want_s1
931D 8D 4E 0E STA want_s2
9320 8D 4F 0E STA want_s3
9323 8D 50 0E STA want_s4
9326 20 14 97 JSR swap_slots
.L9329
9329 A9 00 LDA #&00
932B 8D 9B 0E STA cur_req
932E 4C 11 82 JMP svc_exit_claimed
;-----
; OSWORD &7A &9F: SELECT SLOT 0 AND RE-SHOW IT
; Makes slot 0 current; if it holds a valid page, flag it ARRIVED again so that the next
; TEST ARRIVED returns it. (The BIT req_s3 here has no effect.)
;-----
.osw_9F_reshow_slot0
9331 AD 9B 0E LDA cur_req
9334 A9 00 LDA #&00
9336 8D 9B 0E STA cur_req
9339 2C A5 0E BIT req_s1
933C 50 0B BVC L9349
933E 2C A7 0E BIT req_s3
9341 AD A3 0E LDA req_tens
9344 09 40 ORA #&40
9346 8D A3 0E STA req_tens
.L9349
9349 A9 00 LDA #&00
934B A0 01 LDY #&01
934D 91 F0 STA (osw_x),Y
934F 4C 11 82 JMP svc_exit_claimed
9352 .db A9 01 4C 4B 93 ; ..LK.
;-----

```

```

; OSWORD &7A &9B: ALLOCATE PAGE BUFFERS
; Carve page buffers out of user memory, starting at OSHWM (PAGE), and give one to each
; request slot from 1 upwards until the next would pass &7100 (keeping clear of the MODE 7
; screen). Each buffer is &602 bytes: 2 header bytes + 32 row records of 48 bytes.
; Used by terminal mode, which as a language owns that memory.
;-----
.osw_9B_alloc_buffers
9357 A9 83 LDA #&83
9359 20 F4 FF JSR OSBYTE ; OSBYTE &83: read OSHWM (PAGE)
935C 86 B0 STX zp_b0
935E 84 B1 STY zp_b1
9360 A0 00 LDY #&00
.L9362
9362 A9 3A LDA #&3A
9364 85 BC STA ptr
9366 A9 0E LDA #&0E
9368 85 BD STA ptr+1
936A C8 INY
936B A5 B0 LDA zp_b0
936D 8D 3C 0E STA osw_blk+2
9370 18 CLC
9371 69 02 ADC #&02
9373 85 B0 STA zp_b0
9375 A5 B1 LDA zp_b1
9377 8D 3D 0E STA osw_blk+3
937A 69 06 ADC #&06
937C 85 B1 STA zp_b1
937E C9 71 CMP #&71 ; stop below &7100
9380 B0 50 BCS L93D2
9382 8A TXA
9383 48 PHA
9384 98 TYA
9385 48 PHA
9386 8C 3B 0E STY osw_blk+1 ; osw_blk is reused as a parameter block for the slot set-up below
9389 A0 01 LDY #&01
938B B1 BC LDA (ptr),Y
938D AA TAX
938E A9 00 LDA #&00
.L9390
9390 CA DEX
9391 30 06 BMI L9399
9393 18 CLC
9394 69 0A ADC #&0A
9396 4C 90 93 JMP L9390
.L9399
9399 AA TAX
939A A9 00 LDA #&00
939C 9D A9 0E STA req_state,X
939F C8 INY
93A0 B1 BC LDA (ptr),Y
93A2 9D A0 0E STA req_buf,X
93A5 C8 INY
93A6 B1 BC LDA (ptr),Y
93A8 9D A1 0E STA req_buf+1,X
93AB A9 00 LDA #&00
93AD A0 07 LDY #&07
.L93AF
93AF 9D A2 0E STA req_mag,X
93B2 E8 INX
93B3 88 DEY
93B4 D0 F9 BNE L93AF
93B6 A9 40 LDA #&40 ; new slot is active
93B8 9D A2 0E STA req_mag,X
93BB BD 99 0E LDA prev_language,X
93BE 85 BC STA ptr
93C0 BD 9A 0E LDA prev_fs,X
93C3 85 BD STA ptr+1
93C5 A0 00 LDY #&00
93C7 A9 20 LDA #&20 ; buffer capacity = 32 records
93C9 91 BC STA (ptr),Y
93CB 68 PLA
93CC A8 TAY
93CD 68 PLA
93CE AA TAX
93CF 4C 62 93 JMP L9362
.L93D2
93D2 4C 11 82 JMP svc_exit_claimed
;-----
; Two 24-byte tables for links 0-23:
; &93D5 row 27 designation code that carries the link (links 0-5 in 27/0, 6-11 in 27/1, ...)
; &93ED offset of the 6-byte link within that row record
;-----
.link_tables
93D5 .db 00 00 00 00 00 00 00 01 01 01 01 01 01 02 02 02 02 ; .....
93E5 .db 02 02 03 03 03 03 03 03 09 0F 15 1B 21 27 09 0F ; .....!'.
93F5 .db 15 1B 21 27 09 0F 15 1B 21 27 09 0F 15 1B 21 27 ; ..!'.!'.!'.!'.
;-----
; OSWORD &7A &96: SELECT LINK [&96, link 0-23, result]
; Find link n in the current page's row 27 and request the page it points to.
; Result: 0 ok, 1 no such link, 2 no links present, 3 link number out of range.

```


; Returns through &989B (pushed as &989A for RTS) to set up the new request.

;osw_96_select_link

```
9405 A0 01 LDY #&01
9407 B1 F0 LDA (osw_x),Y
9409 C9 18 CMP #&18
940B 90 09 BCC L9416
940D A9 03 LDA #&03
940F A0 02 LDY #&02
9411 91 F0 STA (osw_x),Y
9413 4C 11 82 JMP svc_exit_claimed
```

.L9416

```
9416 A8 TAY
9417 A9 98 LDA #&98
9419 48 PHA
941A A9 9A LDA #&9A
941C 48 PHA
941D 4C 27 94 JMP L9427
```

; Entry used by RELOAD / CHAINED / HEAD OF CHAIN: as above, continuing at &980C.

.L9420

```
9420 A8 TAY
9421 A9 98 LDA #&98
9423 48 PHA
9424 A9 0B LDA #&0B
9426 48 PHA
```

.L9427

```
9427 B9 D5 93 LDA link_tables,Y ; designation code wanted
942A 85 B0 STA zp_b0
942C B9 ED 93 LDA &93ED,Y ; offset of the link in the record
942F 85 B1 STA zp_b1
9431 AD A0 0E LDA req_buf ; slot 0's page buffer
9434 85 BC STA ptr
9436 AD A1 0E LDA req_buf+1
9439 85 BD STA ptr+1
943B A0 02 LDY #&02
943D A9 02 LDA #&02 ; assume "no links present"
943F 91 F0 STA (osw_x),Y
9441 A0 01 LDY #&01
9443 B1 BC LDA (ptr),Y ; number of extra rows + 26
9445 38 SEC
9446 E9 1B SBC #&1B
9448 48 PHA
9449 18 CLC
944A A5 BC LDA ptr
944C 69 E2 ADC #&E2 ; first extra-row record (26*48 + 2)
944E 85 BC STA ptr
9450 A5 BD LDA ptr+1
9452 69 04 ADC #&04
9454 85 BD STA ptr+1
```

.L9456

```
9456 68 PLA
9457 49 00 EOR #&00
9459 10 05 BPL L9460
945B 68 PLA
945C 68 PLA
945D 4C 11 82 JMP svc_exit_claimed
```

.L9460

```
9460 38 SEC
9461 E9 01 SBC #&01
9463 48 PHA
9464 A0 00 LDY #&00
9466 B1 BC LDA (ptr),Y ; a row 27 record?
9468 C9 1B CMP #&1B
946A F0 03 BEQ L946F
946C 4C 00 95 JMP L9500
```

.L946F

```
946F A0 02 LDY #&02 ; links present, but not this one (yet)
9471 A9 01 LDA #&01
9473 91 F0 STA (osw_x),Y
9475 A0 08 LDY #&08 ; designation code
9477 B1 BC LDA (ptr),Y
9479 A8 TAY
947A B9 20 80 LDA hamming_table,Y
947D 30 06 BMI L9485
947F 29 0F AND #&0F
9481 C5 B0 CMP zp_b0
9483 F0 03 BEQ L9488
```

.L9485

```
9485 4C 00 95 JMP L9500
```

.L9488

```
9488 A5 B1 LDA zp_b1 ; decode the 6 link bytes: units, tens, S1, S2(+mag bit), S3, S4(+mag bits)
948A A8 TAY
948B B1 BC LDA (ptr),Y
948D AA TAX
948E 20 BB 90 JSR hamming_decode
9491 30 F2 BMI L9485
9493 8D 4C 0E STA want_units
9496 C8 INY
```

```

9497 B1 BC LDA (ptr),Y
9499 AA TAX
949A 20 BB 90 JSR hamming_decode
949D 30 E6 BMI L9485
949F 8D 4B 0E STA want_tens
94A2 C8 INY
94A3 B1 BC LDA (ptr),Y
94A5 AA TAX
94A6 20 BB 90 JSR hamming_decode
94A9 30 DA BMI L9485
94AB 8D 50 0E STA want_s4
94AE C8 INY
94AF B1 BC LDA (ptr),Y
94B1 AA TAX
94B2 20 BB 90 JSR hamming_decode
94B5 30 CE BMI L9485
94B7 48 PHA
94B8 29 07 AND #07
94BA 8D 4F 0E STA want_s3
94BD 68 PLA
94BE 4A LSR A
94BF 4A LSR A
94C0 4A LSR A
94C1 48 PHA
94C2 C8 INY
94C3 B1 BC LDA (ptr),Y
94C5 AA TAX
94C6 20 BB 90 JSR hamming_decode
94C9 10 03 BPL L94CE
.L94CB
94CB 4C FF 94 JMP L94FF
.L94CE
94CE 8D 4E 0E STA want_s2
94D1 C8 INY
94D2 B1 BC LDA (ptr),Y
94D4 AA TAX
94D5 20 BB 90 JSR hamming_decode
94D8 30 F1 BMI L94CB
94DA 48 PHA
94DB 29 03 AND #03
94DD 8D 4D 0E STA want_s1
94E0 68 PLA
94E1 4A LSR A
94E2 29 06 AND #06
94E4 85 BC STA ptr
94E6 68 PLA
94E7 05 BC ORA ptr
94E9 4D A2 0E EOR req_mag ; magazine is sent relative to the current page's magazine
94EC 29 07 AND #07
94EE 8D 4A 0E STA want_mag
94F1 A9 00 LDA #00 ; result 0: found
94F3 A0 02 LDY #02
94F5 91 F0 STA (osw_x),Y
94F7 68 PLA ; discard the counter; RTS goes to the address pushed by the caller
94F8 60 RTS
94F9 .db 4C 0C 98 4C 11 82 ; L..L..
.L94FF
94FF 68 PLA
.L9500
9500 A5 BC LDA ptr ; next extra-row record (+48)
9502 18 CLC
9503 69 30 ADC #30
9505 85 BC STA ptr
9507 A5 BD LDA ptr+1
9509 69 00 ADC #00
950B 85 BD STA ptr+1
950D 4C 56 94 JMP L9456
;-----
; BGET CHANNEL 15: READ THE NEXT CHARACTER OF THE CURRENT PAGE
; Waits for the page (with *OPT2 checksum handling) the first time, then returns
; characters in the order the page check word covers them: row 0 from column &11
; (skipping the clock) then rows 1-25. Returns C set with A=2 at the limit or the end of
; the page, or A=&FE if ESCAPE was pressed. Missing rows read as spaces.
;-----
.L9510
9510 98 TYA
9511 48 PHA
9512 8A TXA
9513 48 PHA
9514 24 FF BIT escape_flag
9516 30 1F BMI L9537
9518 A5 EF LDA osw_a
951A 48 PHA
951B A5 F0 LDA osw_x
951D 48 PHA
951E A5 F1 LDA osw_y
9520 48 PHA
9521 08 PHP
9522 58 CLI
9523 AE 9B 0E LDX cur_req

```

```

9526 20 BF 8C JSR L8CBF ; wait for the page and check it
9529 28      PLP
952A 68      PLA
952B 85 F1    STA osw_y
952D 68      PLA
952E 85 F0    STA osw_x
9530 68      PLA
9531 85 EF    STA osw_a
9533 24 FF    BIT escape_flag
9535 10 05    BPL L953C

.L9537
9537 A9 FE    LDA #&FE
9539 4C 91 95 JMP L9591

.L953C
953C AD 8F 0E LDA rd_row ; skip the clock in row 0
953F D0 0C    BNE L954D
9541 AE 90 0E LDX rd_col
9544 E0 29    CPX #&29 ; ')'
9546 D0 05    BNE L954D
9548 A2 31    LDX #&31
954A 8E 90 0E STX rd_col

.L954D
954D AE 90 0E LDX rd_col ; next column; wrap to the next row after column &31
9550 E8      INX
9551 E0 32    CPX #&32 ; '2'
9553 D0 05    BNE L955A
9555 18      CLC
9556 69 01    ADC #&01
9558 A2 0A    LDX #&0A

.L955A
955A C9 1A    CMP #&1A ; past row 25: end of page
955C 90 3C    BCC L959A
955E A9 02    LDA #&02
9560 4C 91 95 JMP L9591
9563 .db CA F0 2F A9 01 20 1C 8D A0 02 B1 BC C9 1B D0 F0 ; ../. ....
9573 .db A0 0A B1 BC A8 20 BB 90 30 54 D0 E4 A0 2F B1 BC ; .... ..0T.../..
9583 .db A8 20 BB 90 30 48 29 03 C9 02 D0 06 A9 01 ; ..0H).....

.L9591
9591 38      SEC
9592 4C E7 8A JMP L8AE7
9595 .db A9 03 4C 91 95 ; ..L..

.L959A
959A CD 95 0E CMP rd_limit_row ; at the limit set by OSWORD &83?
959D D0 0A    BNE L95A9
959F EC 96 0E CPX rd_limit_col
95A2 D0 05    BNE L95A9
95A4 A9 02    LDA #&02
95A6 4C 91 95 JMP L9591

.L95A9
95A9 8E 90 0E STX rd_col
95AC 8D 8F 0E STA rd_row
95AF AC A0 0E LDY req_buf
95B2 84 BC    STY ptr
95B4 AC A1 0E LDY req_buf+1
95B7 84 BD    STY ptr+1
95B9 20 1C 8D JSR row_offset ; find the row record
95BC A0 02    LDY #&02
95BE B1 BC    LDA (ptr),Y
95C0 C9 FF    CMP #&FF ; row missing: space
95C2 D0 05    BNE L95C9
95C4 A9 20    LDA #&20
95C6 4C CD 95 JMP L95CD

.L95C9
95C9 8A      TXA
95CA A8      TAY
95CB B1 BC    LDA (ptr),Y

.L95CD
95CD 18      CLC
95CE 4C E7 8A JMP L8AE7
95D1 .db A9 01 4C 91 95 ; ..L..

;-----
; OSWORD &7A &94: SET UP REQUEST SLOT [&94, slot, buffer lo, buffer hi]
; Give slot n a buffer, clear its page number and status, and mark it active.
;-----

.osw_94_setup_request
95D6 A0 01    LDY #&01
95D8 B1 F0    LDA (osw_x),Y
95DA AA      TAX
95DB A9 00    LDA #&00

.L95DD
95DD CA      DEX
95DE 30 05    BMI L95E5
95E0 18      CLC
95E1 69 0A    ADC #&0A
95E3 D0 F8    BNE L95DD

.L95E5
95E5 AA      TAX
95E6 A9 00    LDA #&00
95E8 9D A9 0E STA req_state,X
95EB C8      INY

```

```

95EC B1 F0 LDA (osw_x),Y
95EE 9D A0 0E STA req_buf,X
95F1 C8 INY
95F2 B1 F0 LDA (osw_x),Y
95F4 9D A1 0E STA req_buf+1,X
95F7 A9 00 LDA #000
95F9 A0 07 LDY #07
.L95FB
95FB 9D A2 0E STA req_mag,X
95FE E8 INX
95FF 88 DEY
9600 D0 F9 BNE L95FB
9602 A9 40 LDA #40
9604 9D A2 0E STA req_mag,X
9607 4C 11 82 JMP svc_exit_claimed
;
; OSWORD &7A &A1: DEACTIVATE SLOTS 1-24 and clear KEPT on slot 0.
;
.osw_A1_reset_requests
960A A0 00 LDY #000
960C A2 0A LDX #0A
.L960E
960E 98 TYA
960F 9D A9 0E STA req_state,X
9612 8A TXA
9613 18 CLC
9614 69 0A ADC #0A
9616 AA TAX
9617 E0 FA CPX #FA
9619 D0 F3 BNE L960E
961B AD A8 0E LDA req_s4
961E 29 BF AND #BF
9620 8D A8 0E STA req_s4
9623 4C 11 82 JMP svc_exit_claimed
;
; OSWORD &7A &A2: MOVE SLOT 0 BACK TO THE PRIVATE-WORKSPACE BUFFER
; If slot 0's buffer is not the one in private workspace, swap buffers with whichever slot
; has it, abandoning both captures. Used when terminal mode exits and user memory is no
; longer available for page buffers.
;
.osw_A2_restore_slot0_buffer
9626 A6 F4 LDX romsel_copy
9628 BD F0 0D LDA rom_wksp_table,X
962B 29 7F AND #7F
962D 85 BC STA ptr
962F A0 00 LDY #000
9631 AD A1 0E LDA req_buf+1
9634 C5 BC CMP ptr
9636 F0 36 BEQ L966E
.L9638
9638 98 TYA
9639 18 CLC
963A 69 0A ADC #0A
963C A8 TAY
963D C0 FA CPY #FA
963F F0 2D BEQ L966E
9641 B9 A1 0E LDA req_buf+1,Y
9644 C5 BC CMP ptr
9646 D0 F0 BNE L9638
9648 B9 A0 0E LDA req_buf,Y
964B 48 PHA
964C B9 A1 0E LDA req_buf+1,Y
964F 48 PHA
9650 AD A0 0E LDA req_buf
9653 99 A0 0E STA req_buf,Y
9656 AD A1 0E LDA req_buf+1
9659 99 A1 0E STA req_buf+1,Y
965C 68 PLA
965D 8D A1 0E STA req_buf+1
9660 68 PLA
9661 8D A0 0E STA req_buf
9664 A2 00 LDX #000
9666 20 A2 90 JSR clear_capture_flags
9669 98 TYA
966A AA TAX
966B 20 A2 90 JSR clear_capture_flags
.L966E
966E 4C 11 82 JMP svc_exit_claimed
;
; OSWORD &7A &A4: SUSPEND ALL REQUESTS. Shifts each slot's state right, so "active" (&40)
; becomes &20 and the interrupt handler ignores every slot.
;
.osw_A4_suspend_all
9671 A2 00 LDX #000
.L9673
9673 5E A9 0E LSR req_state,X
9676 8A TXA
9677 18 CLC
9678 69 0A ADC #0A
967A AA TAX

```

```

967B C9 FA    CMP #&FA
967D D0 F4    BNE L9673
967F 4C 11 82 JMP svc_exit_claimed
; OSWORD &7A &A5: RESUME ALL REQUESTS (the reverse of &A4).
.osw_A5_resume_all
9682 A2 00    LDX #&00
.L9684
9684 1E A9 0E ASL req_state,X
9687 8A       TXA
9688 18       CLC
9689 69 0A    ADC #&0A
968B AA       TAX
968C C9 FA    CMP #&FA
968E D0 F4    BNE L9684
9690 4C 11 82 JMP svc_exit_claimed
;-----
; OSWORD &7A &A0: REQUEST THE INITIAL PAGE
; Decode the "initial page" carried in the broadcast service data packet (bytes 1-6 of
; tsdp) and request it, as a FETCH PAGE would. If there is no valid packet, request page
; x00 of any magazine with any sub-code instead. Used for the index page (SHIFT f1) and
; when Terminal mode starts.
;-----
.osw_A0_request_initial_page
9693 AD 5D 0E LDA tsdp+1
9696 AA       TAX
9697 20 BB 90 JSR hamming_decode
969A 30 5C    BMI L96F8
969C 8D 4C 0E STA want_units
969F AD 5E 0E LDA tsdp+2
96A2 AA       TAX
96A3 20 BB 90 JSR hamming_decode
96A6 30 50    BMI L96F8
96A8 8D 4B 0E STA want_tens
96AB AD 5F 0E LDA tsdp+3
96AE AA       TAX
96AF 20 BB 90 JSR hamming_decode
96B2 30 44    BMI L96F8
96B4 8D 50 0E STA want_s4
96B7 AD 60 0E LDA tsdp+4
96BA AA       TAX
96BB 20 BB 90 JSR hamming_decode
96BE 30 38    BMI L96F8
96C0 48       PHA
96C1 29 07    AND #&07
96C3 8D 4F 0E STA want_s3
96C6 68       PLA
96C7 4A       LSR A
96C8 4A       LSR A
96C9 4A       LSR A
96CA 48       PHA
96CB AD 61 0E LDA tsdp+5
96CE AA       TAX
96CF 20 BB 90 JSR hamming_decode
96D2 30 23    BMI L96F7
96D4 8D 4E 0E STA want_s2
96D7 AD 62 0E LDA tsdp+6
96DA AA       TAX
96DB 20 BB 90 JSR hamming_decode
96DE 30 17    BMI L96F7
96E0 48       PHA
96E1 29 03    AND #&03
96E3 8D 4D 0E STA want_s1
96E6 68       PLA
96E7 4A       LSR A
96E8 29 06    AND #&06
96EA 85 BC    STA ptr
96EC 68       PLA
96ED 05 BC    ORA ptr
96EF 29 07    AND #&07
96F1 8D 4A 0E STA want_mag
96F4 4C 0C 98 JMP L980C
.L96F7
96F7 68       PLA
.L96F8
96F8 A9 FF    LDA #&FF
96FA 8D 4A 0E STA want_mag
96FD 8D 4D 0E STA want_s1
9700 8D 4E 0E STA want_s2
9703 8D 4F 0E STA want_s3
9706 8D 50 0E STA want_s4
9709 A9 00    LDA #&00
970B 8D 4B 0E STA want_tens
970E 8D 4C 0E STA want_units
9711 4C 0C 98 JMP L980C
;-----
; Swap request slot req_scan (X) with slot 0 (Y): buffer pointers and page digits.
; Any capture in progress in either is abandoned. Slot 0 then gets the page number in
; want_mag..want_s4 (wild digits kept). If slot 0 already holds a valid, checked page it
; is flagged ARRIVED so that it is shown at once.
;-----

```

.swap_slots

```
9714 A0 00 LDY #000
9716 B9 A9 0E LDA req_state,Y
9719 48 PHA
971A 29 BF AND #0BF
971C 99 A9 0E STA req_state,Y
971F B9 A4 0E LDA req_units,Y
9722 2A ROL A
9723 10 18 BPL L973D
9725 B9 A5 0E LDA req_s1,Y
9728 29 BF AND #0BF
972A 99 A5 0E STA req_s1,Y
972D B9 A3 0E LDA req_tens,Y
9730 29 BF AND #0BF
9732 99 A3 0E STA req_tens,Y
9735 B9 A4 0E LDA req_units,Y
9738 29 BF AND #0BF
973A 99 A4 0E STA req_units,Y
```

.L973D

```
973D AE 1E 0E LDX req_scan
9740 BD A9 0E LDA req_state,X
9743 48 PHA
9744 29 BF AND #0BF
9746 9D A9 0E STA req_state,X
9749 BD A4 0E LDA req_units,X
974C 2A ROL A
974D 10 18 BPL L9767
974F BD A5 0E LDA req_s1,X
9752 29 BF AND #0BF
9754 9D A5 0E STA req_s1,X
9757 BD A3 0E LDA req_tens,X
975A 29 BF AND #0BF
975C 9D A3 0E STA req_tens,X
975F BD A4 0E LDA req_units,X
9762 29 BF AND #0BF
9764 9D A4 0E STA req_units,X
```

.L9767

```
9767 B9 A0 0E LDA req_buf,Y
976A 48 PHA
976B BD A0 0E LDA req_buf,X
976E 99 A0 0E STA req_buf,Y
9771 68 PLA
9772 9D A0 0E STA req_buf,X
9775 E8 INX
9776 C8 INY
9777 C0 09 CPY #009
9779 D0 EC BNE L9767
977B 8A TXA
977C 48 PHA
977D 98 TYA
977E 48 PHA
977F A0 00 LDY #000
```

.L9781

```
9781 B9 A2 0E LDA req_mag,Y
9784 29 7F AND #07F
9786 99 A2 0E STA req_mag,Y
9789 B9 4A 0E LDA want_mag,Y
978C 10 08 BPL L9796
978E A9 80 LDA #080
9790 19 A2 0E ORA req_mag,Y
9793 99 A2 0E STA req_mag,Y
```

.L9796

```
9796 C8 INY
9797 C0 07 CPY #007
9799 D0 E6 BNE L9781
979B 2C A5 0E BIT req_s1
979E 50 12 BVC L97B2
97A0 2C A7 0E BIT req_s3
97A3 50 00 BVC L97B2
97A5 2C A6 0E BIT req_s2
97A8 50 08 BVC L97B2
97AA AD A3 0E LDA req_tens
97AD 09 40 ORA #040
97AF 8D A3 0E STA req_tens
```

.L97B2

```
97B2 68 PLA
97B3 A8 TAY
97B4 68 PLA
97B5 AA TAX
97B6 68 PLA
97B7 99 A0 0E STA req_buf,Y
97BA 68 PLA
97BB 9D A0 0E STA req_buf,X
97BE 60 RTS
```

```
;-----
; Convert the 7 requested digits in want_mag..want_s4 to internal form:
; ASCII hex or binary digits are masked to their field width (mask table at &97E1:
; magazine 3 bits, tens 4, units 4, S1 2, S2 4, S3 3, S4 4); "#", &FF or any value with
; bit 7 set becomes a wild digit (bit 7 set).
```

```

.normalise_page_number
97BF A2 06 LDX #&06
.L97C1
97C1 BD 4A 0E LDA want_mag,X
97C4 30 0F BMI L97D5
97C6 C9 23 CMP #&23 ; '#'
97C8 F0 0B BEQ L97D5
97CA C9 41 CMP #&41 ; 'A'
97CC 90 02 BCC L97D0
97CE E9 37 SBC #&37
.L97D0
97D0 3D E1 97 AND &97E1,X
97D3 10 05 BPL L97DA
.L97D5
97D5 BD E1 97 LDA &97E1,X
97D8 09 80 ORA #&80
.L97DA
97DA 9D 4A 0E STA want_mag,X
97DD CA DEX
97DE 10 E1 BPL L97C1
97E0 60 RTS
97E1 .db 07 0F 0F 03 0F 07 0F ; .....
;
; -----
; Reset the page read position to row 0 column &11, the limit to row &19 column &32
; (the end of the page), and make slot 0 the current request.
;
; -----
.reset_read_position
97E8 A9 11 LDA #&11
97EA 8D 90 0E STA rd_col
97ED A9 00 LDA #&00
97EF 8D 8F 0E STA rd_row
97F2 8D 9B 0E STA cur_req
97F5 A9 19 LDA #&19
97F7 8D 95 0E STA rd_limit_row
97FA A9 32 LDA #&32
97FC 8D 96 0E STA rd_limit_col
97FF 60 RTS
;
; -----
; OSWORD &7A &85: FETCH PAGE [&85, mag, tens, units, s1, s2, s3, s4]
; Make the given page (digits may be wild) the current page, in slot 0:
; - if another slot already has a request for it (a kept or prefetched page), swap that
; slot into slot 0, so it can be shown at once
; - otherwise, if the page in slot 0 is KEPT, move it to a free slot first
; (result 1 if there is none: "Buffers full")
; - then slot 0's request becomes the new page number
;
; -----
.osw_85_fetch_page
9800 A0 01 LDY #&01
.L9802
9802 B1 F0 LDA (osw_x),Y
9804 99 49 0E STA &0E49,Y
9807 C8 INY
9808 C0 08 CPY #&08
980A D0 F6 BNE L9802
.L980C
980C 20 E8 97 JSR reset_read_position
980F 20 BF 97 JSR normalise_page_number
9812 A2 F6 LDX #&F6 ; scan slots 1-24
9814 8E 1E 0E STX req_scan
.L9817
9817 AD 1E 0E LDA req_scan
981A 18 CLC
981B 69 0A ADC #&0A
981D 8D 1E 0E STA req_scan
9820 C9 F0 CMP #&F0
9822 D0 42 BNE L9866
9824 2C A8 0E BIT req_s4 ; slot 0 kept?
9827 50 22 BVC L984B
9829 A0 F0 LDY #&F0 ; look for a slot that is active and not kept, from the top
.L982B
982B B9 A9 0E LDA req_state,Y
982E 2A ROL A
982F 10 06 BPL L9837
9831 B9 A8 0E LDA req_s4,Y
9834 2A ROL A
9835 10 0E BPL L9845
.L9837
9837 98 TYA
9838 38 SEC
9839 E9 0A SBC #&0A
983B A8 TAY
983C D0 ED BNE L982B
983E A9 01 LDA #&01 ; result 1: no free buffer
9840 91 F0 STA (osw_x),Y
9842 4C 11 82 JMP svc_exit_claimed
.L9845
9845 8C 1E 0E STY req_scan ; move the kept page out of slot 0
9848 20 14 97 JSR swap_slots
.L984B
984B A2 00 LDX #&00 ; slot 0 := new request, inactive while it is written

```

```

984D 8E 9B 0E STX cur_req
9850 8E A9 0E STX req_state
.L9853
9853 BD 4A 0E LDA want_mag,X
9856 9D A2 0E STA req_mag,X
9859 E8      INX
985A E0 07   CPX #&07
985C D0 F5   BNE L9853
985E A9 40   LDA #&40          ; active
9860 8D A9 0E STA req_state
9863 4C 11 82 JMP svc_exit_claimed
.L9866
9866 A0 00   LDY #&00          ; does slot X already ask for this page?
9868 AE 1E 0E LDX req_scan
986B BD A9 0E LDA req_state,X
986E 85 BC   STA ptr
9870 24 BC   BIT ptr
9872 50 10   BVC L9884
.L9874
9874 B9 4A 0E LDA want_mag,Y
9877 30 0E   BMI L9887
9879 BD A2 0E LDA req_mag,X
987C 39 E1 97 AND &97E1,Y
987F D9 4A 0E CMP want_mag,Y
9882 F0 03   BEQ L9887
.L9884
9884 4C 17 98 JMP L9817
.L9887
9887 E8      INX
9888 C8      INY
9889 C0 07   CPY #&07
988B D0 E7   BNE L9874
988D 20 14 97 JSR swap_slots    ; yes: bring it into slot 0, unlocked
9890 AD A2 0E LDA req_mag
9893 29 BF   AND #&BF
9895 8D A2 0E STA req_mag
9898 4C 11 82 JMP svc_exit_claimed
;
; OSWORD &7A &96 continued (entered by RTS from L9420 after decoding a link).
; Uses the link's page number (want_mag..want_s4) as a FETCH PAGE would; slot 0's KEPT
; page is moved aside if necessary.
;
989B .db 20 E8 97 20 BF 97 A2 00 A2 F6 8E 1E 0E AD 1E 0E ; .. .....
98AB .db 18 69 0A 8D 1E 0E C9 F0 D0 3E A0 F0 B9 A9 0E 2A ; .i.....>.....*
98BB .db 10 06 B9 A8 0E 2A 10 12 98 38 E9 0A A8 C9 0A D0 ; .....*...8.....
98CB .db EB A0 00 A9 01 91 F0 4C 11 82 8C 9B 0E A9 00 99 ; .....L.....
98DB .db A9 0E A2 00 BD 4A 0E 99 A2 0E E8 C8 E0 07 D0 F4 ; ....J.....
98EB .db A9 40 99 A2 0E 4C 11 82 A0 00 AE 1E 0E BD A9 0E ; .@...L.....
98FB .db 2A 10 10 B9 4A 0E 30 0E BD A2 0E 39 E1 97 D9 4A ; *...J.0...9...J
990B .db 0E F0 03 4C A8 98 E8 C8 C0 07 D0 E7 AD 1E 0E 8D ; ...L.....
991B .db 9B 0E AA BD A5 0E 2A 10 08 BD A3 0E 09 40 9D A3 ; .....*.....@..
992B .db 0E 4C 11 82 ; .L..
;
; LINKED PAGES: after a page with row 27 arrives in slot 0, point request slots at the
; pages its links name, so they are captured in the background and appear at once when
; selected. Only row 27 designation codes 0-3 are used; the table at &9982 gives the first
; slot for each. Slots that are KEPT, inactive or locked are left alone. A slot whose page
; number changes has its status reset so the new page is captured from scratch.
;
.prefetch_links
992F AD A0 0E LDA req_buf
9932 85 C4   STA irq_ptr
9934 AD A1 0E LDA req_buf+1
9937 85 C5   STA irq_ptr+1
9939 A0 01   LDY #&01
993B B1 C4   LDA (irq_ptr),Y
993D 38      SEC
993E E9 1B   SBC #&1B
9940 48      PHA
9941 18      CLC
9942 A5 C4   LDA irq_ptr
9944 69 E2   ADC #&E2
9946 85 C4   STA irq_ptr
9948 A5 C5   LDA irq_ptr+1
994A 69 04   ADC #&04
994C 85 C5   STA irq_ptr+1
.L994E
994E 68      PLA
994F 49 00   EOR #&00
9951 30 33   BMI L9986
9953 38      SEC
9954 E9 01   SBC #&01
9956 48      PHA
9957 A0 00   LDY #&00
9959 B1 C4   LDA (irq_ptr),Y
995B C9 1B   CMP #&1B
995D D0 13   BNE L9972
995F A0 08   LDY #&08
9961 B1 C4   LDA (irq_ptr),Y
9963 A8      TAY

```



```

9964 B9 20 80 LDA hamming_table,Y
9967 30 09 BMI L9972
9969 29 0F AND #0F
996B C9 04 CMP #04
996D B0 03 BCS L9972
996F 20 87 99 JSR L9987
.L9972
9972 A5 C4 LDA irq_ptr
9974 18 CLC
9975 69 30 ADC #30
9977 85 C4 STA irq_ptr
9979 A5 C5 LDA irq_ptr+1
997B 69 00 ADC #00
997D 85 C5 STA irq_ptr+1
997F 4C 4E 99 JMP L994E
9982 .db 0A 46 82 BE ; .F..
.L9986
9986 60 RTS
.L9987
9987 A8 TAY
9988 B9 82 99 LDA &9982,Y
998B A8 TAY
998C A2 09 LDX #09
.L998E
998E B9 A8 0E LDA req_s4,Y
9991 2A ROL A
9992 30 17 BMI L99AB
9994 B9 A9 0E LDA req_state,Y
9997 2A ROL A
9998 10 11 BPL L99AB
999A B9 A2 0E LDA req_mag,Y
999D 2A ROL A
999E 30 0B BMI L99AB
99A0 8A TXA
99A1 48 PHA
99A2 98 TYA
99A3 48 PHA
99A4 4C BA 99 JMP L99BA
.L99A7
99A7 68 PLA
99A8 A8 TAY
99A9 68 PLA
99AA AA TAX
.L99AB
99AB 98 TYA
99AC 18 CLC
99AD 69 0A ADC #0A
99AF A8 TAY
99B0 8A TXA
99B1 18 CLC
99B2 69 06 ADC #06
99B4 AA TAX
99B5 E0 2D CPX #2D ; '-'
99B7 D0 D5 BNE L998E
99B9 60 RTS
.L99BA
99BA 68 PLA
99BB AA TAX
99BC 68 PLA
99BD A8 TAY
99BE 48 PHA
99BF 8A TXA
99C0 48 PHA
99C1 A9 00 LDA #00
99C3 8D 09 0E STA row_tmp_hi
99C6 B1 C4 LDA (irq_ptr),Y
99C8 20 01 9B JSR L9B01
99CB 8D 08 0E STA row_tmp
99CE BD A4 0E LDA req_units,X
99D1 29 0F AND #0F
99D3 CD 08 0E CMP row_tmp
99D6 F0 03 BEQ L99DB
99D8 EE 09 0E INC row_tmp_hi
.L99DB
99DB BD A4 0E LDA req_units,X
99DE 29 F0 AND #F0
99E0 0D 08 0E ORA row_tmp
99E3 9D A4 0E STA req_units,X
99E6 C8 INY
99E7 B1 C4 LDA (irq_ptr),Y
99E9 20 01 9B JSR L9B01
99EC 8D 08 0E STA row_tmp
99EF BD A3 0E LDA req_tens,X
99F2 29 0F AND #0F
99F4 CD 08 0E CMP row_tmp
99F7 F0 03 BEQ L99FC
99F9 EE 09 0E INC row_tmp_hi
.L99FC
99FC BD A3 0E LDA req_tens,X
99FF 29 F0 AND #F0

```

```

9A01 0D 08 0E ORA row_tmp
9A04 9D A3 0E STA req_tens,X
9A07 C8      INY
9A08 B1 C4      LDA (irq_ptr),Y
9A0A 20 01 9B JSR L9B01
9A0D 8D 08 0E STA row_tmp
9A10 BD A8 0E LDA req_s4,X
9A13 29 0F      AND #0F
9A15 CD 08 0E CMP row_tmp
9A18 F0 03      BEQ L9A1D
9A1A EE 09 0E INC row_tmp_hi
.L9A1D
9A1D BD A8 0E LDA req_s4,X
9A20 29 F0      AND #0F
9A22 0D 08 0E ORA row_tmp
9A25 9D A8 0E STA req_s4,X
9A28 C8      INY
9A29 B1 C4      LDA (irq_ptr),Y
9A2B 20 01 9B JSR L9B01
9A2E 48      PHA
9A2F 29 07      AND #07
9A31 8D 08 0E STA row_tmp
9A34 BD A7 0E LDA req_s3,X
9A37 29 07      AND #07
9A39 CD 08 0E CMP row_tmp
9A3C F0 03      BEQ L9A41
9A3E EE 09 0E INC row_tmp_hi
.L9A41
9A41 BD A7 0E LDA req_s3,X
9A44 29 F8      AND #0F8
9A46 0D 08 0E ORA row_tmp
9A49 9D A7 0E STA req_s3,X
9A4C 68      PLA
9A4D 4A      LSR A
9A4E 4A      LSR A
9A4F 4A      LSR A
9A50 48      PHA
9A51 C8      INY
9A52 B1 C4      LDA (irq_ptr),Y
9A54 20 01 9B JSR L9B01
9A57 8D 08 0E STA row_tmp
9A5A BD A6 0E LDA req_s2,X
9A5D 29 0F      AND #0F
9A5F CD 08 0E CMP row_tmp
9A62 F0 03      BEQ L9A67
9A64 EE 09 0E INC row_tmp_hi
.L9A67
9A67 BD A6 0E LDA req_s2,X
9A6A 29 F0      AND #0F0
9A6C 0D 08 0E ORA row_tmp
9A6F 9D A6 0E STA req_s2,X
9A72 C8      INY
9A73 B1 C4      LDA (irq_ptr),Y
9A75 20 01 9B JSR L9B01
9A78 48      PHA
9A79 29 03      AND #03
9A7B 8D 08 0E STA row_tmp
9A7E BD A5 0E LDA req_s1,X
9A81 29 03      AND #03
9A83 CD 08 0E CMP row_tmp
9A86 F0 03      BEQ L9A8B
9A88 EE 09 0E INC row_tmp_hi
.L9A8B
9A8B BD A5 0E LDA req_s1,X
9A8E 29 FC      AND #0FC
9A90 0D 08 0E ORA row_tmp
9A93 9D A5 0E STA req_s1,X
9A96 68      PLA
9A97 4A      LSR A
9A98 29 06      AND #06
9A9A 8D 08 0E STA row_tmp
9A9D 68      PLA
9A9E 0D 08 0E ORA row_tmp
9AA1 4D A2 0E EOR req_mag
9AA4 29 07      AND #07
9AA6 8D 08 0E STA row_tmp
9AA9 BD A2 0E LDA req_mag,X
9AAC 29 07      AND #07
9AAE CD 08 0E CMP row_tmp
9AB1 F0 03      BEQ L9AB6
9AB3 EE 09 0E INC row_tmp_hi
.L9AB6
9AB6 BD A2 0E LDA req_mag,X
9AB9 29 F8      AND #0F8
9ABB 0D 08 0E ORA row_tmp
9ABE 9D A2 0E STA req_mag,X
9AC1 AD 09 0E LDA row_tmp_hi
9AC4 F0 38      BEQ L9AFE
9AC6 BD A4 0E LDA req_units,X
9AC9 29 0F      AND #0F

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9ACB 9D A4 0E STA req_units,X
9ACE BD A3 0E LDA req_tens,X
9AD1 29 0F AND #&0F
9AD3 9D A3 0E STA req_tens,X
9AD6 BD A8 0E LDA req_s4,X
9AD9 29 0F AND #&0F
9ADB 9D A8 0E STA req_s4,X
9ADE BD A7 0E LDA req_s3,X
9AE1 29 07 AND #&07
9AE3 9D A7 0E STA req_s3,X
9AE6 BD A6 0E LDA req_s2,X
9AE9 29 0F AND #&0F
9AEB 9D A6 0E STA req_s2,X
9AEE BD A5 0E LDA req_s1,X
9AF1 29 03 AND #&03
9AF3 9D A5 0E STA req_s1,X
9AF6 BD A2 0E LDA req_mag,X
9AF9 29 07 AND #&07
9AFB 9D A2 0E STA req_mag,X
.L9AFE
9AFE 4C A7 99 JMP L99A7
;-----
; Hamming-decode the byte in A (Y preserved). Returns the 4-bit value.
;-----
.L9B01
9B01 8C 08 0E STY row_tmp
9B04 A8 TAY
9B05 B9 20 80 LDA hamming_table,Y
9B08 AC 08 0E LDY row_tmp
9B0B 29 0F AND #&0F
9B0D 60 RTS
;-----
; Default status message for the *TUNE screen: "No status broadcast."
;-----
9B0E .db 4E 6F 20 73 74 61 74 75 73 20 62 72 6F 61 64 63 ; No status broadcast
9B1E .db 61 73 74 2E ; ast.
;-----
; Issue OSWORD &7A &92 to send rolling headers to page A, offset &2C (A=0: off).
;-----
.set_headers
9B22 8D 3C 0E STA osw_blk+2
9B25 A9 2C LDA #&2C
9B27 8D 3B 0E STA osw_blk+1
9B2A A9 92 LDA #&92
9B2C 8D 3A 0E STA osw_blk
9B2F A2 3A LDX #&3A
9B31 A0 0E LDY #&0E
9B33 A9 7A LDA #&7A
9B35 4C F1 FF JMP OSWORD ; OSWORD &7A: Teletext system call
;-----
; *TUNE and SHIFT f4: SIGNAL-QUALITY TUNING SCREEN
; Clears control bit 4 so the TDA2541's AFC doesn't fight the manual tuning, then
; repeatedly measures reception over 25 fields:
; ham_count / ham_errors are incremented by the interrupt handler for every
; Hamming byte decoded / found in error.
; The bar length is 32 * (1 - errors / bytes). With no bytes at all, "No teletext being
; received" is shown. The 20-character status message from the broadcast service data is
; shown when available. SPACE or ESCAPE leaves (AFC back on); any other key re-measures.
;-----
.cmd_TUNE
9B38 A2 14 LDX #&14
9B3A A0 00 LDY #&00
.L9B3C
9B3C B9 0E 9B LDA &9B0E,Y ; preload "No status broadcast."
9B3F 99 70 0E STA tsdp_status,Y
9B42 C8 INY
9B43 CA DEX
9B44 D0 F6 BNE L9B3C
9B46 AD 8E 0E LDA ctrl_shadow ; control bit 4 off: AFC off while hand tuning
9B49 29 EF AND #&EF
9B4B 8D 8E 0E STA ctrl_shadow
9B4E 8D 10 FC STA TT_STATUS_CTRL
9B51 20 AF A2 JSR print_inline
9B54 .str '\x16\x07\nCh\x00'
9B5A AD 8E 0E LDA ctrl_shadow
9B5D 29 03 AND #&03
9B5F 18 CLC
9B60 69 31 ADC #&31
9B62 20 EE FF JSR OSWRCH
9B65 20 5E B5 JSR LB55E
.L9B68
9B68 A9 7C LDA #&7C ; rolling headers to &7C2C (row 1 of the MODE 7 screen)
9B6A 20 22 9B JSR set_headers
9B6D A9 00 LDA #&00
9B6F 8D 54 0E STA tune_err_sum
9B72 8D 55 0E STA tune_err_sum+1
9B75 8D 56 0E STA tune_byte_sum
9B78 8D 57 0E STA tune_byte_sum+1
9B7B 8D 52 0E STA ham_count
9B7E 8D 53 0E STA ham_errors

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9B81 A9 19 LDA #&19 ; 25 fields
9B83 8D 5A 0E STA tune_frames
.L9B86
9B86 A9 13 LDA #&13 ; wait for the next field
9B88 20 F4 FF JSR OSBYTE ; OSBYTE &13: wait for vertical sync
9B8B AC 53 0E LDY ham_errors
9B8E AE 52 0E LDX ham_count
9B91 A9 00 LDA #&00
9B93 8D 52 0E STA ham_count
9B96 8D 53 0E STA ham_errors
9B99 18 CLC
9B9A 98 TYA
9B9B 6D 54 0E ADC tune_err_sum
9B9E 8D 54 0E STA tune_err_sum
9BA1 90 03 BCC L9BA6
9BA3 EE 55 0E INC tune_err_sum+1
.L9BA6
9BA6 8A TXA
9BA7 6D 56 0E ADC tune_byte_sum
9BAA 8D 56 0E STA tune_byte_sum
9BAD 90 03 BCC L9BB2
9BAF EE 57 0E INC tune_byte_sum+1
.L9BB2
9BB2 CE 5A 0E DEC tune_frames
9BB5 D0 CF BNE L9B86
9BB7 AD 56 0E LDA tune_byte_sum ; any teletext at all?
9BBA D0 6D BNE L9C29
9BBC AD 57 0E LDA tune_byte_sum+1
9BBF D0 68 BNE L9C29
9BC1 20 AF A2 JSR print_inline
9BC4 .str '\x1f\x07\x14No teletext being received\x1f\x04\x01 \x1f\n\x08 \x00'
9C1C A9 20 LDA #&20
9C1E 8D 54 0E STA tune_err_sum
9C21 A9 00 LDA #&00
9C23 8D 55 0E STA tune_err_sum+1
9C26 4C 63 9C JMP L9C63
.L9C29
9C29 20 AF A2 JSR print_inline ; show the broadcast status message
9C2C .str '\x1f\x07\x14 Receiving teletext \x1f\n\x08\x00'
9C4D A2 14 LDX #&14
9C4F A0 14 LDY #&14
.L9C51
9C51 BD 5C 0E LDA tsdp,X
9C54 09 80 ORA #&80
9C56 20 EE FF JSR OSWRCH
9C59 E8 INX
9C5A 88 DEY
9C5B D0 F4 BNE L9C51
9C5D 20 23 9E JSR L9E23 ; error ratio * 32 ...
9C60 20 58 9E JSR L9E58 ; ... / bytes decoded
.L9C63
9C63 20 8A 9C JSR L9C8A
9C66 A9 20 LDA #&20 ; bar length = 32 - that
9C68 38 SEC
9C69 ED 54 0E SBC tune_err_sum
9C6C 85 BD STA ptr+1
9C6E A9 FF LDA #&FF ; draw the bar with character &FF (solid block)
9C70 A2 00 LDX #&00
.L9C72
9C72 E4 BD CPX ptr+1
9C74 F0 07 BEQ L9C7D
9C76 20 EE FF JSR OSWRCH
9C79 E8 INX
9C7A 4C 72 9C JMP L9C72
.L9C7D
9C7D A9 20 LDA #&20
9C7F E0 20 CPX #&20
9C81 F0 60 BEQ L9CE3
9C83 20 EE FF JSR OSWRCH
9C86 E8 INX
9C87 4C 7D 9C JMP L9C7D
;-----
; Draw the static parts of the tuning screen: prompt and 0% / 50% / 100% scale.
;-----
.L9C8A
9C8A 20 AF A2 JSR print_inline
9C8D .str '\x1f\x01\x17*** Space to
continue\x1f\x03\x0e\x97\xea.....7.....7.....7.....\xeb\x1f\x04\x0f0%\x1f\x13\x0f50%\x1f#\x0f100%\x1f\x03\r\x93j\x00'
9CE2 60 RTS
.L9CE3
9CE3 A9 81 LDA #&81 ; poll the keyboard (OSBYTE &81 with no delay)
9CE5 A2 00 LDX #&00
9CE7 A0 00 LDY #&00
9CE9 20 F4 FF JSR OSBYTE
9CEC 98 TYA
9CED D0 08 BNE L9CF7
9CEF 8A TXA
9CF0 C9 20 CMP #&20
9CF2 F0 0A BEQ L9CFE
9CF4 4C E3 9C JMP L9CE3

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.L9CF7
9CF7 C9 1B    CMP #&1B
9CF9 F0 03    BEQ L9CFE
9CFB 4C 68 9B  JMP L9B68

.L9CFE
9CFE 20 AF A2  JSR print_inline
9D01 .str '\x1f\x01\x18\r\x00'
9D06 A9 00     LDA #&00
9D08 20 22 9B  JSR set_headers           ; rolling headers off
9D0B AD 8E 0E   LDA ctrl_shadow
9D0E 09 10     ORA #&10             ; AFC back on
9D10 8D 8E 0E   STA ctrl_shadow
9D13 8D 10 FC   STA TT_STATUS_CTRL
9D16 20 59 B5  JSR LB559
9D19 38        SEC
9D1A 60        RTS
; GBPbv: not supported ("Data not available").
.gbpbv
9D1B 4C 36 A0  JMP error_not_available
;-----
; *TELETEXT: ENTER TERMINAL MODE
; Select Telesoft as the filing system, remember the current language (for f9), and ask
; the MOS to start this ROM as the language. The MOS prints the title and jumps to &8000.
;-----
.cmd_TELETEXT
9D1E 20 34 9D  JSR cmd_TELESOFT
9D21 A9 FC     LDA #&FC
9D23 A2 00     LDX #&00
9D25 A0 FF     LDY #&FF
9D27 20 F4 FF  JSR OSBYTE           ; OSBYTE &FC: read/write current language ROM number
9D2A 8E 99 0E  STX prev_language
9D2D A9 8E     LDA #&8E
9D2F A6 F4     LDX romsel_copy
9D31 20 F4 FF  JSR OSBYTE           ; OSBYTE &8E: enter language ROM X
;-----
; *TELESOFT (and service &12 with Y=6): SELECT THE TELESOFT FILING SYSTEM
; - remember the current filing system number and tell it (FSCV A=6) that it is being
;   replaced
; - point FILEV..FSCV at the extended vector entries and fill those with this ROM's
;   handlers (table at &9E15)
; - issue service &0F (vectors changed)
; - if TFS's state was stashed: issue service &0A so the current owner saves its own
;   state, then restore TFS's 1K from the adapter RAM and 2K from private workspace,
;   clear the adapter RAM, and restart all captures
; - switch capture and interrupts on
; NOTE: the restore writes 3K from &0E8E, i.e. up to &1A8D, so 142 bytes at &1A00-&1A8D
; (normally the start of user memory) are overwritten with what was there when the state
; was stashed. This looks like a bug; it has not been tested.
;-----
.cmd_TELESOFT
9D34 A9 00     LDA #&00
9D36 A8        TAY
9D37 20 DA FF  JSR OSARGS           ; OSARGS A=0 Y=0: read current filing system number
9D3A 48        PHA
9D3B A9 06     LDA #&06
9D3D 20 2F 9E  JSR L9E2F           ; FSCV 6: new filing system starting
9D40 A9 A8     LDA #&A8
9D42 A2 00     LDX #&00
9D44 A0 FF     LDY #&FF
9D46 20 F4 FF  JSR OSBYTE           ; OSBYTE &A8: read address of extended vector table
9D49 86 BC     STX ptr
9D4B 84 BD     STY ptr+1
9D4D A2 0C     LDX #&0C             ; 7 vectors, 3 bytes each in the extended vector table
9D4F A0 2F     LDY #&2F

.L9D51
9D51 A5 F4     LDA romsel_copy      ; extended vector: address lo, hi, ROM number
9D53 91 BC     STA (ptr),Y
9D55 88        DEY
9D56 BD 16 9E  LDA &9E16,X
9D59 91 BC     STA (ptr),Y
9D5B 88        DEY
9D5C BD 15 9E  LDA fs_vectors,X
9D5F 91 BC     STA (ptr),Y
9D61 A9 FF     LDA #&FF           ; main vector → &FF00+offset (the MOS's extended vector entry)
9D63 9D 13 02  STA FILEV+1,X
9D66 98        TYA
9D67 9D 12 02  STA FILEV,X
9D6A 88        DEY
9D6B CA        DEX
9D6C CA        DEX
9D6D 10 E2     BPL L9D51
9D6F A9 8F     LDA #&8F           ; service &0F: vectors claimed
9D71 A2 0F     LDX #&0F
9D73 20 F4 FF  JSR OSBYTE           ; OSBYTE &8F: issue paged ROM service call X
9D76 A6 F4     LDX romsel_copy
9D78 BD F0 0D  LDA rom_wksp_table,X ; state stashed?
9D7B 10 7F     BPL L9DFC
9D7D 29 7F     AND #&7F
9D7F 48        PHA
9D80 A9 8F     LDA #&8F           ; service &0A: tell the current owner to save its workspace

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9D82 A2 0A LDX #0A
9D84 A0 FF LDY #0FF
9D86 20 F4 FF JSR OSBYTE ; OSBYTE &8F: issue paged ROM service call X
9D89 A6 F4 LDX romsel_copy
9D8B 68 PLA
9D8C 48 PHA
9D8D 9D F0 0D STA rom_wksp_table,X
9D90 08 PHP
9D91 78 SEI
9D92 A9 8E LDA #8E
9D94 85 BC STA ptr
9D96 A9 0E LDA #0E
9D98 85 BD STA ptr+1
9D9A A2 0F LDX #0F
.L9D9C
9D9C A0 3F LDY #3F ; read 1K back from the 16 adapter RAM slots
9D9E 8E 11 FC STX TT_SLOT
.L9DA1
9DA1 AD 12 FC LDA TT_DATA
9DA4 91 BC STA (ptr),Y
9DA6 88 DEY
9DA7 10 F8 BPL L9DA1
9DA9 A5 BC LDA ptr
9DAB 18 CLC
9DAC 69 40 ADC #40
9DAE 85 BC STA ptr
9DB0 90 02 BCC L9DB4
9DB2 E6 BD INC ptr+1
.L9DB4
9DB4 CA DEX
9DB5 10 E5 BPL L9D9C
9DB7 A9 00 LDA #00 ; clear the adapter RAM slots (first byte 0 = empty)
9DB9 A2 0F LDX #0F
.L9DBB
9DBB 8E 11 FC STX TT_SLOT
9DBE 8D 12 FC STA TT_DATA
9DC1 CA DEX
9DC2 10 F7 BPL L9DBB
9DC4 68 PLA
9DC5 85 C0 STA ptr2
9DC7 68 PLA
9DC8 85 C1 STA ptr2+1
9DCA A5 C0 LDA ptr2
9DCC 48 PHA
9DCD A9 00 LDA #00
9DCF 85 C0 STA ptr2
9DD1 A2 07 LDX #07
.L9DD3
9DD3 A0 00 LDY #00 ; copy 2K back from private workspace
.L9DD5
9DD5 B1 C0 LDA (ptr2),Y
9DD7 91 BC STA (ptr),Y
9DD9 C8 INY
9DDA D0 F9 BNE L9DD5
9DDC E6 BD INC ptr+1
9DDE E6 C1 INC ptr2+1
9DE0 CA DEX
9DE1 10 F0 BPL L9DD3
9DE3 28 PLP
9DE4 A2 07 LDX #07
9DE6 A9 FF LDA #0FF ; no page being captured in any magazine
.L9DE8
9DE8 9D 15 0E STA mag_capture,X
9DEB CA DEX
9DEC 10 FA BPL L9DE8
9DEE A2 00 LDX #00
.L9DF0
9DF0 20 A2 90 JSR clear_capture_flags ; abandon all captures (their buffers may be stale)
9DF3 8A TXA
9DF4 18 CLC
9DF5 69 0A ADC #0A
9DF7 AA TAX
9DF8 E0 FA CPX #FA
9DFA D0 F4 BNE L9DF0
.L9DFC
9DFC 20 5A A2 JSR capture_on ; capture and IRQ on
9DFF A0 00 LDY #00
9E01 B9 A0 0E LDA req_buf,Y
9E04 85 BC STA ptr
9E06 B9 A1 0E LDA req_buf+1,Y
9E09 85 BD STA ptr+1
9E0B A9 20 LDA #20 ; slot 0's buffer capacity = 32 records
9E0D 91 BC STA (ptr),Y
9E0F 68 PLA
9E10 8D 9A 0E STA prev_fs ; remember the previous filing system for f9
9E13 38 SEC
.L9E14
9E14 60 RTS
;-----
; Filing system vector table: FILEV ARGSV BGETV BPUTV GBPBV FINDV FSCV.

```

```

;-----
.fs_vectors
9E15 .db 9D 81 32 9E A9 83 E6 81 1B 9D 84 AC AF A1 ; ..2.....
; tune_err_sum *= 32
.L9E23
9E23 A2 05 LDX #&05
.L9E25
9E25 0E 54 0E ASL tune_err_sum
9E28 2E 55 0E ROL tune_err_sum+1
9E2B CA DEX
9E2C D0 F7 BNE L9E25
9E2E 60 RTS
; Call the current filing system's FSCV.
.L9E2F
9E2F 6C 1E 02 JMP (FSCV)
;-----
; ARGSV: only A=0, Y=0 (read filing system number) is supported: returns 6 for TFS.
;-----
.argvsv
9E32 C9 00 CMP #&00
9E34 D0 DE BNE L9E14
9E36 C0 00 CPY #&00
9E38 D0 DA BNE L9E14
9E3A A9 06 LDA #&06
9E3C 60 RTS
; Unused: print A as two hex digits.
9E3D .db 98 20 42 9E 8A 48 4A 4A 4A 4A 20 4D 9E 68 29 0F ; . B..HJJJJ M.h).
9E4D .db C9 0A 90 02 69 06 69 30 4C EE FF ; ....i.i0L..
;-----
; tune_err_sum = tune_err_sum / tune_byte_sum (16-bit unsigned division).
;-----
.L9E58
9E58 A9 00 LDA #&00
9E5A 85 BC STA ptr
9E5C 85 BD STA ptr+1
9E5E 18 CLC
9E5F A2 10 LDX #&10
.L9E61
9E61 2E 54 0E ROL tune_err_sum
9E64 2E 55 0E ROL tune_err_sum+1
9E67 26 BC ROL ptr
9E69 26 BD ROL ptr+1
9E6B 38 SEC
9E6C A5 BC LDA ptr
9E6E ED 56 0E SBC tune_byte_sum
9E71 A8 TAY
9E72 A5 BD LDA ptr+1
9E74 ED 57 0E SBC tune_byte_sum+1
9E77 90 04 BCC L9E7D
9E79 84 BC STY ptr
9E7B 85 BD STA ptr+1
.L9E7D
9E7D CA DEX
9E7E D0 E1 BNE L9E61
9E80 2E 54 0E ROL tune_err_sum
9E83 2E 55 0E ROL tune_err_sum+1
9E86 60 RTS
;-----
; MATCH A * COMMAND
; Compare the word at (ptr)+Y with the * command table (&9EE9). The table holds each
; command name, &80, then the handler address. Abbreviations ending in "." are accepted.
; On a match, jump to the handler; carry clear on return if nothing matched.
;-----
.match_star_command
9E87 84 A8 STY &A8
.L9E89
9E89 B1 BC LDA (ptr),Y
9E8B 29 5F AND #&5F
9E8D E8 INX
9E8E C8 INY
9E8F 5D E5 9E EOR &9EE5,X
9E92 F0 F5 BEQ L9E89
9E94 30 1E BMI L9EB4
9E96 88 DEY
9E97 B1 BC LDA (ptr),Y
9E99 C9 2E CMP #&2E ; '.'
9E9B D0 08 BNE L9EA5
.L9E9D
9E9D E8 INX
9E9E BD E5 9E LDA &9EE5,X
9EA1 10 FA BPL L9E9D
9EA3 30 1A BMI L9EBF
.L9EA5
9EA5 E8 INX
9EA6 BD E4 9E LDA &9EE4,X
9EA9 10 FA BPL L9EA5
9EAB E8 INX
9EAC A4 A8 LDY &A8
9EAE DD E6 9E CMP &9EE6,X
9EB1 B0 D6 BCS L9E89

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9EB3 60      RTS
.L9EB4
9EB4 88      DEY
9EB5 B1 BC    LDA (ptr),Y
9EB7 C9 41    CMP #&41          ; 'A'
9EB9 90 04    BCC L9EBF
9EBB C9 7B    CMP #&7B          ; '{'
9EBD 90 E6    BCC L9EA5
.L9EBF
9EBF BD E6 9E LDA &9EE6,X
9EC2 85 B0    STA zp_b0
9EC4 BD E7 9E LDA &9EE7,X
9EC7 85 B1    STA zp_b1
9EC9 6C B0 00 JMP (zp_b0)
;-----
; *HON: rolling headers on, at &7C08 (row 0 of the MODE 7 screen, after the page number).
;-----
.cmd_HON
9ECC A9 7C    LDA #&7C
9ECE 48      PHA
9ECF 4C D5 9E JMP L9ED5
; *HOFF: rolling headers off (address 0).
.cmd_HOFF
9ED2 A9 00    LDA #&00
9ED4 48      PHA
.L9ED5
9ED5 A9 92    LDA #&92          ; OSWORD &7A &92 HEADERS
9ED7 8D 3A 0E STA osw_blk
9EDA 68      PLA
9EDB 8D 3C 0E STA osw_blk+2
9EDE A9 08    LDA #&08
9EE0 8D 3B 0E STA osw_blk+1
9EE3 4C BF A0 JMP LA0BF
9EE6 .db 54 55 4E 45 80 38 9B 44 49 53 50 4C 41 59 80 5C ; TUNE.8.DISPLAY.\
9EF6 .db A1 54 52 41 4E 53 46 45 52 80 4E A1 44 41 54 45 ; .TRANSFER.N.DATE
9F06 .db 80 D2 A4 54 49 4D 45 80 4B A3 42 42 43 80 55 A0 ; ...TIME.K.BBC.U.
9F16 .db 49 54 56 80 4D A0 43 48 80 55 A0 50 41 47 45 80 ; ITV.M.CH.U.PAGE.
9F26 .db 6C A0 48 4F 4E 80 CC 9E 48 4F 46 46 80 D2 9E 54 ; L.HON...HOFF...T
9F36 .db 45 4C 45 54 45 58 54 80 1E 9D 54 45 4C 45 53 4F ; ELETXT...TELES0
9F46 .db 46 54 80 34 9D FF          ; FT.4..
;-----
; OSFILE A=&FF: LOAD A TELES0FTWARE FILE (X/Y = OSFILE parameter block)
; - copy the file name into page 1 (&0100) so that it is safe from being overwritten
; - open the named file on the current page (LAC94: find its first block, apply the
;   load address from the parameter block if one was given)
; - read decoded bytes from channel 14 and store them, through the Tube if necessary,
;   at the load address; a new block may change the address (<DLA>/<DLR>)
; - finally write the length into the parameter block
; run_flag is set when the file is to be run (*RUN, *): "Cannot execute" is given if
; the parameter block asked for its own load address in that case.
;-----
.L9F4C
9F4C A9 00    LDA #&00
9F4E 8D 9A 19 STA run_flag
.L9F51
9F51 86 B4    STX fblk          ; fblk = parameter block
9F53 84 B5    STY fblk+1
9F55 A0 00    LDY #&00
9F57 B1 B4    LDA (fblk),Y      ; zp_b0/zp_b1 = address of the file name
9F59 85 B0    STA zp_b0
9F5B C8      INY
9F5C B1 B4    LDA (fblk),Y
9F5E 85 B1    STA zp_b1
9F60 A9 00    LDA #&00          ; point the block's name field at &0100 and copy the name there
9F62 A8      TAY
9F63 85 B6    STA count
9F65 91 B4    STA (fblk),Y
9F67 C8      INY
9F68 98      TYA
9F69 85 B7    STA count+1
9F6B 91 B4    STA (fblk),Y
9F6D A0 00    LDY #&00
.L9F6F
9F6F B1 B0    LDA (zp_b0),Y
9F71 91 B6    STA (count),Y
9F73 C9 21    CMP #&21          ; '!'
9F75 90 03    BCC L9F7A
9F77 C8      INY
9F78 D0 F5    BNE L9F6F
.L9F7A
9F7A A9 00    LDA #&00
9F7C 91 B6    STA (count),Y
9F7E A9 07    LDA #&07
9F80 A6 B4    LDX fblk
9F82 A4 B5    LDY fblk+1
9F84 20 94 AC JSR LAC94          ; start reading the named file (OSFILE 7 is used internally for this)
9F87 AD 9A 19 LDA run_flag
9F8A F0 19    BEQ L9FA5
9F8C A0 06    LDY #&06
9F8E B1 B4    LDA (fblk),Y

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9F90 F0 13      BEQ L9FA5
9F92 20 98 A1   JSR error_inline
9F95 .str '\xfeCannot execute\x00'
.L9FA5
9FA5 A9 00      LDA #&00
9FA7 85 B6      STA count
9FA9 85 B7      STA count+1
.L9FAB
9FAB A0 0E      LDY #&0E                ; BGET channel 14: next decoded byte of the file
9FAD 20 A9 83   JSR bgetv
9FB0 90 20      BCC L9FD2
9FB2 C9 FE      CMP #&FE                ; &FE: end of file (or ESCAPE)
9FB4 F0 36      BEQ L9FEC
9FB6 C9 02      CMP #&02                ; 2: a new block started: re-read the file information
9FB8 D0 15      BNE L9FCF
9FBA A9 05      LDA #&05
9FBC A6 B4      LDX fblk
9FBE A4 B5      LDY fblk+1
9FC0 20 1A AD   JSR LA01A
9FC3 A0 02      LDY #&02                ; and take its load address
.L9FC5
9FC5 B1 B4      LDA (fblk),Y
9FC7 99 AE 00   STA &00AE,Y
9FCA C8         INY
9FCB C0 06      CPY #&06
9FCD D0 F6      BNE L9FC5
.L9FCF
9FCF 18         CLC
9FD0 90 D9      BCC L9FAB
.L9FD2
9FD2 85 BA      STA xfer_byte            ; store the byte at the load address (Tube aware)
9FD4 A9 01      LDA #&01
9FD6 A2 B0      LDX #&B0
9FD8 A0 00      LDY #&00
9FDA 20 CE A2   JSR LA2CE
9FDD E6 B0      INC zp_b0
9FDF D0 02      BNE L9FE3
9FE1 E6 B1      INC zp_b1
.L9FE3
9FE3 E6 B6      INC count                ; count the bytes loaded
9FE5 D0 02      BNE L9FE9
9FE7 E6 B7      INC count+1
.L9FE9
9FE9 18         CLC
9FEA 90 BF      BCC L9FAB
;-----
; End of file: store the length (count) into the parameter block, and the attributes.
;-----
.L9FEC
9FEC A0 0A      LDY #&0A
9FEE B1 B4      LDA (fblk),Y
9FF0 48         PHA
9FF1 A5 B6      LDA count
9FF3 91 B4      STA (fblk),Y
9FF5 A5 B7      LDA count+1
9FF7 C8         INY
9FF8 91 B4      STA (fblk),Y
9FFA A9 00      LDA #&00
9FFC C8         INY
9FFD 91 B4      STA (fblk),Y
9FFF C8         INY
A000 91 B4      STA (fblk),Y            ; (TFS103.ROM has a corrupt byte here: &00 instead of &91)
A002 68         PLA
A003 C8         INY
A004 91 B4      STA (fblk),Y
A006 60         RTS
;-----
; FSC 2 and 4: */ and *RUN. Load the file with run_flag set, then jump to its execution
; address, through the Tube if a second processor is present.
;-----
.LA007
A007 A9 01      LDA #&01
A009 8D 9A 19   STA run_flag
A00C 20 51 9F   JSR L9F51
A00F A0 06      LDY #&06
.LA011
A011 B1 B4      LDA (fblk),Y            ; copy the execution address to &199A..&199D (overwriting run_flag)
A013 99 94 19   STA &1994,Y
A016 C8         INY
A017 C0 0A      CPY #&0A
A019 D0 F6      BNE LA011
A01B 20 3F A3   JSR LA33F                ; Tube present?
A01E F0 03      BEQ LA023
A020 6C 9A 19   JMP (run_flag)          ; no: just jump to it
.LA023
A023 20 31 A3   JSR LA331
A026 08         PHP
A027 78         SEI
A028 A9 04      LDA #&04                ; Tube: claim it and tell the second processor to start at the address in &199A
A02A A2 9A      LDX #&9A

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A02C A0 19 LDY #&19
A02E 20 06 04 JSR TUBE_CODE
A031 20 39 A3 JSR LA339
A034 28 PLP
A035 60 RTS
;-----
; Error &CF "Data not available" (also used for OSGBPB and reserved OSWORD calls).
;-----
.error_not_available
A036 20 98 A1 JSR error_inline
A039 .str '\xcfdData not available\x00'
;-----
; *ITV1 / *ITV2: channel 3 / 4. The digit after the command is in (ptr),Y.
;-----
.cmd_ITV
A04D B1 BC LDA (ptr),Y
A04F 18 CLC
A050 69 02 ADC #&02
A052 4C 57 A0 JMP LA057
;-----
; *BBC1 / *BBC2 / *CH1-*CH4: channel = digit - 1.
; Three dummy bytes are pushed as a service-call frame so that the SELECT CHANNEL code,
; which ends with JMP svc_exit_claimed, returns here cleanly.
;-----
.cmd_BBC_CH
A055 B1 BC LDA (ptr),Y
.LA057
A057 38 SEC
A058 E9 31 SBC #&31
A05A 48 PHA
A05B 48 PHA
A05C 48 PHA
A05D 38 SEC
A05E 4C 95 8A JMP LA8A95
; Fill osw_blk+1..+7 with &FF (all seven page digits wild).
.LA061
A061 A2 07 LDX #&07
A063 A9 FF LDA #&FF
.LA065
A065 9D 3A 0E STA osw_blk,X
A068 CA DEX
A069 D0 FA BNE LA065
A06B 60 RTS
;-----
; *PAGE <digits>: up to 7 characters 0-9, A-F or "#" (wild). Digits not given stay wild,
; so *PAGE 100 means page 100 with any sub-code. Issues OSWORD &7A &85 FETCH PAGE.
;-----
.cmd_PAGE
A06C 20 61 A0 JSR LA061
A06F B1 BC LDA (ptr),Y
A071 C9 2E CMP #&2E ; '.'
A073 D0 01 BNE LA076
A075 C8 INY
.LA076
A076 B1 BC LDA (ptr),Y
A078 C8 INY
A079 C9 0D CMP #&0D
A07B F0 3D BEQ LA0BA
A07D C9 20 CMP #&20
A07F F0 F5 BEQ LA076
A081 C9 23 CMP #&23 ; '#'
A083 F0 1B BEQ LA0A0
A085 C9 30 CMP #&30 ; '0'
A087 90 1F BCC LA0A8
A089 C9 3A CMP #&3A ; ':'
A08B 90 13 BCC LA0A0
A08D C9 41 CMP #&41 ; 'A'
A08F 90 17 BCC LA0A8
A091 C9 47 CMP #&47 ; 'G'
A093 90 0B BCC LA0A0
A095 C9 61 CMP #&61 ; 'a'
A097 90 0F BCC LA0A8
A099 C9 67 CMP #&67 ; 'g'
A09B 90 03 BCC LA0A0
A09D 4C A8 A0 JMP LA0A8
.LA0A0
A0A0 E8 INX
A0A1 9D 3A 0E STA osw_blk,X
A0A4 E0 08 CPX #&08
A0A6 D0 CE BNE LA076
.LA0A8
A0A8 20 98 A1 JSR error_inline ; anything else: "Bad parameter"
A0AB .str '\xccBad parameter\x00'
.LA0BA
A0BA A9 85 LDA #&85 ; FETCH PAGE
A0BC 8D 3A 0E STA osw_blk
; Issue OSWORD &7A with the block at osw_blk.
.LA0BF
A0BF A9 7A LDA #&7A
A0C1 A2 3A LDX #&3A

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A0C3 A0 0E LDY #0E
A0C5 20 F1 FF JSR OSWORD
A0C8 38 SEC
A0C9 60 RTS
;-----
; Parse a hexadecimal address (3 to 8 digits) into osw_blk+1..+4, low byte first.
;-----
.LA0CA
A0CA A2 04 LDX #04
A0CC A9 00 LDA #00
.LA0CE
A0CE 9D 3A 0E STA osw_blk,X
A0D1 CA DEX
A0D2 D0 FA BNE LA0CE
A0D4 B1 BC LDA (ptr),Y
A0D6 C9 2E CMP #2E ; '.'
A0D8 F0 04 BEQ LA0DE
A0DA C9 20 CMP #20
A0DC D0 CA BNE LA0A8
.LA0DE
A0DE C8 INY
A0DF B1 BC LDA (ptr),Y
A0E1 C9 20 CMP #20
A0E3 F0 F9 BEQ LA0DE
A0E5 A2 00 LDX #00
.LA0E7
A0E7 B1 BC LDA (ptr),Y
A0E9 C9 00 CMP #00
A0EB F0 08 BEQ LA0F5
A0ED C9 20 CMP #20
A0EF F0 B7 BEQ LA0A8
A0F1 E8 INX
A0F2 C8 INY
A0F3 D0 F2 BNE LA0E7
.LA0F5
A0F5 E0 03 CPX #03
A0F7 90 AF BCC LA0A8
A0F9 88 DEY
A0FA A2 00 LDX #00
.LA0FC
A0FC E8 INX
A0FD E0 06 CPX #06
A0FF F0 A7 BEQ LA0A8
A101 20 1F A1 JSR LA11F
A104 B0 15 BCS LA11B
A106 9D 3A 0E STA osw_blk,X
A109 20 1F A1 JSR LA11F
A10C B0 0D BCS LA11B
A10E 0A ASL A
A10F 0A ASL A
A110 0A ASL A
A111 0A ASL A
A112 1D 3A 0E ORA osw_blk,X
A115 9D 3A 0E STA osw_blk,X
A118 4C FC A0 JMP LA0FC
.LA11B
A11B 60 RTS
.LA11C
A11C 4C A8 A0 JMP LA0A8
;-----
; Read one hex digit (working backwards from the end of the address). C set at the start
; of the address (space or ".").
;-----
.LA11F
A11F B1 BC LDA (ptr),Y
A121 C9 2E CMP #2E ; '.'
A123 F0 04 BEQ LA129
A125 C9 20 CMP #20
A127 D0 02 BNE LA12B
.LA129
A129 38 SEC
A12A 60 RTS
.LA12B
A12B 88 DEY
A12C 38 SEC
A12D E9 30 SBC #30
A12F 30 EB BMI LA11C
A131 C9 0A CMP #0A
A133 90 0A BCC LA13F
A135 E9 11 SBC #11
A137 30 E3 BMI LA11C
A139 C9 06 CMP #06
A13B B0 04 BCS LA141
.LA13D
A13D 69 0A ADC #0A
.LA13F
A13F 18 CLC
A140 60 RTS
.LA141
A141 E9 20 SBC #20

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A143 30 D7 BMI LA11C
A145 C9 06 CMP #006
A147 B0 D3 BCS LA11C
A149 90 F2 BCC LA13D
; FSC 5 (*CAT): not supported: "Bad command".
.LA14B
A14B 4C FA A1 JMP LA1FA
;-----
; *TRANSFER <addr>: OSWORD &7A &8F TRANSFER the current page to <addr>.
;-----
.cmd_TRANSFER
A14E 20 61 A0 JSR LA061
A151 20 CA A0 JSR LA0CA
A154 A9 8F LDA #8F
A156 8D 3A 0E STA osw_blk
A159 4C BF A0 JMP LA0BF
;-----
; *DISPLAY <addr>: MODE 7, then print the 960 bytes stored at <addr> by *TRANSFER
; (bit 7 set, zero shown as a space) until the cursor reaches the bottom row.
;-----
.cmd_DISPLAY
A15C 20 CA A0 JSR LA0CA
A15F 20 AF A2 JSR print_inline
A162 .str '\x16\x07\x00'
.LA165
A165 A2 3B LDX #3B
A167 A0 0E LDY #0E
A169 A9 00 LDA #00
A16B 20 CE A2 JSR LA2CE ; read a byte (Tube aware)
A16E C9 00 CMP #00
A170 D0 02 BNE LA174
A172 A9 20 LDA #20
.LA174
A174 09 80 ORA #80
A176 20 EE FF JSR OSWRCH
A179 A9 86 LDA #86 ; stop at row 24
A17B 20 F4 FF JSR OSBYTE ; OSBYTE &86: read text cursor position
A17E C0 18 CPY #18
A180 F0 15 BEQ LA197
A182 EE 3B 0E INC osw_blk+1
A185 D0 DE BNE LA165
A187 EE 3C 0E INC osw_blk+2
A18A D0 D9 BNE LA165
A18C EE 3D 0E INC osw_blk+3
A18F D0 D4 BNE LA165
A191 EE 3E 0E INC OSBYTE
A194 4C 65 A1 JMP LA165
.LA197
A197 60 RTS
;-----
; RAISE AN ERROR: the bytes after the JSR are the error number and message,
; zero-terminated. They are copied to &0100 behind a BRK (&00) which is then executed.
;-----
.error_inline
A198 68 PLA
A199 85 BC STA ptr
A19B 68 PLA
A19C 85 BD STA ptr+1
A19E A9 00 LDA #00
A1A0 A8 TAY
A1A1 8D 00 01 STA stack_page
.LA1A4
A1A4 C8 INY
A1A5 B1 BC LDA (ptr),Y
A1A7 99 00 01 STA stack_page,Y
A1AA D0 F8 BNE LA1A4
A1AC 4C 00 01 JMP stack_page
;-----
; FSCV: FILING SYSTEM CONTROL, A = reason code (table at &A266):
; 0 *OPT 1 EOF 2 */ 3 unrecognised * command
; 4 *RUN 5 *CAT (Bad command) 6 another filing system starting: stop capture
; 7 report the range of file handles (&0E-&0F)
;-----
.fscv
A1AF 85 BC STA ptr
A1B1 48 PHA
A1B2 8A TXA
A1B3 48 PHA
A1B4 A5 BC LDA ptr
A1B6 C9 08 CMP #08
A1B8 90 04 BCC LA1BE
A1BA 68 PLA
A1BB AA TAX
A1BC 68 PLA
A1BD 60 RTS
.LA1BE
A1BE 0A ASL A
A1BF AA TAX
A1C0 BD 66 A2 LDA fsc_table,X
A1C3 85 BC STA ptr

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A1C5 BD 67 A2 LDA &A267,X
A1C8 85 BD STA ptr+1
A1CA 68 PLA
A1CB AA TAX
A1CC 68 PLA
A1CD 6C BC 00 JMP (ptr)
;-----
; FSC 7: TFS uses handles &0E (decoded Telesoftware file) and &0F (raw page).
;-----
.LA1D0
A1D0 A2 0E LDX #&0E
A1D2 A0 0F LDY #&0F
A1D4 60 RTS
;-----
; FSC 1: EOF#X. Returns the end-of-file flag for handle &0E or &0F.
;-----
.LA1D5
A1D5 E0 0E CPX #&0E
A1D7 D0 04 BNE LA1DD
A1D9 AE 9E 0E LDX &0E9E
A1DC 60 RTS
.LA1DD
A1DD E0 0F CPX #&0F
A1DF D0 04 BNE LA1E5
A1E1 AE 9F 0E LDX &0E9F
A1E4 60 RTS
.LA1E5
A1E5 4C B7 83 JMP L83B7
;-----
; FSC 3: an unrecognised * command while TFS is the filing system: look it up in the
; * command table, otherwise "Bad command".
;-----
.LA1E8
A1E8 48 PHA
A1E9 8A TXA
A1EA 48 PHA
A1EB 85 BC STA ptr
A1ED 98 TYA
A1EE 48 PHA
A1EF 85 BD STA ptr+1
A1F1 A2 00 LDX #&00
A1F3 A0 00 LDY #&00
A1F5 20 87 9E JSR match_star_command
A1F8 B0 10 BCS LA20A
.LA1FA
A1FA 20 98 A1 JSR error_inline
A1FD .str '\xfeBad command\x00'
.LA20A
A20A 4C 11 82 JMP svc_exit_claimed
;-----
; FSC 0: *OPT X,Y
; *OPT 0 defaults (*OPT1,128 *OPT2,1 *OPT3,0)
; *OPT 1,n Telesoftware messages (bit 7: "Searching..." etc.)
; *OPT 2,n checksum errors: 0 ignore, 1 retry, 2 error
; *OPT 3,n 1 = keep name-table changes from one file to the next
;-----
.LA20D
A20D E0 00 CPX #&00
A20F D0 0F BNE LA220
A211 A9 01 LDA #&01
A213 8D 98 0E STA opt2_errors
A216 A9 80 LDA #&80
A218 20 7C AC JSR LAC7C
A21B A9 00 LDA #&00
A21D 4C 80 AC JMP LAC80
.LA220
A220 E0 01 CPX #&01
A222 D0 05 BNE LA229
A224 98 TYA
A225 20 7C AC JSR LAC7C
A228 60 RTS
.LA229
A229 E0 02 CPX #&02
A22B D0 05 BNE LA232
A22D 98 TYA
A22E 8D 98 0E STA opt2_errors
A231 60 RTS
.LA232
A232 E0 03 CPX #&03
A234 D0 04 BNE LA23A
A236 98 TYA
A237 4C 80 AC JMP LAC80
.LA23A
A23A 20 98 A1 JSR error_inline
A23D .str '\xcbBad option\x00'
;-----
; Stop the adapter: clear b3 (IRQ enable) in the latch and the shadow, then also clear b2
; (TTXEN, capture enable) in the latch only. The CPU then owns the adapter RAM. The shadow
; keeps b2 set so that capture_on restores it.
;-----

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.capture_off
A249 AD 8E 0E LDA ctrl_shadow
A24C 29 F7 AND #&F7
A24E 8D 10 FC STA TT_STATUS_CTRL
A251 8D 8E 0E STA ctrl_shadow
A254 29 FB AND #&FB
A256 8D 10 FC STA TT_STATUS_CTRL
A259 60 RTS
;-----
; Restart the adapter: b3 (IRQ enable) back on; b2 (capture) comes back from the shadow.
;-----
.capture_on
A25A AD 8E 0E LDA ctrl_shadow
A25D 09 08 ORA #&08
A25F 8D 8E 0E STA ctrl_shadow
A262 8D 10 FC STA TT_STATUS_CTRL
A265 60 RTS
; FSC reason code dispatch table (codes 0-7).
.fsc_table
A266 .db 0D 0A 25 A1 07 A0 E8 A1 07 A0 4B A1 49 A2 D0 A1 ; .....K,I...
; Copy the 4-byte address at (ptr) to ptr2..&C3.
.LA276
A276 A0 00 LDY #&00
A278 B1 BC LDA (ptr),Y
A27A 85 C0 STA ptr2
A27C C8 INY
A27D B1 BC LDA (ptr),Y
A27F 85 C1 STA ptr2+1
A281 C8 INY
A282 B1 BC LDA (ptr),Y
A284 85 C2 STA &C2
A286 C8 INY
A287 B1 BC LDA (ptr),Y
A289 85 C3 STA &C3
A28B 60 RTS
;-----
; Store A at the 32-bit address ptr2..&C3 and step the address (Tube aware).
;-----
.LA28C
A28C 8D 1D 0E STA &0E1D
A28F 98 TYA
A290 48 PHA
A291 8A TXA
A292 48 PHA
A293 A2 C0 LDX #&C0
A295 A0 00 LDY #&00
A297 AD 1D 0E LDA &0E1D
A29A 85 BA STA xfer_byte
A29C A9 01 LDA #&01
A29E 20 CE A2 JSR LA2CE
A2A1 E6 C0 INC ptr2
A2A3 D0 02 BNE LA2A7
A2A5 E6 C1 INC ptr2+1
.LA2A7
A2A7 68 PLA
A2A8 AA TAX
A2A9 68 PLA
A2AA A8 TAY
A2AB AD 1D 0E LDA &0E1D
A2AE 60 RTS
;-----
; PRINT INLINE STRING: prints the zero-terminated bytes that follow the JSR and returns
; to the instruction after the terminator.
;-----
.print_inline
A2AF 68 PLA
A2B0 85 BC STA ptr
A2B2 68 PLA
A2B3 85 BD STA ptr+1
A2B5 A0 01 LDY #&01
A2B7 08 PHP
A2B8 88 DEY
.LA2B9
A2B9 E6 BC INC ptr
A2BB D0 02 BNE LA2BF
A2BD E6 BD INC ptr+1
.LA2BF
A2BF 28 PLP
A2C0 F0 09 BEQ LA2CB
A2C2 B1 BC LDA (ptr),Y
A2C4 08 PHP
A2C5 20 EE FF JSR OSWRCH
A2C8 4C B9 A2 JMP LA2B9
.LA2CB
A2CB 6C BC 00 JMP (ptr)
;-----
; READ OR WRITE ONE BYTE ANYWHERE: X/Y point to a 4-byte address in zero page,
; A=0 read (result in A), A=1 write xfer_byte. Addresses &FFFxxxx, or any address when
; there is no Tube, are in this machine. Others are reached through the Tube: claim it,
; set up a single-byte transfer, move the byte through Tube register 3, release it.

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;-----
.LA2CE
A2CE 48 PHA
A2CF 86 B8 STX xfer_blk
A2D1 84 B9 STY xfer_blk+1
A2D3 A0 03 LDY #03 ; top two bytes &FFFF: I/O processor memory
A2D5 B1 B8 LDA (xfer_blk),Y
A2D7 C9 FF CMP #&FF
A2D9 D0 1E BNE LA2F9
A2DB 88 DEY
A2DC B1 B8 LDA (xfer_blk),Y
A2DE C9 FF CMP #&FF
A2E0 D0 17 BNE LA2F9
.LA2E2
A2E2 88 DEY
A2E3 B1 B8 LDA (xfer_blk),Y
A2E5 48 PHA
A2E6 88 DEY
A2E7 B1 B8 LDA (xfer_blk),Y
A2E9 85 B8 STA xfer_blk
A2EB 68 PLA
A2EC 85 B9 STA xfer_blk+1
A2EE 68 PLA
A2EF F0 05 BEQ LA2F6
A2F1 A5 BA LDA xfer_byte
A2F3 91 B8 STA (xfer_blk),Y
A2F5 60 RTS
.LA2F6
A2F6 B1 B8 LDA (xfer_blk),Y
A2F8 60 RTS
.LA2F9
A2F9 20 3F A3 JSR LA33F ; no Tube: also local
A2FC 08 PHP
A2FD A0 02 LDY #02
A2FF 28 PLP
A300 F0 E0 BEQ LA2E2
A302 20 31 A3 JSR LA331 ; claim the Tube
A305 68 PLA
A306 08 PHP
A307 48 PHA
A308 78 SEI
A309 A6 B8 LDX xfer_blk
A30B A4 B9 LDY xfer_blk+1
A30D 20 06 04 JSR TUBE_CODE ; start a transfer of type A (0 = parasite to host, 1 = host to parasite)
A310 68 PLA
A311 F0 0F BEQ LA322
.LA313
A313 2C E4 FE BIT TUBE_R3_STATUS
A316 50 FB BVC LA313
A318 A5 BA LDA xfer_byte
A31A 8D E5 FE STA TUBE_R3_DATA
A31D 20 39 A3 JSR LA339
A320 28 PLP
A321 60 RTS
.LA322
A322 2C E4 FE BIT TUBE_R3_STATUS
A325 10 FB BPL LA322
A327 AD E5 FE LDA TUBE_R3_DATA
A32A 48 PHA
A32B 20 39 A3 JSR LA339
A32E 68 PLA
A32F 28 PLP
A330 60 RTS
; Claim the Tube (claim ID &C1); loop until granted.
.LA331
A331 A9 C1 LDA #&C1
A333 20 06 04 JSR TUBE_CODE
A336 90 F9 BCC LA331
A338 60 RTS
; Release the Tube.
.LA339
A339 A9 81 LDA #&81
A33B 20 06 04 JSR TUBE_CODE
A33E 60 RTS
; Is a second processor present? (OSBYTE &EA). Z set if not.
.LA33F
A33F A9 EA LDA #&EA
A341 A2 00 LDX #00
A343 A0 FF LDY #&FF
A345 20 F4 FF JSR OSBYTE
A348 E0 00 CPX #00
A34A 60 RTS
;-----
; *TIME: print local time as HH:MM:SS from the broadcast service data packet
; (magazine 8, row 30), or "Time not available" / "Time reception error".
;-----
.cmd_TIME
A34B A9 98 LDA #&98
A34D 8D 21 0E STA time_buf
A350 20 B5 A3 JSR read_time

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A353 A0 0B LDY #00B
A355 B9 21 0E LDA time_buf,Y
A358 C9 F0 CMP #0F0
A35A D0 19 BNE LA375
A35C 20 AF A2 JSR print_inline
A35F .str 'Time not available\x00'
A372 4C B0 A3 JMP LA3B0
.LA375
A375 C9 FF CMP #0FF
A377 D0 1B BNE LA394
A379 20 AF A2 JSR print_inline
A37C .str 'Time reception error\x00'
A391 4C B0 A3 JMP LA3B0
.LA394
A394 A0 05 LDY #005
.LA396
A396 B9 21 0E LDA time_buf,Y
A399 20 EE FF JSR OSWRCH
A39C C8 INY
A39D B9 21 0E LDA time_buf,Y
A3A0 20 EE FF JSR OSWRCH
A3A3 C8 INY
A3A4 C0 0B CPY #00B
A3A6 F0 08 BEQ LA3B0
A3A8 A9 3A LDA #3A
A3AA 20 EE FF JSR OSWRCH ; print ':'
A3AD 4C 96 A3 JMP LA396
.LA3B0
A3B0 20 E7 FF JSR OSNEWL
A3B3 38 SEC
A3B4 60 RTS
;-----
; Wait up to 60 fields for a fresh service data packet (tsdp+9, the time offset byte, is
; cleared first and reappears when a new packet arrives). Then:
; - UTC (tsdp+13..15) is sent as BCD with every digit increased by 1: subtract &11
; - check the hours are 00-23, else "reception error" (&FF in time_buf+11)
; - apply the local time offset (tsdp+9): bit 6 sign, then hours and a half-hour bit
; - note a day carry in time_buf+11 for *DATE, and format HH MM SS as ASCII
; time_buf+11 = &F0 if no packet arrived.
;-----
.read_time
A3B5 08 PHP
A3B6 A2 09 LDX #009
A3B8 A9 00 LDA #000
A3BA 9D 5C 0E STA tsdp,X
A3BD A9 3C LDA #3C ; 60 fields
A3BF 8D 1F 0E STA tsdp_timeout
.LA3C2
A3C2 A9 13 LDA #13
A3C4 20 F4 FF JSR OSBYTE ; OSBYTE &13: wait for vertical sync
A3C7 A2 09 LDX #009
A3C9 BD 5C 0E LDA tsdp,X
A3CC D0 0F BNE LA3DD
A3CE CE 1F 0E DEC tsdp_timeout
A3D1 D0 EF BNE LA3C2
A3D3 A0 0B LDY #00B ; no packet: "not available"
A3D5 A9 F0 LDA #0F0
A3D7 99 21 0E STA time_buf,Y
A3DA 4C 45 A4 JMP LA445
.LA3DD
A3DD 78 SEI
A3DE A0 04 LDY #004
A3E0 99 21 0E STA time_buf,Y
A3E3 A2 0F LDX #00F ; copy UTC hours, minutes, seconds
A3E5 A0 03 LDY #003
.LA3E7
A3E7 BD 5C 0E LDA tsdp,X
A3EA 99 21 0E STA time_buf,Y
A3ED CA DEX
A3EE 88 DEY
A3EF D0 F6 BNE LA3E7
A3F1 C9 11 CMP #11 ; hours in range (&11-&34 before the -&11 adjustment)?
A3F3 90 04 BCC LA3F9
A3F5 C9 35 CMP #35 ; '5'
A3F7 90 0A BCC LA403
.LA3F9
A3F9 A0 0B LDY #00B
A3FB A9 FF LDA #0FF
A3FD 99 21 0E STA time_buf,Y
A400 4C 45 A4 JMP LA445
.LA403
A403 A9 00 LDA #000
A405 A0 0B LDY #00B
A407 99 21 0E STA time_buf,Y
A40A F8 SED ; decimal mode for the digit arithmetic
A40B A0 01 LDY #001
.LA40D
A40D B9 21 0E LDA time_buf,Y
A410 38 SEC
A411 E9 11 SBC #11

```



```

A413 99 21 0E STA time_buf,Y
A416 C8      INY
A417 C0 04    CPY #004
A419 D0 F2    BNE LA40D
A41B A2 00    LDX #000
A41D B9 21 0E LDA time_buf,Y      ; time offset byte
A420 0A      ASL A
A421 0A      ASL A
A422 90 02    BCC LA426
A424 A2 FF    LDX #&FF
.LA426
A426 4A      LSR A
A427 4A      LSR A
A428 4A      LSR A
A429 4A      LSR A
A42A 90 07    BCC LA433
A42C 48      PHA
A42D A9 30    LDA #030
A42F 20 47 A4 JSR LA447
A432 68      PLA
.LA433
A433 F0 0C    BEQ LA441
A435 C9 0A    CMP #00A
A437 90 05    BCC LA43E
A439 38      SEC
A43A E9 0A    SBC #00A
A43C 09 10    ORA #010
.LA43E
A43E 20 64 A4 JSR LA464
.LA441
A441 D8      CLD
A442 20 A2 A4 JSR LA4A2
.LA445
A445 28      PLP
A446 60      RTS
;
; -----
; Add or subtract 30 minutes (X=&FF: subtract), carrying into the hours.
; -----
.LA447
A447 A0 02    LDY #002
A449 20 83 A4 JSR LA483
A44C B0 04    BCS LA452
A44E C9 60    CMP #060      ; ''
A450 90 11    BCC LA463
.LA452
A452 8A      TXA
A453 48      PHA
A454 49 FF    EOR #&FF
A456 AA      TAX
A457 A9 60    LDA #060
A459 20 83 A4 JSR LA483
A45C 68      PLA
A45D AA      TAX
A45E A9 01    LDA #001
A460 20 64 A4 JSR LA464
.LA463
A463 60      RTS
;
; -----
; Add or subtract hours, wrapping at 24 and recording the day carry.
; -----
.LA464
A464 A0 01    LDY #001
A466 20 83 A4 JSR LA483
A469 C9 24    CMP #024      ; '$'
A46B 90 15    BCC LA482
A46D 8A      TXA
A46E 48      PHA
A46F 49 FF    EOR #&FF
A471 AA      TAX
A472 A9 24    LDA #024
A474 20 83 A4 JSR LA483
A477 68      PLA
A478 AA      TAX
A479 29 08    AND #008
A47B 09 01    ORA #001
A47D A0 0B    LDY #00B
A47F 99 21 0E STA time_buf,Y
.LA482
A482 60      RTS
.LA483
A483 8D 1F 0E STA tsdp_timeout
A486 B9 21 0E LDA time_buf,Y
A489 E0 00    CPX #000
A48B D0 07    BNE LA494
A48D 18      CLC
A48E 6D 1F 0E ADC tsdp_timeout
A491 4C 9E A4 JMP LA49E
.LA494
A494 38      SEC
A495 ED 1F 0E SBC tsdp_timeout

```

```

A498 90 03 BCC LA49D
A49A 18 CLC
A49B 90 01 BCC LA49E
.LA49D
A49D 38 SEC
.LA49E
A49E 99 21 0E STA time_buf,Y
A4A1 60 RTS
; Convert the BCD hours, minutes and seconds to ASCII digits.
.LA4A2
A4A2 A0 05 LDY #05
A4A4 8C 1F 0E STY tsdp_timeout
A4A7 A0 01 LDY #01
.LA4A9
A4A9 B9 21 0E LDA time_buf,Y
A4AC 48 PHA
A4AD 4A LSR A
A4AE 4A LSR A
A4AF 4A LSR A
A4B0 4A LSR A
A4B1 09 30 ORA #30
A4B3 8C 20 0E STY &0E20
A4B6 AC 1F 0E LDY tsdp_timeout
A4B9 99 21 0E STA time_buf,Y
A4BC 68 PLA
A4BD 29 0F AND #0F
A4BF 09 30 ORA #30
A4C1 C8 INY
A4C2 99 21 0E STA time_buf,Y
A4C5 C8 INY
A4C6 8C 1F 0E STY tsdp_timeout
A4C9 AC 20 0E LDY &0E20
A4CC C8 INY
A4CD C0 04 CPY #04
A4CF D0 D8 BNE LA4A9
A4D1 60 RTS
;-----
; *DATE: print the date as DD/MM/YYYY from the Modified Julian Date in the broadcast
; service data packet, adjusted for the local time offset.
;-----
.cmd_DATE
A4D2 A0 0C LDY #0C
A4D4 A9 00 LDA #00
A4D6 99 3A 0E STA osw_blk,Y
A4D9 A9 99 LDA #99
A4DB 8D 3A 0E STA osw_blk
A4DE 20 47 A5 JSR read_date
A4E1 A0 0C LDY #0C
A4E3 B9 3A 0E LDA osw_blk,Y
A4E6 C9 F0 CMP #F0
A4E8 D0 19 BNE LA503
A4EA 20 AF A2 JSR print_inline
A4ED .str 'Date not available\x00'
A500 4C 42 A5 JMP LA542
.LA503
A503 C9 FF CMP #FF
A505 D0 1B BNE LA522
A507 20 AF A2 JSR print_inline
A50A .str 'Date reception error\x00'
A51F 4C 42 A5 JMP LA542
.LA522
A522 A0 04 LDY #04
.LA524
A524 B9 3A 0E LDA osw_blk,Y
A527 20 EE FF JSR OSWRCH
A52A C8 INY
A52B B9 3A 0E LDA osw_blk,Y
A52E 20 EE FF JSR OSWRCH
A531 C8 INY
A532 C0 0C CPY #0C
A534 F0 0C BEQ LA542
A536 C0 09 CPY #09
A538 B0 EA BCS LA524
A53A A9 2F LDA #2F
A53C 20 EE FF JSR OSWRCH ; print '/'
A53F 4C 24 A5 JMP LA524
.LA542
A542 20 E7 FF JSR OSNEWL
A545 38 SEC
A546 60 RTS
;-----
; CONVERT THE MODIFIED JULIAN DATE TO A CALENDAR DATE
; The MJD (tsdp+10..12, five BCD digits each sent +1) counts days. The routine works in
; decimal mode throughout:
; - subtract MJD 44969, so that 1 January 1982 (MJD 44970) is day 1; 44968 or 44970
; is used instead when the local time offset moved the date by a day (see read_time)
; - subtract whole years (365, or 366 when the binary year in &0E35 is divisible by 4)
; counting the BCD year up from 1982
; - then subtract whole months using the days-per-month table at &A6E3 (February gets
; an extra day in a leap year); what is left is the day of the month

```

; Dates before 1982 cannot be shown.

```
;-----
.read_date
A547 A9 98 LDA #&98
A549 8D 21 0E STA time_buf
A54C 20 B5 A3 JSR read_time
A54F A0 0B LDY #&0B
A551 B9 21 0E LDA time_buf,Y
A554 10 07 BPL LA55D
A556 C8 INY
A557 99 3A 0E STA osw_blk,Y
A55A 4C 61 A6 JMP LA661
.LA55D
A55D A2 0C LDX #&0C
A55F A0 03 LDY #&03
.LA561
A561 BD 5C 0E LDA tsdp,X
A564 99 3A 0E STA osw_blk,Y
A567 CA DEX
A568 88 DEY
A569 D0 F6 BNE LA561
A56B 08 PHP
A56C 78 SEI
A56D F8 SED
A56E A0 03 LDY #&03
.LA570
A570 B9 3A 0E LDA osw_blk,Y
A573 38 SEC
A574 E9 11 SBC #&11
A576 99 3A 0E STA osw_blk,Y
A579 99 21 0E STA time_buf,Y
A57C 88 DEY
A57D D0 F1 BNE LA570
A57F C8 INY
A580 29 0F AND #&0F
A582 99 3A 0E STA osw_blk,Y
A585 A9 BE LDA #&BE ; binary year 1982 (&07BE), used for the leap-year test
A587 8D 35 0E STA &0E35
A58A A9 07 LDA #&07
A58C 8D 36 0E STA &0E36
A58F 20 62 A6 JSR LA662
A592 A9 82 LDA #&82 ; BCD year &1982
A594 8D 32 0E STA &0E32
A597 A9 19 LDA #&19
A599 8D 33 0E STA &0E33
A59C A9 04 LDA #&04
A59E 8D 37 0E STA &0E37
A5A1 A9 49 LDA #&49
A5A3 8D 38 0E STA &0E38
A5A6 A0 0B LDY #&0B
A5A8 B9 21 0E LDA time_buf,Y
A5AB D0 08 BNE LA5B5
A5AD A9 69 LDA #&69
A5AF 8D 39 0E STA &0E39
A5B2 4C C6 A5 JMP LA5C6
.LA5B5
A5B5 C9 01 CMP #&01
A5B7 D0 08 BNE LA5C1
A5B9 A9 68 LDA #&68
A5BB 8D 39 0E STA &0E39
A5BE 4C C6 A5 JMP LA5C6
.LA5C1
A5C1 A9 70 LDA #&70
A5C3 8D 39 0E STA &0E39
.LA5C6
A5C6 20 AE A6 JSR LA6AE
A5C9 A9 00 LDA #&00
A5CB 8D 37 0E STA &0E37
A5CE A9 03 LDA #&03
A5D0 8D 38 0E STA &0E38
A5D3 A9 65 LDA #&65
A5D5 8D 39 0E STA &0E39
.LA5D8
A5D8 20 72 A6 JSR LA672
A5DB 90 25 BCC LA602
A5DD 20 AE A6 JSR LA6AE
A5E0 AD 32 0E LDA &0E32
A5E3 18 CLC
A5E4 69 01 ADC #&01
A5E6 8D 32 0E STA &0E32
A5E9 90 09 BCC LA5F4
A5EB AD 33 0E LDA &0E33
A5EE 18 CLC
A5EF 69 01 ADC #&01
A5F1 8D 33 0E STA &0E33
.LA5F4
A5F4 EE 35 0E INC &0E35
A5F7 D0 03 BNE LA5FC
A5F9 EE 36 0E INC &0E36
.LA5FC
```

```

A5FC 20 62 A6 JSR LA662
A5FF 4C D8 A5 JMP LA5D8
.LA602
A602 A9 00 LDA #&00
A604 8D 38 0E STA &0E38
A607 A2 00 LDX #&00
A609 A9 01 LDA #&01
A60B 8D 34 0E STA &0E34
.LA60E
A60E E0 01 CPX #&01
A610 F0 03 BEQ LA615
A612 0E 31 0E ASL &0E31
.LA615
A615 BD E3 A6 LDA month\_days,X
A618 8D 39 0E STA &0E39
A61B 20 72 A6 JSR LA672
A61E 90 17 BCC LA637
A620 20 AE A6 JSR LA6AE
A623 E0 01 CPX #&01
A625 F0 03 BEQ LA62A
A627 4E 31 0E LSR &0E31
.LA62A
A62A E8 INX
A62B AD 34 0E LDA &0E34
A62E 18 CLC
A62F 69 01 ADC #&01
A631 8D 34 0E STA &0E34
A634 4C 0E A6 JMP LA60E
.LA637
A637 28 PLP
A638 A0 03 LDY #&03
A63A B9 3A 0E LDA osw\_blk,Y
A63D C8 INY
A63E 20 CF A6 JSR LA6CF
A641 AD 34 0E LDA &0E34
A644 C8 INY
A645 20 CF A6 JSR LA6CF
A648 AD 33 0E LDA &0E33
A64B C8 INY
A64C 20 CF A6 JSR LA6CF
A64F AD 32 0E LDA &0E32
A652 C8 INY
A653 20 CF A6 JSR LA6CF
A656 A0 03 LDY #&03
.LA658
A658 B9 21 0E LDA time\_buf,Y
A65B 99 3A 0E STA osw\_blk,Y
A65E 88 DEY
A65F D0 F7 BNE LA658
.LA661
A661 60 RTS
.LA662
A662 A9 00 LDA #&00
A664 8D 31 0E STA &0E31
A667 A9 03 LDA #&03
A669 2C 35 0E BIT &0E35
A66C D0 03 BNE LA671
A66E EE 31 0E INC &0E31
.LA671
A671 60 RTS
.LA672
A672 8A TXA
A673 48 PHA
A674 A2 00 LDX #&00
A676 A0 01 LDY #&01
.LA678
A678 B9 3A 0E LDA osw\_blk,Y
A67B DD 37 0E CMP &0E37,X
A67E 90 28 BCC LA6AB
A680 D0 29 BNE LA6AB
A682 C8 INY
A683 E8 INX
A684 E0 01 CPX #&01
A686 F0 F0 BEQ LA678
A688 B9 3A 0E LDA osw\_blk,Y
A68B DD 37 0E CMP &0E37,X
A68E 90 18 BCC LA6AB
A690 D0 03 BNE LA695
A692 18 CLC
A693 90 16 BCC LA6AB
.LA695
A695 A9 01 LDA #&01
A697 2C 31 0E BIT &0E31
A69A F0 0F BEQ LA6AB
A69C B9 3A 0E LDA osw\_blk,Y
A69F FE 37 0E INC &0E37,X
A6A2 DD 37 0E CMP &0E37,X
A6A5 D0 01 BNE LA6A8
A6A7 18 CLC
.LA6A8

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A6A8 DE 37 0E DEC &0E37,X
.LA6AB
A6AB 68 PLA
A6AC AA TAX
A6AD 60 RTS
.LA6AE
A6AE 8A TXA
A6AF 48 PHA
A6B0 A2 02 LDX #&02
A6B2 A0 03 LDY #&03
A6B4 A9 01 LDA #&01
A6B6 2C 31 0E BIT &0E31
A6B9 F0 03 BEQ LA6BE
A6BB 18 CLC
A6BC 90 01 BCC LA6BF
.LA6BE
A6BE 38 SEC
.LA6BF
A6BF B9 3A 0E LDA osw_blk,Y
A6C2 FD 37 0E SBC &0E37,X
A6C5 99 3A 0E STA osw_blk,Y
A6C8 CA DEX
A6C9 88 DEY
A6CA D0 F3 BNE LA6BF
A6CC 68 PLA
A6CD AA TAX
A6CE 60 RTS
.LA6CF
A6CF 48 PHA
A6D0 4A LSR A
A6D1 4A LSR A
A6D2 4A LSR A
A6D3 4A LSR A
A6D4 09 30 ORA #&30
A6D6 99 3A 0E STA osw_blk,Y
A6D9 C8 INY
A6DA 68 PLA
A6DB 29 0F AND #&0F
A6DD 09 30 ORA #&30
A6DF 99 3A 0E STA osw_blk,Y
A6E2 60 RTS
; Days in each month, BCD (January to December).
.month_days
A6E3 .db 31 28 31 30 31 30 ; 1(1010
;-----
; (Tail of the date code's "1(1010110101" table and the start of the command table.)
;-----
A6E9 .db 31 31 30 31 30 31 ; 110101
;-----
; TELESOFTWARE COMMAND SUBROUTINE TABLE: 42 six-byte entries
; 3-letter name, flags, handler address (lo, hi)
; Flags: bit 7 = install at start-up; bits 0-3 = which pass the command runs in, bit 4 =
; "is a command". Entries using &AA00 are commands this decoder doesn't implement: they
; raise "Telesoftware error". The table is followed by the letters giving each command's
; default escaped name (&A7EC: "A v j i c ...").
;-----
.ts_commands
A6EF .db 43 45 42 11 0D AA 43 45 54 90 0D AA 43 53 42 10 ; CEB...CET...CSB.
A6FF .db 0D AA 43 45 30 12 0D AA 43 45 31 12 0D AA 43 45 ; ..CE0...CE1...CE
A70F .db 32 12 0D AA 43 45 33 12 0D AA 43 45 34 12 0D AA ; 2...CE3...CE4...
A71F .db 43 45 35 12 0D AA 44 41 43 92 B7 AF 44 43 4F 92 ; CE5...DAC...DC0.
A72F .db AA B0 44 44 54 92 B7 AF 44 45 42 91 6A B2 44 45 ; ..DDT...DEB.j.DE
A73F .db 43 92 EC B1 44 45 46 92 A3 AF 44 45 53 92 D7 AF ; C...DEF...DES...
A74F .db 44 45 54 92 AD AF 44 49 47 92 B7 AF 44 49 52 92 ; DET...DIG...DIR.
A75F .db AF B1 44 4C 41 92 ED AE 44 4C 43 92 DB B1 44 4C ; ..DLA...DLC...DL
A76F .db 52 92 20 AF 44 4C 53 92 34 B0 44 4E 43 92 B7 AF ; R. .DLS.4.DNC...
A77F .db 44 4E 44 92 B2 AF 44 53 41 90 0D AA 44 53 42 90 ; DND...DSA...DSB.
A78F .db 2F AE 44 53 4C 92 0D AA 44 53 4E 90 0D AA 44 53 ; /.DSL...DSN...DS
A79F .db 54 92 0D AA 44 54 4C 92 0C B1 44 58 41 92 0E AF ; T...DTL...DXA...
A7AF .db 44 58 52 92 44 AF 45 53 43 00 00 00 45 4D 50 90 ; DXR.D.ESC...EMP.
A7BF .db 0D AA 54 58 54 90 99 AD 54 58 38 90 0D AA 55 45 ; ..TXT...TX8...UE
A7CF .db 52 1F 00 00 55 4C 42 12 B4 B1 55 52 42 12 BD B1 ; R...ULB...URB...
A7DF .db 56 44 58 90 0D AA 54 58 4F 90 9C AD 00 ; VDX...TX0....
;-----
; Default escaped-name letter for each installed command (in table order).
;-----
A7EC .db 41 76 6A 69 63 72 79 71 64 6B 70 6C 74 6D 73 67 ; Avjicryqdkplmtsg
A7FC .db 78 75 62 65 66 68 61 6E 6F 45 42 43 44 46 ; xubefhanoEBCDF
;-----
; INITIALISE THE DECODER: default name tables, empty heap at &169A, clear the decoder
; state &161A-&1691, reset the output queue.
;-----
.ts_init
A80A 20 45 A8 JSR ts_default_names
A80D A9 9A LDA #&9A
A80F 8D 92 16 STA heap
A812 A9 16 LDA #&16
A814 8D 93 16 STA heap+1
A817 A0 77 LDY #&77
A819 4C 1E A8 JMP LA81E

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```

; Reset the decoder state (but not the name tables).
.ts_reset
    A81C A0 63    LDY #&63
.LA81E
    A81E A9 00    LDA #&00
.LA820
    A820 99 1A 16 STA frame_sp,Y
    A823 88        DEY
    A824 10 FA     BPL LA820
    A826 8C 23 16 STY ts_blocks_total    ; no blocks expected yet
    A829 8C 6E 16 STY ts_total_save
    A82C 8C 5F 16 STY ts_attr
    A82F A0 01     LDY #&01                ; block 1 next
    A831 8C 67 16 STY ts_block_no
    A834 8C 25 16 STY ts_block_map
    A837 20 BE AB JSR ts_reset_queue
    A83A A9 9A     LDA #&9A                ; frame stack at &159A
    A83C 8D 1A 16 STA frame_sp
    A83F A9 15     LDA #&15
    A841 8D 1B 16 STA frame_sp+1
    A844 60        RTS
;-----
; DEFAULT NAME TABLES (the general Telesoftware format, manual table 9.2):
; every lone name decodes to itself (kind 1, value = name)
; every escaped name means <ESC> (kind &1B) until a command is installed below
; lone &1B and &9B, escaped &7C and &FC are the escape operator
; Each installed command from the table at &A6EF is given the escaped name from &A7EC
; and the same name with bit 7 set (both parities).
;-----
.ts_default_names
    A845 20 DF A8 JSR use_lone_tables
    A848 A0 00    LDY #&00
.LA84A
    A84A A9 01    LDA #&01
    A84C 91 C6    STA (nm_kind),Y
    A84E 98        TYA
    A84F 91 C8    STA (nm_lo),Y
    A851 A9 00    LDA #&00
    A853 91 CA    STA (nm_hi),Y
    A855 C8        INY
    A856 D0 F2    BNE LA84A
    A858 A9 1B    LDA #&1B
    A85A 8D B5 0F STA &0FB5
    A85D 8D 16 10 STA &1016
    A860 8D 35 10 STA &1035
    A863 8D 96 10 STA &1096
    A866 20 FA A8 JSR use_escaped_tables
.LA869
    A869 A9 1B    LDA #&1B
    A86B 91 C6    STA (nm_kind),Y
    A86D A9 00    LDA #&00
    A86F 91 C8    STA (nm_lo),Y
    A871 91 CA    STA (nm_hi),Y
    A873 C8        INY
    A874 D0 F3    BNE LA869
    A876 A0 FF    LDY #&FF
    A878 A2 FF    LDX #&FF
.LA87A
    A87A C8        INY
    A87B B9 EF A6 LDA ts_commands,Y
    A87E F0 40    BEQ LA8C0
    A880 C8        INY
    A881 C8        INY
    A882 C8        INY
    A883 B9 EF A6 LDA ts_commands,Y
    A886 85 CE    STA ts_tmp
    A888 29 80    AND #&80
    A88A F0 2F    BEQ LA8BB
    A88C A5 CE    LDA ts_tmp
    A88E 29 7F    AND #&7F
    A890 48        PHA
    A891 C8        INY
    A892 B9 EF A6 LDA ts_commands,Y
    A895 85 CC    STA ts_ptr
    A897 C8        INY
    A898 B9 EF A6 LDA ts_commands,Y
    A89B 85 CD    STA ts_ptr+1
    A89D 84 CE    STY ts_tmp
    A89F E8        INX
    A8A0 BD EC A7 LDA &A7EC,X
.LA8A3
    A8A3 A8        TAY
    A8A4 68        PLA
    A8A5 48        PHA
    A8A6 91 C6    STA (nm_kind),Y
    A8A8 A5 CC    LDA ts_ptr
    A8AA 91 C8    STA (nm_lo),Y
    A8AC A5 CD    LDA ts_ptr+1
    A8AE 91 CA    STA (nm_hi),Y
    A8B0 98        TYA

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```

A8B1 49 80    EOR #&80
A8B3 30 EE    BMI LA8A3
A8B5 68       PLA
A8B6 A4 CE    LDY ts_tmp
A8B8 4C 7A A8 JMP LA87A
.LA8BB
A8BB C8       INY
A8BC C8       INY
A8BD 4C 7A A8 JMP LA87A
.LA8C0
A8C0 A9 01    LDA #&01
A8C2 A0 1B    LDY #&1B
A8C4 91 C6    STA (nm_kind),Y
A8C6 8C B5 13 STY esc_kind+&1B
A8C9 A0 7C    LDY #&7C
A8CB 91 C6    STA (nm_kind),Y
A8CD 8C 16 14 STY esc_kind+&7C
A8D0 A0 9B    LDY #&9B
A8D2 91 C6    STA (nm_kind),Y
A8D4 8C 35 14 STY esc_kind+&9B
A8D7 A0 FC    LDY #&FC
A8D9 91 C6    STA (nm_kind),Y
A8DB 8C 96 14 STY esc_kind+&FC
A8DE 60       RTS
; Point nm_kind/nm_lo/nm_hi at the lone-name tables.
.use_lone_tables
A8DF 48       PHA
A8E0 A9 0F    LDA #&0F
A8E2 85 C7    STA nm_kind+1
A8E4 A9 9A    LDA #&9A
A8E6 85 C6    STA nm_kind
A8E8 A9 10    LDA #&10
A8EA 85 C9    STA nm_lo+1
A8EC A9 9A    LDA #&9A
A8EE 85 C8    STA nm_lo
A8F0 A9 11    LDA #&11
A8F2 85 CB    STA nm_hi+1
A8F4 A9 9A    LDA #&9A
A8F6 85 CA    STA nm_hi
A8F8 68       PLA
A8F9 60       RTS
; Point them at the escaped-name tables.
.use_escaped_tables
A8FA 48       PHA
A8FB A9 12    LDA #&12
A8FD 85 C7    STA nm_kind+1
A8FF A9 9A    LDA #&9A
A901 85 C6    STA nm_kind
A903 A9 13    LDA #&13
A905 85 C9    STA nm_lo+1
A907 A9 9A    LDA #&9A
A909 85 C8    STA nm_lo
A90B A9 14    LDA #&14
A90D 85 CB    STA nm_hi+1
A90F A9 9A    LDA #&9A
A911 85 CA    STA nm_hi
A913 68       PLA
A914 60       RTS
;-----
; Save A, X, Y and ts_ptr/ts_tmp on the decoder's own frame stack (6 bytes per frame,
; at frame_sp). Error &C8 if it overflows into the heap.
;-----
.push_frame
A915 48       PHA
A916 8A       TXA
A917 48       PHA
A918 98       TYA
A919 48       PHA
A91A A5 CC    LDA ts_ptr
A91C 48       PHA
A91D A5 CD    LDA ts_ptr+1
A91F 48       PHA
A920 A5 CE    LDA ts_tmp
A922 48       PHA
A923 AD 1A 16 LDA frame_sp
A926 85 CC    STA ts_ptr
A928 AD 1B 16 LDA frame_sp+1
A92B 85 CD    STA ts_ptr+1
A92D A0 00    LDY #&00
.LA92F
A92F 68       PLA
A930 91 CC    STA (ts_ptr),Y
A932 C8       INY
A933 C0 06    CPY #&06
A935 D0 F8    BNE LA92F
A937 98       TYA
A938 A4 CD    LDY ts_ptr+1
A93A 18       CLC
A93B 65 CC    ADC ts_ptr
A93D 8D 1A 16 STA frame_sp

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```

A940 90 01 BCC LA943
A942 C8 INY
.LA943
A943 8C 1B 16 STY frame_sp+1
A946 C0 16 CPY #&16
A948 30 0B BMI LA955
A94A D0 04 BNE LA950
A94C C9 19 CMP #&19
A94E 90 05 BCC LA955
.LA950
A950 A9 C8 LDA #&C8
A952 4C 0D AA JMP ts_error
.LA955
A955 60 RTS
; Restore A, X, Y and ts_ptr/ts_tmp from the frame stack.
.pop_frame
A956 AD 1A 16 LDA frame_sp
A959 AC 1B 16 LDY frame_sp+1
A95C 38 SEC
A95D E9 06 SBC #&06
A95F B0 01 BCS LA962
A961 88 DEY
.LA962
A962 8D 1A 16 STA frame_sp
A965 85 CC STA ts_ptr
A967 8C 1B 16 STY frame_sp+1
A96A 84 CD STY ts_ptr+1
A96C A0 05 LDY #&05
.LA96E
A96E B1 CC LDA (ts_ptr),Y
A970 48 PHA
A971 88 DEY
A972 10 FA BPL LA96E
A974 68 PLA
A975 85 CE STA ts_tmp
A977 68 PLA
A978 85 CD STA ts_ptr+1
A97A 68 PLA
A97B 85 CC STA ts_ptr
A97D 68 PLA
A97E A8 TAY
A97F 68 PLA
A980 AA TAX
A981 68 PLA
A982 60 RTS
;-----
; NEXT DECODED BYTE: from the output queue if it has anything; otherwise decode more
; names from the page (ts_decode_name) until it does. C set at the end of a block.
; Error &C9 if decoding stops unexpectedly.
;-----
.ts_getbyte
A983 20 15 A9 JSR push_frame
.LA986
A986 AD 1C 16 LDA outq_wr
A989 CD 1E 16 CMP outq_rd
A98C AD 1D 16 LDA outq_wr+1
A98F ED 1F 16 SBC outq_rd+1
A992 90 20 BCC LA9B4
A994 20 BE AB JSR ts_reset_queue
A997 20 6E AA JSR ts_decode_name
A99A 90 EA BCC LA986
A99C 68 PLA
A99D A8 TAY
A99E 68 PLA
A99F C0 F9 CPY #&F9
A9A1 D0 0C BNE LA9AF
A9A3 C9 AB CMP #&AB
A9A5 D0 08 BNE LA9AF
A9A7 48 PHA
A9A8 98 TYA
A9A9 48 PHA
A9AA 20 56 A9 JSR pop_frame
A9AD 38 SEC
A9AE 60 RTS
.LA9AF
A9AF A9 C9 LDA #&C9
A9B1 4C 0D AA JMP ts_error
.LA9B4
A9B4 AD 1E 16 LDA outq_rd
A9B7 08 PHP
A9B8 85 CC STA ts_ptr
A9BA AD 1F 16 LDA outq_rd+1
A9BD 85 CD STA ts_ptr+1
A9BF A0 00 LDY #&00
A9C1 B1 CC LDA (ts_ptr),Y
A9C3 CE 1E 16 DEC outq_rd
A9C6 28 PLP
A9C7 D0 03 BNE LA9CC
A9C9 CE 1F 16 DEC outq_rd+1
.LA9CC

```



```

A9CC 48      PHA
A9CD 20 56 A9 JSR pop_frame
A9D0 68      PLA
A9D1 18      CLC
A9D2 60      RTS
;
;-----
; Issue OSWORD &7A sub-command A for the decoder (block at ts_osw). For FETCH, CHAINED
; and HEAD OF CHAIN, "Searching..." is printed when *OPT1 messages are on.
;-----
.ts_osword
A9D3 8D 6F 16 STA ts_osw
A9D6 48      PHA
A9D7 20 15 A9 JSR push_frame
A9DA 68      PLA
A9DB C9 95    CMP #&95
A9DD F0 08    BEQ LA9E7
A9DF C9 86    CMP #&86
A9E1 F0 04    BEQ LA9E7
A9E3 C9 88    CMP #&88
A9E5 D0 19    BNE LAA00
.LA9E7
A9E7 AD 94 16 LDA opt1_msgs
A9EA 10 14    BPL LAA00
A9EC 20 AF A2 JSR print_inline
A9EF .str '\r\nSearching...\r\n\x00'
.LAA00
AA00 A2 6F    LDX #&6F
AA02 A0 16    LDY #&16
AA04 A9 7A    LDA #&7A
AA06 20 F1 FF JSR OSWORD          ; OSWORD &7A: Teletext system call
AA09 20 56 A9 JSR pop_frame
AA0C 60      RTS
;
;-----
; RAISE "Telesoftware error" with error number A, after resetting the decoder:
; &C8 frame stack full &C9 unexpected end &CA bad name kind &CB unknown command
; &CC bad digit &CD output queue full &CE page data error &CF block out of sequence
; Commands this decoder doesn't implement point here, entered with A=&CB.
;-----
.ts_error
AA0D 48      PHA
AA0E 20 1C A8 JSR ts_reset
AA11 A0 16    LDY #&16
.LAA13
AA13 B9 23 AA LDA &AA23,Y
AA16 99 00 01 STA stack_page,Y
AA19 88      DEY
AA1A 10 F7    BPL LAA13
AA1C 68      PLA
AA1D 8D 01 01 STA &0101
AA20 4C 00 01 JMP stack_page
;
;-----
; Error template: BRK, number (filled in), "Telesoftware error", 0.
;-----
AA23 .db 00 EA 54 65 6C 65 73 6F 66 74 77 61 72 65 20 65 ; ..Telesoftware e
AA33 .db 72 72 6F 72 00                                ; rror.
;
;-----
; Read an operand string of A decoded bytes into the heap (length byte first).
;-----
.ts_read_string
AA38 48      PHA
AA39 20 15 A9 JSR push_frame
AA3C AD 92 16 LDA heap
AA3F 85 CC    STA ts_ptr
AA41 AD 93 16 LDA heap+1
AA44 85 CD    STA ts_ptr+1
AA46 68      PLA
AA47 EE 4B 16 INC ts_operand
AA4A A0 00    LDY #&00
AA4C 91 CC    STA (ts_ptr),Y
AA4E AA      TAX
AA4F F0 09    BEQ LAA5A
.LAA51
AA51 C8      INY
AA52 20 83 A9 JSR ts_getbyte
AA55 91 CC    STA (ts_ptr),Y
AA57 CA      DEX
AA58 D0 F7    BNE LAA51
.LAA5A
AA5A 98      TYA
AA5B 38      SEC
AA5C 6D 92 16 ADC heap
AA5F 8D 92 16 STA heap
AA62 90 03    BCC LAA67
AA64 EE 93 16 INC heap+1
.LAA67
AA67 CE 4B 16 DEC ts_operand
AA6A 20 56 A9 JSR pop_frame
AA6D 60      RTS
;
;-----
; DECODE ONE NAME FROM THE PAGE

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```

; Remember the read position (OSWORD &84), fetch the next raw byte, and look it up:
; kind 0      ignore
; kind 1      emit one byte (nm_lo)
; kind 2      emit two bytes
; kind 3+     emit the string at (nm_lo, nm_hi)
; kind &1B    escape operator: look the next name up in the escaped tables
; kind &1F    <UER> transmission error: reload the page and restart the block
; kind &1x    a command: run it if its pass number matches the current pass
; kind >&1F    error &CA
; Data is only emitted inside a block (ts_state 2) or while a command reads an operand.
;-----
.ts_decode_name
AA6E 20 15 A9 JSR push_frame
AA71 A9 84 LDA #&84 ; OSWORD &84: mark the position of this name
AA73 20 D3 A9 JSR ts_osword
.LAA76
AA76 20 DF A8 JSR use_lone_tables
.LAA79
AA79 20 7F AB JSR ts_rawbyte ; next raw byte from the page
AA7C 90 05 BCC LAA83
AA7E 20 56 A9 JSR pop_frame
AA81 38 SEC
AA82 60 RTS
.LAA83
AA83 A8 TAY
AA84 B1 C6 LDA (nm_kind),Y ; kind of this name
AA86 AA TAX
AA87 F0 62 BEQ LAAEB
AA89 C9 1B CMP #&1B ; escape operator?
AA8B F0 58 BEQ LAAE5
AA8D C9 1F CMP #&1F ; <UER>?
AA8F F0 44 BEQ LAA05
AA91 B0 3D BCS LAA00
AA93 29 10 AND #&10 ; a command?
AA95 D0 59 BNE LAAF0
AA97 A9 02 LDA #&02
AA99 CD 20 16 CMP ts_state
AA9C F0 05 BEQ LAAA3
AA9E AD 4B 16 LDA ts_operand
AAA1 F0 48 BEQ LAAEB
.LAAA3
AAA3 E0 02 CPX #&02
AAA5 30 21 BMI LAAC8
AAA7 F0 1A BEQ LAAC3
AAA9 B1 C8 LDA (nm_lo),Y
AAB 85 CC STA ts_ptr
AAB AD B1 CA LDA (nm_hi),Y
AAAF 85 CD STA ts_ptr+1
AAB1 A0 00 LDY #&00
AAB3 B1 CC LDA (ts_ptr),Y
AAB5 F0 34 BEQ LAAEB
AAB7 AA TAX
.LAAB8
AAB8 C8 INY
AAB9 B1 CC LDA (ts_ptr),Y
AABB 20 4D AB JSR ts_emit
AABE CA DEX
AABF D0 F7 BNE LAAB8
AAC1 F0 28 BEQ LAAEB
.LAAC3
AAC3 B1 CA LDA (nm_hi),Y
AAC5 20 4D AB JSR ts_emit
.LAAC8
AAC8 B1 C8 LDA (nm_lo),Y
AACA 20 4D AB JSR ts_emit
AACD 4C EB AA JMP LAAEB
.LAAD0
AAD0 A9 CA LDA #&CA
AAD2 4C 0D AA JMP ts_error
.LAAD5
AAD5 20 BE AB JSR ts_reset_queue ; <UER>: OSWORD &80 RELOAD PAGE and start the block again
AAD8 A9 00 LDA #&00
AADA 8D 20 16 STA ts_state
AADD A9 80 LDA #&80
AADF 20 D3 A9 JSR ts_osword
AAE2 4C 76 AA JMP LAA76
.LAAE5
AAE5 20 FA A8 JSR use_escaped_tables
AAE8 4C 79 AA JMP LAA79
.LAAEB
AAEB 20 56 A9 JSR pop_frame
AAEE 18 CLC
AAEF 60 RTS
.LAAF0
AAF0 8A TXA ; run the command if it belongs to this pass
AAF1 29 0F AND #&0F
AAF3 CD 20 16 CMP ts_state
AAF6 D0 F3 BNE LAAEB
AAF8 B1 C8 LDA (nm_lo),Y
AAFA 85 CC STA ts_ptr

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AAFC B1 CA LDA (nm_hi),Y
AAFE 85 CD STA ts_ptr+1
AB00 20 0D AB JSR LAB0D
AB03 B0 03 BCS LAB08
AB05 4C EB AA JMP LAABEB
.LAB08
AB08 20 56 A9 JSR pop_frame
AB0B 38 SEC
AB0C 60 RTS
;-----
; Call a command subroutine at (ts_ptr), with A=&CB for unimplemented ones.
;-----
.LAB0D
AB0D A9 CB LDA #&CB
AB0F 6C CC 00 JMP (ts_ptr)
;-----
; READ A NUMBER OPERAND: one decoded character. After <TXT>, numbers are ASCII hex:
; "0"- "9", "A"- "F", or "X" followed by two hex digits. Error &CC for anything else.
;-----
.ts_number
AB12 EE 4B 16 INC ts_operand
AB15 20 83 A9 JSR ts_getbyte
AB18 48 PHA
AB19 AD 7E 16 LDA ts_hexmode
AB1C D0 04 BNE LAB22
AB1E 68 PLA
AB1F 4C 49 AB JMP LAB49
.LAB22
AB22 68 PLA
AB23 29 4F AND #&4F
AB25 C9 0A CMP #&0A
AB27 30 20 BMI LAB49
AB29 38 SEC
AB2A E9 37 SBC #&37
AB2C C9 10 CMP #&10
AB2E 30 19 BMI LAB49
AB30 C9 11 CMP #&11
AB32 F0 05 BEQ LAB39
AB34 A9 CC LDA #&CC
AB36 4C 0D AA JMP ts_error
.LAB39
AB39 20 12 AB JSR ts_number
AB3C 0A ASL A
AB3D 0A ASL A
AB3E 0A ASL A
AB3F 0A ASL A
AB40 8D 21 16 STA ts_hex
AB43 20 12 AB JSR ts_number
AB46 0D 21 16 ORA ts_hex
.LAB49
AB49 CE 4B 16 DEC ts_operand
AB4C 60 RTS
;-----
; Put a decoded byte on the output queue (which grows down from &1999 towards the heap).
; Error &CD if it meets the heap.
;-----
.ts_emit
AB4D 48 PHA
AB4E 20 15 A9 JSR push_frame
AB51 68 PLA
AB52 AC 1C 16 LDY outq_wr
AB55 84 CC STY ts_ptr
AB57 CE 1C 16 DEC outq_wr
AB5A AE 1D 16 LDX outq_wr+1
AB5D 86 CD STX ts_ptr+1
AB5F C0 00 CPY #&00
AB61 D0 03 BNE LAB66
AB63 CE 1D 16 DEC outq_wr+1
.LAB66
AB66 EC 93 16 CPX heap+1
AB69 90 07 BCC LAB72
AB6B D0 0A BNE LAB77
AB6D CC 92 16 CPY heap
AB70 B0 05 BCS LAB77
.LAB72
AB72 A9 CD LDA #&CD
AB74 4C 0D AA JMP ts_error
.LAB77
AB77 A0 00 LDY #&00
AB79 91 CC STA (ts_ptr),Y
AB7B 20 56 A9 JSR pop_frame
AB7E 60 RTS
;-----
; NEXT RAW BYTE FROM THE PAGE: BGET channel 15. At the end of the page, unlock it and ask
; for the next page of the file: HEAD OF CHAIN (&88) when blocks may come in any order,
; otherwise CHAINED PAGE (&86). ESCAPE or a page error gives error &CE.
;-----
.ts_rawbyte
AB7F 8C 66 16 STY ts_savey
.LAB82

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AB82 A0 0F LDY #&0F
AB84 20 D7 FF JSR OSBGET
AB87 90 29 BCC LABB2
AB89 C9 FE CMP #&FE
AB8B F0 20 BEQ LABAD
AB8D C9 01 CMP #&01
AB8F 30 1C BMI LABAD
AB91 D0 03 BNE LAB96
AB93 EE 24 16 INC ts_eof ; count the page

.LAB96
AB96 A9 9A LDA #&9A ; OSWORD &9A: release this page
AB98 20 D3 A9 JSR ts_osword
AB9B AD 68 16 LDA ts_random
AB9E F0 05 BEQ LABA5
ABA0 A9 88 LDA #&88
ABA2 4C A7 AB JMP LABA7

.LABA5
ABA5 A9 86 LDA #&86

.LABA7
ABA7 20 D3 A9 JSR ts_osword
ABAA 4C 82 AB JMP LAB82

.LABAD
ABAD A9 CE LDA #&CE
ABAF 4C 0D AA JMP ts_error

.LABB2
ABB2 AC 20 16 LDY ts_state
ABB5 C0 01 CPY #&01
ABB7 D0 00 BNE LABB9

.LABB9
ABB9 AC 66 16 LDY ts_savey
ABBC 18 CLC
ABBD 60 RTS

; Empty the output queue (both pointers to &1999).
.ts_reset_queue
ABBE 48 PHA
ABBF A9 99 LDA #&99
ABC1 8D 1C 16 STA outq_wr
ABC4 8D 1E 16 STA outq_rd
ABC7 A9 19 LDA #&19
ABC9 8D 1D 16 STA outq_wr+1
ABCC 8D 1F 16 STA outq_rd+1
ABCF 68 PLA
ABD0 60 RTS

;-----
; BGET CHANNEL 14: NEXT BYTE OF THE DECODED FILE
; Returns C clear with the byte; C set with A=1 or 2 when a new block changes the load
; address (2 = before any data); C set with A=&FE at the end of the file, when "Loaded."
; is printed and the file is closed.
;-----

.LABD1
ABD1 AD 55 16 LDA ts_name
ABD4 D0 05 BNE LABDB
ABD6 AD 56 16 LDA ts_name+1
ABD9 F0 62 BEQ LAC3D

.LABDB
ABDB AD 63 16 LDA ts_to_memory
ABDE F0 11 BEQ LABF1
ABE0 AD 61 16 LDA ts_newaddr
ABE3 F0 0C BEQ LABF1
ABE5 A9 00 LDA #&00
ABE7 8D 61 16 STA ts_newaddr
ABEA AD 62 16 LDA ts_pending
ABED 48 PHA
ABEE 4C 07 AC JMP LAC07

.LABF1
ABF1 8D 49 16 STA ts_comment
ABF4 8D 4A 16 STA ts_comment+1
ABF7 20 83 A9 JSR ts_getbyte
ABFA B0 36 BCS LAC32
ABFC 48 PHA
ABFD AD 63 16 LDA ts_to_memory
AC00 F0 05 BEQ LAC07
AC02 AD 61 16 LDA ts_newaddr
AC05 D0 15 BNE LAC1C

.LAC07
AC07 EE 50 16 INC ts_offset
AC0A D0 0D BNE LAC19
AC0C EE 51 16 INC ts_offset+1
AC0F D0 08 BNE LAC19
AC11 EE 52 16 INC ts_offset+2
AC14 D0 03 BNE LAC19
AC16 EE 53 16 INC ts_offset+3

.LAC19
AC19 68 PLA
AC1A 18 CLC
AC1B 60 RTS

.LAC1C
AC1C 68 PLA
AC1D 8D 62 16 STA ts_pending
AC20 AD 49 16 LDA ts_comment

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AC23 D0 09 BNE LAC2E
AC25 AD 4A 16 LDA ts_comment+1
AC28 D0 04 BNE LAC2E
AC2A A9 02 LDA #&02
AC2C 38 SEC
AC2D 60 RTS
.LAC2E
AC2E A9 01 LDA #&01
AC30 38 SEC
AC31 60 RTS
.LAC32
AC32 20 41 AC JSR LAC41
AC35 A9 00 LDA #&00
AC37 8D 55 16 STA ts_name
AC3A 8D 56 16 STA ts_name+1
.LAC3D
AC3D A9 FE LDA #&FE
AC3F 38 SEC
AC40 60 RTS
;-----
; End of file: "Loaded. Press RETURN to continue" (if *OPT1 messages on), headers off.
;-----
.LAC41
AC41 AD 94 16 LDA opt1_msgs
AC44 10 35 BPL LAC7B
AC46 20 AF A2 JSR print_inline
AC49 .str '\r\n\x07Loaded.\r\nPress RETURN to continue\x00'
AC6E A9 00 LDA #&00
AC70 8D 70 16 STA &1670
AC73 8D 71 16 STA &1671
AC76 A9 92 LDA #&92
AC78 20 D3 A9 JSR ts_osword
.LAC7B
AC7B 60 RTS
; *OPT1: store the message level.
.LAC7C
AC7C 8D 94 16 STA opt1_msgs
AC7F 60 RTS
; *OPT3: store the "keep name tables" flag.
.LAC80
AC80 8D 60 16 STA opt3_keep
AC83 60 RTS
;-----
; FINDV (OSFIND). A=0 close: nothing to do. Otherwise open the file named at X/Y for
; reading: reset the decoder, wait for the page ("Searching..."), remember the name.
; Returns handle &0E. Used by *EXEC.
;-----
.findv
AC84 C9 00 CMP #&00
AC86 D0 01 BNE LAC89
AC88 60 RTS
.LAC89
AC89 20 C3 AC JSR LACC3
AC8C 20 DF AC JSR LACDF
AC8F A9 0E LDA #&0E
AC91 A0 00 LDY #&00
AC93 60 RTS
;-----
; OSFILE 7 (internal): start reading a named file for loading. If the parameter block
; gives a load address (byte 6 = 0) it overrides the file's own. Returns handle &0E.
;-----
.LAC94
AC94 20 C3 AC JSR LACC3
AC97 86 CC STX ts_ptr
AC99 84 CD STY ts_ptr+1
AC9B A0 06 LDY #&06
AC9D B1 CC LDA (ts_ptr),Y
AC9F D0 0D BNE LACAE
.LACA1
ACA1 88 DEY
ACA2 B1 CC LDA (ts_ptr),Y
ACA4 99 4A 16 STA ts_comment+1,Y
ACA7 C0 02 CPY #&02
ACA9 D0 F6 BNE LACA1
ACAB EE 54 16 INC ts_user_load
.LACAE
ACAE EE 63 16 INC ts_to_memory
ACB1 EE 68 16 INC ts_random
ACB4 A0 00 LDY #&00
ACB6 B1 CC LDA (ts_ptr),Y
ACB8 AA TAX
ACB9 C8 INY
ACBA B1 CC LDA (ts_ptr),Y
ACBC A8 TAY
ACBD 20 DF AC JSR LACDF
ACC0 A9 0E LDA #&0E
ACC2 60 RTS
;-----
; Reset the decoder (keeping the name tables if *OPT3,1) and wait for the current page
; with OSWORD &95.

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```

;-----
.LACC3
ACC3 98      TYA
ACC4 48      PHA
ACC5 8A      TXA
ACC6 48      PHA
ACC7 AD 60 16 LDA opt3_keep
ACCA D0 06    BNE LACD2
ACCC 20 0A A8 JSR ts_init
ACCF 4C D5 AC JMP LACD5
.LACD2
ACD2 20 1C A8 JSR ts_reset
.LACD5
ACD5 A9 95    LDA #&95
ACD7 20 D3 A9 JSR ts_osword
ACDA 68      PLA
ACDB AA      TAX
ACDC 68      PLA
ACDD A8      TAY
ACDE 60      RTS
;-----
; Copy the file name at X/Y (quotes removed) to the heap and make it the wanted name.
;-----
.LACDF
ACDF 86 CC    STX ts_ptr
ACE1 84 CD    STY ts_ptr+1
ACE3 A0 00    LDY #&00
ACE5 A2 00    LDX #&00
.LACE7
ACE7 B1 CC    LDA (ts_ptr),Y
ACE9 C9 22    CMP #&22          ; ""
ACEB F0 02    BEQ LACEF
ACED E8      INX
ACEE 48      PHA
.LACEF
ACEF C8      INY
ACF0 C9 0D    CMP #&0D
ACF2 D0 F3    BNE LACE7
ACF4 8A      TXA
ACF5 A8      TAY
ACF6 AE 92 16 LDX heap
ACF9 86 CC    STX ts_ptr
ACFB 8E 55 16 STX ts_name
ACFE AE 93 16 LDX heap+1
AD01 86 CD    STX ts_ptr+1
AD03 8E 56 16 STX ts_name+1
AD06 18      CLC
AD07 6D 92 16 ADC heap
AD0A 8D 92 16 STA heap
AD0D 90 03    BCC LAD12
AD0F EE 93 16 INC heap+1
.LAD12
AD12 88      DEY
.LAD13
AD13 68      PLA
AD14 91 CC    STA (ts_ptr),Y
AD16 88      DEY
AD17 10 FA    BPL LAD13
AD19 60      RTS
;-----
; OSFILE 5: READ CATALOGUE INFORMATION: copy the name pointer, load address (base +
; offset), execution address and attributes into the parameter block.
;-----
.LAD1A
AD1A 98      TYA
AD1B 48      PHA
AD1C 8A      TXA
AD1D 48      PHA
AD1E 20 15 A9 JSR push_frame
AD21 68      PLA
AD22 85 CC    STA ts_ptr
AD24 68      PLA
AD25 85 CD    STA ts_ptr+1
AD27 AD 4C 16 LDA ts_load_base
AD2A 18      CLC
AD2B 6D 50 16 ADC ts_offset
AD2E 8D 57 16 STA ts_load
AD31 AD 40 16 LDA &164D
AD34 6D 51 16 ADC ts_offset+1
AD37 8D 58 16 STA &1658
AD3A AD 4E 16 LDA &164E
AD3D 6D 52 16 ADC ts_offset+2
AD40 8D 59 16 STA &1659
AD43 AD 4F 16 LDA &164F
AD46 6D 53 16 ADC ts_offset+3
AD49 8D 5A 16 STA &165A
AD4C A0 0A    LDY #&0A
.LAD4E
AD4E B9 55 16 LDA ts_name,Y
AD51 91 CC    STA (ts_ptr),Y

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AD53 88      DEY
AD54 10 F8   BPL LAD4E
AD56 20 56 A9 JSR pop_frame
AD59 60      RTS
;-----
; OSWORD &7A with a sub-command below &80: decoder information.
; Only codes 0 (returns 0) and 1 (returns the title pointer ts_title) can be reached:
; the BCS after CMP #1 catches every code from 1 up, so the code for 1-3 at &AD78-&AD8A
; is never used, and &167F-&1682 are never written anyway.
;-----
.LAD5A
AD5A C9 01   CMP #&01
AD5C B0 11   BCS LAD6F
AD5E F0 18   BEQ LAD78
AD60 C9 03   CMP #&03
AD62 B0 1D   BCS LAD81
AD64 F0 24   BEQ LAD8A
AD66 A0 00   LDY #&00
AD68 98      TYA
AD69 91 F0   STA (osw_x),Y
AD6B C8      INY
AD6C 91 F0   STA (osw_x),Y
AD6E 60      RTS
.LAD6F
AD6F AE 64 16 LDX ts_title
AD72 AD 65 16 LDA ts_title+1
AD75 4C 90 AD JMP LAD90
.LAD78
AD78 AE 49 16 LDX ts_comment
AD7B AD 4A 16 LDA ts_comment+1
AD7E 4C 90 AD JMP LAD90
.LAD81
AD81 AE 7F 16 LDX &167F
AD84 AD 80 16 LDA &1680
AD87 4C 90 AD JMP LAD90
.LAD8A
AD8A AE 81 16 LDX &1681
AD8D AD 82 16 LDA &1682
.LAD90
AD90 A0 01   LDY #&01
AD92 91 F0   STA (osw_x),Y
AD94 8A      TXA
AD95 88      DEY
AD96 91 F0   STA (osw_x),Y
AD98 60      RTS
;-----
; <TXT> TELETEXT FORMAT
; Sets up the name tables for data received from teletext, then starts a block (<DSB>):
; - every lone name with correct (odd) parity decodes to itself with bit 7 removed
; - lone &FE becomes <ULB> and lone &E0 <URB> (the manual says the other way round)
; - escaped &FE and &E0 decode to &7E and &60
; - every command moves to the single pass used here
; - escaped &C2 ("B" with its parity bit) and &46 become <DSB>; escaped &E3 ("c"
;   with parity) becomes <DEB> without its operand
; - numbers are now ASCII hex
; <TXT> also sets ts_random: the next block is found with HEAD OF CHAIN.
;-----
.ts_TXT
AD99 EE 68 16 INC ts_random      ; <TXT>: blocks may arrive in any order
; <TX0>: as <TXT> without that.
.ts_TX0
AD9C A0 00   LDY #&00
.LAD9E
AD9E B9 9A 0F LDA lone_kind,Y
ADA1 C9 01   CMP #&01
ADA3 D0 06   BNE LADAB
ADA5 98      TYA
ADA6 29 7F   AND #&7F
ADA8 99 9A 10 STA lone_lo,Y
.LADAB
ADAB C8      INY
ADAC D0 F0   BNE LAD9E
ADAE A2 FE   LDX #&FE
ADB0 A0 E0   LDY #&E0
ADB2 A9 12   LDA #&12
ADB4 9D 9A 0F STA lone_kind,X
ADB7 99 9A 0F STA lone_kind,Y
ADBA A9 B4   LDA #&B4
ADBC 9D 9A 10 STA lone_lo,X
ADBF A9 B1   LDA #&B1
ADC1 9D 9A 11 STA lone_hi,X
ADC4 A9 BD   LDA #&BD
ADC6 99 9A 10 STA lone_lo,Y
ADC9 A9 B1   LDA #&B1
ADCB 99 9A 11 STA lone_hi,Y
ADCE A9 01   LDA #&01
ADD0 9D 9A 12 STA esc_kind,X
ADD3 99 9A 12 STA esc_kind,Y
ADD6 A9 7E   LDA #&7E
ADD8 9D 9A 13 STA esc_lo,X

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    ADDB A9 60    LDA #&60
    ADDD 99 9A 13 STA esc_lo,Y
    ADE0 A0 00    LDY #&00
.LADE2
    ADE2 B9 9A 12 LDA esc_kind,Y
    ADE5 C9 11    CMP #&11
    ADE7 D0 05    BNE LADEE
    ADE9 A9 12    LDA #&12
    ADEB 99 9A 12 STA esc_kind,Y
.LADEE
    ADEE C8      INY
    ADEF D0 F1    BNE LADE2
    ADF1 EE 7E 16 INC ts_hexmode
    ADF4 A0 C2    LDY #&C2
    ADF6 A9 10    LDA #&10
    ADF8 99 9A 12 STA esc_kind,Y
    ADFB A0 46    LDY #&46
    ADFD 99 9A 12 STA esc_kind,Y
    AE00 A9 20    LDA #&20
    AE02 99 9A 13 STA esc_lo,Y
    AE05 A0 C2    LDY #&C2
    AE07 99 9A 13 STA esc_lo,Y
    AE0A A9 AE    LDA #&AE
    AE0C 99 9A 14 STA esc_hi,Y
    AE0F A0 46    LDY #&46
    AE11 99 9A 14 STA esc_hi,Y
    AE14 A0 E3    LDY #&E3
    AE16 A9 9A    LDA #&9A
    AE18 99 9A 13 STA esc_lo,Y
    AE1B A9 B2    LDA #&B2
    AE1D 99 9A 14 STA esc_hi,Y
; Escaped &C2 / &46 enter here: <DSB> within <TXT> data.
.LAE20
    AE20 20 2F AE JSR ts_DSB
    AE23 AD 20 16 LDA ts_state
    AE26 F0 05    BEQ LAE2D
    AE28 A9 02    LDA #&02
    AE2A 8D 20 16 STA ts_state
.LAE2D
    AE2D 18      CLC
    AE2E 60      RTS
;-----
; <DSB> START OF BLOCK [block number] [number of blocks]
; Ordered files: a block other than the one expected is skipped (ask for the next page).
; Random-order files: a block already received (bit set in ts_block_map) is skipped.
;-----
.ts_DSB
    AE2F A9 01    LDA #&01
    AE31 8D 20 16 STA ts_state
    AE34 A9 00    LDA #&00
    AE36 8D 69 16 STA ts_blk_check
    AE39 8D 6A 16 STA &166A
    AE3C 20 12 AB JSR ts_number      ; number of operands
    AE3F C9 01    CMP #&01
    AE41 08      PHP
    AE42 30 48    BMI LAE8C
    AE44 A9 67    LDA #&67          ; block number into ts_block_no
    AE46 85 CC    STA ts_ptr
    AE48 A9 16    LDA #&16
    AE4A 85 CD    STA ts_ptr+1
    AE4C 20 6D AF JSR ts_read_value
    AE4F E0 00    CPX #&00
    AE51 F0 39    BEQ LAE8C
    AE53 AE 68 16 LDX ts_random
    AE56 D0 34    BNE LAE8C
    AE58 AD 67 16 LDA ts_block_no    ; random order: has this block been seen?
    AE5B 4A      LSR A
    AE5C 4A      LSR A
    AE5D 4A      LSR A
    AE5E A8      TAY
    AE5F AA      TAX
.LAE60
    AE60 C0 00    CPY #&00
    AE62 F0 0C    BEQ LAE70
    AE64 88      DEY
    AE65 B9 25 16 LDA ts_block_map,Y
    AE68 C9 FF    CMP #&FF
    AE6A F0 F4    BEQ LAE60
.LAE6C
    AE6C 68      PLA
    AE6D 4C D3 AE JMP LAED3
.LAE70
    AE70 8A      TXA
    AE71 A8      TAY
    AE72 AD 67 16 LDA ts_block_no
    AE75 29 07    AND #&07
    AE77 AA      TAX
    AE78 A9 00    LDA #&00
    AE7A E0 00    CPX #&00
    AE7C F0 05    BEQ LAE83

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.LAE7E
AE7E 38      SEC
AE7F 2A      ROL A
AE80 CA      DEX
AE81 D0 FB   BNE LAE7E

.LAE83
AE83 85 CE   STA ts_tmp
AE85 39 25 16 AND ts_block_map,Y
AE88 C5 CE   CMP ts_tmp
AE8A D0 E0   BNE LAE6C

.LAE8C
AE8C AD 23 16 LDA ts_blocks_total ; block beyond the total?
AE8F 38      SEC
AE90 ED 67 16 SBC ts_block_no
AE93 90 D7   BCC LAE6C
AE95 28      PLP
AE96 F0 13   BEQ LAEAB
AE98 30 11   BMI LAEAB
AE9A AD 23 16 LDA ts_blocks_total ; number of blocks in the file
AE9D 8D 6E 16 STA ts_total_save
AEA0 A9 23   LDA #&23
AEA2 85 CC   STA ts_ptr
AEA4 A9 16   LDA #&16
AEA6 85 CD   STA ts_ptr+1
AEA8 20 6D AF JSR ts_read_value

.LAEAB
AEAB AD 67 16 LDA ts_block_no
AEAE 4A      LSR A
AEAF 4A      LSR A
AEB0 4A      LSR A
AEB1 A8      TAY
AEB2 AD 67 16 LDA ts_block_no
AEB5 29 07   AND #&07
AEB7 AA      TAX
AEB8 A9 01   LDA #&01
AEBA E0 00   CPX #&00
AEBC F0 04   BEQ LAEC2

.LAEBE
AEBE 0A      ASL A
AEBF CA      DEX
AEC0 D0 FC   BNE LAEBE

.LAEC2
AEC2 39 25 16 AND ts_block_map,Y
AEC5 D0 0C   BNE LAED3
AEC7 A9 01   LDA #&01 ; a wanted block: start decoding it
AEC9 8D 20 16 STA ts_state
AECB A9 81   LDA #&81

.LAECE
AECE 20 D3 A9 JSR ts_osword
AED1 18      CLC
AED2 60      RTS

.LAED3
AED3 AD 6E 16 LDA ts_total_save ; not wanted: go and get the next page
AED6 8D 23 16 STA ts_blocks_total
AED9 A9 00   LDA #&00
AEDB 8D 20 16 STA ts_state
AEDF AD 68 16 LDA ts_random
AEE1 F0 05   BEQ LAEE8
AEE3 A9 88   LDA #&88
AEE5 4C CE AE JMP LAECE

.LAEE8
AEE8 A9 86   LDA #&86
AEEA 4C CE AE JMP LAECE
;-----
; <DLA> LOAD ADDRESS (absolute), unless the user gave a load address. Resets the offset.
;-----
.ts_DLA
AEED AD 54 16 LDA ts_user_load
AEF0 D0 1A   BNE LAF0C
AEF2 A9 4C   LDA #&4C
AEF4 85 CC   STA ts_ptr
AEF6 A9 16   LDA #&16
AEF8 85 CD   STA ts_ptr+1
AEFA A9 00   LDA #&00
AEFC A0 03   LDY #&03

.LAEFE
AEFE 91 CC   STA (ts_ptr),Y
AF00 99 50 16 STA ts_offset,Y
AF03 88      DEY
AF04 10 F8   BPL LAEFE
AF06 20 6D AF JSR ts_read_value
AF09 EE 61 16 INC ts_newaddr

.LAF0C
AF0C 18      CLC
AF0D 60      RTS
; <DXA> EXECUTION ADDRESS (absolute).
.ts_DXA
AF0E A9 5B   LDA #&5B
AF10 85 CC   STA ts_ptr
AF12 A9 16   LDA #&16

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AF14 85 CD STA ts_ptr+1
AF16 20 6D AF JSR ts_read_value
AF19 A9 00 LDA #000
AF1B 8D 5F 16 STA ts_attr
AF1E 18 CLC
AF1F 60 RTS
;-----
; <DLR> LOAD ADDRESS RELATIVE: set the offset from the base. With no operand the data is
; unordered (passed to *EXEC as it arrives), so blocks may be fetched in any order.
;-----
.ts_DLR
AF20 A9 50 LDA #50
AF22 85 CC STA ts_ptr
AF24 A9 16 LDA #16
AF26 85 CD STA ts_ptr+1
AF28 A9 00 LDA #000
AF2A A0 03 LDY #03
.LAF2C
AF2C 91 CC STA (ts_ptr),Y
AF2E 88 DEY
AF2F 10 FB BPL LAF2C
AF31 20 6D AF JSR ts_read_value
AF34 8A TXA
AF35 F0 05 BEQ LAF3C
AF37 AD 63 16 LDA ts_to_memory
AF3A F0 03 BEQ LAF3F
.LAF3C
AF3C EE 68 16 INC ts_random
.LAF3F
AF3F EE 61 16 INC ts_newaddr
AF42 18 CLC
AF43 60 RTS
; <DXR> EXECUTION ADDRESS relative to the load address.
.ts_DXR
AF44 20 0E AF JSR ts_DXA
AF47 AD 4C 16 LDA ts_load_base
AF4A 6D 5B 16 ADC ts_exec
AF4D 8D 5B 16 STA ts_exec
AF50 AD 4D 16 LDA &164D
AF53 6D 5C 16 ADC ts_exec+1
AF56 8D 5C 16 STA ts_exec+1
AF59 AD 4E 16 LDA &164E
AF5C 6D 5D 16 ADC ts_exec+2
AF5F 8D 5D 16 STA ts_exec+2
AF62 AD 4F 16 LDA &164F
AF65 6D 5E 16 ADC ts_exec+3
AF68 8D 5E 16 STA ts_exec+3
AF6B 18 CLC
AF6C 60 RTS
;-----
; Read a variable-length numeric operand into (ts_ptr), most significant byte first:
; a length then that many hex digit pairs (or raw bytes outside hex mode).
;-----
.ts_read_value
AF6D 20 12 AB JSR ts_number
AF70 AA TAX
AF71 F0 24 BEQ LAF97
AF73 AC 7E 16 LDY ts_hexmode
AF76 F0 20 BEQ LAF98
AF78 4A LSR A
AF79 A8 TAY
AF7A 90 05 BCC LAF81
AF7C 20 12 AB JSR ts_number
AF7F 91 CC STA (ts_ptr),Y
.LAF81
AF81 88 DEY
AF82 30 13 BMI LAF97
.LAF84
AF84 20 12 AB JSR ts_number
AF87 0A ASL A
AF88 0A ASL A
AF89 0A ASL A
AF8A 0A ASL A
AF8B 85 CE STA ts_tmp
AF8D 20 12 AB JSR ts_number
AF90 05 CE ORA ts_tmp
AF92 91 CC STA (ts_ptr),Y
AF94 88 DEY
AF95 10 ED BPL LAF84
.LAF97
AF97 60 RTS
.LAF98
AF98 A8 TAY
AF99 88 DEY
.LAF9A
AF9A 20 12 AB JSR ts_number
AF9D 91 CC STA (ts_ptr),Y
AF9F 88 DEY
AFA0 10 F8 BPL LAF9A
AFA2 60 RTS

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; <DEF> revert to the default name tables.
.ts_DEF
    AFA3 20 45 A8 JSR ts_default_names
    AFA6 A9 00 LDA #000
    AFA8 8D 7E 16 STA ts_hexmode
    AFAB 18 CLC
    AFAC 60 RTS
; <DET> end of file: this is the last block.
.ts_DET
    AFAD EE 24 16 INC ts_eof
    AFB0 38 SEC
    AFB1 60 RTS
; <DND> don't reset the name tables before the next file.
.ts_DND
    AFB2 EE 60 16 INC opt3_keep
    AFB5 18 CLC
    AFB6 60 RTS
; <DAC> <DDT> <DIG> <DNC>: read and ignore all the operands.
.ts_ignore
    AFB7 EE 4B 16 INC ts_operand
    AFBA 20 12 AB JSR ts_number
    AFB0 AA TAX
    AFBE F0 0F BEQ LAFCF
.LAFC0
    AFC0 20 12 AB JSR ts_number
    AFC3 A8 TAY
    AFC4 F0 06 BEQ LAFCC
.LAFC6
    AFC6 20 83 A9 JSR ts_getbyte
    AFC9 88 DEY
    AFCA D0 FA BNE LAFCF
.LAFCC
    AFCC CA DEX
    AFCD D0 F1 BNE LAFCF
.LAFCF
    AFCF 20 BE AB JSR ts_reset_queue
    AFD2 CE 4B 16 DEC ts_operand
    AFD5 18 CLC
    AFD6 60 RTS
;-----
; <DES> give an escaped name a new meaning: nothing, one byte, two bytes or a string.
;-----
.ts_DES
    AFD7 EE 4B 16 INC ts_operand
    AFDA 20 83 A9 JSR ts_getbyte
    AFDD 20 91 B0 JSR LB091
    AFE0 A8 TAY
    AFE1 20 12 AB JSR ts_number
    AFE4 AA TAX
    AFE5 F0 29 BEQ LB010
    AFE7 E0 03 CPX #03
    AFE9 B0 2B BCS LB016
    AFEB CA DEX
    AFEC F0 14 BEQ LB002
    AFEE 20 83 A9 JSR ts_getbyte
    AFF1 99 9A 14 STA esc_hi,Y
    AFF4 20 83 A9 JSR ts_getbyte
    AFF7 99 9A 13 STA esc_lo,Y
    AFFA A9 02 LDA #02
    AFFC 99 9A 12 STA esc_kind,Y
    AFFF 4C 2F B0 JMP LB02F
.LB002
    B002 20 83 A9 JSR ts_getbyte
    B005 99 9A 13 STA esc_lo,Y
    B008 A9 01 LDA #01
    B00A 99 9A 12 STA esc_kind,Y
    B00D 4C 2F B0 JMP LB02F
.LB010
    B010 99 9A 12 STA esc_kind,Y
    B013 4C 2F B0 JMP LB02F
.LB016
    B016 AD 92 16 LDA heap
    B019 48 PHA
    B01A AD 93 16 LDA heap+1
    B01D 48 PHA
    B01E A9 0F LDA #0F
    B020 99 9A 12 STA esc_kind,Y
    B023 8A TXA
    B024 20 38 AA JSR ts_read_string
    B027 68 PLA
    B028 99 9A 14 STA esc_hi,Y
    B02B 68 PLA
    B02C 99 9A 13 STA esc_lo,Y
.LB02F
    B02F CE 4B 16 DEC ts_operand
    B032 18 CLC
    B033 60 RTS
;-----
; <DLS> give a lone name a new meaning (as <DES> for the lone tables).
;-----

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```

.ts_DLS
B034 EE 4B 16 INC ts_operand
B037 20 83 A9 JSR ts_getbyte
B03A 20 91 B0 JSR LB091
B03D A8 TAY
B03E 20 12 AB JSR ts_number
B041 AA TAX
B042 F0 29 BEQ LB06D
B044 E0 03 CPX #03
B046 B0 2B BCS LB073
B048 CA DEX
B049 F0 14 BEQ LB05F
B04B 20 83 A9 JSR ts_getbyte
B04E 99 9A 11 STA lone_hi,Y
B051 20 83 A9 JSR ts_getbyte
B054 99 9A 10 STA lone_lo,Y
B057 A9 02 LDA #02
B059 99 9A 0F STA lone_kind,Y
B05C 4C 8C B0 JMP LB08C

.LB05F
B05F 20 83 A9 JSR ts_getbyte
B062 99 9A 10 STA lone_lo,Y
B065 A9 01 LDA #01
B067 99 9A 0F STA lone_kind,Y
B06A 4C 8C B0 JMP LB08C

.LB06D
B06D 99 9A 0F STA lone_kind,Y
B070 4C 8C B0 JMP LB08C

.LB073
B073 AD 92 16 LDA heap
B076 48 PHA
B077 AD 93 16 LDA heap+1
B07A 48 PHA
B07B A9 0F LDA #0F
B07D 99 9A 0F STA lone_kind,Y
B080 8A TXA
B081 20 38 AA JSR ts_read_string
B084 68 PLA
B085 99 9A 11 STA lone_hi,Y
B088 68 PLA
B089 99 9A 10 STA lone_lo,Y

.LB08C
B08C CE 4B 16 DEC ts_operand
B08F 18 CLC
B090 60 RTS
; Add odd parity to the name in A (when in teletext hex mode).
.LB091
B091 AE 7E 16 LDX ts_hexmode
B094 F0 13 BEQ LB0A9
B096 0A ASL A
B097 48 PHA
B098 09 01 ORA #01
B09A 85 CE STA ts_tmp
B09C A2 08 LDX #08
B09E 0A ASL A

.LB09F
B09F 06 CE ASL ts_tmp
B0A1 69 00 ADC #00
B0A3 CA DEX
B0A4 D0 F9 BNE LB09F
B0A6 6A ROR A
B0A7 68 PLA
B0A8 6A ROR A

.LB0A9
B0A9 60 RTS
;
;-----
; <DC0> COMMENT: store up to *OPT1 operands (title, language, computer, date, source)
; and print them when loading into memory.
;-----
.ts_DC0
B0AA 20 12 AB JSR ts_number
B0AD 48 PHA
B0AE CD 94 16 CMP opt1_msgs
B0B1 90 03 BCC LB0B6
B0B3 AD 94 16 LDA opt1_msgs

.LB0B6
B0B6 AC 92 16 LDY heap
B0B9 84 CC STY ts_ptr
B0BB 8C 49 16 STY ts_comment
B0BE AC 93 16 LDY heap+1
B0C1 84 CD STY ts_ptr+1
B0C3 8C 4A 16 STY ts_comment+1
B0C6 20 E6 B0 JSR LB0E6
B0C9 EE 61 16 INC ts_newaddr
B0CC 68 PLA
B0CD 38 SEC
B0CE ED 94 16 SBC opt1_msgs
B0D1 90 06 BCC LB0D9
B0D3 AA TAX
B0D4 F0 03 BEQ LB0D9

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    B0D6 4C C0 AF JMP LAF00
.LB0D9
    B0D9 20 BE AB JSR ts_reset_queue
    B0DC AD 63 16 LDA ts_to_memory
    B0DF D0 03 BNE LB0E4
    B0E1 20 10 B3 JSR LB310
.LB0E4
    B0E4 18 CLC
    B0E5 60 RTS
;-----
; Read X string operands into (ts_ptr), each ended with CR; byte 0 = total length.
;-----
.LB0E6
    B0E6 A0 00 LDY #00
    B0E8 AA TAX
    B0E9 F0 1B BEQ LB106
.LB0EB
    B0EB 20 12 AB JSR ts_number
    B0EE 85 CE STA ts_tmp
    B0F0 C9 00 CMP #00
    B0F2 F0 0A BEQ LB0FE
.LB0F4
    B0F4 20 83 A9 JSR ts_getbyte
    B0F7 C8 INY
    B0F8 91 CC STA (ts_ptr),Y
    B0FA C6 CE DEC ts_tmp
    B0FC D0 F6 BNE LB0F4
.LB0FE
    B0FE A9 0D LDA #0D
    B100 C8 INY
    B101 91 CC STA (ts_ptr),Y
    B103 CA DEX
    B104 D0 E5 BNE LB0EB
.LB106
    B106 98 TYA
    B107 A0 00 LDY #00
    B109 91 CC STA (ts_ptr),Y
    B10B 60 RTS
;-----
; <DTL> TITLE OF FILE: compare the title with the wanted file name (ignoring case).
; A match (or no name wanted, *EXEC "") records the title; otherwise skip to the next
; file with HEAD OF CHAIN.
;-----
.ts_DTL
    B10C 20 12 AB JSR ts_number
    B10F C9 00 CMP #00
    B111 F0 48 BEQ LB15B
    B113 AC 92 16 LDY heap
    B116 84 CC STY ts_ptr
    B118 AC 93 16 LDY heap+1
    B11B 84 CD STY ts_ptr+1
    B11D 20 E6 B0 JSR LB0E6
    B120 85 CE STA ts_tmp
    B122 A0 01 LDY #01
.LB124
    B124 B1 CC LDA (ts_ptr),Y
    B126 48 PHA
    B127 C8 INY
    B128 C9 00 CMP #00
    B12A D0 F8 BNE LB124
    B12C 88 DEY
    B12D 88 DEY
    B12E AD 55 16 LDA ts_name
    B131 85 CC STA ts_ptr
    B133 AD 56 16 LDA ts_name+1
    B136 85 CD STA ts_ptr+1
    B138 98 TYA
    B139 48 PHA
    B13A A0 00 LDY #00
    B13C B1 CC LDA (ts_ptr),Y
    B13E C9 00 CMP #00
    B140 F0 34 BEQ LB176
    B142 68 PLA
    B143 A8 TAY
.LB144
    B144 68 PLA
    B145 51 CC EOR (ts_ptr),Y
    B147 29 DF AND #DF
    B149 D0 12 BNE LB15D
    B14B 88 DEY
    B14C 10 F6 BPL LB144
    B14E 20 A0 B1 JSR LB1A0
    B151 A5 CC LDA ts_ptr
    B153 8D 64 16 STA ts_title
    B156 A5 CD LDA ts_ptr+1
    B158 8D 65 16 STA ts_title+1
.LB15B
    B15B 18 CLC
    B15C 60 RTS
.LB15D

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B15D 88      DEY
B15E 30 04   BMI LB164
B160 68      PLA
B161 4C 5D B1 JMP LB15D
.LB164
B164 AD 6E 16 LDA ts_total_save
B167 8D 23 16 STA ts_blocks_total
B16A A9 00   LDA #&00
B16C 8D 20 16 STA ts_state
B16F A9 88   LDA #&88
B171 20 D3 A9 JSR ts_osword
B174 18      CLC
B175 60      RTS
.LB176
B176 20 A0 B1 JSR LB1A0
B179 AD 92 16 LDA heap
B17C 8D 55 16 STA ts_name
B17F 85 CC   STA ts_ptr
B181 AD 93 16 LDA heap+1
B184 8D 56 16 STA ts_name+1
B187 85 CD   STA ts_ptr+1
B189 68      PLA
B18A A8      TAY
B18B 38      SEC
B18C 6D 92 16 ADC heap
B18F 8D 92 16 STA heap
B192 90 03   BCC LB197
B194 EE 93 16 INC heap+1
.LB197
B197 68      PLA
B198 91 CC   STA (ts_ptr),Y
B19A 88      DEY
B19B 10 FA   BPL LB197
B19D 4C 5B B1 JMP LB15B
.LB1A0
B1A0 A5 CE   LDA ts_tmp
B1A2 38      SEC
B1A3 6D 92 16 ADC heap
B1A6 8D 92 16 STA heap
B1A9 90 03   BCC LB1AE
B1AB EE 93 16 INC heap+1
.LB1AE
B1AE 60      RTS
; <DIR> inhibit running the file after loading.
.ts_DIR
B1AF EE 6D 16 INC ts_no_run
B1B2 18      CLC
B1B3 60      RTS
;-----
; <ULB> next byte - &58 (gives &C8-&FF and &00-&27 from 7-bit data).
;-----
.ts_ULB
B1B4 20 83 A9 JSR ts_getbyte
B1B7 38      SEC
B1B8 E9 58   SBC #&58
B1BA 4C C3 B1 JMP LB1C3
; <URB> next byte + &58.
.ts_URB
B1BD 20 83 A9 JSR ts_getbyte
B1C0 18      CLC
B1C1 69 58   ADC #&58
.LB1C3
B1C3 EE 1E 16 INC outq_rd
B1C6 D0 03   BNE LB1CB
B1C8 EE 1F 16 INC outq_rd+1
.LB1CB
B1CB AE 1E 16 LDX outq_rd
B1CE 86 CC   STX ts_ptr
B1D0 AE 1F 16 LDX outq_rd+1
B1D3 86 CD   STX ts_ptr+1
B1D5 A0 00   LDY #&00
B1D7 91 CC   STA (ts_ptr),Y
B1D9 18      CLC
B1DA 60      RTS
;-----
; <DLC> give a lone name a command meaning (3-letter name looked up in the table).
;-----
.ts_DLC
B1DB 20 F5 B1 JSR LB1F5
B1DE 20 DF A8 JSR use_lone_tables
.LB1E1
B1E1 91 CA   STA (nm_hi),Y
B1E3 8A      TXA
B1E4 91 C8   STA (nm_lo),Y
B1E6 A5 CE   LDA ts_tmp
B1E8 91 C6   STA (nm_kind),Y
B1EA 18      CLC
B1EB 60      RTS
; <DEC> give an escaped name a command meaning.
.ts_DEC

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```

B1EC 20 F5 B1 JSR LB1F5
B1EF 20 FA A8 JSR use_escaped_tables
B1F2 4C E1 B1 JMP LB1E1
;-----
; Read a command name operand and look it up in the command table (and a second table
; at ts_cmdtab2, which is never set). Error &CB if unknown.
;-----
.LB1F5
B1F5 A0 04 LDY #&04
.LB1F7
B1F7 20 83 A9 JSR ts_getbyte
B1FA 48 PHA
B1FB 88 DEY
B1FC D0 F9 BNE LB1F7
B1FE A9 EF LDA #&EF
B200 85 CC STA ts_ptr
B202 A9 A6 LDA #&A6
B204 85 CD STA ts_ptr+1
B206 68 PLA
B207 A8 TAY
B208 68 PLA
B209 AA TAX
B20A 68 PLA
B20B 20 2F B2 JSR LB22F
B20E 90 13 BCC LB223
B210 48 PHA
B211 AD 83 16 LDA ts_cmdtab2
B214 F0 14 BEQ LB22A
B216 85 CD STA ts_ptr+1
B218 AD 83 16 LDA ts_cmdtab2
B21B 85 CC STA ts_ptr
B21D 68 PLA
B21E 20 2F B2 JSR LB22F
B221 B0 07 BCS LB22A
.LB223
B223 85 CD STA ts_ptr+1
B225 68 PLA
B226 A8 TAY
B227 A5 CD LDA ts_ptr+1
B229 60 RTS
.LB22A
B22A A9 CB LDA #&CB
B22C 4C 0D AA JMP ts_error
;-----
; Search a command table at (ts_ptr) for the 3 letters in A, X, Y.
;-----
.LB22F
B22F 85 CE STA ts_tmp
B231 98 TYA
B232 48 PHA
B233 A0 00 LDY #&00
.LB235
B235 B1 CC LDA (ts_ptr),Y
B237 F0 2B BEQ LB264
B239 C5 CE CMP ts_tmp
B23B D0 1E BNE LB25B
B23D C8 INY
B23E 8A TXA
B23F D1 CC CMP (ts_ptr),Y
B241 D0 19 BNE LB25C
B243 C8 INY
B244 68 PLA
B245 48 PHA
B246 D1 CC CMP (ts_ptr),Y
B248 D0 13 BNE LB25D
B24A 68 PLA
B24B C8 INY
B24C B1 CC LDA (ts_ptr),Y
B24E 29 7F AND #&7F
B250 85 CE STA ts_tmp
B252 C8 INY
B253 B1 CC LDA (ts_ptr),Y
B255 AA TAX
B256 C8 INY
B257 B1 CC LDA (ts_ptr),Y
B259 18 CLC
B25A 60 RTS
.LB25B
B25B C8 INY
.LB25C
B25C C8 INY
.LB25D
B25D C8 INY
B25E C8 INY
B25F C8 INY
B260 C8 INY
B261 4C 35 B2 JMP LB235
.LB264
B264 68 PLA
B265 A8 TAY

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```

B266 A5 CE LDA ts_tmp
B268 38 SEC
B269 60 RTS
;-----
; <DEB> END OF BLOCK [check value]
; ts_blk_check is cleared by <DSB> and never updated, so any non-zero check value is
; treated as a mismatch: the page is reloaded and the block restarted.
; Otherwise the block is complete: mark it in ts_block_map, and unless it was the last,
; ask for the next page (HEAD OF CHAIN or CHAINED PAGE).
;-----
.ts_DEB
B26A A9 83 LDA #&83
B26C 20 D3 A9 JSR ts_osword
B26F A9 6B LDA #&6B
B271 85 CC STA ts_ptr
B273 A9 16 LDA #&16
B275 85 CD STA ts_ptr+1
B277 20 6D AF JSR ts_read_value
B27A E0 00 CPX #&00
B27C F0 21 BEQ LB29F
B27E AD 69 16 LDA ts_blk_check
B281 CD 6B 16 CMP ts_deb_check
B284 D0 08 BNE LB28E
B286 AD 6A 16 LDA &166A
B289 CD 6C 16 CMP &166C
B28C F0 11 BEQ LB29F
.LB28E
B28E A9 00 LDA #&00
B290 8D 20 16 STA ts_state
B293 A9 80 LDA #&80
B295 20 D3 A9 JSR ts_osword
B298 18 CLC
B299 60 RTS
; <DEB> as installed by <TXT> (escaped &E3): no operand.
.LB29A
B29A A9 83 LDA #&83
B29C 20 D3 A9 JSR ts_osword
.LB29F
B29F AD 20 16 LDA ts_state
B2A2 C9 01 CMP #&01
B2A4 30 0D BMI LB2B3
B2A6 D0 10 BNE LB2B8
B2A8 A9 82 LDA #&82 ; block just started: set the read limit and begin decoding
B2AA 20 D3 A9 JSR ts_osword
B2AD EE 20 16 INC ts_state
B2B0 4C 0C B3 JMP LB30C
.LB2B3
B2B3 A9 CF LDA #&CF ; <DEB> outside a block: error &CF
B2B5 4C 0D AA JMP ts_error
.LB2B8
B2B8 A9 9A LDA #&9A ; release the page
B2BA 20 D3 A9 JSR ts_osword
B2BD AD 24 16 LDA ts_eof
B2C0 F0 0B BEQ LB2CD
B2C2 AD 67 16 LDA ts_block_no
B2C5 8D 23 16 STA ts_blocks_total
B2C8 A9 00 LDA #&00
B2CA 8D 24 16 STA ts_eof
.LB2CD
B2CD EE 22 16 INC ts_blocks_done ; count the block
B2D0 AD 23 16 LDA ts_blocks_total
B2D3 CD 22 16 CMP ts_blocks_done
B2D6 F0 36 BEQ LB30E ; all blocks received?
B2D8 A9 00 LDA #&00
B2DA 8D 20 16 STA ts_state
B2DD AD 68 16 LDA ts_random
B2E0 F0 05 BEQ LB2E7
B2E2 A9 88 LDA #&88
B2E4 4C E9 B2 JMP LB2E9
.LB2E7
B2E7 A9 86 LDA #&86
.LB2E9
B2E9 20 D3 A9 JSR ts_osword
B2EC AD 67 16 LDA ts_block_no
B2EF 4A LSR A
B2F0 4A LSR A
B2F1 4A LSR A
B2F2 A8 TAY
B2F3 AD 67 16 LDA ts_block_no
B2F6 29 07 AND #&07
B2F8 AA TAX
B2F9 A9 01 LDA #&01
B2FB E0 00 CPX #&00
B2FD F0 04 BEQ LB303
.LB2FF
B2FF 0A ASL A
B300 CA DEX
B301 D0 FC BNE LB2FF
.LB303
B303 19 25 16 ORA ts_block_map,Y

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B306 99 25 16 STA ts_block_map,Y
B309 EE 67 16 INC ts_block_no
.LB30C
B30C 18      CLC
B30D 60      RTS
.LB30E
B30E 38      SEC
B30F 60      RTS
; Print the stored comment text.
.LB310
B310 20 15 A9 JSR push_frame
B313 AD 49 16 LDA ts_comment
B316 85 CC    STA ts_ptr
B318 AD 4A 16 LDA ts_comment+1
B31B 85 CD    STA ts_ptr+1
B31D A0 00    LDY #000
B31F B1 CC    LDA (ts_ptr),Y
B321 AA      TAX
.LB322
B322 C8      INY
B323 B1 CC    LDA (ts_ptr),Y
B325 20 E3 FF JSR OSASCI
B328 CA      DEX
B329 D0 F7    BNE LB322
B32B 20 56 A9 JSR pop_frame
B32E 60      RTS
;-----
; TERMINAL MODE ENTRY (language entry, via the JMP at &8000)
; Set up MODE 7, the function keys and BRKV, make Telesoft the filing system, start
; rolling headers on the top row, allocate page buffers in user memory, request the
; index page, then run the main loop.
;-----
.LB32F
B32F 58      CLI
B330 20 07 BC JSR LBC07          ; (does nothing)
B333 A9 16    LDA #&16
B335 20 EE FF JSR OSWRCH        ; VDU 22: MODE
B338 A9 07    LDA #&07
B33A 20 EE FF JSR OSWRCH        ; VDU 7: BEL
B33D 20 1B B4 JSR LB41B          ; function and cursor keys produce codes; cursor on
B340 20 02 B4 JSR LB402          ; current filing system
B343 A5 32    LDA t_fs
B345 C9 06    CMP #&06
B347 F0 03    BEQ LB34C
B349 20 7B B4 JSR LB47B          ; make TFS the filing system if it isn't
.LB34C
B34C 20 22 B4 JSR LB422          ; channel 0, buffers, index page, rolling headers
B34F 20 0B B4 JSR LB40B          ; remember the previous language and filing system
B352 20 32 B5 JSR LB532          ; errors come to terminal_brk_handler
B355 4C 8E B3 JMP main_loop
; Held page: show "Press SPACE to continue" and wait.
.LB358
B358 20 3A B7 JSR status_line_inline
B35B .str ' *** Press SPACE to continue '
B382 EA      NOP
;-----
; Count down the delay that stops wild-card pages replacing each other too fast.
;-----
.LB383
B383 A9 13    LDA #&13
B385 20 F4 FF JSR OSBYTE        ; OSBYTE &13: wait for vertical sync
B388 C6 35    DEC t_delay
B38A 10 02    BPL main_loop
B38C E6 35    INC t_delay
;-----
; MAIN LOOP: handle a key if one was pressed, else show a new copy of the page when it
; arrives. Pages are fetched into the buffer at &7800 and copied to the screen from there.
;-----
.LB38E
B38E 20 E3 B3 JSR read_key
B391 B0 1F    BCS LB382          ; ESCAPE
B393 F0 03    BEQ LB398
B395 4C 3B B8 JMP terminal_dispatch_key; a key: dispatch it
.LB398
B398 24 0B    BIT t_hold          ; held?
B39A 70 BC    BVS LB358
B39C A5 35    LDA t_delay          ; still in the wild-card delay?
B39E D0 E3    BNE LB383
B3A0 20 79 B5 JSR LB579          ; TEST ARRIVED
B3A3 10 E9    BPL main_loop
B3A5 20 9B B5 JSR LB59B          ; new page: fetch its flags and number, TRANSFER it to &7800, unlock
.LB3A8
B3A8 20 DA B5 JSR draw_page        ; draw it, then wait ~2.5 s before showing another page
B3AB A9 7F    LDA #&7F
B3AD 85 35    STA t_delay
B3AF 4C 8E B3 JMP main_loop
; ESCAPE: acknowledge it and redraw.
.LB3B2
B3B2 A9 7E    LDA #&7E
B3B4 20 F4 FF JSR OSBYTE        ; OSBYTE &7E: acknowledge ESCAPE

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    B3B7 4C 86 BB JMP LBB86
; After an error: "*** Press ESCAPE to continue".
.LB3BA
    B3BA 20 37 B7 JSR clear_status_inline
    B3BD .str '\x88*** Press ESCAPE to continue'
    B3DA EA      NOP
.LB3DB
    B3DB 20 E3 B3 JSR read_key
    B3DE 90 FB    BCC LB3DB
    B3E0 4C B2 B3 JMP LB3B2
;-----
; Poll the keyboard (no wait). C set: ESCAPE. Z set: no key. Otherwise A = key.
;-----
.read_key
    B3E3 A9 81    LDA #&81
    B3E5 A2 00    LDX #&00
    B3E7 A0 00    LDY #&00
    B3E9 20 F4 FF JSR OSBYTE
    B3EC 8A       TXA
    B3ED 90 08    BCC LB3F7
    B3EF C0 1B    CPY #&1B
    B3F1 38       SEC
    B3F2 F0 03    BEQ LB3F7
    B3F4 18       CLC
    B3F5 A9 00    LDA #&00
.LB3F7
    B3F7 60       RTS
; OSWORD &7A with the block at t_osw (zero page &0C).
.t_osword
    B3F8 A9 7A    LDA #&7A
    B3FA A2 0C    LDX #&0C
    B3FC A0 00    LDY #&00
    B3FE 20 F1 FF JSR OSWORD
    B401 60       RTS
; t_fs = current filing system number.
.LB402
    B402 A9 00    LDA #&00
    B404 A8       TAY
    B405 20 DA FF JSR OSARGS
    B408 85 32    STA t_fs
    B40A 60       RTS
;-----
; Read the previous language and filing system (OSWORD &7A &91).
;-----
.LB40B
    B40B A9 91    LDA #&91
    B40D 85 0C    STA t_osw
    B40F 20 F8 B3 JSR t_osword          ; OSWORD &7A &91: read previous language
    B412 A5 00    LDA t_osw+1
    B414 85 31    STA t_prev_lang
    B416 A5 0E    LDA t_osw+2
    B418 85 32    STA t_fs
    B41A 60       RTS
;-----
; Function keys &90+, cursor keys as codes, SHIFT+function keys &9A+; cursor off.
;-----
.LB41B
    B41B 20 3D B5 JSR LB53D
    B41E 20 5E B5 JSR LB55E
    B421 60       RTS
;-----
; Start-up: channel 1, allocate page buffers (OSWORD &9B), request the index page,
; rolling headers on, clear the state and the page buffer at &7800-&7BFF.
;-----
.LB422
    B422 A9 8B    LDA #&8B
    B424 85 0C    STA t_osw
    B426 A9 00    LDA #&00
    B428 85 0D    STA t_osw+1
    B42A 20 F8 B3 JSR t_osword          ; OSWORD &7A &8B: SELECT CHANNEL
    B42D 20 5F B7 JSR LB75F
    B430 20 CF B4 JSR LB4CF
    B433 20 18 B5 JSR LB518
    B436 A9 01    LDA #&01
    B438 85 0A    STA t_link
    B43A A9 00    LDA #&00
    B43C 85 35    STA t_delay
    B43E 85 09    STA t_reveal
    B440 85 0B    STA t_hold
    B442 85 34    STA t_flags
    B444 A9 57    LDA #&57              ; "Waiting" shown as the page number until the first page arrives
    B446 85 2A    STA &2A
    B448 A9 61    LDA #&61
    B44A 85 2B    STA &2B
    B44C A9 69    LDA #&69
    B44E 85 2C    STA &2C
    B450 A9 74    LDA #&74
    B452 85 2D    STA &2D
    B454 A9 69    LDA #&69
    B456 85 2E    STA &2E

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B458 A9 6E LDA #&6E
B45A 85 2F STA &2F
B45C A9 67 LDA #&67
B45E 85 30 STA &30
B460 A9 00 LDA #&00
B462 85 00 STA t_ptr
B464 A9 78 LDA #&78
B466 85 01 STA t_ptr+1
B468 A0 00 LDY #&00
.LB46A
B46A A9 20 LDA #&20
B46C 91 00 STA (t_ptr),Y
B46E E6 00 INC t_ptr
B470 D0 02 BNE LB474
B472 E6 01 INC t_ptr+1
.LB474
B474 A5 01 LDA t_ptr+1
B476 C9 7C CMP #&7C ; '|'
B478 D0 F0 BNE LB46A
B47A 60 RTS
; Service &12 with Y=6: make TFS the filing system.
.LB47B
B47B A9 8F LDA #&8F
B47D A2 12 LDX #&12
B47F A0 06 LDY #&06
B481 20 F4 FF JSR OSBYTE
B484 60 RTS
;
; Reselect the previous filing system (t_fs): 1/2 tape at 1200/300 baud, 3 ROM,
; 4 disc (*DISK), others by service &12.
;
.LB485
B485 A4 32 LDY t_fs
B487 C0 04 CPY #&04
B489 10 1C BPL LB4A7
B48B C0 03 CPY #&03
B48D D0 06 BNE LB495
B48F A9 8D LDA #&8D
B491 20 F4 FF JSR OSBYTE ; OSBYTE &8D: select ROM filing system
B494 60 RTS
.LB495
B495 A9 8C LDA #&8C
B497 C0 02 CPY #&02
B499 D0 06 BNE LB4A1
B49B A2 03 LDX #&03
B49D 20 F4 FF JSR OSBYTE
B4A0 60 RTS
.LB4A1
B4A1 A2 0C LDX #&0C
B4A3 20 F4 FF JSR OSBYTE
B4A6 60 RTS
.LB4A7
B4A7 C0 04 CPY #&04
B4A9 D0 1C BNE LB4C7
B4AB A9 44 LDA #&44
B4AD 85 0C STA t_osw
B4AF A9 49 LDA #&49
B4B1 85 0D STA t_osw+1
B4B3 A9 53 LDA #&53
B4B5 85 0E STA t_osw+2
B4B7 A9 4B LDA #&4B
B4B9 85 0F STA t_osw+3
B4BB A9 0D LDA #&0D
B4BD 85 10 STA t_osw+4
B4BF A2 0C LDX #&0C
B4C1 A0 00 LDY #&00
B4C3 20 F7 FF JSR OSCLI
B4C6 60 RTS
.LB4C7
B4C7 A9 8F LDA #&8F
B4C9 A2 12 LDX #&12
B4CB 20 F4 FF JSR OSBYTE
B4CE 60 RTS
;
; SHIFT f1, f4 and start-up: request the index page (OSWORD &A0: the initial page from
; the service data, else x00) and show "Waiting for page". The template at &B50A is a
; FETCH of page x00 with any sub-code.
;
.LB4CF
B4CF A9 FF LDA #&FF
B4D1 85 0A STA t_link
B4D3 A2 07 LDX #&07
.LB4D5
B4D5 BD 0A B5 LDA &B50A,X
B4D8 95 0C STA t_osw,X
B4DA E0 00 CPX #&00
B4DC F0 02 BEQ LB4E0
B4DE 95 1F STA &1F,X
.LB4E0
B4E0 CA DEX

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B4E1 10 F2 BPL LB4D5
B4E3 A9 A0 LDA #&A0
B4E5 85 0C STA t_osw
B4E7 20 F8 B3 JSR t_osword ; OSWORD &7A &A0: decode date/time
B4EA 20 A9 B6 JSR LB6A9
; "*** Waiting for page " then the page number.
.LB4ED
B4ED 20 37 B7 JSR clear_status_inline
B4F0 .str ' *** Waiting for page '
B506 EA NOP
B507 4C AA B6 JMP LB6AA
B50A .db 85 FF 00 00 FF FF FF FF ; .....
; Rolling headers off.
.LB512
B512 A9 00 LDA #&00
B514 48 PHA
B515 4C 1B B5 JMP LB51B
;-----
; Rolling headers on: OSWORD &92 to &7C08, the top row of the MODE 7 screen.
;-----
.LB518
B518 A9 7C LDA #&7C
B51A 48 PHA
.LB51B
B51B A9 92 LDA #&92
B51D 85 0C STA t_osw
B51F 68 PLA
B520 85 0E STA t_osw+2
B522 A9 08 LDA #&08
B524 85 0D STA t_osw+1
B526 A9 FF LDA #&FF
B528 85 10 STA t_osw+4
B52A A9 FF LDA #&FF
B52C 85 0F STA t_osw+3
B52E 20 F8 B3 JSR t_osword
B531 60 RTS
; BRKV := terminal_brk_handler
.LB532
B532 A9 67 LDA #&67
B534 8D 02 02 STA BRKV
B537 A9 B7 LDA #&B7
B539 8D 03 02 STA BRKV+1
B53C 60 RTS
;-----
; Function keys give &90-&99, cursor keys &87-&8B, SHIFT+function keys &9A-&A3.
;-----
.LB53D
B53D A2 90 LDX #&90
B53F A0 00 LDY #&00
B541 A9 E1 LDA #&E1
B543 20 F4 FF JSR OSBYTE ; OSBYTE &E1: set base code for function keys
B546 A9 04 LDA #&04
B548 A2 01 LDX #&01
B54A A0 00 LDY #&00
B54C 20 F4 FF JSR OSBYTE ; OSBYTE &04: cursor key mode (X=1: cursor keys return codes &87-&8B)
B54F A9 E2 LDA #&E2
B551 A2 9A LDX #&9A
B553 A0 00 LDY #&00
B555 20 F4 FF JSR OSBYTE ; OSBYTE &E2: set base code for SHIFT+function keys
B558 60 RTS
; Cursor on (VDU 23,1,1,0;0;0;).
.LB559
B559 A0 01 LDY #&01
B55B 4C 60 B5 JMP LB560
; Cursor off (VDU 23,1,0;0;0;0;).
.LB55E
B55E A0 00 LDY #&00
.LB560
B560 A9 17 LDA #&17
B562 20 EE FF JSR OSWRCH ; VDU 23: VDU 23 (define/cursor)
B565 A9 01 LDA #&01
B567 20 EE FF JSR OSWRCH ; VDU 1
B56A 98 TYA
B56B 20 EE FF JSR OSWRCH
B56E A9 00 LDA #&00
B570 A0 07 LDY #&07
.LB572
B572 20 EE FF JSR OSWRCH ; VDU 0
B575 88 DEY
B576 D0 FA BNE LB572
B578 60 RTS
;-----
; OSWORD &7A &89 TEST ARRIVED: N set (from BIT) if a new page has arrived.
;-----
.LB579
B579 A9 89 LDA #&89
B57B 85 0C STA t_osw
B57D 20 F8 B3 JSR t_osword ; OSWORD &7A &89: TEST ARRIVED
B580 24 0D BIT t_osw+1
B582 60 RTS
```

```

B583 .db A9 8C 85 0C 20 F8 B3 24 0D 50 03 A9 FF 60 24 34 ; ....$.P...`$4
B593 .db 70 03 A9 FF 60 A9 00 60 ; p...`

;-----
; Get the new page: its checksum flags, its page number (shown with COPY), then TRANSFER
; it to &7800 (in this machine: high address from OSBYTE &82) and UNLOCK it.
;-----
.LB59B
B59B A9 8C LDA #&8C
B59D 85 0C STA t_osw
B59F 20 F8 B3 JSR t_osword ; OSWORD &7A &8C: read page flags
B5A2 A5 0D LDA t_osw+1
B5A4 85 34 STA t_flags
B5A6 A9 A3 LDA #&A3
B5A8 85 0C STA t_osw
B5AA 20 F8 B3 JSR t_osword ; OSWORD &7A &A3: read page number
B5AD A0 01 LDY #&01

.LB5AF
B5AF B9 0C 00 LDA t_osw,Y
B5B2 99 29 00 STA t_shown_page,Y
B5B5 C8 INY
B5B6 C0 08 CPY #&08
B5B8 D0 F5 BNE LB5AF
B5BA A9 8F LDA #&8F
B5BC 85 0C STA t_osw
B5BE A9 78 LDA #&78
B5C0 85 0E STA t_osw+2
B5C2 A9 00 LDA #&00
B5C4 85 0D STA t_osw+1
B5C6 A9 82 LDA #&82
B5C8 20 F4 FF JSR OSBYTE ; OSBYTE &82: read machine high order address
B5CB 84 10 STY t_osw+4
B5CD 86 0F STX t_osw+3
B5CF 20 F8 B3 JSR t_osword
B5D2 A9 8A LDA #&8A
B5D4 85 0C STA t_osw
B5D6 4C F8 B3 JMP t_osword ; OSWORD &7A &8A: UNLOCK PAGE
B5D9 .db 60 ; `

;-----
; DRAW THE PAGE: rows 1-23 from the buffer at &7800 (row 0 shows the rolling header).
; When revealing (t_reveal), the conceal code &98 is replaced by a space so the hidden
; text shows. Then report the checksum state at the bottom right:
; check word present + matched : nothing
; check word present, mismatch : "Bad data"
; no check word : "No check"
;-----
.draw_page
B5DA A9 1E LDA #&1E
B5DC 20 EE FF JSR OSWRCH ; VDU 30: cursor home
B5DF 20 AA B6 JSR LB6AA
B5E2 20 3A B7 JSR status_line_inline
B5E5 .str '\x1e\n'
B5E7 EA NOP
B5E8 A9 28 LDA #&28 ; start at row 1 of the buffer
B5EA 85 00 STA t_ptr
B5EC A9 78 LDA #&78
B5EE 85 01 STA t_ptr+1
B5F0 A0 00 LDY #&00

.LB5F2
B5F2 B1 00 LDA (t_ptr),Y
B5F4 C9 98 CMP #&98 ; conceal code?
B5F6 D0 06 BNE LB5FE
B5F8 A6 09 LDX t_reveal
B5FA F0 02 BEQ LB5FE
B5FC A9 20 LDA #&20

.LB5FE
B5FE 09 80 ORA #&80 ; MODE 7 code
B600 20 EE FF JSR OSWRCH
B603 E6 00 INC t_ptr
B605 A5 00 LDA t_ptr
B607 C9 C0 CMP #&C0
B609 D0 06 BNE LB611
B60B A6 01 LDX t_ptr+1
B60D E0 7B CPX #&7B ; '{'
B60F F0 08 BEQ LB619

.LB611
B611 AA TAX
B612 D0 DE BNE LB5F2
B614 E6 01 INC t_ptr+1
B616 4C F2 B5 JMP LB5F2

.LB619
B619 24 34 BIT t_flags
B61B 30 32 BMI LB64F
B61D 70 5E BVS LB67D
B61F 20 37 B7 JSR clear_status_inline
B622 .str ' No check'
B649 EA NOP
B64A 60 RTS

.LB64B
B64B 20 0B B7 JSR clear_status_line
B64E 60 RTS

```

```

.LB64F
B64F 70 FA BVS LB64B
B651 20 37 B7 JSR clear_status_inline
B654 .str ' Bad data'
B67B EA NOP
B67C 60 RTS

.LB67D
B67D 20 37 B7 JSR clear_status_inline
B680 .str ' No check'
B6A7 EA NOP
B6A8 60 RTS

.LB6A9
B6A9 60 RTS
; Print the requested page number from t_last_page.
.LB6AA
B6AA A2 06 LDX #&06
.LB6AC
B6AC B5 20 LDA t_last_page,X
B6AE 95 0D STA t_osw+1,X
B6B0 CA DEX
B6B1 10 F9 BPL LB6AC
;-----
; Print the 7 digits at t_osw+1: "#" for a wild digit, a space before the sub-code, and
; the sub-code left out if it is all wild.
;-----
.LB6B3
B6B3 A0 00 LDY #&00
B6B5 A2 23 LDX #&23
B6B7 20 F6 B6 JSR LB6F6
B6BA C8 INY
B6BB 20 F6 B6 JSR LB6F6
B6BE C8 INY
B6BF 20 F6 B6 JSR LB6F6
B6C2 C8 INY
B6C3 A2 2F LDX #&2F
B6C5 B9 10 00 LDA t_osw+4,Y
B6C8 10 11 BPL LB6DB
B6CA B9 0F 00 LDA t_osw+3,Y
B6CD 10 0C BPL LB6DB
B6CF B9 0E 00 LDA t_osw+2,Y
B6D2 10 07 BPL LB6DB
B6D4 B9 0D 00 LDA t_osw+1,Y
B6D7 10 02 BPL LB6DB
B6D9 A2 20 LDX #&20

.LB6DB
B6DB 20 F6 B6 JSR LB6F6
B6DE E0 20 CPX #&20
B6E0 F0 02 BEQ LB6E4
B6E2 A2 23 LDX #&23

.LB6E4
B6E4 C8 INY
B6E5 20 F6 B6 JSR LB6F6
B6E8 C8 INY
B6E9 20 F6 B6 JSR LB6F6
B6EC C8 INY
B6ED 20 F6 B6 JSR LB6F6
B6F0 A9 20 LDA #&20
B6F2 20 EE FF JSR OSWRCH ; print ' '
B6F5 60 RTS

.LB6F6
B6F6 B9 0D 00 LDA t_osw+1,Y
B6F9 10 04 BPL LB6FF
B6FB 8A TXA
B6FC 4C EE FF JMP OSWRCH

.LB6FF
B6FF 18 CLC
B700 69 30 ADC #&30
B702 C9 3A CMP #&3A ; ':'
B704 90 02 BCC LB708
B706 69 06 ADC #&06

.LB708
B708 4C EE FF JMP OSWRCH
; Blank row 24.
.clear_status_line
B70B 20 3A B7 JSR status_line_inline
B70E .str '
B735 EA NOP
B736 60 RTS
; Blank row 24, then print the following string there.
.clear_status_inline
B737 20 0B B7 JSR clear_status_line
;-----
; PRINT INLINE TEXT ON ROW 24: TAB(0,24), then print the bytes after the JSR up to an &EA
; byte, which is then executed as a NOP.
;-----
.status_line_inline
B73A A9 1F LDA #&1F
B73C 20 EE FF JSR OSWRCH ; VDU 31: TAB(x,y)
B73F A9 00 LDA #&00
B741 A8 TAY

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```

B742 20 EE FF JSR OSWRCH          ; VDU 0
B745 68      PLA
B746 85 00    STA t_ptr
B748 68      PLA
B749 85 01    STA t_ptr+1
B74B A9 18    LDA #18             ; 24 is the row for the VDU 31 sequence
.LB74D
B74D 20 EE FF JSR OSWRCH          ; (first time round: the row number)
B750 E6 00    INC t_ptr
B752 D0 02    BNE LB756
B754 E6 01    INC t_ptr+1
.LB756
B756 B1 00    LDA (t_ptr),Y
B758 C9 EA    CMP #EA
B75A D0 F1    BNE LB74D
B75C 6C 00 00 JMP (t_ptr)
; OSWORD &7A &9B: allocate page buffers from PAGE.
.LB75F
B75F A9 9B    LDA #9B
B761 85 0C    STA t_osw
B763 20 F8 B3 JSR t_osword        ; OSWORD &7A &9B
B766 60      RTS
;
;-----
; TERMINAL MODE BRK HANDLER: reset the stack, reselect TFS, print the error message,
; then wait for ESCAPE.
;-----
.terminal_brk_handler
B767 A2 FF    LDX #FF
B769 9A      TXS
B76A 20 7B B4 JSR LB47B
B76D 20 1B B4 JSR LB41B
B770 A9 0D    LDA #0D
B772 20 E3 FF JSR OSASCI          ; VDU 13: carriage return
B775 A0 01    LDY #01
.LB777
B777 B1 FD    LDA (brk_ptr),Y
B779 F0 06    BEQ LB781
B77B 20 E3 FF JSR OSASCI
B77E C8      INY
B77F D0 F6    BNE LB777
.LB781
B781 A9 0D    LDA #0D
B783 20 E3 FF JSR OSASCI          ; VDU 13: carriage return
B786 4C BA B3 JMP LB3BA
;
;-----
; f0 SELECT A PAGE: type up to 7 characters (0-9, A-F, # for wild), DELETE, RETURN.
; Unspecified digits are wild. Issues FETCH PAGE and remembers it as the last explicit page.
;-----
.LB789
B789 A9 FF    LDA #FF
B78B 85 0A    STA t_link
B78D 20 0B B7 JSR cclear_status_line
B790 A2 06    LDX #06
B792 A9 FF    LDA #FF
.LB794
B794 95 0D    STA t_osw+1,X
B796 CA      DEX
B797 10 FB    BPL LB794
B799 A9 00    LDA #00
B79B 85 00    STA t_ptr
B79D F0 05    BEQ LB7A4
.LB79F
B79F A9 07    LDA #07
B7A1 20 EE FF JSR OSWRCH          ; VDU 7: BEL
.LB7A4
B7A4 A5 00    LDA t_ptr
B7A6 48      PHA
B7A7 20 3A B7 JSR status_line_inline
B7AA .str 'Page '
B7AF EA      NOP
B7B0 20 B3 B6 JSR LB6B3
B7B3 68      PLA
B7B4 85 00    STA t_ptr
B7B6 20 E3 B3 JSR read_key
B7B9 B0 44    BCS LB7FF
B7BB F0 E7    BEQ LB7A4
B7BD C9 0D    CMP #0D
B7BF F0 2B    BEQ LB7EC
B7C1 C9 7F    CMP #7F
B7C3 D0 12    BNE LB7D7
B7C5 C6 00    DEC t_ptr
B7C7 10 05    BPL LB7CE
B7C9 E6 00    INC t_ptr
B7CB 4C A4 B7 JMP LB7A4
.LB7CE
B7CE A6 00    LDX t_ptr
B7D0 A9 FF    LDA #FF
B7D2 95 0D    STA t_osw+1,X
B7D4 4C A4 B7 JMP LB7A4
.LB7D7

```

```

B7D7 A6 00 LDX t_ptr
B7D9 E0 07 CPX #&07
B7DB F0 C2 BEQ LB79F
B7DD 20 02 B8 JSR LB802
B7E0 F0 B0 BEQ LB79F
B7E2 A6 00 LDX t_ptr
B7E4 95 00 STA t_osw+1,X
B7E6 E8 INX
B7E7 86 00 STX t_ptr
B7E9 4C A4 B7 JMP LB7A4
.LB7EC
B7EC A9 85 LDA #&85
B7EE 85 0C STA t_osw
B7F0 20 F8 B3 JSR t_osword ; OSWORD &7A &85: FETCH PAGE
B7F3 A2 07 LDX #&07
.LB7F5
B7F5 B5 00 LDA t_osw+1,X
B7F7 95 20 STA t_last_page,X
B7F9 CA DEX
B7FA 10 F9 BPL LB7F5
B7FC 4C F4 B9 JMP LB9F4
.LB7FF
B7FF 4C B2 B3 JMP LB3B2
;-----
; Convert a typed character to a digit: "#" → &FF, 0-9, A-F/a-f → 10-15; 0 if invalid.
;-----
.LB802
B802 C9 23 CMP #&23 ; '#'
B804 D0 03 BNE LB809
B806 A9 FF LDA #&FF
B808 60 RTS
.LB809
B809 38 SEC
B80A E9 30 SBC #&30
B80C 30 19 BMI LB827
B80E C9 0A CMP #&0A
B810 B0 01 BCS LB813
B812 60 RTS
.LB813
B813 E9 11 SBC #&11
B815 30 10 BMI LB827
B817 C9 06 CMP #&06
B819 B0 03 BCS LB81E
B81B 69 0A ADC #&0A
B81D 60 RTS
.LB81E
B81E E9 20 SBC #&20
B820 C9 06 CMP #&06
B822 B0 03 BCS LB827
B824 69 0A ADC #&0A
B826 60 RTS
.LB827
B827 A9 00 LDA #&00
B829 60 RTS
; Invalid key: beep.
.key_invalid
B82A A9 07 LDA #&07
B82C 20 EE FF JSR OSWRCH ; VDU 7: BEL
B82F 4C 98 B3 JMP LB398
.LB832
B832 4C CE B9 JMP LB9CE
.LB835
B835 4C 22 BB JMP key_f5_star_cmd
.LB838
B838 4C 86 BB JMP LBB86
;-----
; KEY DISPATCH: SPACE redraws, "*" is f5, digits start a link number. The cursor and
; function key codes &87-&A4 go through the table at &B871.
;-----
.terminal_dispatch_key
B83B C9 20 CMP #&20
B83D F0 F9 BEQ LB838
B83F C9 2A CMP #&2A ; '*'
B841 F0 F2 BEQ LB835
B843 38 SEC
B844 E9 30 SBC #&30
B846 30 E2 BMI key_invalid
B848 C9 0A CMP #&0A
B84A 90 E6 BCC LB832
B84C E9 11 SBC #&11
B84E 30 DA BMI key_invalid
B850 C9 18 CMP #&18
B852 90 D6 BCC key_invalid
B854 E9 20 SBC #&20
B856 30 D2 BMI key_invalid
B858 C9 18 CMP #&18
B85A 90 CE BCC key_invalid
B85C E9 26 SBC #&26
B85E C9 1E CMP #&1E
B860 B0 C8 BCS key_invalid

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B862 0A      ASL A
B863 AA      TAX
B864 BD 71 B8 LDA key_table,X
B867 85 00    STA t_ptr
B869 BD 72 B8 LDA &B872,X
B86C 85 01    STA t_ptr+1
B86E 6C 00 00 JMP (t_ptr)
;-----
; Terminal-mode key table, codes &87-&A4: COPY, cursors, f0-f9 (&90), SHIFT f0-f9 (&9A).
;-----
.key_table
B871 .db AD B8 C7 B8 EA B8 0E B9 1C B9 2A B8 2A B8 2A B8 ; .....*.*.*.
B881 .db 2A B8 D9 B9 EB B9 78 BB 7D B9 FA B9 22 BB 57 BA ; *....x.}...".W.
B891 .db 6C BA 7F BB A0 BB 2A B8 51 BB 99 BB 46 B9 57 BB ; \.....Q...F.W.
B8A1 .db 2A B8 2A B8 2A B8 2A B8 B7 BB 2A B8 ; *.*.*.*.*.
; COPY: show the number of the page on screen.
.key_COPY
B8AD 20 37 B7 JSR clear_status_inline
B8B0 .str 'Page: '
B8B6 EA      NOP
B8B7 A0 00    LDY #000
.LB8B9
B8B9 B9 2A 00 LDA &002A,Y
B8BC 20 EE FF JSR OSWRCH
B8BF C8      INY
B8C0 C0 07    CPY #007
B8C2 D0 F5    BNE LB8B9
B8C4 4C 98 B3 JMP LB398
; CURSOR LEFT: previous link (t_link - 1).
.key_left
B8C7 A5 0A    LDA t_link
B8C9 30 02    BMI LB8CD
B8CB D0 15    BNE LB8E2
.LB8CD
B8CD 20 37 B7 JSR clear_status_inline
B8D0 .str 'Cannot go left'
B8DE EA      NOP
B8DF 4C 98 B3 JMP LB398
.LB8E2
B8E2 38      SEC
B8E3 E9 01    SBC #001
B8E5 85 0A    STA t_link
B8E7 4C C3 B9 JMP LB9C3
; CURSOR RIGHT: next link (up to 23).
.key_right
B8EA A5 0A    LDA t_link
B8EC C9 18    CMP #018
B8EE 90 16    BCC LB906
B8F0 20 37 B7 JSR clear_status_inline
B8F3 .str 'Cannot go right'
B902 EA      NOP
B903 4C 98 B3 JMP LB398
.LB906
B906 18      CLC
B907 69 01    ADC #001
B909 85 0A    STA t_link
B90B 4C C3 B9 JMP LB9C3
; CURSOR DOWN: follow link 1 (the next page of a story).
.key_down
B90E A9 9E    LDA #09E
B910 85 0C    STA t_osw
B912 20 F8 B3 JSR t_osword ; OSWORD &7A &9E
B915 A9 01    LDA #001
B917 85 0A    STA t_link
B919 4C C3 B9 JMP LB9C3
;-----
; CURSOR UP: leave link navigation (t_link = &FF) and show the current page again.
; "Cannot go up" if no link is being followed.
;-----
.key_up
B91C A9 9F    LDA #09F
B91E 85 0C    STA t_osw
B920 20 F8 B3 JSR t_osword ; OSWORD &7A &9F
B923 A5 00    LDA t_osw+1
B925 A5 0A    LDA t_link
B927 10 13    BPL LB93C
B929 20 37 B7 JSR clear_status_inline
B92C .str 'Cannot go up'
B938 EA      NOP
B939 4C 98 B3 JMP LB398
.LB93C
B93C A9 FF    LDA #0FF
B93E 85 0A    STA t_link
B940 20 ED B4 JSR LB4ED
B943 4C 98 B3 JMP LB398
; SHIFT f3: reRelease a kept page (OSWORD &9D).
.key_sf3_release
B946 A9 9D    LDA #09D
B948 85 0C    STA t_osw
B94A 20 F8 B3 JSR t_osword ; OSWORD &7A &9D: test page changed

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```

B94D A5 0D LDA t_osw+1
B94F D0 14 BNE LB965
B951 20 37 B7 JSR clear_status_inline
B954 .str 'Page released'
B961 EA NOP
B962 4C 98 B3 JMP LB398
.LB965
B965 20 37 B7 JSR clear_status_inline
B968 .str 'Page was not kept'
B979 EA NOP
B97A 4C 98 B3 JMP LB398
; f3: keep the page (OSWORD &9C).
.key_f3_keep
B97D A9 9C LDA #&9C
B97F 85 0C STA t_osw
B981 20 F8 B3 JSR t_osword ; OSWORD &7A &9C: find free buffer
B984 A5 0D LDA t_osw+1
B986 D0 10 BNE LB998
B988 20 37 B7 JSR clear_status_inline
B98B .str 'Page kept'
B994 EA NOP
B995 4C 98 B3 JMP LB398
.LB998
B998 20 37 B7 JSR clear_status_inline
B99B .str 'Buffers full - Please release a page'
B9BF EA NOP
B9C0 4C 98 B3 JMP LB398
;
; Select link t_link as though it had been typed: put RETURN in the keyboard buffer and
; enter the link number.
;
.LB9C3
B9C3 A0 0D LDY #&0D
B9C5 A2 00 LDX #&00
B9C7 A9 8A LDA #&8A
B9C9 20 F4 FF JSR OSBYTE ; OSBYTE &8A: insert character into buffer
B9CC A5 0A LDA t_link
.LB9CE
B9CE 85 0A STA t_link
B9D0 18 CLC
B9D1 69 30 ADC #&30
B9D3 20 32 BC JSR LBC32
B9D6 4C 98 B3 JMP LB398
; f0: select a page.
.key_f0_select_page
B9D9 4C 89 B7 JMP LB789
; FETCH PAGE using the digits at t_last_page.
.LB9DC
B9DC B5 20 LDA t_last_page,X
B9DE 95 0D STA t_osw+1,X
B9E0 CA DEX
B9E1 10 F9 BPL LB9DC
B9E3 A9 85 LDA #&85
B9E5 85 0C STA t_osw
B9E7 20 F8 B3 JSR t_osword ; OSWORD &7A &85: FETCH PAGE
B9EA 60 RTS
; f1: last explicitly selected page.
.key_f1_last_explicit
B9EB A9 FF LDA #&FF
B9ED 85 0A STA t_link
B9EF A2 06 LDX #&06
B9F1 20 DC B9 JSR LB9DC
.LB9F4
B9F4 20 ED B4 JSR LB4ED
B9F7 4C 98 B3 JMP LB398
;
; f4: SELECT CHANNEL 1-4 then RETURN; then request that channel's index page.
;
.key_f4_channel
B9FA 20 37 B7 JSR clear_status_inline
B9FD .str 'Channel '
BA05 EA NOP
.LBA06
BA06 20 E0 FF JSR OSRDCH
BA09 B0 49 BCS LBA54
BA0B C9 7F CMP #&7F
BA0D D0 02 BNE LBA11
BA0F A9 20 LDA #&20
.LBA11
BA11 20 EE FF JSR OSWRCH ; print ' '
BA14 20 02 B8 JSR LB802
BA17 D0 0D BNE LBA26
.LBA19
BA19 A9 08 LDA #&08
BA1B 20 EE FF JSR OSWRCH ; VDU 8
BA1E A9 07 LDA #&07
.LBA20
BA20 20 EE FF JSR OSWRCH ; VDU 7: BEL
BA23 4C 06 BA JMP LBA06
.LBA26

```

```

BA26 38      SEC
BA27 E9 01    SBC #&01
BA29 C9 04    CMP #&04
BA2B B0 EC    BCS LBA19
BA2D 85 0D    STA t_osw+1
BA2F A9 8B    LDA #&8B
BA31 85 0C    STA t_osw
.LBA33
BA33 20 E0 FF JSR OSRDCH
BA36 B0 1C    BCS LBA54
BA38 C9 7F    CMP #&7F
BA3A F0 E4    BEQ LBA20
BA3C C9 0D    CMP #&0D
BA3E F0 08    BEQ LBA48
BA40 A9 07    LDA #&07
BA42 20 EE FF JSR OSWRCH          ; VDU 7: BEL
BA45 4C 33 BA JMP LBA33
.LBA48
BA48 20 F8 B3 JSR t_osword
BA4B 20 0B B7 JSR clear_status_line
BA4E 20 CF B4 JSR LB4CF
BA51 4C 98 B3 JMP LB398
.LBA54
BA54 4C B2 B3 JMP LB3B2
; f6: LOAD A PAGE from a file into &7800, show it and hold it.
.key_f6_load
BA57 20 37 B7 JSR clear_status_inline
BA5A .str 'Load file '
BA64 EA      NOP
BA65 20 FA BA JSR LBAFA
BA68 A9 FF    LDA #&FF
BA6A 30 13    BMI LBA7F
;
; f7: SAVE THE PAGE to a file, from &7800 to &7C41.
; The previous filing system is selected for the load or save, and TFS afterwards.
;
.key_f7_save
BA6C 20 37 B7 JSR clear_status_inline
BA6F .str 'Save file '
BA79 EA      NOP
BA7A 20 FA BA JSR LBAFA
BA7D A9 00    LDA #&00
.LBA7F
BA7F 48      PHA
BA80 20 12 B5 JSR LB512
BA83 20 85 B4 JSR LB485
BA86 A2 37    LDX #&37
BA88 86 0C    STX t_osw
BA8A A2 00    LDX #&00
BA8C 86 0D    STX t_osw+1
BA8E A9 00    LDA #&00
BA90 A2 00    LDX #&00
.LBA92
BA92 95 0E    STA t_osw+2,X
BA94 E8      INX
BA95 E0 10    CPX #&10
BA97 D0 F9    BNE LBA92
BA99 A2 78    LDX #&78
BA9B 86 0F    STX t_osw+3
BA9D 86 13    STX &13
BA9F 86 17    STX &17
BAA1 A2 42    LDX #&42
BAA3 86 1A    STX &1A
BAA5 A2 7C    LDX #&7C
BAA7 86 1B    STX &1B
BAA9 68      PLA
BAAA A2 0C    LDX #&0C
BAAC A0 00    LDY #&00
BAAE 20 DD FF JSR OSFILE
BAB1 20 DA B5 JSR draw_page
BAB4 20 7B B4 JSR LB47B
BAB7 20 5E B5 JSR LB55E
BABA 20 3A B7 JSR status_line_inline
BABD .str '\x1e'
BAE7 EA      NOP
BAE8 A2 00    LDX #&00
.LBAEA
BAEA B5 37    LDA t_line,X
BAEC 20 EE FF JSR OSWRCH
BAEF E8      INX
BAF0 C9 0D    CMP #&0D
BAF2 D0 F6    BNE LBAEA
BAF4 20 DC B9 JSR LB9DC
BAF7 4C 7F BB JMP key_f8_hold
;
; Read a line (up to 32 characters) into t_line with OSWORD 0. ESCAPE abandons the command.
;
.LBAFA
BAFA 20 1E BC JSR LBC1E
BAFD A9 26    LDA #&26

```

```

BAFF 85 0E STA t_osw+2
BB01 A9 37 LDA #&37
BB03 85 0C STA t_osw
BB05 A9 00 LDA #&00
BB07 85 0D STA t_osw+1
BB09 A9 20 LDA #&20
BB0B 85 0F STA t_osw+3
BB0D A9 7F LDA #&7F
BB0F 85 10 STA t_osw+4
BB11 A2 0C LDX #&0C
BB13 A0 00 LDY #&00
BB15 A9 00 LDA #&00
BB17 20 F1 FF JSR OSWORD ; OSWORD &00: read line from keyboard
BB1A 90 05 BCC LBB21
BB1C 68 PLA
BB1D 68 PLA
BB1E 4C 54 BA JMP LBA54
.LBB21
BB21 60 RTS
;-----
; f5: * COMMAND. Suspend all page requests while the command runs (it may use the memory
; holding the page buffers), run it with the previous filing system selected, then
; reselect TFS and resume.
;-----
.key_f5_star_cmd
BB22 20 12 B5 JSR LB512
BB25 20 37 B7 JSR clear_status_inline
BB28 .str '*'
BB29 EA NOP
BB2A 20 FA BA JSR LBAFA
BB2D A9 A4 LDA #&A4
BB2F 85 0C STA t_osw
BB31 20 F8 B3 JSR t_osword ; OSWORD &7A &A4: suspend all requests
BB34 20 85 B4 JSR LB485
BB37 A2 37 LDX #&37
BB39 A0 00 LDY #&00
BB3B 20 F7 FF JSR OSCLI
BB3E 20 02 B4 JSR LB402
BB41 20 7B B4 JSR LB47B
BB44 A9 A5 LDA #&A5
BB46 85 0C STA t_osw
BB48 20 F8 B3 JSR t_osword ; OSWORD &7A &A5: resume all requests
BB4B 20 1B B4 JSR LB41B
BB4E 4C 7F BB JMP key_f8_hold
; SHIFT f1: the index page.
.key_sf1_index
BB51 20 CF B4 JSR LB4CF
BB54 4C 98 B3 JMP LB398
; SHIFT f4: *TUNE (the command string is at &BB73).
.key_sf4_tune
BB57 A9 73 LDA #&73
BB59 85 00 STA t_ptr
BB5B A9 BB LDA #&BB
BB5D 85 01 STA t_ptr+1
BB5F A0 04 LDY #&04
.LBB61
BB61 B1 00 LDA (t_ptr),Y
BB63 99 37 00 STA t_line,Y
BB66 88 DEY
BB67 10 F8 BPL LBB61
BB69 A2 37 LDX #&37
BB6B A0 00 LDY #&00
BB6D 20 F7 FF JSR OSCLI
BB70 4C 86 BB JMP LBB86
; "TUNE",CR
BB73 .db 54 55 4E 45 0D ; TUNE.
; f2: REVEAL concealed text, and hold the page.
.key_f2_reveal
BB78 A9 FF LDA #&FF
BB7A 85 09 STA t_reveal
BB7C 20 DA B5 JSR draw_page
; f8: HOLD the page.
.key_f8_hold
BB7F A9 FF LDA #&FF
BB81 85 0B STA t_hold
BB83 4C 98 B3 JMP LB398
;-----
; SPACE / ESCAPE: release the hold, redraw (MODE 7), rolling headers on.
;-----
.LBB86
BB86 A9 00 LDA #&00
BB88 85 0B STA t_hold
BB8A 20 3A B7 JSR status_line_inline
BB8D .str '\x16\x07'
BB8F EA NOP
BB90 20 1B B4 JSR LB41B
BB93 20 18 B5 JSR LB518
BB96 20 A9 B6 JSR LB6A9
; SHIFT f2: conceal again, keeping the hold.
.key_sf2_conceal

```

```

BB99 A9 00 LDA #&00
BB9B 85 09 STA t_reveal
BB9D 4C A8 B3 JMP LB3A8
;-----
; f9: EXIT TO THE PREVIOUS LANGUAGE (page buffers released, previous filing system back).
;-----
.key_f9_exit_language
BBA0 20 08 BC JSR LBC08
BBA3 20 12 B5 JSR LB512
BBA6 20 37 B7 JSR clear_status_inline
BBA9 EA NOP
BBAA 20 85 B4 JSR LB485
BBAD 4C D1 BB JMP LBBD1
; "EXEC","",CR: *EXEC the first Telesoftware file on the page.
BBB0 .db 45 58 45 43 22 22 0D ; EXEC"".
;-----
; SHIFT f9: EXIT TO TELESOFT AND EXECUTE: a text window below the header row, then
; *EXEC "" so the program on the page is typed into the language, which is entered next.
;-----
.key_sf9_tesoft
BBB7 20 08 BC JSR LBC08
BBBA 20 3A B7 JSR status_line_inline
BBBD .str "\x16\x07\x1c\x00\x18'\x01"
BBC4 EA NOP
BBC5 A9 00 LDA #&00
BBC7 20 E3 FF JSR OSASCI ; VDU 13: carriage return
BBCA A2 B0 LDX #&B0
BBCC A0 BB LDY #&BB
BBCE 20 F7 FF JSR OSCLI
;-----
; Enter the previous language. If that was TFS itself (after BREAK in terminal mode),
; use BASIC instead when there is one.
;-----
.LBBD1
BBD1 20 1E BC JSR LBC1E
BBD4 A2 F4 LDX #&F4
BBD6 86 5E STX &5E
BBD8 A2 00 LDX #&00
BBDA 86 5F STX &5F
BBDC A2 FF LDX #&FF
BBDE 86 60 STX &60
BBE0 86 61 STX &61
BBE2 A2 5E LDX #&5E
BBE4 A0 00 LDY #&00
BBE6 A9 05 LDA #&05 ; read the current ROM number (&F4) from the I/O processor
BBE8 20 F1 FF JSR OSWORD ; OSWORD &05: read I/O processor memory
BBEB A6 31 LDX t_prev_lang
BBED E4 62 CPX &62
BBEF D0 0F BNE LBC00
BBF1 A9 BB LDA #&BB
BBF3 A2 00 LDX #&00
BBF5 A0 FF LDY #&FF
BBF7 20 F4 FF JSR OSBYTE ; OSBYTE &BB: read BASIC ROM number
BBFA E0 FF CPX #&FF
BBFC D0 02 BNE LBC00
BBFE A6 31 LDX t_prev_lang
.LBC00
BC00 A0 12 LDY #&12
BC02 A9 8E LDA #&8E
BC04 4C F4 FF JMP OSBYTE ; OSBYTE &8E: enter language ROM X
.LBC07
BC07 60 RTS
;-----
; Leaving terminal mode: slot 0 back to its private-workspace buffer, other slots off.
;-----
.LBC08
BC08 A9 9E LDA #&9E
BC0A 85 0C STA t_osw
BC0C 20 F8 B3 JSR t_osword ; OSWORD &7A &9E
BC0F A9 A2 LDA #&A2
BC11 85 0C STA t_osw
BC13 20 F8 B3 JSR t_osword ; OSWORD &7A &A2
BC16 A9 A1 LDA #&A1
BC18 85 0C STA t_osw
BC1A 20 F8 B3 JSR t_osword ; OSWORD &7A &A1: reset requests
BC1D 60 RTS
;-----
; Keys back to normal: cursor editing, function keys as strings, cursor on.
;-----
.LBC1E
BC1E A9 04 LDA #&04
BC20 A2 00 LDX #&00
BC22 A0 00 LDY #&00
BC24 20 F4 FF JSR OSBYTE
BC27 A9 E1 LDA #&E1
BC29 A2 01 LDX #&01
BC2B 20 F4 FF JSR OSBYTE ; OSBYTE &E1: set base code for function keys
BC2E 20 59 B5 JSR LB559
BC31 60 RTS
;-----

```

; ENTER A LINK NUMBER (0-23), started by the digit in A. RETURN selects it (OSWORD &96):
; "Waiting for link n", "No such link" or "No links present".

```
-----  
.LBC32  
BC32 48 PHA  
BC33 20 37 B7 JSR clear_status_inline  
BC36 .str 'Link '  
BC3B EA NOP  
BC3C 68 PLA  
BC3D AA TAX  
BC3E 20 EE FF JSR OSWRCH  
BC41 A0 FF LDY #&FF  
.LBC43  
BC43 20 E0 FF JSR OSRDCH  
BC46 90 07 BCC LBC4F  
BC48 C9 1B CMP #&1B  
BC4A D0 03 BNE LBC4F  
BC4C 4C 22 BD JMP LBD22  
.LBC4F  
BC4F C9 00 CMP #&00  
BC51 F0 45 BEQ LBC98  
BC53 C9 7F CMP #&7F  
BC55 F0 2E BEQ LBC85  
BC57 C9 30 CMP #&30 ; '0'  
BC59 90 04 BCC LBC5F  
BC5B C9 3A CMP #&3A ; ':'  
BC5D 90 08 BCC LBC67  
.LBC5F  
BC5F A9 07 LDA #&07  
BC61 20 EE FF JSR OSWRCH ; VDU 7: BEL  
BC64 4C 43 BC JMP LBC43  
.LBC67  
BC67 E0 FF CPX #&FF  
BC69 D0 03 BNE LBC6E  
BC6B AA TAX  
BC6C 10 11 BPL LBC7F  
.LBC6E  
BC6E C0 FF CPY #&FF  
BC70 D0 ED BNE LBC5F  
BC72 E0 33 CPX #&33 ; '3'  
BC74 B0 E9 BCS LBC5F  
BC76 E0 32 CPX #&32 ; '2'  
BC78 D0 04 BNE LBC7E  
BC7A C9 34 CMP #&34 ; '4'  
BC7C B0 E1 BCS LBC5F  
.LBC7E  
BC7E A8 TAY  
.LBC7F  
BC7F 20 EE FF JSR OSWRCH  
BC82 4C 43 BC JMP LBC43  
.LBC85  
BC85 E0 FF CPX #&FF  
BC87 F0 D6 BEQ LBC5F  
BC89 20 EE FF JSR OSWRCH  
BC8C A9 FF LDA #&FF  
BC8E C0 FF CPY #&FF  
BC90 F0 03 BEQ LBC95  
BC92 A8 TAY  
BC93 D0 AE BNE LBC43  
.LBC95  
BC95 AA TAX  
BC96 D0 AB BNE LBC43  
.LBC98  
BC98 E0 FF CPX #&FF  
BC9A D0 03 BNE LBC9F  
BC9C 4C 27 BD JMP LBD27  
.LBC9F  
BC9F A9 00 LDA #&00  
BCA1 85 00 STA t_osw+1  
BCA3 85 29 STA t_shown_page  
BCA5 C0 FF CPY #&FF  
BCA7 F0 14 BEQ LBC8D  
BCA9 8A TXA  
BCAA 85 28 STA &28  
BCAC 38 SEC  
BCAD E9 30 SBC #&30  
BCAF 0A ASL A  
BCB0 85 00 STA t_osw+1  
BCB2 0A ASL A  
BCB3 0A ASL A  
BCB4 65 00 ADC t_osw+1  
BCB6 85 00 STA t_osw+1  
BCB8 98 TYA  
BCB9 85 29 STA t_shown_page  
BCBB 10 03 BPL LBCC0  
.LBCC0  
BCBD 8A TXA  
BCBE 85 28 STA &28  
.LBCC0  
BCC0 38 SEC
```

[illegible]

Appendix B: ATS250.ROM (ATS 2.50) listing

```
; =====
; ATS 2.50 - BBC Advanced Teletext System ROM, for the BBC Micro / Master
; Commented disassembly of ATS250.ROM (16K, &8000-&BFFF)
; =====
;
; Generated by dis.py from ats250.json + ats250.ann. Labels and comments are
; an analyst's reconstruction, not the BBC's source. See the Advanced Teletext
; System User Guide (1986) for the documented interface.
;
; WHAT THE ROM DOES
; A complete rewrite of Acorn's TFS 1.03 for the same Teletext Adapter:
; - a "page grabber" in the interrupt handler with 20 records, each naming a
;   page (magazine, page, sub-code) and a buffer to store it in
; - OSWORD &7A calls 0-27 (numbered, not &80+ as in TFS)
; - * commands: TELETXT TTXON/OFF HON HOFF MAGAZINE BBC CH ITV S4C NOTTX
;   TUNE TIME DATE PAGE TRANSFER DATA DISPLAY TELESOFT (any command may
;   take extra leading T's: *TTIME)
; - a terminal with Fastext-style coloured links, kept pages, previous page,
;   sub-page entry, status page, live subtitles and CRC colouring
; - a Telesoftware terminal that reads the broadcast catalogue and downloads
;   files to disc, network or tape, block by block
; - support for the Master 128 (hidden-RAM workspace, service calls &22-&27)
; It is not a filing system: Telesoftware is downloaded with OSWORD 24.
;
; ADAPTER REGISTERS (as for TFS 1.03)
; &FC10 r: b7 INT, b6 DOR, b4 FSYN, b3-b0 links      w: b0-1 channel,
;           b2 capture, b3 IRQ, b4 AFC
; &FC11 w: adapter RAM slot   &FC12 r/w: data   &FC13 w: STATCLR
;
; MEMORY
;   private workspace: one page (see init_workspace for the layout); on a
;                           Master it is in hidden RAM
;   terminal:  &4A00 page buffer, &4F00-&6D00 link buffers, kept pages from
;               OSHWM, &7000-&79FF Telesoftware, &7A00-&7B5F workspace
;   the page grabber stores pages in I/O processor memory only
;
; CONVENTIONS
;   &xx hexadecimal; Lxxxx an address with no better name.
;
; =====
; SYMBOLS (RAM, zero page and I/O addresses named in this listing)
; =====
; &00A8 wk
; &00A9 wk+1
; &00EF osw_a
; &00F0 osw_x
; &00F1 osw_y
; &00F2 text_ptr
; &00F3 text_ptr+1
; &00F4 romsel_copy
; &00FD brk_ptr
; &00FE brk_ptr+1
; &00FF escape_flag
; &0100 stack_page
; &0200 USERV
; &0202 BRKV
; &0203 BRKV+1
; &0212 FILEV
; &0213 FILEV+1
; &021E FSCV
; &0406 TUBE_CODE
; &0DF0 rom_wksp_table
; &4A00 page_buf
; &4A02 page_buf+2
; &4A03 page_buf+3
; &4A04 page_buf+4
; &7097 dl_rowptr
; &7098 dl_rowptr+1
; &7099 dl_col
; &709A dl_blockchar
; &709B dl_to_file
; &709C dl_save_each
; &709D dl_mag
; &709E dl_page
; &709F dl_sub_hi
; &70A0 dl_sub_lo
; &70AB dl_blocks_rx
; &70AC dl_hiwater
; &70AD dl_hiwater+1
; &70B2 dl_keep_tables
; &70B3 dl_ordered
; &70B4 dl_load
; &70B8 dl_exec
; &70BC dl_handle
; &70BD dl_out
; &70BE dl_out+1
```



```

; &70DA dl_name
; &70EE dl_skip
; &70EF dl_nblocks
; &70F1 dl_next_block
; &70F2 dl_block
; &70F9 dl_txt_mode
; &70FA dl_adjust
; &70FD dl_kind
; &70FE dl_value
; &70FF dl_raw
; &7100 dl_block_map
; &7A00 t_osw
; &7A01 t_osw+1
; &7A02 t_osw+2
; &7A03 t_osw+3
; &7A04 t_osw+4
; &7A05 t_osw+5
; &7A06 t_osw+6
; &7A07 t_osw+7
; &7A08 t_osw+8
; &7A09 t_osw+9
; &7A12 t_osw2
; &7A13 t_osw2+1
; &7A14 t_osw2+2
; &7A15 t_osw2+3
; &7A25 t_ndigits
; &7A26 t_digits
; &7A27 t_digits+1
; &7A28 t_digits+2
; &7A29 t_digits+3
; &7A2A t_entry_len
; &7A2B t_hold
; &7A2C t_line_len
; &7A2D t_line
; &7A4B crc_rx_hi
; &7A4C crc_rx_lo
; &7A4D crc_lo
; &7A4E crc_hi
; &7A53 link_ctrl
; &7A55 t_prev_state
; &7A56 t_dbl_flag
; &7A57 t_saved_sp
; &7A58 t_saved_stack
; &7B58 t_kept_base
; &7B59 t_kept_base+1
; &7B5C t_nrecords
; &7B5D inline_count
; &7B5E t_hdr_colour
; &FC10 TT_STATUS_CTRL
; &FC11 TT_SLOT
; &FC12 TT_DATA
; &FC13 TT_STATCLR
; &FEE4 TUBE_R3_STATUS
; &FEE5 TUBE_R3_DATA
; &FFB9 OSRDRM
; &FFCE OSFIND
; &FFD1 OSGBPB
; &FFD4 OSBPUT
; &FFD7 OSBGET
; &FFDA OSARGS
; &FFDD OSFILE
; &FFE0 OSRDCH
; &FFE3 OSASCI
; &FFE7 OSNEWL
; &FFEE OSWRCH
; &FFF1 OSWORD
; &FFF4 OSBYTE
; &FFF7 OSCLI
;
;-----
; SIDEWAYS ROM HEADER
; &8000 JMP &9810      language entry (terminal mode)
; &8003 JMP &8024      service entry
; &8006 &C2           service + language ROM, 6502 code
; &8007 &17           copyright offset
; &8008 &02           version
; &8009 "TELETEXT",0,"2.50",CR,0,"(C)BBC 1986",0
;-----
.L8000
8000 4C 10 98 JMP terminal_entry
.L8003
8003 4C 24 80 JMP service_entry
;-----
; ROM type, copyright offset, version, title "TELETEXT", version "2.50", copyright.
;-----
8006 .db C2 17 02 54 45 4C 45 54 45 58 54 00 32 2E 35 30 ; ...TELETEXT.2.50
8016 .db 00 00 28 43 29 42 42 43 20 31 39 38 36 00 ; ..(C)BBC 1986.
;-----
; PAGED ROM SERVICE ENTRY (A = service call, X = ROM number, Y = parameter)
; Unlike TFS 1.03 there is no dispatch table: each call is tested in turn.

```

```

; &FF      Tube initialisation: hide the language flag (see below)
; 4        unrecognised * command
; 9        *HELP
; 8        OSWORD (&7A)
; 5        interrupt: the page grabber
; 2        private workspace (model B / B+)
; &22, &24 private workspace in hidden RAM (Master 128)
; &27      reset (Master)
; After *NOTTX (bit 7 of the workspace byte set) only service 3 and &FF do anything.
; There is no auto-boot handling: ATS is not a filing system.
;-----
.service_entry
8024 08      PHP                      ; P and A are saved; every exit restores them (L81DF)
8025 48      PHA
8026 C9 FF   CMP #&FF                  ; Tube initialisation?
8028 D0 2C   BNE L8056

.L802A
802A 98      TYA                      ; clear bit 6 (language) of this ROM's entry in the MOS ROM type table, so the MOS
802B 48      PHA
802C 8A      TXA                      ; never copies the ROM across the Tube or enters it at BREAK; the terminal then runs
802D 48      PHA
802E A5 84   LDA &84                  ; as an OS command instead (see the guide: "does not start up as a language")
8030 48      PHA
8031 A5 85   LDA &85
8033 48      PHA
8034 A9 AA   LDA #&AA
8036 A2 00   LDX #&00
8038 A0 FF   LDY #&FF
803A 20 F4 FF JSR OSBYTE              ; OSBYTE &AA
803D 86 84   STX &84
803F 84 85   STY &85
8041 A4 F4   LDY romsel_copy
8043 B1 84   LDA (&84),Y
8045 29 BF   AND #&BF
8047 91 84   STA (&84),Y
8049 68      PLA
804A 85 85   STA &85
804C 68      PLA
804D 85 84   STA &84
804F 68      PLA
8050 AA      TAX
8051 68      PLA
8052 A8      TAY
8053 4C DF 81 JMP L81DF

.L8056
8056 BD F0 0D LDA rom_wksp_table,X    ; ROM switched off by *NOTTX?
8059 10 09   BPL L8064
805B 68      PLA
805C 48      PHA
805D C9 03   CMP #&03
805F F0 C9   BEQ L802A
8061 4C DF 81 JMP L81DF

.L8064
8064 68      PLA
8065 48      PHA
8066 C9 04   CMP #&04                  ; service 4
8068 D0 03   BNE L806D
806A 4C 82 81 JMP star_command

.L806D
806D C9 09   CMP #&09                  ; service 9
806F D0 03   BNE L8074
8071 4C E2 81 JMP help

.L8074
8074 C9 08   CMP #&08                  ; service 8
8076 D0 03   BNE L807B
8078 4C 74 82 JMP osword

.L807B
807B C9 05   CMP #&05                  ; service 5
807D D0 03   BNE L8082
807F 4C 8F 8B JMP irq

.L8082
8082 C9 02   CMP #&02                  ; service 2: only claim main-memory workspace on OS 1.2 / B+ (OSBYTE 0 < 3)
8084 D0 30   BNE L80B6
8086 98      TYA
8087 48      PHA
8088 8A      TXA
8089 48      PHA
808A A9 00   LDA #&00
808C A2 01   LDX #&01
808E A0 00   LDY #&00
8090 20 F4 FF JSR OSBYTE              ; OSBYTE &00
8093 E0 03   CPX #&03
8095 B0 18   BCS L80AF
8097 68      PLA
8098 AA      TAX
8099 68      PLA
809A 48      PHA
809B 9D F0 0D STA rom_wksp_table,X
809E 20 F4 80 JSR init_or_disable    ; initialise the workspace
80A1 68      PLA

```

```

80A2 A8      TAY
80A3 A6 F4   LDX romsel_copy
80A5 BD F0 0D LDA rom_wksp_table,X      ; claim one page (Y+1) if the ROM is enabled
80A8 F0 02   BEQ L80AC
80AA A8      TAY
80AB C8      INY
.L80AC
80AC 4C DF 81 JMP L81DF
.L80AF
80AF 68      PLA
80B0 AA      TAX
80B1 68      PLA
80B2 A8      TAY
80B3 4C DF 81 JMP L81DF
.L80B6
80B6 C9 27   CMP #&27                  ; service &27 (reset)
80B8 D0 0E   BNE L80C8
80BA 98      TYA
80BB 48      PHA
80BC 8A      TXA
80BD 48      PHA
80BE 20 F4 80 JSR init_or_disable
80C1 68      PLA
80C2 AA      TAX
80C3 68      PLA
80C4 A8      TAY
80C5 4C DF 81 JMP L81DF
.L80C8
80C8 C9 24   CMP #&24                  ; service &24: one page of hidden workspace wanted
80CA D0 04   BNE L80D0
80CC 88      DEY
80CD 4C DF 81 JMP L81DF
.L80D0
80D0 C9 22   CMP #&22                  ; service &22: claim hidden workspace
80D2 D0 1D   BNE L80F1
80D4 98      TYA
80D5 9D F0 0D STA rom_wksp_table,X
80D8 48      PHA
80D9 20 F4 80 JSR init_or_disable
80DC 68      PLA
80DD A8      TAY
80DE A6 F4   LDX romsel_copy
80E0 BD F0 0D LDA rom_wksp_table,X
80E3 F0 09   BEQ L80EE
80E5 29 7F   AND #&7F
80E7 9D F0 0D STA rom_wksp_table,X
80EA 09 80   ORA #&80
80EC A8      TAY
80ED C8      INY
.L80EE
80EE 4C DF 81 JMP L81DF
.L80F1
80F1 4C DF 81 JMP L81DF
;-----
; Check the adaptor (adaptor_status). If it is off or missing, switch the ROM off for
; this session: no workspace (the &0DF0 byte is cleared) and no language flag, so
; PAGE is not raised. Otherwise initialise the private workspace.
;-----
.init_or_disable
80F4 20 38 90 JSR adaptor_status      ; 0 on, 1 no signal, 2 off, &FF none
80F7 C9 02   CMP #&02
80F9 90 2D   BCC init_workspace
80FB A5 84   LDA &84
80FD 48      PHA
80FE A5 85   LDA &85
8100 48      PHA
8101 A9 AA   LDA #&AA                  ; clear the language bit as at &802A
8103 A2 00   LDX #&00
8105 A0 FF   LDY #&FF
8107 20 F4 FF JSR OSBYTE              ; OSBYTE &AA
810A 86 84   STX &84
810C 84 85   STY &85
810E A4 F4   LDY romsel_copy
8110 B1 84   LDA (&84),Y
8112 29 BF   AND #&BF
8114 91 84   STA (&84),Y
8116 68      PLA
8117 85 85   STA &85
8119 68      PLA
811A 85 84   STA &84
811C A6 F4   LDX romsel_copy
811E BD F0 0D LDA rom_wksp_table,X      ; workspace byte → 0: nothing claimed
8121 A8      TAY
8122 A9 00   LDA #&00
8124 9D F0 0D STA rom_wksp_table,X
8127 60      RTS
;-----
; INITIALISE THE PRIVATE WORKSPACE PAGE (pointer in &84/&85):
; +&00-&9F 20 page grabber records of 8 bytes: magazine, page, sub-code hi, lo,
;          buffer address lo, hi (+2 spare) - all &FF: not in use

```

```
; +&A0-&B3 20 record flags: b0 HOLD, b1 RECEIVED, b2 BEING UPDATED, b3/b5 row 27 seen
; +&B4      rows to capture per page          +&B5/&B6 record being grabbed / its magazine
; +&B7      header status (0 off, 1 clock, &FF no clock) +&B8 header magazine (&FF all)
; +&B9/&BA  write pointer for the next row      +&BD      control latch shadow
; +&BE      adapter RAM slot being read        +&BF-&D2 saved zero page &70-&83
; +&D4-&E0  TSDP (initial page, network id, offset, MJD, UTC)
; +&E7      subtitles (bit 7 set = off)        +&EE/&EF subtitle magazine / page
; +&E8      rows captured so far              +&E9      interrupt mode (0 off, &FF on, &FE tune)
; +&EA-&ED  error / byte counters for *TUNE    +&F0      magazines seen with page &FF headers
; +&F1      row length: &2A (42, ATS format) or &28 (40, TFS compatible)
; +&F2      Last double-height row written
; On a Master (OS 3+) bit 7 is put back into the page number: the workspace is in the
; hidden RAM at &C000-&DFFF.
```

.init_workspace

```
8128 A5 84 LDA &84
812A 48 PHA
812B A5 85 LDA &85
812D 48 PHA
812E A6 F4 LDX romsel_copy
8130 BD F0 0D LDA rom_wksp_table,X
8133 85 85 STA &85
8135 A9 00 LDA #&00
8137 85 84 STA &84
8139 A9 00 LDA #&00
813B 20 B7 A0 JSR L80B7
813E E0 03 CPX #&03
8140 90 06 BCC L8148
8142 A5 85 LDA &85
8144 09 80 ORA #&80
8146 85 85 STA &85
```

.L8148

```
8148 A0 00 LDY #&00
814A A9 FF LDA #&FF
```

.L814C

```
814C 91 84 STA (&84),Y
814E C8 INY
814F D0 FB BNE L814C
8151 A0 A0 LDY #&A0
8153 A9 00 LDA #&00
```

.L8155

```
8155 91 84 STA (&84),Y
8157 C8 INY
8158 C0 B4 CPY #&B4
815A D0 F9 BNE L8155
815C A9 00 LDA #&00
815E 91 84 STA (&84),Y
8160 A0 B7 LDY #&B7 ; workspace +&B7: header status
8162 91 84 STA (&84),Y
8164 A0 F0 LDY #&F0 ; workspace +&F0: magazines seen
8166 91 84 STA (&84),Y
8168 A0 E9 LDY #&E9 ; workspace +&E9: interrupt mode
816A 91 84 STA (&84),Y
816C A0 F1 LDY #&F1
816E A9 2A LDA #&2A
8170 91 84 STA (&84),Y
8172 A0 BD LDY #&BD
8174 A9 10 LDA #&10
8176 91 84 STA (&84),Y
8178 8D 10 FC STA TT_STATUS_CTRL ; control latch &10: AFC on, capture and IRQ off
817B 68 PLA
817C 85 85 STA &85
817E 68 PLA
817F 85 84 STA &84
8181 60 RTS
```

```
; SERVICE 4: * COMMANDS. Any number of leading "T"s are skipped, so *TTIME = *TIME and
; *TELETEXT is stored as "ELETEXT". Abbreviations with "." are accepted (the "." AND &DF
; is &0E). Table at &84BD: name, 0, address.
```

.star_command

```
8182 98 TYA
8183 48 PHA
8184 8A TXA
8185 48 PHA
8186 A6 F4 LDX romsel_copy
8188 BD F0 0D LDA rom_wksp_table,X
818B F0 4E BEQ L81DB
818D A2 00 LDX #&00
```

.L818F

```
818F B1 F2 LDA (text_ptr),Y ; skip leading T's
8191 29 DF AND #&DF
8193 C9 54 CMP #&54 ; 'T'
8195 D0 03 BNE L819A
8197 C8 INY
8198 D0 F5 BNE L818F
```

.L819A

```
819A 84 A8 STY wk
```

.L819C

```
819C B1 F2 LDA (text_ptr),Y
```

```

819E 29 DF AND #&DF
81A0 DD BD 84 CMP command_table,X ; compare with the table
81A3 D0 16 BNE L81BB
81A5 C8 INY
81A6 E8 INX
81A7 BD BD 84 LDA command_table,X
81AA D0 F0 BNE L819C
.L81AC
81AC E8 INX
81AD BD BD 84 LDA command_table,X
81B0 85 A9 STA wk+1
81B2 E8 INX
81B3 BD BD 84 LDA command_table,X
81B6 85 AA STA &AA
81B8 6C A9 00 JMP (wk+1)
.L81BB
81BB C9 0E CMP #&0E ; "." abbreviation
81BD D0 09 BNE L81C8
81BF C8 INY
.L81C0
81C0 E8 INX
81C1 BD BD 84 LDA command_table,X
81C4 D0 FA BNE L81C0
81C6 F0 E4 BEQ L81AC
.L81C8
81C8 E8 INX
81C9 BD BD 84 LDA command_table,X
81CC D0 FA BNE L81C8
81CE E8 INX
81CF E8 INX
81D0 E8 INX
81D1 BD BD 84 LDA command_table,X
81D4 30 05 BMI L81DB ; end of table (bit 7)
81D6 A4 A8 LDY wk
81D8 4C 9C 81 JMP L819C
; Service exit: restore X, Y, A and P (not claimed).
.L81DB
81DB 68 PLA
81DC AA TAX
81DD 68 PLA
81DE A8 TAY
.L81DF
81DF 68 PLA
81E0 28 PLP
81E1 60 RTS
;-----
; SERVICE 9: *HELP with no keyword: "TELETEXT 2.50" and the adaptor state:
; "No Power", "No Signal", "No TTX Unit", or the channel name in brackets.
;-----
.help
81E2 B1 F2 LDA (text_ptr),Y
81E4 C9 00 CMP #&00
81E6 F0 03 BEQ L81EB
81E8 4C DF 81 JMP L81DF
.L81EB
81EB 98 TYA
81EC 48 PHA
81ED 8A TXA
81EE 48 PHA
81EF 20 E7 FF JSR OSNEWL
81F2 A2 00 LDX #&00
.L81F4
81F4 BD 09 80 LDA &8009,X
81F7 D0 02 BNE L81FB
81F9 A9 20 LDA #&20
.L81FB
81FB C9 00 CMP #&00
81FD F0 06 BEQ L8205
81FF 20 E3 FF JSR OSASCI
8202 E8 INX
8203 D0 EF BNE L81F4
.L8205
8205 20 38 90 JSR adaptor_status
8208 F0 33 BEQ L823D
820A 30 22 BMI L822E
820C C9 02 CMP #&02
820E D0 0F BNE L821F
8210 A2 00 LDX #&00
.L8212
8212 BD 55 82 LDA &8255,X
8215 20 EE FF JSR OSWRCH
8218 E8 INX
8219 E0 09 CPX #&09
821B D0 F5 BNE L8212
821D F0 30 BEQ L824F
.L821F
821F A2 00 LDX #&00
.L8221
8221 BD 5E 82 LDA &825E,X
8224 20 EE FF JSR OSWRCH

```

```

8227 E8      INX
8228 E0 0A    CPX #&0A
822A D0 F5    BNE L8221
822C F0 0F    BEQ L823D
.L822E
822E A2 00    LDX #&00
.L8230
8230 BD 68 82 LDA &8268,X
8233 20 EE FF JSR OSWRCH
8236 E8      INX
8237 E0 0C    CPX #&0C
8239 D0 F5    BNE L8230
823B F0 12    BEQ L824F
.L823D
823D A9 20    LDA #&20
823F 20 EE FF JSR OSWRCH          ; print ' '
8242 A9 28    LDA #&28
8244 20 EE FF JSR OSWRCH          ; print '('
8247 20 D0 9D JSR L9D0D0
824A A9 29    LDA #&29
824C 20 EE FF JSR OSWRCH          ; print ')'
.L824F
824F 20 E7 FF JSR OSNEWL
8252 4C DB 81 JMP L81DB
8255 .db 20 4E 6F 20 50 6F 77 65 72 20 4E 6F 20 53 69 67 ; No Power No Sig
8265 .db 6E 61 6C 20 4E 6F 20 54 54 58 20 55 6E 69 74 ; na! No TTX Unit
;-----
; SERVICE 8: OSWORD &7A. Codes 0-27 dispatch through the table at &82EA.
; Code 17 (adaptor status) works even when the ROM is switched off.
; wk/wk+1 (&A8/&A9) point at the private workspace for the handlers.
;-----
.osword
8274 A5 EF    LDA osw_a
8276 C9 7A    CMP #&7A          ; 'z'
8278 F0 03    BEQ L827D
827A 4C DF 81 JMP L81DF
.L827D
827D 98      TYA
827E 48      PHA
827F 8A      TXA
8280 48      PHA
8281 A0 00    LDY #&00
8283 B1 F0    LDA (osw_x),Y
8285 C9 11    CMP #&11          ; 17 = read adaptor status
8287 F0 0A    BEQ L8293
8289 A6 F4    LDX romsel_copy
828B BD F0 0D LDA rom_wksp_table,X
828E D0 03    BNE L8293
.L8290
8290 4C DB 81 JMP L81DB
.L8293
8293 A0 00    LDY #&00
8295 B1 F0    LDA (osw_x),Y
8297 C9 1C    CMP #&1C          ; codes 28 and up are not claimed
8299 B0 F5    BCS L8290
829B A9 00    LDA #&00
829D 85 A8    STA wk
829F A6 F4    LDX romsel_copy
82A1 BD F0 0D LDA rom_wksp_table,X
82A4 29 7F    AND #&7F
82A6 85 A9    STA wk+1
82A8 A5 EF    LDA osw_a
82AA 48      PHA
82AB A5 F0    LDA osw_x
82AD 48      PHA
82AE A5 F1    LDA osw_y
82B0 48      PHA
82B1 A9 00    LDA #&00          ; Master: workspace in hidden RAM
82B3 A2 01    LDX #&01
82B5 A0 00    LDY #&00
82B7 20 F4 FF JSR OSBYTE          ; OSBYTE &00
82BA E0 03    CPX #&03
82BC 90 06    BCC L82C4
82BE A5 A9    LDA wk+1
82C0 09 80    ORA #&80
82C2 85 A9    STA wk+1
.L82C4
82C4 68      PLA
82C5 85 F1    STA osw_y
82C7 68      PLA
82C8 85 F0    STA osw_x
82CA 68      PLA
82CB 85 EF    STA osw_a
82CD A0 00    LDY #&00
82CF B1 F0    LDA (osw_x),Y
82D1 0A      ASL A
82D2 A8      TAY
82D3 B9 EA 82 LDA osword_table,Y
82D6 85 AA    STA &AA
82D8 C8      INY

```

```

82D9 B9 EA 82 LDA osword_table,Y
82DC 85 AB STA &AB
82DE 6C AA 00 JMP (&00AA) ; jump to the handler
; OSWORD exit: claim the call.
.L82E1
82E1 68 PLA
82E2 AA TAX
82E3 68 PLA
82E4 A8 TAY
82E5 68 PLA
82E6 A9 00 LDA #&00
82E8 28 PLP
82E9 60 RTS
;-----
; OSWORD &7A dispatch table, codes 0-27 (26 and 27 are undocumented).
;-----
osword_table
82EA .db 38 83 22 83 45 83 55 83 68 83 79 83 B1 83 B6 83 ; 8.".E.U.h.y.....
82FA .db C0 83 C5 83 DA 83 E5 83 EF 83 03 84 19 84 4E 84 ; .....N.
830A .db 6B 84 79 84 8B 84 A2 84 AF 92 4D 94 46 95 DF 96 ; k.y.....M.F...
831A .db B6 97 F7 97 AC 84 B3 84 ; .....
;-----
; OSWORD 1: SET CHANNEL [1, channel 0-3]
; Capture off, select the channel (latch bits 0-1), then clear all grabber records.
;-----
.L8322
8322 A9 10 LDA #&10
8324 A0 BD LDY #&BD ; workspace +&BD: latch shadow
8326 91 A8 STA (wk),Y
8328 8D 10 FC STA TT_STATUS_CTRL
832B A0 01 LDY #&01
832D 11 F0 ORA (osw_x),Y
832F A0 BD LDY #&BD
8331 09 0C ORA #&0C ; capture + IRQ enable bits
8333 91 A8 STA (wk),Y
8335 4C 19 84 JMP L8419
; OSWORD 0: READ CHANNEL
.L8338
8338 A0 BD LDY #&BD ; workspace +&BD: latch shadow
833A B1 A8 LDA (wk),Y
833C 29 03 AND #&03
833E A0 01 LDY #&01
8340 91 F0 STA (osw_x),Y
8342 4C E1 82 JMP L82E1
;-----
; OSWORD 2: READ RECORD FLAG [2, record] → flag, record being grabbed
;-----
.L8345
8345 20 B8 84 JSR L84B8
8348 18 CLC
8349 69 A0 ADC #&A0
834B A8 TAY
834C B1 A8 LDA (wk),Y
834E A0 02 LDY #&02
8350 91 F0 STA (osw_x),Y
8352 4C E1 82 JMP L82E1
;-----
; OSWORD 3: SET RECORD FLAG [3, record, flag] (HOLD, or clear RECEIVED)
;-----
.L8355
8355 20 B8 84 JSR L84B8
8358 18 CLC
8359 69 A0 ADC #&A0
835B AA TAX
835C C8 INY
835D B1 F0 LDA (osw_x),Y
835F 48 PHA
8360 8A TXA
8361 A8 TAY
8362 68 PLA
8363 91 A8 STA (wk),Y
8365 4C E1 82 JMP L82E1
;-----
; OSWORD 4: READ GRABBER RECORD [4, record] → magazine, page, sub hi, sub lo, address
;-----
.L8368
8368 20 94 83 JSR L8394
836B A0 02 LDY #&02
.L836D
836D B1 AA LDA (&AA),Y
836F 91 F0 STA (osw_x),Y
8371 C8 INY
8372 C0 0A CPY #&0A
8374 D0 F7 BNE L836D
8376 4C E1 82 JMP L82E1
;-----
; OSWORD 5: SET GRABBER RECORD [5, record, magazine, page, sub hi, sub lo, address]
; Interrupts and capture are stopped while the record is changed, the adapter RAM is
; cleared, then interrupts come back on (L8403).
;-----

```

```

.L8379
8379 78 SEI
837A A9 10 LDA #&10
837C 8D 10 FC STA TT_STATUS_CTRL
837F 20 94 83 JSR L8394
8382 A0 02 LDY #&02

.L8384
8384 B1 F0 LDA (osw_x),Y
8386 91 AA STA (&AA),Y
8388 C8 INY
8389 C0 0A CPY #&0A
838B D0 F7 BNE L8384
838D 20 E2 90 JSR L90E2
8390 58 CLI
8391 4C 03 84 JMP L8403

;-----
; &AA/&AB = workspace + record*8 - 2 (so that block offsets 2-9 line up with the record).
;-----

.L8394
8394 20 B8 84 JSR L84B8
8397 0A ASL A
8398 0A ASL A
8399 0A ASL A
839A 18 CLC
839B 65 A8 ADC wk
839D 85 AA STA &AA
839F A5 A9 LDA wk+1
83A1 85 AB STA &AB
83A3 38 SEC
83A4 A5 AA LDA &AA
83A6 E9 02 SBC #&02
83A8 85 AA STA &AA
83AA A5 AB LDA &AB
83AC E9 00 SBC #&00
83AE 85 AB STA &AB
83B0 60 RTS
; OSWORD 6: READ ROWS PER PAGE

.L83B1
83B1 A0 B4 LDY #&B4
83B3 4C DC 83 JMP L83DC

;-----
; OSWORD 7: SET ROWS PER PAGE (24 = displayable, 30 with links and Telesoftware rows)
;-----

.L83B6
83B6 20 B8 84 JSR L84B8
83B9 A0 B4 LDY #&B4 ; workspace +&B4: rows per page
83BB 91 A8 STA (wk),Y
83BD 4C E1 82 JMP L82E1
; OSWORD 8: READ HEADER STATUS

.L83C0
83C0 A0 B7 LDY #&B7
83C2 4C DC 83 JMP L83DC

;-----
; OSWORD 9: SET HEADER STATUS (0 off, 1 with clock, &FF without). Turns capture on.
;-----

.L83C5
83C5 20 B8 84 JSR L84B8
83C8 A0 B7 LDY #&B7 ; workspace +&B7: header status
83CA 91 A8 STA (wk),Y
83CC A0 BD LDY #&BD ; workspace +&BD: latch shadow
83CE B1 A8 LDA (wk),Y
83D0 09 1C ORA #&1C
83D2 91 A8 STA (wk),Y
83D4 8D 10 FC STA TT_STATUS_CTRL
83D7 4C E1 82 JMP L82E1
; OSWORD 10: READ HEADER MAGAZINE

.L83DA
83DA A0 B8 LDY #&B8 ; workspace +&B8: header magazine

.L83DC
83DC B1 A8 LDA (wk),Y
83DE A0 01 LDY #&01
83E0 91 F0 STA (osw_x),Y
83E2 4C E1 82 JMP L82E1
; OSWORD 11: SET HEADER MAGAZINE (&FF = all)

.L83E5
83E5 20 B8 84 JSR L84B8
83E8 A0 B8 LDY #&B8 ; workspace +&B8: header magazine
83EA 91 A8 STA (wk),Y
83EC 4C E1 82 JMP L82E1

;-----
; OSWORD 12: TELETXT INTERRUPTS OFF (*TTXOFF): latch bits 2-3 off, mode 0.
;-----

.L83EF
83EF A0 BD LDY #&BD ; workspace +&BD: latch shadow
83F1 B1 A8 LDA (wk),Y
83F3 29 F3 AND #&F3
83F5 91 A8 STA (wk),Y
83F7 8D 10 FC STA TT_STATUS_CTRL
83FA A0 E9 LDY #&E9
83FC A9 00 LDA #&00

```



```

83FE 91 A8 STA (wk),Y
8400 4C E1 82 JMP L82E1
;-----
; OSWORD 13: TELETXT INTERRUPTS ON (*TTXON): mode &FF, latch bits 2-3 on.
;-----
.L8403
8403 78 SEI
8404 A0 E9 LDY #&E9
8406 A9 FF LDA #&FF
8408 91 A8 STA (wk),Y
840A A0 BD LDY #&BD ; workspace +&BD: latch shadow
840C B1 A8 LDA (wk),Y
840E 09 0C ORA #&0C
8410 91 A8 STA (wk),Y
8412 8D 10 FC STA TT_STATUS_CTRL
8415 58 CLI
8416 4C E1 82 JMP L82E1
;-----
; OSWORD 14: CLEAR ALL GRABBER RECORDS (and flags, TSDP and magazine bitmap).
;-----
.L8419
8419 A9 10 LDA #&10
841B 8D 10 FC STA TT_STATUS_CTRL
841E A0 00 LDY #&00
8420 A9 FF LDA #&FF
.L8422
8422 91 A8 STA (wk),Y
8424 C8 INY
8425 C0 A0 CPY #&A0
8427 D0 F9 BNE L8422
8429 A9 00 LDA #&00
.L842B
842B 91 A8 STA (wk),Y
842D C8 INY
842E C0 B4 CPY #&B4
8430 D0 F9 BNE L842B
8432 C8 INY
8433 A9 FF LDA #&FF
8435 91 A8 STA (wk),Y
8437 C8 INY
8438 91 A8 STA (wk),Y
843A A0 D4 LDY #&D4 ; workspace +&D4: TSDP initial magazine
843C 91 A8 STA (wk),Y
843E A9 00 LDA #&00
8440 A0 F0 LDY #&F0 ; workspace +&F0: magazines seen
8442 91 A8 STA (wk),Y
8444 A0 BD LDY #&BD ; workspace +&BD: latch shadow
8446 B1 A8 LDA (wk),Y
8448 8D 10 FC STA TT_STATUS_CTRL
844B 4C E1 82 JMP L82E1
;-----
; OSWORD 15: READ TSDP: 13 bytes from workspace +&D4 to the block.
;-----
.L844E
844E A9 01 LDA #&01
8450 85 AA STA &AA
8452 A9 D4 LDA #&D4
8454 85 AB STA &AB
.L8456
8456 A4 AB LDY &AB
8458 B1 A8 LDA (wk),Y
845A A4 AA LDY &AA
845C 91 F0 STA (osw_x),Y
845E E6 AA INC &AA
8460 E6 AB INC &AB
8462 A5 AA LDA &AA
8464 C9 0E CMP #&0E
8466 D0 EE BNE L8456
8468 4C E1 82 JMP L82E1
; OSWORD 16: STOP THE GRABBER: no record is being grabbed.
.L846B
846B 78 SEI
846C A0 B5 LDY #&B5
846E A9 FF LDA #&FF
8470 91 A8 STA (wk),Y
8472 C8 INY
8473 91 A8 STA (wk),Y
8475 58 CLI
8476 4C E1 82 JMP L82E1
;-----
; OSWORD 17: READ ADAPTOR STATUS (0 ok, 1 no signal, 2 off, &FF none) and the
; interrupt mode byte.
;-----
.L8479
8479 20 38 90 JSR adaptor_status
847C A0 01 LDY #&01
847E 91 F0 STA (osw_x),Y
8480 A0 E9 LDY #&E9 ; workspace +&E9: interrupt mode
8482 B1 A8 LDA (wk),Y
8484 A0 02 LDY #&02

```

```

8486 91 F0 STA (osw_x),Y
8488 4C E1 82 JMP L82E1
; OSWORD 18: SUBTITLES [18, enable, magazine, page]
.L848B
848B 20 B8 84 JSR L84B8
848E A0 E7 LDY #E7 ; workspace +E7: subtitle enable
8490 91 A8 STA (wk),Y
8492 A0 02 LDY #02
8494 B1 F0 LDA (osw_x),Y
8496 48 PHA
8497 C8 INY
8498 B1 F0 LDA (osw_x),Y
849A A0 EF LDY #EF ; workspace +EF: subtitle page
849C 91 A8 STA (wk),Y
849E 68 PLA
849F 88 DEY
84A0 D0 05 BNE L84A7
; OSWORD 19: INTERRUPT HANDLING FLAG (stored in the mode byte)
.L84A2
84A2 20 B8 84 JSR L84B8
84A5 A0 E9 LDY #E9 ; workspace +E9: interrupt mode
.L84A7
84A7 91 A8 STA (wk),Y
84A9 4C E1 82 JMP L82E1
;-----
; OSWORD 26 (undocumented): SET ROW LENGTH: &2A ATS format, &28 TFS format.
;-----
.L84AC
84AC 20 B8 84 JSR L84B8
84AF A0 F1 LDY #F1
84B1 D0 F4 BNE L84A7
; OSWORD 27 (undocumented): READ ROW LENGTH.
.L84B3
84B3 A0 F1 LDY #F1
84B5 4C DC 83 JMP L83DC
; A = byte 1 of the OSWORD block.
.L84B8
84B8 A0 01 LDY #01
84BA B1 F0 LDA (osw_x),Y
84BC 60 RTS
;-----
; * command table: name (leading T removed), 0, address; ends with a byte ≥ &80.
;-----
.command_table
84BD .db 45 4C 45 54 45 58 54 00 42 85 48 4F 4E 00 92 85 ; ELETXT.B.HON...
84CD .db 48 4F 46 46 00 F4 85 42 42 43 00 45 86 49 54 56 ; HOFF...BBC.E.ITV
84DD .db 00 73 86 43 48 00 45 86 53 14 43 00 8B 86 4E 4F ; .s.CH.E.S.C...NO
84ED .db 54 54 58 00 68 85 58 00 98 86 55 4E 45 00 F4 86 ; TTX.h.X...UNE...
84FD .db 49 4D 45 00 14 87 44 41 54 45 00 5B 87 4D 41 47 ; IME...DATE.[.MAG
850D .db 41 5A 49 4E 45 00 0B 86 50 41 47 45 00 95 87 52 ; AZINE...PAGE...R
851D .db 41 4E 53 46 45 52 00 3E 88 44 41 54 41 00 35 88 ; ANSFER.>.DATA.5.
852D .db 44 49 53 50 4C 41 59 00 96 89 45 4C 45 53 4F 46 ; DISPLAY...ELESOF
853D .db 54 00 F0 86 FF ; T....
;-----
; *TELETEXT: TFS-format off (row length 42), then start the terminal: as a language
; when there is no Tube, otherwise directly as an OS command (L981A).
;-----
.L8542
8542 20 29 8B JSR skip_spaces
8545 C9 0D CMP #0D
8547 D0 1C BNE L8565
8549 A9 1A LDA #1A
854B 8D 00 01 STA stack_page
854E A9 2A LDA #2A
8550 8D 01 01 STA &0101
8553 20 20 8B JSR own_osword
8556 20 8A 89 JSR L898A
8559 F0 03 BEQ L855E
855B 4C 1A 98 JMP L981A
.L855E
855E A6 F4 LDX romsel_copy
8560 A9 8E LDA #8E
8562 4C F4 FF JMP OSBYTE ; OSBYTE &8E: enter language ROM X
; Not ours: pass the command on.
.L8565
8565 4C DB 81 JMP L81DB
;-----
; *NOTTX: mark the ROM switched off (bit 7 of its workspace byte), print
; "Press CTRL-BREAK" and hang with interrupts off until the user does.
;-----
.L8568
8568 20 29 8B JSR skip_spaces
856B C9 0D CMP #0D
856D D0 F6 BNE L8565
856F A6 F4 LDX romsel_copy
8571 A9 80 LDA #80
8573 9D F0 0D STA rom_wksp_table,X
8576 A9 13 LDA #13
8578 78 SEI
8579 20 1A BB JSR print_inline_count

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```

857C .str '\r\n\nPress CTRL-BREAK'
.L858F
858F 4C 8F 85 JMP L858F ; wait for CTRL-BREAK
;-----
; *H0N [0|1]: interrupts on, header with clock (0) or without (1), all magazines.
;-----
.L8592
8592 20 29 8B JSR skip_spaces
8595 C9 0D CMP #0D
8597 F0 13 BEQ L85AC
8599 C9 30 CMP #30 ; '0'
859B F0 07 BEQ L85A4
859D C9 31 CMP #31 ; '1'
859F F0 33 BEQ L85D4
.L85A1
85A1 4C DB 81 JMP L81DB
.L85A4
85A4 C8 INY
85A5 20 29 8B JSR skip_spaces
85A8 C9 0D CMP #0D
85AA D0 F5 BNE L85A1
.L85AC
85AC 20 E7 85 JSR L85E7
85AF A9 01 LDA #01
85B1 8D 01 01 STA &0101
.L85B4
85B4 A9 09 LDA #09
85B6 8D 00 01 STA stack_page
85B9 20 20 8B JSR own_osword
85BC A9 0B LDA #0B
85BE 8D 00 01 STA stack_page
85C1 A9 FF LDA #FF
85C3 8D 01 01 STA &0101
85C6 20 20 8B JSR own_osword
85C9 A9 10 LDA #10
85CB 8D 00 01 STA stack_page
85CE 20 20 8B JSR own_osword
85D1 4C 33 8B JMP L8B33
.L85D4
85D4 C8 INY
85D5 20 29 8B JSR skip_spaces
85D8 C9 0D CMP #0D
85DA D0 C5 BNE L85A1
85DC 20 E7 85 JSR L85E7
85DF A9 FF LDA #FF
85E1 8D 01 01 STA &0101
85E4 4C B4 85 JMP L85B4
; OSWORD 13: teletext interrupts on.
.L85E7
85E7 98 TYA
85E8 48 PHA
85E9 A9 0D LDA #0D
85EB 8D 00 01 STA stack_page
85EE 20 20 8B JSR own_osword
85F1 68 PLA
85F2 A8 TAY
85F3 60 RTS
; *H0FF
.L85F4
85F4 20 29 8B JSR skip_spaces
85F7 C9 0D CMP #0D
85F9 D0 A6 BNE L85A1
85FB A9 09 LDA #09
85FD 8D 00 01 STA stack_page
8600 A9 00 LDA #00
8602 8D 01 01 STA &0101
8605 20 20 8B JSR own_osword
8608 4C 33 8B JMP L8B33
; *MAGAZINE n (1-8; 8 is sent as 0)
.L860B
860B 20 29 8B JSR skip_spaces
860E C9 0D CMP #0D
8610 D0 03 BNE L8615
8612 A9 31 LDA #31
8614 88 DEY
.L8615
8615 C9 31 CMP #31 ; '1'
8617 90 29 BCC L8642
8619 C9 39 CMP #39 ; '9'
861B B0 25 BCS L8642
861D 38 SEC
861E E9 30 SBC #30
8620 8D 01 01 STA &0101
8623 C9 08 CMP #08
8625 D0 05 BNE L862C
8627 A9 00 LDA #00
8629 8D 01 01 STA &0101
.L862C
862C C8 INY
862D 20 29 8B JSR skip_spaces

```

```

8630 C9 0D    CMP #0D
8632 D0 0E    BNE L8642
8634 A9 0B    LDA #0B
8636 8D 00 01  STA stack_page
8639 20 20 8B  JSR own_oword
863C 20 E7 85  JSR L85E7
863F 4C 33 8B  JMP L8B33
.L8642
8642 4C DB 81  JMP L81DB
; *BBC n / *CH n (1-4, default 1)
.L8645
8645 20 29 8B  JSR skip_spaces
8648 C9 0D    CMP #0D
864A D0 03    BNE L864F
864C A9 31    LDA #31
864E 88      DEY
.L864F
864F C9 31    CMP #31          ; '1'
8651 90 EF    BCC L8642
8653 C9 35    CMP #35          ; '5'
8655 B0 EB    BCS L8642
.L8657
8657 38      SEC
8658 E9 31    SBC #31
865A 8D 01 01 STA &0101
865D C8      INY
865E 20 29 8B  JSR skip_spaces
8661 C9 0D    CMP #0D
8663 D0 DD    BNE L8642
8665 20 E7 85  JSR L85E7
8668 A9 01    LDA #01
866A 8D 00 01  STA stack_page
866D 20 20 8B  JSR own_oword
8670 4C 33 8B  JMP L8B33
; *ITV n (1-2 → channels 3-4)
.L8673
8673 20 29 8B  JSR skip_spaces
8676 C9 0D    CMP #0D
8678 D0 03    BNE L867D
867A A9 31    LDA #31
867C 88      DEY
.L867D
867D C9 31    CMP #31          ; '1'
867F 90 C1    BCC L8642
8681 C9 33    CMP #33          ; '3'
8683 B0 BD    BCS L8642
8685 18      CLC
8686 69 02    ADC #02
8688 4C 57 86  JMP L8657
; *S4C: channel 4
.L8688
8688 20 29 8B  JSR skip_spaces
868E C9 0D    CMP #0D
8690 D0 B0    BNE L8642
8692 A9 34    LDA #34
8694 88      DEY
8695 4C 57 86  JMP L8657
; *TTXON / *TTXOFF
.L8698
8698 20 29 8B  JSR skip_spaces
869B 29 DF    AND #DF
869D C9 4F    CMP #4F          ; '0'
869F D0 0D    BNE L86AE
86A1 C8      INY
86A2 B1 F2    LDA (text_ptr),Y
86A4 29 DF    AND #DF
86A6 C9 4E    CMP #4E          ; 'N'
86A8 F0 07    BEQ L86B1
86AA C9 46    CMP #46          ; 'F'
86AC F0 1E    BEQ L86CC
.L86AE
86AE 4C DB 81  JMP L81DB
.L86B1
86B1 C8      INY
86B2 20 29 8B  JSR skip_spaces
86B5 C9 0D    CMP #0D
86B7 D0 F5    BNE L86AE
86B9 A9 2A    LDA #2A          ; ON: ATS row format
.L86BB
86BB 8D 01 01  STA &0101
86BE A9 1A    LDA #1A
86C0 8D 00 01  STA stack_page
86C3 20 20 8B  JSR own_oword
86C6 20 E7 85  JSR L85E7
86C9 4C 33 8B  JMP L8B33
.L86CC
86CC C8      INY
86CD B1 F2    LDA (text_ptr),Y
86CF 29 DF    AND #DF
86D1 C9 46    CMP #46          ; 'F'

```

```

86D3 D0 D9 BNE L86AE
86D5 C8 INY
86D6 20 29 8B JSR skip_spaces
86D9 C9 0D CMP #&0D
86DB D0 D1 BNE L86AE
86DD A9 0C LDA #&0C ; OFF: interrupts off, records cleared
86DF 8D 00 01 STA stack_page
86E2 20 20 8B JSR own_osword
86E5 A9 0E LDA #&0E
86E7 8D 00 01 STA stack_page
86EA 20 20 8B JSR own_osword
86ED 4C 33 8B JMP L8B33
;
; *TELESOFT: as *TTXON but with 40-byte rows (TFS-compatible *TRANSFER/*DISPLAY).
;
.L86F0
86F0 A9 28 LDA #&28
86F2 D0 C7 BNE L86BB
; *TUNE: OSWORD 20, then clear the screen.
.L86F4
86F4 20 29 8B JSR skip_spaces
86F7 C9 0D CMP #&0D
86F9 D0 16 BNE L8711
86FB 20 E7 85 JSR L85E7
86FE A9 14 LDA #&14
8700 8D 00 01 STA stack_page
8703 20 20 8B JSR own_osword
8706 A9 0C LDA #&0C
8708 20 EE FF JSR OSWRCH ; VDU 12: CLS
870B 20 26 A8 JSR L8826
870E 4C 33 8B JMP L8B33
.L8711
8711 4C DB 81 JMP L81DB
; *TIME: OSWORD 21; "No TSDP" if there is no service packet.
.L8714
8714 20 29 8B JSR skip_spaces
8717 C9 0D CMP #&0D
8719 D0 33 BNE L874E
871B A9 15 LDA #&15
871D 8D 00 01 STA stack_page
8720 20 20 8B JSR own_osword
8723 AD 0A 01 LDA &010A
8726 30 13 BMI L873B
8728 A0 00 LDY #&00
.L872A
872A B9 01 01 LDA &0101,Y
872D 20 EE FF JSR OSWRCH
8730 C8 INY
8731 C0 08 CPY #&08
8733 D0 F5 BNE L872A
8735 20 E7 FF JSR OSNEWL
8738 4C 33 8B JMP L8B33
.L873B
873B A0 00 LDY #&00
.L873D
873D B9 51 87 LDA &8751,Y
8740 20 EE FF JSR OSWRCH
8743 C8 INY
8744 C0 08 CPY #&08
8746 D0 F5 BNE L873D
8748 20 E7 FF JSR OSNEWL
874B 4C 33 8B JMP L8B33
.L874E
874E 4C DB 81 JMP L81DB
8751 .db 4E 6F 20 54 53 44 50 20 20 20 ; No TSDP
; *DATE: OSWORD 22.
.L875B
875B 20 29 8B JSR skip_spaces
875E C9 0D CMP #&0D
8760 D0 EC BNE L874E
8762 A9 16 LDA #&16
8764 8D 00 01 STA stack_page
8767 20 20 8B JSR own_osword
876A AD 0C 01 LDA &010C
876D 30 13 BMI L8782
876F A0 00 LDY #&00
.L8771
8771 B9 01 01 LDA &0101,Y
8774 20 EE FF JSR OSWRCH
8777 C8 INY
8778 C0 0A CPY #&0A
877A D0 F5 BNE L8771
877C 20 E7 FF JSR OSNEWL
877F 4C 33 8B JMP L8B33
.L8782
8782 A0 00 LDY #&00
.L8784
8784 B9 51 87 LDA &8751,Y
8787 20 EE FF JSR OSWRCH
878A C8 INY

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878B C0 0A    CPY #0A
878D D0 F5    BNE L8784
878F 20 E7 FF JSR OSNEWL
8792 4C 33 8B JMP L8B33
;
; *PAGE mpp[S]ssss: hex digits. Without a sub-page (&3F7F) is accepted.
; Sets record 0 with no buffer address yet (&FFFF); *TRANSFER/*DATA give it one.
;
.L8795
8795 20 13 88 JSR L8813
8798 B0 5E    BCS L87F8
879A 29 07    AND #07
879C 8D 02 01 STA &0102
879F 20 FB 87 JSR L87FB
87A2 B0 54    BCS L87F8
87A4 8D 03 01 STA &0103
87A7 A9 3F    LDA #3F
87A9 8D 04 01 STA &0104
87AC A9 7F    LDA #7F
87AE 8D 05 01 STA &0105
87B1 A9 FF    LDA #FF
87B3 8D 06 01 STA &0106
87B6 8D 07 01 STA &0107
87B9 8D 08 01 STA &0108
87BC 8D 09 01 STA &0109
87BF 20 13 88 JSR L8813
87C2 90 11    BCC L87D5
87C4 08       PHP
87C5 C9 0D    CMP #0D
87C7 F0 1E    BEQ L87E7
87C9 29 DF    AND #DF
87CB C9 53    CMP #53           ; 'S'
87CD F0 04    BEQ L87D3
87CF 28       PLP
87D0 4C F8 87 JMP L87F8
.L87D3
87D3 C8       INY
87D4 28       PLP
.L87D5
87D5 88       DEY
87D6 20 FB 87 JSR L87FB
87D9 B0 1D    BCS L87F8
87DB 8D 04 01 STA &0104
87DE 20 FB 87 JSR L87FB
87E1 B0 15    BCS L87F8
87E3 8D 05 01 STA &0105
87E6 08       PHP
.L87E7
87E7 28       PLP
87E8 A9 05    LDA #05
87EA 8D 00 01 STA stack_page
87ED A9 00    LDA #00
87EF 8D 01 01 STA &0101
87F2 20 20 8B JSR own_oword
87F5 4C 33 8B JMP L8B33
.L87F8
87F8 4C DB 81 JMP L81DB
; Two hex digits → A. C set on error.
.L87FB
87FB 20 13 88 JSR L8813
87FE B0 11    BCS L8811
8800 0A       ASL A
8801 0A       ASL A
8802 0A       ASL A
8803 0A       ASL A
8804 8D 00 01 STA stack_page
8807 20 13 88 JSR L8813
880A B0 05    BCS L8811
880C 0D 00 01 ORA stack_page
880F 18       CLC
8810 60       RTS
.L8811
8811 38       SEC
8812 60       RTS
; One hex digit (0-9, A-F) → A. C set on error.
.L8813
8813 20 29 8B JSR skip_spaces
8816 C8       INY
8817 C9 30    CMP #30           ; '0'
8819 90 09    BCC L8824
881B C9 3A    CMP #3A           ; ':'
881D B0 07    BCS L8826
881F 38       SEC
8820 E9 30    SBC #30
8822 18       CLC
8823 60       RTS
.L8824
8824 38       SEC
8825 60       RTS
.L8826

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```

8826 29 DF    AND #&DF
8828 C9 41    CMP #&41          ; 'A'
882A 90 F8    BCC L8824
882C C9 47    CMP #&47          ; 'G'
882E B0 F4    BCS L8824
8830 38      SEC
8831 E9 37    SBC #&37
8833 18      CLC
8834 60      RTS
;-----
; *DATA addr: as *TRANSFER but 48 rows of 42 bytes (all rows, 2K).
;-----
.L8835
8835 98      TYA
8836 48      PHA
8837 20 E7 85 JSR L85E7
883A A9 30    LDA #&30
883C D0 07    BNE L8845
;-----
; *TRANSFER addr: 24 rows into memory, waiting for the page (ESCAPE aborts with
; "Escape"). With a second processor and an address not &FFFFxxx the page is grabbed
; at OSHWM in the I/O processor and then copied across the Tube, 256 bytes at a time.
;-----
.L883E
883E 98      TYA
883F 48      PHA
8840 20 E7 85 JSR L85E7
8843 A9 18    LDA #&18
.L8845
8845 8D 01 01 STA &0101
8848 A9 07    LDA #&07
884A 8D 00 01 STA stack_page
884D 20 20 8B JSR own_osword
8850 A9 04    LDA #&04
8852 8D 00 01 STA stack_page
8855 A9 00    LDA #&00
8857 8D 01 01 STA &0101
885A 20 20 8B JSR own_osword
885D 68      PLA
885E A8      TAY
885F 20 D9 8A JSR L8AD9
8862 AD 06 01 LDA &0106
8865 0D 07 01 ORA &0107
8868 D0 03    BNE L886D
886A 4C DB 81 JMP L81DB
.L886D
886D 20 8A 89 JSR L898A
8870 F0 25    BEQ L8897
8872 AD 06 01 LDA &0106
8875 8D 0A 01 STA &010A
8878 AD 07 01 LDA &0107
887B 8D 0B 01 STA &010B
887E AD 08 01 LDA &0108
8881 C9 FF    CMP #&FF
8883 D0 07    BNE L888C
8885 AD 09 01 LDA &0109
8888 C9 FF    CMP #&FF
888A F0 0B    BEQ L8897
.L888C
888C A9 83    LDA #&83
888E 20 F4 FF JSR OSBYTE          ; OSBYTE &83: read OSHWM (PAGE)
8891 8E 06 01 STX &0106
8894 8C 07 01 STY &0107
.L8897
8897 A9 05    LDA #&05
8899 8D 00 01 STA stack_page
889C 20 20 8B JSR own_osword
889F A9 03    LDA #&03
88A1 8D 00 01 STA stack_page
88A4 A9 00    LDA #&00
88A6 8D 02 01 STA &0102
88A9 20 20 8B JSR own_osword
.L88AC
88AC A9 02    LDA #&02          ; wait for RECEIVED
88AE 8D 00 01 STA stack_page
88B1 20 20 8B JSR own_osword
88B4 AD 02 01 LDA &0102
88B7 29 02    AND #&02
88B9 D0 1C    BNE L88D7
88BB A5 FF    LDA escape_flag
88BD 29 80    AND #&80
88BF D0 03    BNE L88C4
88C1 4C AC 88 JMP L88AC
.L88C4
88C4 20 5F 89 JSR L895F          ; ESCAPE: stop grabbing and raise the error
88C7 A0 00    LDY #&00
.L88C9
88C9 B9 81 89 LDA &8981,Y
88CC 99 00 01 STA stack_page,Y
88CF C8      INY

```

```

88D0 C0 09 CPY #&09
88D2 D0 F5 BNE L88C9
88D4 4C 00 01 JMP stack_page
.L88D7
88D7 20 8A 89 JSR L898A
88DA F0 70 BEQ L894C
88DC AD 08 01 LDA &0108
88DF C9 FF CMP #&FF
88E1 D0 07 BNE L88EA
88E3 AD 09 01 LDA &0109
88E6 C9 FF CMP #&FF
88E8 F0 62 BEQ L894C
.L88EA
88EA A5 77 LDA &77 ; copy to the second processor
88EC 48 PHA
88ED A5 78 LDA &78
88EF 48 PHA
88F0 AD 06 01 LDA &0106
88F3 85 77 STA &77
88F5 AD 07 01 LDA &0107
88F8 85 78 STA &78
88FA AD 0A 01 LDA &010A
88FD 8D 06 01 STA &0106
8900 AD 0B 01 LDA &010B
8903 8D 07 01 STA &0107
8906 20 3C 8B JSR L8B3C
8909 A9 06 LDA #&06
890B 8D 00 01 STA stack_page
890E 20 20 8B JSR own_osword
8911 AD 01 01 LDA &0101
8914 C9 18 CMP #&18
8916 F0 04 BEQ L891C
8918 A2 08 LDX #&08
891A D0 02 BNE L891E
.L891C
891C A2 04 LDX #&04
.L891E
891E 8A TXA
891F 48 PHA
8920 20 4F 8B JSR L8B4F
8923 18 CLC
8924 AD 07 01 LDA &0107
8927 69 01 ADC #&01
8929 8D 07 01 STA &0107
892C AD 08 01 LDA &0108
892F 69 00 ADC #&00
8931 8D 08 01 STA &0108
8934 AD 09 01 LDA &0109
8937 69 00 ADC #&00
8939 8D 09 01 STA &0109
893C E6 78 INC &78
893E 68 PLA
893F AA TAX
8940 CA DEX
8941 D0 DB BNE L891E
8943 68 PLA
8944 85 78 STA &78
8946 68 PLA
8947 85 77 STA &77
8949 20 46 8B JSR L8B46
.L894C
894C 20 5F 89 JSR L895F
894F 4C 33 8B JMP L8B33
.L8952
8952 4C DB 81 JMP L81DB
.L8955
8955 A9 FF LDA #&FF
8957 8D 02 01 STA &0102
895A 8D 03 01 STA &0103
895D D0 00 BNE L896C
;-----
; Stop grabbing into the user's buffer: record 0 gets address &FFFF.
;-----
.L895F
895F A9 04 LDA #&04
8961 8D 00 01 STA stack_page
8964 A9 00 LDA #&00
8966 8D 01 01 STA &0101
8969 20 20 8B JSR own_osword
.L896C
896C A9 FF LDA #&FF
896E 8D 06 01 STA &0106
8971 8D 07 01 STA &0107
8974 A9 05 LDA #&05
8976 8D 00 01 STA stack_page
8979 A9 00 LDA #&00
897B 8D 01 01 STA &0101
897E 4C 20 8B JMP own_osword
8981 .db 00 1B 45 73 63 61 70 65 00 ; ..Escape.
; Tube present? (OSBYTE &EA) Z set if not.

```


.L898A

```
898A A9 EA LDA #&EA
898C A2 00 LDX #&00
898E A0 FF LDY #&FF
8990 20 F4 FF JSR OSBYTE
8993 E0 00 CPX #&00
8995 60 RTS
```

```
;
; *DISPLAY addr: show a page stored by *TRANSFER on the MODE 7 screen. ATS format
; (42-byte rows found by row number, double height handled, missing rows blank) or,
; after *TELESOFT, the 40-byte TFS format copied straight to the screen.
```

.L8996

```
8996 20 D9 8A JSR L8AD9
8999 AD 06 01 LDA &0106
899C 0D 07 01 ORA &0107
899F F0 B1 BEQ L8952
89A1 20 8A 89 JSR L898A
89A4 F0 5D BEQ L8A03
89A6 AD 08 01 LDA &0108
89A9 C9 FF CMP #&FF
89AB D0 07 BNE L89B4
89AD AD 09 01 LDA &0109
89B0 C9 FF CMP #&FF
89B2 F0 4F BEQ L8A03
```

.L89B4

```
89B4 20 3C 8B JSR L8B3C
89B7 A5 77 LDA &77
89B9 48 PHA
89BA A5 78 LDA &78
89BC 48 PHA
89BD A9 83 LDA #&83
89BF 20 F4 FF JSR OSBYTE ; OSBYTE &83: read OSHWM (PAGE)
89C2 86 77 STX &77
89C4 84 78 STY &78
89C6 A2 00 LDX #&00
```

.L89C8

```
89C8 8A TXA
89C9 48 PHA
89CA 20 6F 8B JSR L8B6F
89CD 68 PLA
89CE AA TAX
89CF 18 CLC
89D0 AD 07 01 LDA &0107
89D3 69 01 ADC #&01
89D5 8D 07 01 STA &0107
89D8 AD 08 01 LDA &0108
89DB 69 00 ADC #&00
89DD 8D 08 01 STA &0108
89E0 AD 09 01 LDA &0109
89E3 69 00 ADC #&00
89E5 8D 09 01 STA &0109
89E8 E6 78 INC &78
89EA E8 INX
89EB E0 04 CPX #&04
89ED D0 D9 BNE L89C8
89EF 20 46 8B JSR L8B46
89F2 68 PLA
89F3 85 78 STA &78
89F5 68 PLA
89F6 85 77 STA &77
89F8 A9 83 LDA #&83
89FA 20 F4 FF JSR OSBYTE ; OSBYTE &83: read OSHWM (PAGE)
89FD 8E 06 01 STX &0106
8A00 8C 07 01 STY &0107
```

.L8A03

```
8A03 A5 77 LDA &77
8A05 48 PHA
8A06 A5 78 LDA &78
8A08 48 PHA
8A09 AD 06 01 LDA &0106
8A0C 85 77 STA &77
8A0E AD 07 01 LDA &0107
8A11 85 78 STA &78
8A13 A9 16 LDA #&16
8A15 20 EE FF JSR OSWRCH ; VDU 22: MODE
8A18 A9 07 LDA #&07
8A1A 20 EE FF JSR OSWRCH ; VDU 7: BEL
8A1D A9 1B LDA #&1B ; read the row length
8A1F 8D 00 01 STA stack_page
8A22 20 20 8B JSR own_osword
8A25 AD 01 01 LDA &0101
8A28 C9 28 CMP #&28 ; '('
8A2A D0 03 BNE L8A2F
8A2C 4C B7 8A JMP L8AB7
```

.L8A2F

```
8A2F A2 08 LDX #&08
8A31 A9 A0 LDA #&A0
```

.L8A33

```
8A33 20 EE FF JSR OSWRCH ; VDU 160
```

```

8A36 CA      DEX
8A37 D0 FA   BNE L8A33
8A39 A9 00   LDA #&00
8A3B 8D 09 01 STA &0109
8A3E 8D 08 01 STA &0108
8A41 A0 0A   LDY #&0A      ; workspace +&0A: grabber records (8 bytes each)
.L8A43
8A43 B1 77   LDA (&77),Y
8A45 09 80   ORA #&80
8A47 C9 8D   CMP #&8D
8A49 D0 03   BNE L8A4E
8A4B 8D 09 01 STA &0109
.L8A4E
8A4E 20 EE FF JSR 0SWRCH
8A51 C8      INY
8A52 C0 2A   CPY #&2A      ; '*'
8A54 D0 ED   BNE L8A43
8A56 AD 09 01 LDA &0109
8A59 F0 16   BEQ L8A71
8A5B A0 02   LDY #&02      ; workspace +&02: grabber records (8 bytes each)
.L8A5D
8A5D B1 77   LDA (&77),Y
8A5F 09 80   ORA #&80
8A61 20 EE FF JSR 0SWRCH
8A64 C8      INY
8A65 C0 2A   CPY #&2A      ; '*'
8A67 D0 F4   BNE L8A5D
8A69 A9 00   LDA #&00
8A6B 8D 09 01 STA &0109
8A6E EE 08 01 INC &0108
.L8A71
8A71 18      CLC
8A72 A5 77   LDA &77
8A74 69 2A   ADC #&2A
8A76 85 77   STA &77
8A78 A5 78   LDA &78
8A7A 69 00   ADC #&00
8A7C 85 78   STA &78
.L8A7E
8A7E EE 08 01 INC &0108
8A81 AD 08 01 LDA &0108
8A84 C9 18   CMP #&18
8A86 B0 44   BCS L8A8C
.L8A88
8A88 A0 01   LDY #&01      ; workspace +&01: grabber records (8 bytes each)
8A8A B1 77   LDA (&77),Y
8A8C CD 08 01 CMP &0108
8A8F F0 21   BEQ L8AB2
8A91 B0 10   BCS L8AA3
8A93 18      CLC
8A94 A5 77   LDA &77
8A96 69 2A   ADC #&2A
8A98 85 77   STA &77
8A9A A5 78   LDA &78
8A9C 69 00   ADC #&00
8A9E 85 78   STA &78
8AA0 4C 88 8A JMP L8A88
.L8AA3
8AA3 A0 00   LDY #&00
8AA5 A9 A0   LDA #&A0
.L8AA7
8AA7 20 EE FF JSR 0SWRCH      ; VDU 160
8AAA C8      INY
8AAB C0 28   CPY #&28      ; '('
8AAD D0 F8   BNE L8AA7
8AAF 4C 7E 8A JMP L8A7E
.L8AB2
8AB2 A0 02   LDY #&02
8AB4 4C 43 8A JMP L8A43
.L8AB7
8AB7 A2 03   LDX #&03
.L8AB9
8AB9 A0 00   LDY #&00      ; workspace +&00: grabber records (8 bytes each)
.L8ABB
8ABB B1 77   LDA (&77),Y
8ABD 20 EE FF JSR 0SWRCH
8AC0 C8      INY
8AC1 98      TYA
8AC2 DD D5 8A CMP &8AD5,X
8AC5 D0 F4   BNE L8ABB
8AC7 E6 78   INC &78
8AC9 CA      DEX
8ACA 10 ED   BPL L8AB9
.L8ACC
8ACC 68      PLA
8ACD 85 78   STA &78
8ACF 68      PLA
8AD0 85 77   STA &77
8AD2 4C 33 8B JMP L8B33
8AD5 .db C0 00 00 00      ; ....

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;-----
; Parse a hexadecimal address (up to 8 digits) into &0106-&0109.
;-----
.L8AD9
8AD9 A9 00 LDA #&00
8ADB 8D 06 01 STA &0106
8ADE 8D 07 01 STA &0107
8AE1 8D 08 01 STA &0108
8AE4 8D 09 01 STA &0109
.L8AE7
8AE7 20 29 8B JSR skip_spaces
8AEA C8 INY
8AEB C9 0D CMP #&0D
8AED F0 30 BEQ L8B1F
8AEF A2 00 LDX #&00
.L8AF1
8AF1 DD C0 B7 CMP &B7C0,X
8AF4 F0 0A BEQ L8B00
8AF6 E8 INX
8AF7 E0 10 CPX #&10
8AF9 D0 F6 BNE L8AF1
.L8AFB
8AFB 68 PLA
8AFC 68 PLA
8AFD 4C DB 81 JMP L81DB
.L8B00
8B00 8A TXA
8B01 A2 00 LDX #&00
.L8B03
8B03 0E 06 01 ASL &0106
8B06 2E 07 01 ROL &0107
8B09 2E 08 01 ROL &0108
8B0C 2E 09 01 ROL &0109
8B0F B0 EA BCS L8AFB
8B11 E8 INX
8B12 E0 04 CPX #&04
8B14 D0 ED BNE L8B03
8B16 0D 06 01 ORA &0106
8B19 8D 06 01 STA &0106
8B1C 4C E7 8A JMP L8AE7
.L8B1F
8B1F 60 RTS
; OSWORD &7A with the block at &0100.
.own_osword
8B20 A2 00 LDX #&00
8B22 A0 01 LDY #&01
8B24 A9 7A LDA #&7A
8B26 4C F1 FF JMP OSWORD ; OSWORD &7A: Teletext system call
.skip_spaces
8B29 B1 F2 LDA (text_ptr),Y
8B2B C9 20 CMP #&20
8B2D D0 03 BNE L8B32
8B2F C8 INY
8B30 D0 F7 BNE skip_spaces
.L8B32
8B32 60 RTS
; Command exit: claim it.
.L8B33
8B33 68 PLA
8B34 AA TAX
8B35 68 PLA
8B36 A8 TAY
8B37 68 PLA
8B38 28 PLP
8B39 A9 00 LDA #&00
8B3B 60 RTS
; Claim the Tube.
.L8B3C
8B3C 48 PHA
.L8B3D
8B3D A9 C5 LDA #&C5
8B3F 20 06 04 JSR TUBE_CODE
8B42 90 F9 BCC L8B3D
8B44 68 PLA
8B45 60 RTS
; Release the Tube.
.L8B46
8B46 48 PHA
8B47 A9 85 LDA #&85
8B49 20 06 04 JSR TUBE_CODE
8B4C 68 PLA
8B4D 60 RTS
.L8B4E
8B4E 60 RTS
;-----
; Send 256 bytes from (&77) to the second processor (Tube transfer type 7).
;-----
.L8B4F
8B4F A2 06 LDX #&06
8B51 A0 01 LDY #&01

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8B53 A9 07 LDA #&07
8B55 20 06 04 JSR TUBE_CODE
8B58 20 4E 8B JSR L8B4E
8B5B 20 4E 8B JSR L8B4E
8B5E 20 4E 8B JSR L8B4E
8B61 A0 00 LDY #&00 ; workspace +&00: grabber records (8 bytes each)
.L8B63
8B63 B1 77 LDA (&77),Y
8B65 8D E5 FE STA TUBE_R3_DATA
8B68 EA NOP
8B69 EA NOP
8B6A EA NOP
8B6B C8 INY
8B6C D0 F5 BNE L8B63
8B6E 60 RTS
;-----
; Fetch 256 bytes from the second processor into (&77) (Tube transfer type 6).
;-----
.L8B6F
8B6F A2 06 LDX #&06
8B71 A0 01 LDY #&01
8B73 A9 06 LDA #&06
8B75 20 06 04 JSR TUBE_CODE
8B78 20 4E 8B JSR L8B4E
8B7B 20 4E 8B JSR L8B4E
8B7E 20 4E 8B JSR L8B4E
8B81 A0 00 LDY #&00
.L8B83
8B83 AD E5 FE LDA TUBE_R3_DATA
8B86 91 77 STA (&77),Y
8B88 EA NOP
8B89 EA NOP
8B8A EA NOP
8B8B C8 INY
8B8C D0 F5 BNE L8B83
8B8E 60 RTS
;-----
; SERVICE 5: INTERRUPT - THE PAGE GRABBER
; Only one page is grabbed at a time: the record whose header arrived most recently
; (+&B5) in its magazine (+&B6). Zero page &70-&83 is used as scratch, so the caller's
; values are saved in the workspace and put back at the end.
; For each full adapter slot:
; - decode magazine/row (L90F0)
; - in tuning mode (&FE) just count bytes and Hamming errors for the meter
; - row 0: decode the header (L9126) and
;   * show it as a rolling header at &7C08 (32 characters, or 24 when the clock
;   is replaced by HOLD), filtered by the header magazine and control bits
;   * on the subtitle page: subtitle mode, clearing the screen on "erase"
;   * finish the page being grabbed: flag RECEIVED, clear BEING UPDATED
;   * look for a record (not on HOLD) that wants this page and start grabbing it:
;   clear its buffer (rows of 42 zero bytes, or 40 spaces in TFS mode)
; - other rows of the page being grabbed: append as 42-byte records (magazine, row,
; 40 bytes), or in TFS mode write 40 display bytes to row*40 with double height
; - subtitle rows go straight onto the MODE 7 screen
; - magazine 8 row 30: the service data packet (L8FAC)
; Finally clear STATCLR, save zero page back and re-enable capture.
;-----
.irq
8B8F A5 FC LDA &FC
8B91 48 PHA
8B92 8A TXA
8B93 48 PHA
8B94 98 TYA
8B95 48 PHA
8B96 2C 10 FC BIT TT_STATUS_CTRL ; our adaptor interrupting?
8B99 10 03 BPL L8B9E
8B9B 4C A1 8B JMP L8BA1
.L8B9E
8B9E 4C A1 8F JMP L8FA1
.L8BA1
8BA1 A5 84 LDA &84
8BA3 48 PHA
8BA4 A5 85 LDA &85
8BA6 48 PHA
8BA7 A9 00 LDA #&00
8BA9 85 84 STA &84
8BAB A6 F4 LDX romsel_copy
8BAD BD F0 0D LDA rom_wksp_table,X
8BB0 85 85 STA &85
8BB2 A5 EF LDA osw_a
8BB4 48 PHA
8BB5 A5 F0 LDA osw_x
8BB7 48 PHA
8BB8 A5 F1 LDA osw_y
8BBA 48 PHA
8BBB 20 94 91 JSR L9194
8BBE 90 06 BCC L8BC6
8BC0 A5 85 LDA &85
8BC2 09 80 ORA #&80
8BC4 85 85 STA &85

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.L8BC6
8BC6 68      PLA
8BC7 85 F1    STA osw_y
8BC9 68      PLA
8BCA 85 F0    STA osw_x
8BCC 68      PLA
8BCD 85 EF    STA osw_a
8BCF A0 BD    LDY #&BD          ; interrupts enabled in the latch?
8BD1 B1 84    LDA (&84),Y
8BD3 29 08    AND #&08
8BD5 F0 0C    BEQ L8BE3
8BD7 A0 E9    LDY #&E9          ; mode &FF (normal) or &FE (tuning)?
8BD9 B1 84    LDA (&84),Y
8BDB C9 FF    CMP #&FF
8BDD F0 0D    BEQ L8BEC
8BDF C9 FE    CMP #&FE
8BE1 F0 09    BEQ L8BEC

.L8BE3
8BE3 68      PLA
8BE4 85 85    STA &85
8BE6 68      PLA
8BE7 85 84    STA &84
8BE9 4C A1 8F JMP L8FA1

.L8BEC
8BEC 48      PHA
8BED 20 18 90 JSR L9018          ; capture off while the RAM is read
8BF0 58      CLI
8BF1 A0 BE    LDY #&BE          ; save zero page &70-&83
8BF3 A9 00    LDA #&00
8BF5 91 84    STA (&84),Y
8BF7 A2 00    LDX #&00

.L8BF9
8BF9 B5 70    LDA &70,X
8BFB C8      INY
8BFC 91 84    STA (&84),Y
8BFE E8      INX
8BFF E0 14    CPX #&14
8C01 D0 F6    BNE L8BF9
8C03 68      PLA
8C04 85 82    STA &82
8C06 A0 E7    LDY #&E7          ; workspace +&E7: subtitle enable
8C08 B1 84    LDA (&84),Y
8C0A 85 81    STA &81

.L8C0C
8C0C A0 BE    LDY #&BE          ; next adapter slot
8C0E B1 84    LDA (&84),Y
8C10 A8      TAY
8C11 8C 11 FC STY TT_SLOT
8C14 AD 12 FC LDA TT_DATA
8C17 D0 03    BNE L8C1C
8C19 4C 36 8F JMP L8F36

.L8C1C
8C1C A5 82    LDA &82          ; tuning mode?
8C1E C9 FE    CMP #&FE
8C20 F0 11    BEQ L8C33

.L8C22
8C22 20 F0 90 JSR decode_address
8C25 A5 70    LDA &70
8C27 10 03    BPL L8C2C
8C29 4C 43 8F JMP L8F43

.L8C2C
8C2C A5 71    LDA &71
8C2E F0 42    BEQ L8C72
8C30 4C DD 8D JMP L8DDD

.L8C33
8C33 A0 EC    LDY #&EC          ; tuning: count 2 bytes and their errors
8C35 18      CLC
8C36 B1 84    LDA (&84),Y
8C38 69 02    ADC #&02
8C3A 91 84    STA (&84),Y
8C3C C8      INY
8C3D B1 84    LDA (&84),Y
8C3F 69 00    ADC #&00
8C41 91 84    STA (&84),Y
8C43 AE 12 FC LDX TT_DATA
8C46 BD AF 91 LDA hamming_table,X
8C49 10 03    BPL L8C4E
8C4B 20 61 8C JSR L8C61

.L8C4E
8C4E AE 12 FC LDX TT_DATA
8C51 BD AF 91 LDA hamming_table,X
8C54 10 03    BPL L8C59
8C56 20 61 8C JSR L8C61

.L8C59
8C59 A0 BE    LDY #&BE          ; workspace +&BE: adapter slot
8C5B B1 84    LDA (&84),Y
8C5D A8      TAY
8C5E 4C 22 8C JMP L8C22

.L8C61
8C61 A0 EA    LDY #&EA

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8C63 18      CLC
8C64 B1 84   LDA (&84),Y
8C66 69 01   ADC #&01
8C68 91 84   STA (&84),Y
8C6A C8      INY
8C6B B1 84   LDA (&84),Y
8C6D 69 00   ADC #&00
8C6F 91 84   STA (&84),Y
8C71 60      RTS

.L8C72
8C72 20 26 91 JSR decode_header ; row 0: decode the header
8C75 A5 73   LDA &73
8C77 10 03   BPL L8C7C
8C79 4C 43 8F JMP L8F43

.L8C7C
8C7C A5 72   LDA &72 ; page &FF: time-filling header; note the magazine
8C7E C9 FF   CMP #&FF
8C80 D0 0E   BNE L8C90
8C82 A4 70   LDY &70
8C84 C8      INY
8C85 98      TYA
8C86 0A      ASL A
8C87 A0 F0   LDY #&F0 ; workspace +&F0: magazines seen
8C89 11 84   ORA (&84),Y
8C8B 91 84   STA (&84),Y
8C8D 4C 36 8F JMP L8F36

.L8C90
8C90 A5 81   LDA &81 ; subtitles enabled?
8C92 10 36   BPL L8CCA
8C94 A0 B7   LDY #&B7 ; workspace +&B7: header status
8C96 B1 84   LDA (&84),Y
8C98 F0 2D   BEQ L8CC7
8C9A 30 05   BMI L8CA1
8C9C A2 20   LDX #&20
8C9E 4C A3 8C JMP L8CA3

.L8CA1
8CA1 A2 18   LDX #&18

.L8CA3
8CA3 A5 76   LDA &76 ; header display: skip on C9/C10, other magazines unless serial
8CA5 29 0C   AND #&0C
8CA7 D0 1E   BNE L8CC7
8CA9 A5 76   LDA &76
8CAB 29 10   AND #&10
8CAD D0 0A   BNE L8CB9
8CAF A0 B8   LDY #&B8 ; workspace +&B8: header magazine
8CB1 B1 84   LDA (&84),Y
8CB3 30 04   BMI L8CB9
8CB5 C5 70   CMP &70
8CB7 D0 0E   BNE L8CC7

.L8CB9
8CB9 A0 00   LDY #&00 ; rolling header to the screen

.L8CBB
8CBB AD 12 FC LDA TT_DATA
8CBE 09 80   ORA #&80
8CC0 99 08 7C STA &7C08,Y
8CC3 C8      INY
8CC4 CA      DEX
8CC5 D0 F4   BNE L8CBB

.L8CC7
8CC7 4C F6 8C JMP L8CF6

.L8CCA
8CCA A5 70   LDA &70 ; subtitle page?
8CCC A0 EE   LDY #&EE ; workspace +&EE: subtitle magazine
8CCE D1 84   CMP (&84),Y
8CD0 D0 34   BNE L8D06
8CD2 A5 72   LDA &72
8CD4 C8      INY
8CD5 D1 84   CMP (&84),Y
8CD7 F0 09   BEQ L8CE2
8CD9 A9 FF   LDA #&FF
8CDB A0 B5   LDY #&B5 ; workspace +&B5: record being grabbed
8CDD 91 84   STA (&84),Y
8CDF 4C 06 8D JMP L8D06

.L8CE2
8CE2 A9 00   LDA #&00
8CE4 A0 B5   LDY #&B5 ; workspace +&B5: record being grabbed
8CE6 91 84   STA (&84),Y
8CE8 A5 75   LDA &75
8CEA 29 08   AND #&08
8CEC F0 18   BEQ L8D06
8CEE A9 0C   LDA #&0C
8CF0 20 EE FF JSR OSWRCH ; VDU 12: CLS
8CF3 4C 06 8D JMP L8D06

.L8CF6
8CF6 A0 B5   LDY #&B5 ; header ends the page being grabbed?
8CF8 B1 84   LDA (&84),Y
8CFA 30 33   BMI L8D2F
8CFC 0A      ASL A
8CFD 0A      ASL A
8CFE 0A      ASL A

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8CFF A8      TAY
8D00 B1 84   LDA (&84),Y
8D02 C5 70   CMP &70
8D04 F0 03   BEQ L8D09
.L8D06
8D06 4C 36 8F JMP L8F36
.L8D09
8D09 C8      INY
8D0A B1 84   LDA (&84),Y
8D0C C5 72   CMP &72
8D0E D0 06   BNE L8D16
8D10 A5 75   LDA &75
8D12 29 08   AND #&08
8D14 F0 F0   BEQ L8D06
.L8D16
8D16 A0 B5   LDY #&B5
8D18 B1 84   LDA (&84),Y
8D1A 18      CLC
8D1B 69 A0   ADC #&A0
8D1D A8      TAY
8D1E B1 84   LDA (&84),Y
8D20 29 FB   AND #&FB
8D22 09 02   ORA #&02
8D24 91 84   STA (&84),Y
8D26 A0 B5   LDY #&B5
8D28 A9 FF   LDA #&FF
8D2A 91 84   STA (&84),Y
8D2C C8      INY
8D2D 91 84   STA (&84),Y
.L8D2F
8D2F A5 72   LDA &72
8D31 C9 FF   CMP #&FF
8D33 F0 D1   BEQ L8D06
8D35 A2 A0   LDX #&A0
; search the 20 flags/records for this page
.L8D37
8D37 8A      TXA
8D38 A8      TAY
8D39 B1 84   LDA (&84),Y
8D3B 29 01   AND #&01
8D3D D0 29   BNE L8D68
8D3F 8A      TXA
8D40 38      SEC
8D41 E9 A0   SBC #&A0
8D43 0A      ASL A
8D44 0A      ASL A
8D45 0A      ASL A
8D46 A8      TAY
8D47 B1 84   LDA (&84),Y
8D49 C5 70   CMP &70
8D4B D0 1B   BNE L8D68
8D4D C8      INY
8D4E B1 84   LDA (&84),Y
8D50 C5 72   CMP &72
8D52 D0 14   BNE L8D68
8D54 C8      INY
8D55 B1 84   LDA (&84),Y
8D57 C9 3F   CMP #&3F
8D59 F0 18   BEQ L8D73
8D5B C5 73   CMP &73
8D5D D0 09   BNE L8D68
8D5F C8      INY
8D60 B1 84   LDA (&84),Y
8D62 C5 74   CMP &74
8D64 D0 02   BNE L8D68
8D66 F0 0C   BEQ L8D74
.L8D68
8D68 E8      INX
8D69 E0 B4   CPX #&B4
8D6B F0 03   BEQ L8D70
8D6D 4C 37 8D JMP L8D37
.L8D70
8D70 4C 36 8F JMP L8F36
.L8D73
8D73 C8      INY
.L8D74
8D74 C8      INY
8D75 B1 84   LDA (&84),Y
8D77 48      PHA
8D78 C8      INY
8D79 B1 84   LDA (&84),Y
8D7B A0 BA   LDY #&BA
8D7D 91 84   STA (&84),Y
8D7F 85 80   STA &80
8D81 88      DEY
8D82 68      PLA
8D83 91 84   STA (&84),Y
8D85 85 7F   STA &7F
8D87 A5 80   LDA &80
8D89 30 D0   BMI L8D68
8D8B A0 B5   LDY #&B5
; start grabbing this record

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```

8D8D 8A TXA
8D8E 38 SEC
8D8F E9 A0 SBC #&A0
8D91 91 84 STA (&84),Y
8D93 A5 70 LDA &70
8D95 C8 INY
8D96 91 84 STA (&84),Y
8D98 8A TXA
8D99 A8 TAY
8D9A B1 84 LDA (&84),Y
8D9C 29 DD AND #&DD
8D9E 09 04 ORA #&04
8DA0 91 84 STA (&84),Y
8DA2 A0 E8 LDY #&E8
8DA4 A9 00 LDA #&00
8DA6 91 84 STA (&84),Y
8DA8 A0 B4 LDY #&B4
8DAA B1 84 LDA (&84),Y
8DAC AA TAX
8DAD F0 25 BEQ L8DD4
.L8DAF
8DAF A0 F1 LDY #&F1
8DB1 B1 84 LDA (&84),Y
8DB3 C9 28 CMP #&28
8DB5 D0 04 BNE L8DBB
8DB7 A9 A0 LDA #&A0
8DB9 D0 02 BNE L8DBD
.L8DBB
8DBB A9 00 LDA #&00
.L8DBD
8DBD A0 29 LDY #&29
.L8DBF
8DBF 91 7F STA (&7F),Y
8DC1 88 DEY
8DC2 10 FB BPL L8DBF
8DC4 18 CLC
8DC5 A5 7F LDA &7F
8DC7 69 2A ADC #&2A
8DC9 85 7F STA &7F
8DCB A5 80 LDA &80
8DCD 69 00 ADC #&00
8DCF 85 80 STA &80
8DD1 CA DEX
8DD2 D0 DB BNE L8DAF
.L8DD4
8DD4 A0 F2 LDY #&F2
8DD6 A9 FF LDA #&FF
8DD8 91 84 STA (&84),Y
8DDA 20 E2 90 JSR L90E2
.L8DDD
8DDD A5 81 LDA &81
8DDF 30 11 BMI L8DF2
8DE1 A0 B5 LDY #&B5
8DE3 B1 84 LDA (&84),Y
8DE5 D0 89 BNE L8D70
8DE7 A0 EE LDY #&EE
8DE9 B1 84 LDA (&84),Y
8DEB C5 70 CMP &70
8DED D0 81 BNE L8D70
8DEF 4C D4 8E JMP L8ED4
.L8DF2
8DF2 A0 B5 LDY #&B5
8DF4 B1 84 LDA (&84),Y
8DF6 30 17 BMI L8E0F
8DF8 C8 INY
8DF9 B1 84 LDA (&84),Y
8DFB C5 70 CMP &70
8DFD D0 10 BNE L8E0F
8DFF A5 71 LDA &71
8E01 C9 18 CMP #&18
8E03 90 00 BCC L8E12
8E05 A0 B4 LDY #&B4
8E07 B1 84 LDA (&84),Y
8E09 C9 18 CMP #&18
8E0B F0 02 BEQ L8E0F
8E0D D0 03 BNE L8E12
.L8E0F
8E0F 4C 36 8F JMP L8F36
.L8E12
8E12 A0 E8 LDY #&E8
8E14 B1 84 LDA (&84),Y
8E16 A0 B4 LDY #&B4
8E18 D1 84 CMP (&84),Y
8E1A B0 F3 BCS L8E0F
8E1C 18 CLC
8E1D 69 01 ADC #&01
8E1F A0 E8 LDY #&E8
8E21 91 84 STA (&84),Y
8E23 A0 B9 LDY #&B9
8E25 B1 84 LDA (&84),Y

```

; workspace +&B4: rows per page

; clear the buffer

; '('

; subtitle row?

; workspace +&B5: record being grabbed

; workspace +&EE: subtitle magazine

; a row of the page being grabbed

; workspace +&B4: rows per page

; buffer full (rows per page)?

; workspace +&B4: rows per page

; workspace +&E8: rows captured

; workspace +&B9: row write pointer


```

8E27 85 7F STA &7F
8E29 C8 INY
8E2A B1 84 LDA (&84),Y
8E2C 85 80 STA &80
8E2E 30 DF BMI L8E0F
8E30 20 9C 91 JSR L919C
8E33 A0 F1 LDY #&F1 ; workspace +&F1: row length
8E35 B1 84 LDA (&84),Y
8E37 85 82 STA &82
8E39 C9 28 CMP #&28 ; TFS mode: display format
8E3B D0 16 BNE L8E53
8E3D 20 7E 8F JSR L8F7E
8E40 18 CLC
8E41 A5 7F LDA &7F
8E43 65 77 ADC &77
8E45 85 77 STA &77
8E47 A5 80 LDA &80
8E49 65 78 ADC &78
8E4B 85 78 STA &78
8E4D 20 ED 8E JSR L8EED
8E50 4C 36 8F JMP L8F36
.L8E53
8E53 A0 00 LDY #&00 ; ATS mode: magazine, row, data
8E55 A5 70 LDA &70
8E57 91 7F STA (&7F),Y
8E59 C8 INY
8E5A A5 71 LDA &71
8E5C 91 7F STA (&7F),Y
8E5E C8 INY
8E5F A5 71 LDA &71
8E61 D0 03 BNE L8E66
8E63 20 A7 8E JSR L8EA7
.L8E66
8E66 C9 18 CMP #&1B ; row 27: note links (designation 0) in the flags
8E68 D0 20 BNE L8E8A
8E6A AE 12 FC LDX TT_DATA
8E6D BD AF 91 LDA hamming_table,X
8E70 29 0F AND #&0F
8E72 D0 12 BNE L8E86
8E74 98 TYA
8E75 48 PHA
8E76 A0 B5 LDY #&B5 ; workspace +&B5: record being grabbed
8E78 B1 84 LDA (&84),Y
8E7A 18 CLC
8E7B 69 A0 ADC #&A0
8E7D A8 TAY
8E7E B1 84 LDA (&84),Y
8E80 09 28 ORA #&28
8E82 91 84 STA (&84),Y
8E84 68 PLA
8E85 A8 TAY
.L8E86
8E86 8A TXA
8E87 4C 8D 8E JMP L8E8D
.L8E8A
8E8A AD 12 FC LDA TT_DATA
.L8E8D
8E8D 91 7F STA (&7F),Y
8E8F C8 INY
8E90 C4 82 CPY &82
8E92 D0 F6 BNE L8E8A
8E94 A0 B9 LDY #&B9
8E96 18 CLC
8E97 A5 7F LDA &7F
8E99 65 82 ADC &82
8E9B 91 84 STA (&84),Y
8E9D C8 INY
8E9E A5 80 LDA &80
8EA0 69 00 ADC #&00
8EA2 91 84 STA (&84),Y
8EA4 4C 36 8F JMP L8F36
.L8EA7
8EA7 A2 08 LDX #&08 ; row 0: 8 header bytes replaced by page, sub-code, control bits, 3 zeros
.L8EA9
8EA9 AD 12 FC LDA TT_DATA
8EAC CA DEX
8EAD D0 FA BNE L8EA9
8EAF A5 72 LDA &72
8EB1 91 7F STA (&7F),Y
8EB3 C8 INY
8EB4 A5 73 LDA &73
8EB6 91 7F STA (&7F),Y
8EB8 C8 INY
8EB9 A5 74 LDA &74
8EBB 91 7F STA (&7F),Y
8EBD C8 INY
8EBE A5 75 LDA &75
8EC0 91 7F STA (&7F),Y
8EC2 C8 INY
8EC3 A5 76 LDA &76

```

```

8EC5 91 7F    STA (&7F),Y
8EC7 C8      INY
8EC8 A9 00    LDA #&00
8ECA 91 7F    STA (&7F),Y
8ECC C8      INY
8ECD 91 7F    STA (&7F),Y
8ECF C8      INY
8ED0 91 7F    STA (&7F),Y
8ED2 C8      INY
8ED3 60      RTS
;-----
; Subtitle row: written straight to the screen at &7C00 + row*40.
;-----
.L8ED4
8ED4 A5 71    LDA &71
8ED6 C9 18    CMP #&18
8ED8 90 03    BCC L8EDD
8EDA 4C 36 8F JMP L8F36
.L8EDD
8EDD 20 7E 8F JSR L8F7E
8EE0 18      CLC
8EE1 A5 78    LDA &78
8EE3 69 7C    ADC #&7C
8EE5 85 78    STA &78
8EE7 20 ED 8E JSR L8EED
8EEA 4C 36 8F JMP L8F36
;-----
; Write a display row (bit 7 set). If it contains double height (&8D), copy it to the
; next row as well and ignore the next transmitted row.
;-----
.L8EED
8EED A5 71    LDA &71
8EEF A0 F2    LDY #&F2                ; workspace +&F2: last double-height row
8EF1 D1 84    CMP (&84),Y
8EF3 F0 40    BEQ L8F35
8EF5 20 9C 91 JSR L919C
8EF8 A9 00    LDA #&00
8EFA 85 7B    STA &7B
8EFC A0 00    LDY #&00
.L8EFE
8EFE AD 12 FC LDA TT_DATA
8F01 09 80    ORA #&80
8F03 C9 8D    CMP #&8D
8F05 D0 02    BNE L8F09
8F07 E6 7B    INC &7B
.L8F09
8F09 91 77    STA (&77),Y
8F0B C8      INY
8F0C C0 28    CPY #&28                ; '('
8F0E D0 EE    BNE L8EFE
8F10 A5 7B    LDA &7B
8F12 F0 21    BEQ L8F35
8F14 A0 F2    LDY #&F2
8F16 A5 71    LDA &71
8F18 18      CLC
8F19 69 01    ADC #&01
8F1B 91 84    STA (&84),Y
8F1D 18      CLC
8F1E A5 77    LDA &77
8F20 69 28    ADC #&28
8F22 85 79    STA &79
8F24 A5 78    LDA &78
8F26 69 00    ADC #&00
8F28 85 7A    STA &7A
8F2A A0 00    LDY #&00                ; workspace +&00: grabber records (8 bytes each)
.L8F2C
8F2C B1 77    LDA (&77),Y
8F2E 91 79    STA (&79),Y
8F30 C8      INY
8F31 C0 28    CPY #&28                ; '('
8F33 D0 F7    BNE L8F2C
.L8F35
8F35 60      RTS
; Next slot; after 16, clear STATCLR and finish.
.L8F36
8F36 A5 70    LDA &70
8F38 D0 09    BNE L8F43
8F3A A5 71    LDA &71
8F3C C9 1E    CMP #&1E
8F3E D0 03    BNE L8F43
8F40 20 AC 8F JSR tsdp_capture
.L8F43
8F43 A0 BE    LDY #&BE                ; workspace +&BE: adapter slot
8F45 B1 84    LDA (&84),Y
8F47 18      CLC
8F48 69 01    ADC #&01
8F4A 91 84    STA (&84),Y
8F4C C9 10    CMP #&10
8F4E F0 03    BEQ L8F53
8F50 4C 0C 8C JMP L8C0C

```

```

.L8F53
8F53 A9 00 LDA #&00 ; STATCLR, slot 0
8F55 8D 13 FC STA TT_STATCLR
8F58 8D 11 FC STA TT_SLOT
8F5B A2 00 LDX #&00
8F5D A0 BE LDY #&BE ; restore zero page &70-&83

.L8F5F
8F5F C8 INY
8F60 B1 84 LDA (&84),Y
8F62 95 70 STA &70,X
8F64 E8 INX
8F65 E0 14 CPX #&14
8F67 D0 F6 BNE L8F5F
8F69 20 28 90 JSR L9028 ; capture and IRQ back on
8F6C 68 PLA
8F6D 85 85 STA &85
8F6F 68 PLA
8F70 85 84 STA &84
8F72 68 PLA
8F73 A8 TAY
8F74 68 PLA
8F75 AA TAX
8F76 68 PLA
8F77 85 FC STA &FC
8F79 68 PLA
8F7A 28 PLP
8F7B A9 00 LDA #&00
8F7D 60 RTS
; &77/&78 = row * 40.

.L8F7E
8F7E A5 71 LDA &71
8F80 0A ASL A
8F81 0A ASL A
8F82 0A ASL A
8F83 85 77 STA &77
8F85 85 79 STA &79
8F87 A9 00 LDA #&00
8F89 85 7A STA &7A
8F8B 06 79 ASL &79
8F8D 26 7A ROL &7A
8F8F 06 79 ASL &79
8F91 26 7A ROL &7A
8F93 18 CLC
8F94 A5 77 LDA &77
8F96 65 79 ADC &79
8F98 85 77 STA &77
8F9A A9 00 LDA #&00
8F9C 65 7A ADC &7A
8F9E 85 78 STA &78
8FA0 60 RTS
; Not our interrupt: pass it on.

.L8FA1
8FA1 68 PLA
8FA2 A8 TAY
8FA3 68 PLA
8FA4 AA TAX
8FA5 68 PLA
8FA6 85 FC STA &FC
8FA8 68 PLA
8FA9 28 PLP
8FAA 60 RTS

.L8FAB
8FAB 60 RTS
;-----
; Magazine 8 row 30: decode the initial page (Hamming) and keep the network id and time
; offset. The MJD and UTC digits are each sent increased by 1; that is removed here.
;-----

.tsdp_capture
8FAC 20 9C 91 JSR L919C
8FAF AE 12 FC LDX TT_DATA
8FB2 BD AF 91 LDA hamming_table,X
8FB5 29 0F AND #&0F
8FB7 D0 F2 BNE L8FAB
8FB9 A0 00 LDY #&00

.L8FBB
8FBB AE 12 FC LDX TT_DATA
8FBE BD AF 91 LDA hamming_table,X
8FC1 30 E8 BMI L8FAB
8FC3 29 0F AND #&0F
8FC5 99 77 00 STA &0077,Y
8FC8 C8 INY
8FC9 C0 06 CPY #&06
8FCB D0 EE BNE L8FBB
8FCD 20 55 91 JSR L9155
8FD0 A5 75 LDA &75
8FD2 4A LSR A
8FD3 4A LSR A
8FD4 4A LSR A
8FD5 45 70 EOR &70
8FD7 A0 D4 LDY #&D4 ; workspace +&D4: TSDP initial magazine

```

```

8FD9 91 84    STA (&84),Y
8FDB C8      INY
8FDC A5 72    LDA &72
8FDE 91 84    STA (&84),Y
8FE0 C8      INY
8FE1 A5 73    LDA &73
8FE3 91 84    STA (&84),Y
8FE5 C8      INY
8FE6 A5 74    LDA &74
8FE8 91 84    STA (&84),Y
8FEA AD 12 FC LDA TT_DATA
8FED C8      INY
8FEE 91 84    STA (&84),Y
8FF0 AD 12 FC LDA TT_DATA
8FF3 C8      INY
8FF4 91 84    STA (&84),Y
8FF6 AD 12 FC LDA TT_DATA
8FF9 C8      INY
8FFA 91 84    STA (&84),Y
8FFC AD 12 FC LDA TT_DATA
8FFF 29 0F    AND #&0F
9001 38      SEC
9002 E9 01    SBC #&01
9004 C8      INY
9005 91 84    STA (&84),Y
9007 A2 00    LDX #&00
.L9009
9009 AD 12 FC LDA TT_DATA
900C 38      SEC
900D E9 11    SBC #&11
900F C8      INY
9010 91 84    STA (&84),Y
9012 E8      INX
9013 E0 05    CPX #&05
9015 D0 F2    BNE L9009
9017 60      RTS
;-----
; Latch bits 2-3 off (capture and IRQ) in the latch and shadow.
;-----
.L9018
9018 98      TYA
9019 48      PHA
901A A0 BD    LDY #&BD                ; workspace +&BD: latch shadow
901C B1 84    LDA (&84),Y
901E 29 F3    AND #&F3
9020 8D 10 FC STA TT_STATUS_CTRL
9023 91 84    STA (&84),Y
9025 68      PLA
9026 A8      TAY
9027 60      RTS
; Latch bits 2-3 on.
.L9028
9028 98      TYA
9029 48      PHA
902A A0 BD    LDY #&BD                ; workspace +&BD: latch shadow
902C B1 84    LDA (&84),Y
902E 09 0C    ORA #&0C
9030 91 84    STA (&84),Y
9032 8D 10 FC STA TT_STATUS_CTRL
9035 68      PLA
9036 A8      TAY
9037 60      RTS
;-----
; ADAPTOR STATUS: switch capture off, clear the status, wait three fields, and see
; whether field-sync pulses (status bit 4) arrive and whether the link bits read
; non-zero. Returns 0 ok, 1 on but no signal, 2 off but connected, &FF no adaptor.
; (Different from TFS 1.03, which only looked at the link bits.)
;-----
.adaptor_status
9038 98      TYA
9039 48      PHA
903A A5 77    LDA &77
903C 48      PHA
903D A5 78    LDA &78
903F 48      PHA
9040 A5 EF    LDA osw_a
9042 48      PHA
9043 A5 F0    LDA osw_x
9045 48      PHA
9046 A5 F1    LDA osw_y
9048 48      PHA
9049 A9 00    LDA #&00
904B 85 77    STA &77
904D A6 F4    LDX romsel_copy
904F BD F0 0D LDA rom_wksp_table,X
9052 85 78    STA &78
9054 20 94 91 JSR L9194
9057 90 06    BCC L905F
9059 A5 78    LDA &78
905B 09 80    ORA #&80

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```

905D 85 78 STA &78
.L905F
905F A0 BD LDY #&BD ; workspace +&BD: latch shadow
9061 B1 77 LDA (&77),Y
9063 29 F3 AND #&F3
9065 8D 10 FC STA TT_STATUS_CTRL
9068 A9 00 LDA #&00
906A 8D 13 FC STA TT_STATCLR
906D A9 13 LDA #&13
906F 20 F4 FF JSR OSBYTE ; OSBYTE &13: wait for vertical sync
9072 20 F4 FF JSR OSBYTE
9075 20 F4 FF JSR OSBYTE
9078 AD 10 FC LDA TT_STATUS_CTRL
907B 29 10 AND #&10
907D 85 77 STA &77
907F A2 02 LDX #&02
.L9081
9081 A9 00 LDA #&00
9083 8D 13 FC STA TT_STATCLR
9086 AD 10 FC LDA TT_STATUS_CTRL
9089 29 10 AND #&10
908B F0 03 BEQ L9090
908D CA DEX
908E D0 F1 BNE L9081
.L9090
9090 85 78 STA &78
9092 05 77 ORA &77
9094 D0 0F BNE L90A5
9096 AD 10 FC LDA TT_STATUS_CTRL
9099 29 0F AND #&0F
909B F0 04 BEQ L90A1
909D A9 01 LDA #&01
909F D0 0E BNE L90AF
.L90A1
90A1 A9 02 LDA #&02
90A3 D0 0A BNE L90AF
.L90A5
90A5 A5 78 LDA &78
90A7 F0 04 BEQ L90AD
90A9 A9 FF LDA #&FF
90AB 30 02 BMI L90AF
.L90AD
90AD A9 00 LDA #&00
.L90AF
90AF 48 PHA
90B0 A9 00 LDA #&00
90B2 85 77 STA &77
90B4 A6 F4 LDX romsel_copy
90B6 BD F0 0D LDA rom_wksp_table,X
90B9 85 78 STA &78
90BB 20 94 91 JSR L9194
90BE 90 06 BCC L90C6
90C0 A5 78 LDA &78
90C2 09 80 ORA #&80
90C4 85 78 STA &78
.L90C6
90C6 A0 BD LDY #&BD ; workspace +&BD: latch shadow
90C8 B1 77 LDA (&77),Y
90CA 8D 10 FC STA TT_STATUS_CTRL
90CD 68 PLA
90CE AA TAX
90CF 68 PLA
90D0 85 F1 STA osw_y
90D2 68 PLA
90D3 85 F0 STA osw_x
90D5 68 PLA
90D6 85 EF STA osw_a
90D8 68 PLA
90D9 85 78 STA &78
90DB 68 PLA
90DC 85 77 STA &77
90DE 68 PLA
90DF A8 TAY
90E0 8A TXA
90E1 60 RTS
; Mark all 16 adapter slots empty.
.L90E2
90E2 A0 0F LDY #&0F
90E4 A9 00 LDA #&00
.L90E6
90E6 8C 11 FC STY TT_SLOT
90E9 8D 12 FC STA TT_DATA
90EC 88 DEY
90ED 10 F7 BPL L90E6
90EF 60 RTS
;-----
; Select slot Y, mark it empty, decode the two address bytes: &70 = magazine,
; &71 = row (&FF if uncorrectable).
;-----
.decode_address

```

```

90F0 8C 11 FC STY TT_SLOT
90F3 A9 00 LDA #000
90F5 8D 12 FC STA TT_DATA
90F8 AE 12 FC LDX TT_DATA
90FB BD AF 91 LDA hamming_table,X
90FE 30 1F BMI L911F
9100 29 07 AND #007
9102 85 70 STA &70
9104 BD AF 91 LDA hamming_table,X
9107 29 08 AND #008
9109 4A LSR A
910A 4A LSR A
910B 4A LSR A
910C 85 71 STA &71
910E AE 12 FC LDX TT_DATA
9111 BD AF 91 LDA hamming_table,X
9114 30 09 BMI L911F
9116 29 0F AND #00F
9118 0A ASL A
9119 18 CLC
911A 65 71 ADC &71
911C 85 71 STA &71
911E 60 RTS
.L911F
911F A9 FF LDA #&FF
9121 85 70 STA &70
9123 85 71 STA &71
9125 60 RTS
;-----
; Decode the 8 header bytes into &72 page, &73/&74 sub-code, &75/&76 control bits.
;-----
.decode_header
9126 A0 00 LDY #000
.L9128
9128 AE 12 FC LDX TT_DATA
912B BD AF 91 LDA hamming_table,X
912E 30 1C BMI L914C
9130 29 0F AND #00F
9132 99 77 00 STA &0077,Y
9135 C8 INY
9136 C0 08 CPY #008
9138 D0 EE BNE L9128
913A 20 55 91 JSR L9155
913D A5 7D LDA &7D
913F 85 76 STA &76
9141 A5 7E LDA &7E
9143 0A ASL A
9144 0A ASL A
9145 0A ASL A
9146 0A ASL A
9147 05 76 ORA &76
9149 85 76 STA &76
914B 60 RTS
.L914C
914C A9 FF LDA #&FF
914E 85 72 STA &72
9150 85 73 STA &73
9152 85 74 STA &74
9154 60 RTS
;-----
; Pack decoded header nibbles into page, sub-code and control bytes.
;-----
.L9155
9155 A5 77 LDA &77
9157 85 72 STA &72
9159 A5 78 LDA &78
915B 0A ASL A
915C 0A ASL A
915D 0A ASL A
915E 0A ASL A
915F 05 72 ORA &72
9161 85 72 STA &72
9163 A5 79 LDA &79
9165 85 74 STA &74
9167 A5 7A LDA &7A
9169 29 07 AND #007
916B 0A ASL A
916C 0A ASL A
916D 0A ASL A
916E 0A ASL A
916F 05 74 ORA &74
9171 85 74 STA &74
9173 A5 7B LDA &7B
9175 85 73 STA &73
9177 A5 7C LDA &7C
9179 29 03 AND #003
917B 0A ASL A
917C 0A ASL A
917D 0A ASL A
917E 0A ASL A

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917F 05 73    ORA &73
9181 85 73    STA &73
9183 A5 7A    LDA &7A
9185 29 08    AND #&08
9187 85 75    STA &75
9189 A5 7C    LDA &7C
918B 29 0C    AND #&0C
918D 0A      ASL A
918E 0A      ASL A
918F 05 75    ORA &75
9191 85 75    STA &75
9193 60      RTS
; Is this a Master (OS version ≥ 3)? C set if so.
.L9194
9194 A9 00    LDA #&00
9196 20 B7 A0 JSR LA0B7
9199 E0 03    CPX #&03
919B 60      RTS
;-----
; Re-select the adapter slot and skip its flag and address bytes.
;-----
.L919C
919C 48      PHA
919D A0 BE    LDY #&BE                ; workspace +&BE: adapter slot
919F B1 84    LDA (&84),Y
91A1 8D 11 FC STA TT_SLOT
91A4 AD 12 FC LDA TT_DATA
91A7 AD 12 FC LDA TT_DATA
91AA AD 12 FC LDA TT_DATA
91AD 68      PLA
91AE 60      RTS
;-----
; Hamming 8/4 decode table (same meaning as in TFS: &0x cclean, &4x corrected, &FF bad).
;-----
.hamming_table
91AF .db 41 FF 01 41 FF 40 41 FF FF 42 41 FF 4A FF FF 47 ; A..A.@A..BA.J..G
91BF .db FF 40 41 FF 40 00 FF 40 46 FF FF 4B FF 40 43 FF ; .@A.@..@F..K.@C.
91CF .db FF 4C 41 FF 44 FF FF 47 46 FF FF 47 FF 47 07 ; .LA.D..GF..G.GG.
91DF .db 46 FF FF 45 FF 40 4D FF 06 46 46 FF 46 FF FF 47 ; F..E.@M..FF.F..G
91EF .db FF 42 41 FF 44 FF FF 49 42 02 FF 42 FF 42 43 FF ; .BA.D..IB..B.BC.
91FF .db 48 FF FF 45 FF 40 43 FF FF 42 43 FF 43 FF 03 43 ; H..E.@C..BC.C..C
920F .db 44 FF FF 45 04 44 44 FF FF 42 4F FF 44 FF FF 47 ; D..E.DD..B0.D..G
921F .db FF 45 45 05 44 FF FF 45 46 FF FF 45 FF 4E 43 FF ; .EE.D..EF..E.NC.
922F .db FF 4C 41 FF 4A FF FF 49 4A FF FF 4B 0A 4A 4A FF ; .LA.J..IJ..K.JJ.
923F .db 48 FF FF 4B FF 40 4D FF FF 4B 4B 0B 4A FF FF 4B ; H..K.@M..KK.J..K
924F .db 4C 0C FF 4C FF 4C 4D FF FF 4C 4F FF 4A FF FF 47 ; L..L.LM..LO.J..G
925F .db FF 4C 4D FF 4D FF 0D 4D 46 FF FF 4B FF 4E 4D FF ; .LM.M..MF..K.NM.
926F .db 48 FF FF 49 FF 49 49 09 FF 42 4F FF 4A FF FF 49 ; H..I.II..B0.J..I
927F .db 08 48 48 FF 48 FF FF 49 48 FF FF 4B FF 4E 43 FF ; .HH.H..IH..K.NC.
928F .db FF 4C 4F FF 44 FF 4F 49 4F FF 0F 4F FF 4E 4F FF ; .LO.D..IO..O.NO.
929F .db 48 FF FF 45 FF 4E 4D FF FF 4E 4F FF 4E 0E FF 4E ; H..E.NM..NO.N..N
;-----
; OSWORD 20: TUNING SCREEN (*TUNE, SHIFT f0). AFC off, interrupt mode &FE so the grabber
; only counts bytes and errors; every 13 fields compute the error rate and draw the bar.
; SPACE leaves.
;-----
.L92AF
92AF A0 E9    LDY #&E9                ; workspace +&E9: interrupt mode
92B1 B1 A8    LDA (wk),Y
92B3 48      PHA
92B4 A9 FF    LDA #&FF
92B6 91 A8    STA (wk),Y
92B8 A0 BD    LDY #&BD                ; workspace +&BD: latch shadow
92BA B1 A8    LDA (wk),Y
92BC 48      PHA
92BD 29 EF    AND #&EF
92BF 09 0C    ORA #&0C
92C1 91 A8    STA (wk),Y
92C3 8D 10 FC STA TT_STATUS_CTRL
92C6 A9 72    LDA #&72
92C8 A2 01    LDX #&01
92CA A0 00    LDY #&00
92CC 20 F4 FF JSR OSBYTE                ; OSBYTE &72
92CF A9 16    LDA #&16
92D1 20 EE FF JSR OSWRCH                ; VDU 22: MODE
92D4 A9 07    LDA #&07
92D6 20 EE FF JSR OSWRCH                ; VDU 7: BEL
92D9 20 2A A8 JSR LA82A
92DC 20 BC 9D JSR L9DBC
92DF A0 00    LDY #&00
.L92E1
92E1 B9 22 94 LDA &9422,Y
92E4 C9 FF    CMP #&FF
92E6 F0 0B    BEQ L92F3
92E8 20 EE FF JSR OSWRCH
.L92EB
92EB C8      INY
92EC C9 00    CMP #&00
92EE D0 F1    BNE L92E1
92F0 4C FF 92 JMP L92FF

```

```

.L92F3
92F3 A2 00 LDX #&00
92F5 A9 A3 LDA #&A3
.L92F7
92F7 20 EE FF JSR OSWRCH ; VDU 163
92FA CA DEX
92FB 10 FA BPL L92F7
92FD 30 EC BMI L92EB
.L92FF
92FF A0 00 LDY #&00
.L9301
9301 B9 5E A0 LDA &A05E,Y
9304 99 C0 7F STA &7FC0,Y
9307 C8 INY
9308 C0 28 CPY #&28 ; '('
930A D0 F5 BNE L9301
930C A0 B7 LDY #&B7 ; workspace +&B7: header status
930E B1 A8 LDA (wk),Y
9310 48 PHA
9311 A9 01 LDA #&01
9313 91 A8 STA (wk),Y
9315 C8 INY
9316 B1 A8 LDA (wk),Y
9318 48 PHA
9319 A9 FF LDA #&FF
931B 91 A8 STA (wk),Y
.L931D
931D 78 SEI
931E A0 E9 LDY #&E9
9320 A9 FE LDA #&FE
9322 91 A8 STA (wk),Y
9324 C8 INY
9325 A9 00 LDA #&00
9327 91 A8 STA (wk),Y
9329 C8 INY
932A 91 A8 STA (wk),Y
932C C8 INY
932D 91 A8 STA (wk),Y
932F C8 INY
9330 91 A8 STA (wk),Y
9332 58 CLI
9333 A9 00 LDA #&00
9335 8D 13 FC STA TT_STATCLR
9338 85 AA STA &AA
.L933A
933A A9 13 LDA #&13
933C 20 F4 FF JSR OSBYTE ; OSBYTE &13: wait for vertical sync
933F C6 AA DEC &AA
9341 A5 AA LDA &AA
9343 D0 F5 BNE L933A
9345 78 SEI
9346 A0 EA LDY #&EA ; workspace +&EA: tune error count
.L9348
9348 B1 A8 LDA (wk),Y
934A 99 C0 FF STA &FFC0,Y
934D C8 INY
934E C0 EE CPY #&EE
9350 D0 F6 BNE L9348
9352 58 CLI
9353 A9 1F LDA #&1F
9355 20 EE FF JSR OSWRCH ; VDU 31: TAB(x,y)
9358 A9 00 LDA #&00
935A 20 EE FF JSR OSWRCH ; VDU 13: carriage return
935D A9 11 LDA #&11
935F 20 EE FF JSR OSWRCH ; VDU 17
9362 A5 AC LDA &AC
9364 05 AD ORA &AD
9366 F0 12 BEQ L937A
9368 A9 82 LDA #&82
936A 20 EE FF JSR OSWRCH ; VDU 130
936D A9 20 LDA #&20
936F 20 EE FF JSR OSWRCH ; print ' '
9372 20 EE FF JSR OSWRCH
9375 A2 03 LDX #&03
9377 4C 81 93 JMP L9381
.L937A
937A A9 81 LDA #&81
937C 20 EE FF JSR OSWRCH ; VDU 129
937F A2 00 LDX #&00
.L9381
9381 BD 3F 94 LDA &943F,X
9384 20 EE FF JSR OSWRCH
9387 E8 INX
9388 C9 00 CMP #&00
938A D0 F5 BNE L9381
938C A5 AC LDA &AC
938E 05 AD ORA &AD
9390 F0 15 BEQ L93A7
9392 A2 00 LDX #&00
.L9394

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9394 38      SEC
9395 A5 AC    LDA &AC
9397 E5 AA    SBC &AA
9399 85 AC    STA &AC
939B A5 AD    LDA &AD
939D E5 AB    SBC &AB
939F 85 AD    STA &AD
93A1 30 04    BMI L93A7
93A3 E8      INX
93A4 D0 EE    BNE L9394
93A6 CA      DEX
.L93A7
93A7 8A      TXA
93A8 4A      LSR A
93A9 4A      LSR A
93AA 85 AA    STA &AA
93AC 4A      LSR A
93AD 85 AB    STA &AB
93AF A9 1F    LDA #&1F
93B1 20 EE FF JSR OSWRCH          ; VDU 31: TAB(x,y)
93B4 A9 03    LDA #&03
93B6 20 EE FF JSR OSWRCH          ; VDU 3
93B9 A9 0C    LDA #&0C
93BB 20 EE FF JSR OSWRCH          ; VDU 12: CLS
93BE A9 96    LDA #&96
93C0 20 EE FF JSR OSWRCH          ; VDU 150
93C3 A6 AB    LDX &AB
93C5 F0 08    BEQ L93CF
93C7 A9 FF    LDA #&FF
.L93C9
93C9 20 EE FF JSR OSWRCH          ; VDU 255
93CC CA      DEX
93CD D0 FA    BNE L93C9
.L93CF
93CF A5 AA    LDA &AA
93D1 29 01    AND #&01
93D3 F0 05    BEQ L93DA
93D5 A9 B5    LDA #&B5
93D7 20 EE FF JSR OSWRCH          ; VDU 181
.L93DA
93DA 20 10 94 JSR L9410
93DD A9 80    LDA #&80
93DF A2 FF    LDX #&FF
93E1 A0 FF    LDY #&FF
93E3 20 F4 FF JSR OSBYTE          ; OSBYTE &80
93E6 E0 00    CPX #&00
93E8 D0 03    BNE L93ED
93EA 4C 1D 93 JMP L931D
.L93ED
93ED 20 E0 FF JSR OSRDCH
93F0 C9 20    CMP #&20
93F2 F0 03    BEQ L93F7
93F4 4C 1D 93 JMP L931D
.L93F7
93F7 68      PLA
93F8 A0 B8    LDY #&B8          ; workspace +&B8: header magazine
93FA 91 A8    STA (wk),Y
93FC 88      DEY
93FD 68      PLA
93FE 91 A8    STA (wk),Y
9400 68      PLA
9401 A0 BD    LDY #&BD          ; workspace +&BD: latch shadow
9403 91 A8    STA (wk),Y
9405 8D 10 FC STA TT_STATUS_CTRL
9408 68      PLA
9409 A0 E9    LDY #&E9          ; workspace +&E9: interrupt mode
940B 91 A8    STA (wk),Y
940D 4C E1 82 JMP L82E1
; Pad the line with spaces to column 38.
.L9410
9410 A9 86    LDA #&86
9412 20 F4 FF JSR OSBYTE          ; OSBYTE &86: read text cursor position
9415 E0 26    CPX #&26          ; '&'
9417 F0 08    BEQ L9421
9419 A9 20    LDA #&20
941B 20 EE FF JSR OSWRCH          ; print ' '
941E 4C 10 94 JMP L9410
.L9421
9421 60      RTS
; Tuning screen scale and "No Teletext" message.
9422 .db 1F 03 0D 97 B7 FF EB FF A3 B5 1F 03 0E 30 25 1F ; .....0%.
9432 .db 12 0E 35 30 25 1F 21 0E 31 30 30 25 00 4E 6F 20 ; ..50%.!..100%.No
9442 .db 54 65 6C 65 74 65 78 74 20 20 00              ; Teletext .
-----
; OSWORD 21: TIME "HH:MM:SS" from the TSDP, local time offset applied.
; Byte 10: 0 ok, &FF no TSDP (1/2 internally: day carried forward/back).
-----
.L9440
9440 20 53 94 JSR L9453
9450 4C E1 82 JMP L82E1

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.L9453
9453 A0 0A LDY #&0A
9455 A9 00 LDA #&00
9457 91 F0 STA (osw_x),Y
9459 A0 D4 LDY #&D4 ; workspace +&D4: TSDP initial magazine
945B B1 A8 LDA (wk),Y
945D 10 03 BPL L9462
945F 4C 3F 95 JMP L953F

.L9462
9462 08 PHP
9463 78 SEI
9464 F8 SED
9465 A0 DE LDY #&DE ; workspace +&DE: TSDP
9467 B1 A8 LDA (wk),Y
9469 85 AA STA &AA
946B C8 INY
946C B1 A8 LDA (wk),Y
946E 85 AB STA &AB
9470 C8 INY
9471 B1 A8 LDA (wk),Y
9473 85 AC STA &AC
9475 A9 00 LDA #&00
9477 85 AD STA &AD
9479 85 AE STA &AE
947B A0 DA LDY #&DA ; workspace +&DA: TSDP
947D B1 A8 LDA (wk),Y
947F 0A ASL A
9480 0A ASL A
9481 90 02 BCC L9485
9483 E6 AD INC &AD

.L9485
9485 4A LSR A
9486 4A LSR A
9487 4A LSR A
9488 4A LSR A
9489 90 02 BCC L948D
948B E6 AE INC &AE

.L948D
948D 85 AF STA &AF
948F F0 0B BEQ L949C
9491 C9 0A CMP #&0A
9493 90 07 BCC L949C
9495 38 SEC
9496 D8 CLD
9497 E9 0A SBC #&0A
9499 09 10 ORA #&10
949B F8 SED

.L949C
949C 85 AF STA &AF
949E A5 AE LDA &AE
94A0 F0 37 BEQ L94D9
94A2 A5 AD LDA &AD
94A4 D0 1C BNE L94C2
94A6 18 CLC
94A7 A5 AB LDA &AB
94A9 69 30 ADC #&30
94AB 85 AB STA &AB
94AD C9 60 CMP #&60 ; ''
94AF 90 0E BCC L94BF
94B1 38 SEC
94B2 E9 60 SBC #&60
94B4 85 AB STA &AB
94B6 18 CLC
94B7 D8 CLD
94B8 A5 AF LDA &AF
94BA 69 01 ADC #&01
94BC 85 AF STA &AF
94BE F8 SED

.L94BF
94BF 4C D9 94 JMP L94D9

.L94C2
94C2 38 SEC
94C3 A5 AB LDA &AB
94C5 E9 30 SBC #&30
94C7 85 AB STA &AB
94C9 B0 0E BCS L94D9
94CB 18 CLC
94CC 69 60 ADC #&60
94CE 85 AB STA &AB
94D0 18 CLC
94D1 D8 CLD
94D2 A5 AF LDA &AF
94D4 69 01 ADC #&01
94D6 85 AF STA &AF
94D8 F8 SED

.L94D9
94D9 A5 AD LDA &AD
94DB D0 18 BNE L94F5
94DD 18 CLC
94DE A5 AA LDA &AA

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```

94E0 65 AF      ADC &AF
94E2 85 AA      STA &AA
94E4 C9 24      CMP #&24          ; '$'
94E6 90 0A      BCC L94F2
94E8 E9 24      SBC #&24
94EA 85 AA      STA &AA
94EC A0 0A      LDY #&0A
94EE A9 01      LDA #&01
94F0 91 F0      STA (osw_x),Y
.L94F2
94F2 4C 0A 95   JMP L950A
.L94F5
94F5 38         SEC
94F6 A5 AA      LDA &AA
94F8 E5 AF      SBC &AF
94FA 85 AA      STA &AA
94FC B0 F4      BCS L94F2
94FE 69 24      ADC #&24
9500 85 AA      STA &AA
9502 A0 0A      LDY #&0A
9504 A9 02      LDA #&02
9506 91 F0      STA (osw_x),Y
9508 D0 E8      BNE L94F2
.L950A
950A A0 01      LDY #&01
950C A5 AA      LDA &AA
950E 20 27 95   JSR L9527
9511 A9 3A      LDA #&3A
9513 91 F0      STA (osw_x),Y
9515 C8         INY
9516 A5 AB      LDA &AB
9518 20 27 95   JSR L9527
951B A9 3A      LDA #&3A
951D 91 F0      STA (osw_x),Y
951F C8         INY
9520 A5 AC      LDA &AC
9522 20 27 95   JSR L9527
9525 28         PLP
9526 60         RTS
; Store BCD A as two ASCII digits.
.L9527
9527 D8         CLD
9528 48         PHA
9529 29 F0      AND #&F0
952B 4A         LSR A
952C 4A         LSR A
952D 4A         LSR A
952E 4A         LSR A
952F 18         CLC
9530 69 30      ADC #&30
9532 91 F0      STA (osw_x),Y
9534 C8         INY
9535 68         PLA
9536 29 0F      AND #&0F
9538 18         CLC
9539 69 30      ADC #&30
953B 91 F0      STA (osw_x),Y
953D C8         INY
953E 60         RTS
.L953F
953F A0 0A      LDY #&0A
9541 A9 FF      LDA #&FF
9543 91 F0      STA (osw_x),Y
9545 60         RTS
;
; OSWORD 22: DATE "DD/MM/YYYY": MJD converted by counting from 1982 as in TFS 1.03.
;
.L9546
9546 20 53 94   JSR L9453
9549 78         SEI
954A F8         SED
954B A0 0A      LDY #&0A
954D B1 F0      LDA (osw_x),Y
954F 10 0A      BPL L955B
9551 A0 0C      LDY #&0C
9553 A9 FF      LDA #&FF
9555 91 F0      STA (osw_x),Y
9557 D8         CLD
9558 4C E1 82   JMP L82E1
.L955B
955B 48         PHA
955C A9 69      LDA #&69
955E 85 AC      STA &AC
9560 68         PLA
9561 F0 0B      BEQ L956E
9563 C9 01      CMP #&01
9565 F0 05      BEQ L956C
9567 E6 AC      INC &AC
9569 4C 6E 95   JMP L956E
.L956C

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956C C6 AC      DEC &AC
.L956E
956E A0 DD      LDY #&DD
9570 B1 A8      LDA (wk),Y
9572 38         SEC
9573 E5 AC      SBC &AC
9575 85 AC      STA &AC
9577 88         DEY
9578 B1 A8      LDA (wk),Y
957A E9 49      SBC #&49
957C 85 AB      STA &AB
957E 88         DEY
957F B1 A8      LDA (wk),Y
9581 E9 04      SBC #&04
9583 85 AA      STA &AA
9585 A5 B0      LDA &B0
9587 48         PHA
9588 A9 BE      LDA #&BE
958A 85 AD      STA &AD
958C A9 07      LDA #&07
958E 85 AE      STA &AE
9590 A9 19      LDA #&19
9592 85 AF      STA &AF
9594 A9 82      LDA #&82
9596 85 B0      STA &B0
.L9598
9598 A5 AA      LDA &AA
959A D0 32      BNE L95CE
959C A5 AB      LDA &AB
959E C9 03      CMP #&03
95A0 F0 12      BEQ L95B4
95A2 C9 02      CMP #&02
95A4 F0 0B      BEQ L95B1
95A6 C9 01      CMP #&01
95A8 F0 07      BEQ L95B1
95AA C9 00      CMP #&00
95AC F0 03      BEQ L95B1
95AE 4C CE 95   JMP L95CE
.L95B1
95B1 4C 0F 96   JMP L960F
.L95B4
95B4 20 AB 96   JSR L96AB
95B7 F0 0C      BEQ L95C5
95B9 38         SEC
95BA A5 AC      LDA &AC
95BC E9 66      SBC #&66
95BE 90 F1      BCC L95B1
95C0 F0 EF      BEQ L95B1
95C2 4C CE 95   JMP L95CE
.L95C5
95C5 38         SEC
95C6 A5 AC      LDA &AC
95C8 E9 65      SBC #&65
95CA 90 E5      BCC L95B1
95CC F0 E3      BEQ L95B1
.L95CE
95CE 20 AB 96   JSR L96AB
95D1 F0 0A      BEQ L95DD
95D3 38         SEC
95D4 A5 AC      LDA &AC
95D6 E9 66      SBC #&66
95D8 85 AC      STA &AC
95DA 4C E4 95   JMP L95E4
.L95DD
95DD 38         SEC
95DE A5 AC      LDA &AC
95E0 E9 65      SBC #&65
95E2 85 AC      STA &AC
.L95E4
95E4 A5 AB      LDA &AB
95E6 E9 03      SBC #&03
95E8 85 AB      STA &AB
95EA A5 AA      LDA &AA
95EC E9 00      SBC #&00
95EE 85 AA      STA &AA
95F0 18         CLC
95F1 A5 B0      LDA &B0
95F3 69 01      ADC #&01
95F5 85 B0      STA &B0
95F7 A5 AF      LDA &AF
95F9 69 00      ADC #&00
95FB 85 AF      STA &AF
95FD D8         CLD
95FE 18         CLC
95FF A5 AD      LDA &AD
9601 69 01      ADC #&01
9603 85 AD      STA &AD
9605 A5 AE      LDA &AE
9607 69 00      ADC #&00
9609 85 AE      STA &AE

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; workspace +&DD: TSDP

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960B F8 SED
960C 4C 98 95 JMP L9598
.L960F
960F A2 00 LDX #&00
9611 A9 01 LDA #&01
9613 85 AD STA &AD
.L9615
9615 A5 AD LDA &AD
9617 C9 02 CMP #&02
9619 D0 20 BNE L963B
961B 20 AB 96 JSR L96AB
961E F0 07 BEQ L9627
9620 A9 29 LDA #&29
9622 85 AE STA &AE
9624 4C 2B 96 JMP L962B
.L9627
9627 A9 28 LDA #&28
9629 85 AE STA &AE
.L962B
962B A5 AB LDA &AB
962D D0 22 BNE L9651
962F A5 AC LDA &AC
9631 38 SEC
9632 E5 AE SBC &AE
9634 90 3C BCC L9672
9636 F0 3A BEQ L9672
9638 4C 51 96 JMP L9651
.L963B
963B BD 9F 96 LDA &969F,X
963E 85 AE STA &AE
9640 A5 AB LDA &AB
9642 D0 0D BNE L9651
9644 A5 AC LDA &AC
9646 38 SEC
9647 FD 9F 96 SBC &969F,X
964A 90 26 BCC L9672
964C F0 24 BEQ L9672
964E 4C 51 96 JMP L9651
.L9651
9651 E0 0B CPX #&0B
9653 D0 05 BNE L965A
9655 A9 07 LDA #&07
9657 20 EE FF JSR 0SWRCH ; VDU 7: BEL
.L965A
965A A5 AC LDA &AC
965C 38 SEC
965D E5 AE SBC &AE
965F 85 AC STA &AC
9661 A5 AB LDA &AB
9663 E9 00 SBC #&00
9665 85 AB STA &AB
9667 E8 INX
9668 18 CLC
9669 A5 AD LDA &AD
966B 69 01 ADC #&01
966D 85 AD STA &AD
966F 4C 15 96 JMP L9615
.L9672
9672 A0 01 LDY #&01
9674 A5 AC LDA &AC
9676 20 27 95 JSR L9527
9679 A9 2F LDA #&2F
967B 91 F0 STA (osw_x),Y
967D C8 INY
967E A5 AD LDA &AD
9680 20 27 95 JSR L9527
9683 A9 2F LDA #&2F
9685 91 F0 STA (osw_x),Y
9687 C8 INY
9688 A5 AF LDA &AF
968A 20 27 95 JSR L9527
968D A5 B0 LDA &B0
968F 20 27 95 JSR L9527
9692 A0 0C LDY #&0C
9694 A9 00 LDA #&00
9696 91 F0 STA (osw_x),Y
9698 D8 CLD
9699 68 PLA
969A 85 B0 STA &B0
969C 4C E1 82 JMP L82E1
; Month-lengths table (BCD), shared format with TFS.
969F .db 31 28 31 30 31 30 31 31 30 31 30 31 ; 1(1010110101
.L96AB
96AB A5 AB LDA &AB
96AD 48 PHA
96AE A5 AD LDA &AD
96B0 29 03 AND #&03
96B2 D0 25 BNE L96D9
96B4 A5 B0 LDA &B0
96B6 D0 1B BNE L96D3

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96B8 D8      CLD
96B9 A5 AF    LDA &AF
96BB 29 F0    AND #&F0
96BD 4A      LSR A
96BE 85 AB    STA &AB
96C0 4A      LSR A
96C1 4A      LSR A
96C2 18      CLC
96C3 65 AB    ADC &AB
96C5 85 AB    STA &AB
96C7 A5 AF    LDA &AF
96C9 29 0F    AND #&0F
96CB 18      CLC
96CC 65 AB    ADC &AB
96CE F8      SED
96CF 29 03    AND #&03
96D1 D0 06    BNE L96D9
.L96D3
96D3 68      PLA
96D4 85 AB    STA &AB
96D6 A9 01    LDA #&01
96D8 60      RTS
.L96D9
96D9 68      PLA
96DA 85 AB    STA &AB
96DC A9 00    LDA #&00
96DE 60      RTS
;
; OSWORD 23: CHECK CRC [23, addr lo, hi] → 0 no CRC, 1 OK, &FF failed.
; Uses row 27/0's check word. The CRC covers row 0 characters 8-31 and rows 1-24;
; TFS 1.03 also included row 25.
;
.L96DF
96DF A5 82    LDA &82
96E1 48      PHA
96E2 A5 83    LDA &83
96E4 48      PHA
96E5 A5 84    LDA &84
96E7 48      PHA
96E8 A5 85    LDA &85
96EA 48      PHA
96EB A0 01    LDY #&01
96ED B1 F0    LDA (osw_x),Y
96EF 85 82    STA &82
96F1 C8      INY
96F2 B1 F0    LDA (osw_x),Y
96F4 85 83    STA &83
96F6 20 90 A4 JSR find_row27
96F9 B0 4B    BCS L9746
96FB A9 00    LDA #&00
96FD 80 40 7A STA crc_lo
9700 80 4E 7A STA crc_hi
9703 A2 00    LDX #&00
9705 20 45 A2 JSR find_row
9708 A0 08    LDY #&08 ; workspace +&08: grabber records (8 bytes each)
.L970A
970A B1 84    LDA (&84),Y
970C 20 5B 97 JSR L975B
970F C8      INY
9710 C0 20    CPY #&20
9712 D0 F6    BNE L970A
9714 A2 00    LDX #&00
.L9716
9716 E8      INX
9717 8A      TXA
9718 48      PHA
9719 20 45 A2 JSR find_row
971C A0 00    LDY #&00 ; workspace +&00: grabber records (8 bytes each)
.L971E
971E B1 84    LDA (&84),Y
9720 20 5B 97 JSR L975B
9723 C8      INY
9724 C0 28    CPY #&28 ; '('
9726 D0 F6    BNE L971E
9728 68      PLA
9729 AA      TAX
972A E0 19    CPX #&19
972C D0 E8    BNE L9716
972E AD 40 7A LDA crc_lo
9731 CD 4C 7A CMP crc_rx_lo
9734 D0 0C    BNE L9742
9736 AD 4E 7A LDA crc_hi
9739 CD 4B 7A CMP crc_rx_hi
973C D0 04    BNE L9742
973E A9 01    LDA #&01
9740 D0 06    BNE L9748
.L9742
9742 A9 FF    LDA #&FF
9744 D0 02    BNE L9748
.L9746

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9746 A9 00 LDA #&00
.L9748
9748 A0 03 LDY #&03
974A 91 F0 STA (osw_x),Y
974C 68 PLA
974D 85 85 STA &85
974F 68 PLA
9750 85 84 STA &84
9752 68 PLA
9753 85 83 STA &83
9755 68 PLA
9756 85 82 STA &82
9758 4C E1 82 JMP L82E1
; Add a byte to the CRC (same polynomial as TFS 1.03).
.L975B
975B 48 PHA
975C AD 4D 7A LDA crc_lo
975F 8D 51 7A STA &7A51
9762 AD 4E 7A LDA crc_hi
9765 8D 52 7A STA &7A52
9768 A2 04 LDX #&04
.L976A
976A 0E 4D 7A ASL crc_lo
976D 2E 4E 7A ROL crc_hi
9770 CA DEX
9771 D0 F7 BNE L976A
9773 AD 4E 7A LDA crc_hi
9776 8D 4F 7A STA &7A4F
9779 A2 03 LDX #&03
.L977B
977B 0E 4D 7A ASL crc_lo
977E 2E 4E 7A ROL crc_hi
9781 CA DEX
9782 D0 F7 BNE L977B
9784 AD 4E 7A LDA crc_hi
9787 8D 50 7A STA &7A50
978A A2 02 LDX #&02
.L978C
978C 0E 4D 7A ASL crc_lo
978F 2E 4E 7A ROL crc_hi
9792 CA DEX
9793 D0 F7 BNE L978C
9795 68 PLA
9796 4D 52 7A EOR &7A52
9799 4D 4F 7A EOR &7A4F
979C 4D 50 7A EOR &7A50
979F 4D 4E 7A EOR crc_hi
97A2 8D 52 7A STA &7A52
97A5 2A ROL A
97A6 A9 00 LDA #&00
97A8 2A ROL A
97A9 4D 52 7A EOR &7A52
97AC 8D 4D 7A STA crc_lo
97AF AD 51 7A LDA &7A51
97B2 8D 4E 7A STA crc_hi
97B5 60 RTS
;-----
; OSWORD 24: DOWNLOAD A TELESOFTWARE FILE [name, load, exec, magazine, page, to file,
; keep tables] - copies the parameters into the download workspace and runs the
; downloader (LADAD).
;-----
.L97B6
97B6 A0 01 LDY #&01
97B8 B1 F0 LDA (osw_x),Y
97BA 85 A8 STA wk
97BC C8 INY
97BD B1 F0 LDA (osw_x),Y
97BF 85 A9 STA wk+1
97C1 A0 00 LDY #&00 ; workspace +&00: grabber records (8 bytes each)
.L97C3
97C3 B1 A8 LDA (wk),Y
97C5 99 DA 70 STA dl_name,Y
97C8 C8 INY
97C9 C9 00 CMP #&00
97CB D0 F6 BNE L97C3
97CD A0 03 LDY #&03
.L97CF
97CF B1 F0 LDA (osw_x),Y
97D1 99 B1 70 STA &70B1,Y
97D4 C8 INY
97D5 C0 0B CPY #&0B
97D7 D0 F6 BNE L97CF
97D9 B1 F0 LDA (osw_x),Y
97DB 8D 9D 70 STA dl_mag
97DE C8 INY
97DF B1 F0 LDA (osw_x),Y
97E1 8D 9E 70 STA dl_page
97E4 C8 INY
97E5 B1 F0 LDA (osw_x),Y
97E7 8D 9B 70 STA dl_to_file

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97EA C8      INY
97EB B1 F0    LDA (osw_x),Y
97ED 8D B2 70 STA dl_keep_tables
97F0 58      CLI
97F1 20 AD AD JSR download
97F4 4C E1 82 JMP L82E1
; OSWORD 25: HAMMING DECODE A BYTE.
.L97F7
97F7 A0 01    LDY #&01
97F9 B1 F0    LDA (osw_x),Y
97FB AA      TAX
97FC BD AF 91 LDA hamming_table,X
97FF A0 02    LDY #&02
9801 91 F0    STA (osw_x),Y
9803 4C E1 82 JMP L82E1
;-----
; OSWORD 5 template for the terminal's first request: record 0, magazine 1, page 00,
; any sub-page, buffer &4A00.
;-----
9806 .db 05 00 01 00 3F 7F 00 4A FF FF ; ....?..J..
;-----
; LANGUAGE ENTRY (A=1) or, with a Tube, entered directly at &981A by *TELETEXT.
; Saves BRKV and the whole stack page so that f9 can return to the caller.
; Memory: page buffer &4A00 (5 pages, 30 rows of 42), link buffers &6D00 down to &4F00,
; kept pages in 5-page buffers from OSHWM up to &4A00, previous-page copies at &7200
; and &7600, workspace &7A00-&7B5F, Telesoftware workspace &7000-&79FF.
;-----
.terminal_entry
9810 C9 01    CMP #&01
9812 F0 01    BEQ L9815
9814 60      RTS
.L9815
9815 58      CLI
9816 D8      CLD
9817 A2 FF    LDX #&FF
9819 9A      TXS
.L981A
981A 20 E7 85 JSR L85E7
981D AD 02 02 LDA BRKV
9820 48      PHA
9821 AD 03 02 LDA BRKV+1
9824 48      PHA
9825 BA      TSX
9826 8E 57 7A STX t_saved_sp
9829 A2 00    LDX #&00
.L982B
982B BD 00 01 LDA stack_page,X
982E 9D 58 7A STA t_saved_stack,X
9831 E8      INX
9832 D0 F7    BNE L982B
9834 A9 01    LDA #&01
9836 8D 00 7A STA t_osw
9839 A9 00    LDA #&00
983B 8D 01 7A STA t_osw+1
983E 20 38 A3 JSR LA338
.L9841
9841 A2 FF    LDX #&FF ; (BREAK and errors restart here)
9843 9A      TXS
9844 20 BE A0 JSR terminal_keys
9847 A9 83    LDA #&83
9849 20 F4 FF JSR OSBYTE ; OSBYTE &83: read OSHWM (PAGE)
984C 8E 58 7B STX t_kept_base ; count how many 5-page kept-page buffers fit between OSHWM and &4A00
984F 8E 5A 7B STX &7B5A
9852 8C 59 7B STY t_kept_base+1
9855 8C 5B 7B STY &7B5B
9858 A9 08    LDA #&08
985A 8D 5C 7B STA t_nrecords
.L985D
985D 18      CLC
985E AD 5B 7B LDA &7B5B
9861 69 05    ADC #&05
9863 8D 5B 7B STA &7B5B
9866 C9 4A    CMP #&4A ; 'J'
9868 B0 06    BCS L9870
986A EE 5C 7B INC t_nrecords
986D 4C 5D 98 JMP L985D
.L9870
9870 38      SEC
9871 AD 5B 7B LDA &7B5B
9874 E9 05    SBC #&05
9876 8D 5B 7B STA &7B5B
.L9879
9879 A9 09    LDA #&09 ; header with clock, all magazines, 30 rows per page
987B 8D 00 7A STA t_osw
987E A9 01    LDA #&01
9880 8D 01 7A STA t_osw+1
9883 20 38 A3 JSR LA338
9886 A9 0B    LDA #&0B
9888 8D 00 7A STA t_osw
988B A9 01    LDA #&01

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988D 8D 01 7A STA t_osw+1
9890 20 38 A3 JSR LA338
9893 A9 07 LDA #&07
9895 8D 00 7A STA t_osw
9898 A9 1E LDA #&1E
989A 8D 01 7A STA t_osw+1
989D 20 38 A3 JSR LA338
98A0 20 BC 9D JSR L9DBC
98A3 A9 82 LDA #&82 ; page number in green: searching
98A5 20 5E 9F JSR L9F5E
98A8 A9 00 LDA #&00
98AA 8D 2B 7A STA t_hold
98AD 8D 54 7A STA &7A54
98B0 A9 FF LDA #&FF
98B2 8D 00 4A STA page_buf
98B5 8D 55 7A STA t_prev_state
98B8 A9 87 LDA #&87
98BA 8D 5E 7B STA t_hdr_colour
98BD A9 04 LDA #&04
98BF 8D 2A 7A STA t_entry_len
98C2 A9 03 LDA #&03
98C4 8D 25 7A STA t_ndigits
98C7 A9 31 LDA #&31
98C9 8D 26 7A STA t_digits
98CC A9 30 LDA #&30
98CE 8D 27 7A STA t_digits+1
98D1 8D 28 7A STA t_digits+2
98D4 A9 0E LDA #&0E
98D6 8D 00 7A STA t_osw
98D9 20 38 A3 JSR LA338
98DC A0 09 LDY #&09 ; record 0 := page 100 into &4A00

.L98DE
98DE B9 06 98 LDA &9806,Y
98E1 99 00 7A STA t_osw,Y
98E4 88 DEY
98E5 10 F7 BPL L98DE
98E7 20 38 A3 JSR LA338
98EA A9 03 LDA #&03
98EC 8D 00 7A STA t_osw
98EF A9 00 LDA #&00
98F1 8D 01 7A STA t_osw+1
98F4 8D 02 7A STA t_osw+2
98F7 20 38 A3 JSR LA338
98FA A9 00 LDA #&00
98FC 8D 00 7A STA t_osw
98FF 20 38 A3 JSR LA338
;-----
; MAIN LOOP: if the current page has arrived, show it (header turns white; or
; red/yellow from the CRC check), fetch its links, update the previous-page store.
; Before any page is chosen, the initial page from the TSDP is requested.
; Then poll the keyboard (with ESCAPE as a key) and dispatch.
;-----
.main_loop
9902 A9 0B LDA #&0B
9904 A2 00 LDX #&00
9906 20 B9 A0 JSR LA0B9
9909 A9 02 LDA #&02
990B 8D 00 7A STA t_osw
990E A9 00 LDA #&00
9910 8D 01 7A STA t_osw+1
9913 20 38 A3 JSR LA338
9916 AD 02 7A LDA t_osw+2 ; RECEIVED?
9919 29 02 AND #&02
991B F0 3D BEQ L995A
991D A9 83 LDA #&83
991F 8D 5E 7B STA t_hdr_colour
9922 A9 87 LDA #&87
9924 20 5E 9F JSR L9F5E
9927 A9 17 LDA #&17 ; check the CRC: white ok, red failed, yellow no check word
9929 8D 12 7A STA t_osw2
992C A9 00 LDA #&00
992E 8D 13 7A STA t_osw2+1
9931 A9 4A LDA #&4A
9933 8D 14 7A STA t_osw2+2
9936 20 3F A3 JSR LA33F
9939 AD 15 7A LDA t_osw2+3
993C 30 04 BMI L9942
993E D0 09 BNE L9949
9940 F0 0C BEQ L994E

.L9942
9942 A9 81 LDA #&81
9944 8D 5E 7B STA t_hdr_colour
9947 D0 05 BNE L994E

.L9949
9949 A9 87 LDA #&87
994B 8D 5E 7B STA t_hdr_colour

.L994E
994E 20 8B 9C JSR L9C8B
9951 20 9C AA JSR LAA9C
9954 20 50 A1 JSR show_page

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9957 20 C1 A6 JSR LA6C1
.L995A
995A AD 00 4A LDA page_buf ; waiting for the initial page?
995D 30 03 BMI L9962
995F 4C E5 99 JMP L99E5
.L9962
9962 A9 0F LDA #&0F
9964 8D 12 7A STA t_osw2
9967 20 3F A3 JSR LA33F
996A AD 14 7A LDA t_osw2+2
996D C9 FF CMP #&FF
996F F0 74 BEQ L99E5
9971 AD 13 7A LDA t_osw2+1
9974 30 6F BMI L99E5
9976 A9 0B LDA #&0B
9978 8D 12 7A STA t_osw2
997B 20 3F A3 JSR LA33F
997E A9 04 LDA #&04
9980 8D 00 7A STA t_osw
9983 A9 00 LDA #&00
9985 8D 01 7A STA t_osw+1
9988 20 38 A3 JSR LA338
998B AD 13 7A LDA t_osw2+1
998E 8D 02 7A STA t_osw+2
9991 18 CLC
9992 69 30 ADC #&30
9994 C9 30 CMP #&30 ; '0'
9996 D0 02 BNE L999A
9998 A9 38 LDA #&38
.L999A
999A 8D 26 7A STA t_digits
999D AD 14 7A LDA t_osw2+2
99A0 8D 03 7A STA t_osw+3
99A3 29 F0 AND #&F0
99A5 4A LSR A
99A6 4A LSR A
99A7 4A LSR A
99A8 4A LSR A
99A9 18 CLC
99AA 69 30 ADC #&30
99AC 8D 27 7A STA t_digits+1
99AF AD 14 7A LDA t_osw2+2
99B2 29 0F AND #&0F
99B4 18 CLC
99B5 69 30 ADC #&30
99B7 8D 28 7A STA t_digits+2
99BA A9 3F LDA #&3F
99BC 8D 04 7A STA t_osw+4
99BF A9 7F LDA #&7F
99C1 8D 05 7A STA t_osw+5
99C4 A9 05 LDA #&05
99C6 8D 00 7A STA t_osw
99C9 20 38 A3 JSR LA338
99CC AD 02 7A LDA t_osw+2
99CF 8D 01 7A STA t_osw+1
99D2 8D 00 4A STA page_buf
99D5 A9 0B LDA #&0B
99D7 8D 00 7A STA t_osw
99DA 20 38 A3 JSR LA338
99DD 20 3C 9D JSR L9D3C
99E0 A9 03 LDA #&03
99E2 8D 25 7A STA t_ndigits
.L99E5
99E5 A9 80 LDA #&80 ; key waiting?
99E7 A2 FF LDX #&FF
99E9 A0 FF LDY #&FF
99EB 20 F4 FF JSR OSBYTE
99EE E0 00 CPX #&00
99F0 D0 03 BNE L99F5
99F2 4C 02 99 JMP main_loop
.L99F5
99F5 20 E0 FF JSR OSRDCH
99F8 48 PHA
99F9 A9 0F LDA #&0F
99FB A2 01 LDX #&01
99FD 20 B9 A0 JSR LA0B9
9A00 68 PLA
9A01 C9 A0 CMP #&A0 ; function keys &A0-&B9 via the table at &9D08
9A03 90 17 BCC L9A1C
9A05 C9 BA CMP #&BA
9A07 B0 13 BCS L9A1C
9A09 38 SEC
9A0A E9 A0 SBC #&A0
9A0C 0A ASL A
9A0D A8 TAY
9A0E B9 08 9D LDA key_table,Y
9A11 85 A8 STA wk
9A13 C8 INY
9A14 B9 08 9D LDA key_table,Y
9A17 85 A9 STA wk+1

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9A19 6C A8 00 JMP (wk)
.L9A1C
9A1C C9 30 CMP #&30 ; digits enter a page or sub-page number
9A1E 90 04 BCC L9A24
9A20 C9 3A CMP #&3A ; ':'
9A22 90 03 BCC L9A27
.L9A24
9A24 4C BC 9B JMP L9BBC
;-----
; A DIGIT: add it to the page (3 digits) or sub-page (4 digits) being typed. When
; complete, request it: if a kept or link record already holds it, show that copy at
; once; otherwise point record 0 at it.
;-----
.L9A27
9A27 48 PHA
9A28 AD 00 4A LDA page_buf
9A2B 29 7F AND #&7F
9A2D 8D 00 4A STA page_buf
9A30 A9 87 LDA #&87
9A32 8D 5E 7B STA t_hdr_colour
9A35 AD 25 7A LDA t_ndigits
9A38 18 CLC
9A39 69 01 ADC #&01
9A3B CD 2A 7A CMP t_entry_len
9A3E D0 02 BNE L9A42
9A40 A9 01 LDA #&01
.L9A42
9A42 8D 25 7A STA t_ndigits
9A45 A8 TAY
9A46 68 PLA
9A47 99 25 7A STA t_ndigits,Y
9A4A 20 8B 9C JSR L9C8B
9A4D AE 25 7A LDX t_ndigits
9A50 E8 INX
9A51 EC 2A 7A CPX t_entry_len
9A54 F0 03 BEQ L9A59
9A56 4C B9 9B JMP L9BB9
.L9A59
9A59 A9 82 LDA #&82
9A5B 20 5E 9F JSR L9F5E
9A5E AD 2A 7A LDA t_entry_len
9A61 C9 04 CMP #&04
9A63 F0 03 BEQ L9A68
9A65 4C 49 9C JMP L9C49
.L9A68
9A68 A9 0B LDA #&0B ; page number complete
9A6A 8D 00 7A STA t_osw
9A6D AD 26 7A LDA t_digits
9A70 38 SEC
9A71 E9 30 SBC #&30
9A73 29 07 AND #&07
9A75 8D 01 7A STA t_osw+1
9A78 20 38 A3 JSR LA338
9A7B A9 09 LDA #&09
9A7D 8D 00 7A STA t_osw
9A80 A9 01 LDA #&01
9A82 8D 01 7A STA t_osw+1
9A85 20 38 A3 JSR LA338
9A88 A9 0C LDA #&0C
9A8A 8D 00 7A STA t_osw
9A8D 20 38 A3 JSR LA338
9A90 A9 10 LDA #&10
9A92 8D 00 7A STA t_osw
9A95 20 38 A3 JSR LA338
9A98 A9 00 LDA #&00
9A9A 8D 01 7A STA t_osw+1
9A9D AD 26 7A LDA t_digits
9AA0 38 SEC
9AA1 E9 30 SBC #&30
9AA3 29 07 AND #&07
9AA5 8D 02 7A STA t_osw+2
9AA8 8D 14 7A STA t_osw+2+2
9AAB AD 27 7A LDA t_digits+1
9AAE 38 SEC
9AAF E9 30 SBC #&30
9AB1 0A ASL A
9AB2 0A ASL A
9AB3 0A ASL A
9AB4 0A ASL A
9AB5 8D 03 7A STA t_osw+3
9AB8 AD 28 7A LDA t_digits+2
9ABB 38 SEC
9ABC E9 30 SBC #&30
9ABE 0D 03 7A ORA t_osw+3
9AC1 8D 03 7A STA t_osw+3
9AC4 18 CLC
9AC5 69 01 ADC #&01
9AC7 8D 15 7A STA t_osw+3+3
9ACA A9 3F LDA #&3F
9ACC 8D 04 7A STA t_osw+4

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9ACF 8D 16 7A STA &7A16
9AD2 A9 7F LDA #&7F
9AD4 8D 05 7A STA t_osw+5
9AD7 8D 17 7A STA &7A17
9ADA A9 05 LDA #&05
9ADC 8D 12 7A STA t_osw2
9ADF A9 07 LDA #&07
9AE1 8D 13 7A STA t_osw2+1
9AE4 A9 00 LDA #&00
9AE6 8D 18 7A STA &7A18
9AE9 A9 4F LDA #&4F
9AEB 8D 19 7A STA &7A19
9AEE A9 FF LDA #&FF
9AF0 8D 1A 7A STA &7A1A
9AF3 8D 1B 7A STA &7A1B
9AF6 20 3F A3 JSR LA33F
.L9AF9
9AF9 A9 04 LDA #&04
9AFB 8D 12 7A STA t_osw2
9AFE A9 01 LDA #&01
9B00 8D 13 7A STA t_osw2+1
.L9B03
9B03 20 3F A3 JSR LA33F
9B06 A0 02 LDY #&02
; is it in one of the 20 records already?
.L9B08
9B08 B9 00 7A LDA t_osw,Y
9B0B D9 12 7A CMP t_osw2,Y
9B0E D0 5A BNE L9B6A
9B10 C8 INY
9B11 C0 06 CPY #&06
9B13 D0 F3 BNE L9B08
9B15 A9 02 LDA #&02
9B17 8D 12 7A STA t_osw2
9B1A 20 3F A3 JSR LA33F
9B1D AD 14 7A LDA t_osw2+2
9B20 29 02 AND #&02
9B22 F0 50 BEQ L9B74
9B24 AD 14 7A LDA t_osw2+2
9B27 48 PHA
9B28 A9 04 LDA #&04
9B2A 8D 12 7A STA t_osw2
9B2D 20 3F A3 JSR LA33F
9B30 AD 18 7A LDA &7A18
9B33 85 82 STA &82
9B35 AD 19 7A LDA &7A19
9B38 85 83 STA &83
9B3A A9 00 LDA #&00
9B3C 85 84 STA &84
9B3E A9 4A LDA #&4A
9B40 85 85 STA &85
9B42 A2 00 LDX #&00
; copy that record's page into the display buffer
.L9B44
9B44 A0 00 LDY #&00
.L9B46
9B46 B1 82 LDA (&82),Y
9B48 91 84 STA (&84),Y
9B4A C8 INY
9B4B D0 F9 BNE L9B46
9B4D E6 83 INC &83
9B4F E6 85 INC &85
9B51 E8 INX
9B52 E0 05 CPX #&05
9B54 D0 EE BNE L9B44
9B56 68 PLA
9B57 8D 14 7A STA t_osw2+2
9B5A A9 00 LDA #&00
9B5C 8D 13 7A STA t_osw2+1
9B5F A9 03 LDA #&03
9B61 8D 12 7A STA t_osw2
9B64 20 3F A3 JSR LA33F
9B67 4C 84 9B JMP L9B84
.L9B6A
9B6A EE 13 7A INC t_osw2+1
9B6D AD 13 7A LDA t_osw2+1
9B70 C9 14 CMP #&14
9B72 D0 8F BNE L9B03
.L9B74
9B74 A9 03 LDA #&03
9B76 8D 12 7A STA t_osw2
9B79 A9 00 LDA #&00
9B7B 8D 13 7A STA t_osw2+1
9B7E 8D 14 7A STA t_osw2+2
9B81 20 3F A3 JSR LA33F
.L9B84
9B84 A9 10 LDA #&10
9B86 8D 12 7A STA t_osw2
9B89 20 3F A3 JSR LA33F
9B8C A9 00 LDA #&00
9B8E 8D 06 7A STA t_osw+6
9B91 A9 4A LDA #&4A

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9B93 8D 07 7A STA t_osw+7
9B96 A9 FF LDA #&FF
9B98 8D 08 7A STA t_osw+8
9B9B 8D 09 7A STA t_osw+9
9B9E A9 05 LDA #&05
9BA0 8D 00 7A STA t_osw
9BA3 20 38 A3 JSR LA338
9BA6 20 3C 9D JSR L9D3C
9BA9 A9 0D LDA #&0D
9BAB 8D 00 7A STA t_osw
9BAE 20 38 A3 JSR LA338
9BB1 A9 00 LDA #&00
9BB3 8D 54 7A STA &A54
9BB6 8D 2B 7A STA t_hold

.L9BB9
9BB9 4C 02 99 JMP main_loop
;-----
; Other keys: DELETE, TAB (previous page), "*" (OS command), SPACE, COPY (keep /
; SHIFT-COPY release all), RETURN (status).
;-----
.L9BBC
9BBC C9 7F CMP #&7F
9BBE D0 11 BNE L9BD1
9BC0 AD 25 7A LDA t_ndigits
9BC3 F0 0C BEQ L9BD1
9BC5 38 SEC
9BC6 E9 01 SBC #&01
9BC8 8D 25 7A STA t_ndigits
9BCB 20 8B 9C JSR L9C8B
9BCE 4C 02 99 JMP main_loop

.L9BD1
9BD1 C9 09 CMP #&09 ; TAB
9BD3 D0 0E BNE L9BE3
9BD5 20 38 A7 JSR LA738
9BD8 AD 55 7A LDA t_prev_state
9BD0 49 01 EOR #&01
9BDD 8D 55 7A STA t_prev_state
9BE0 4C 02 99 JMP main_loop

.L9BE3
9BE3 C9 2A CMP #&2A ; *
9BE5 D0 03 BNE L9BEA
9BE7 4C 76 A7 JMP LA776

.L9BEA
9BEA C9 20 CMP #&20
9BEC D0 03 BNE L9BF1
9BEE 4C 23 A8 JMP LA823

.L9BF1
9BF1 C9 87 CMP #&87 ; COPY: is SHIFT held (INKEY -1)?
9BF3 D0 3A BNE L9C2F
9BF5 A9 81 LDA #&81
9BF7 A2 FF LDX #&FF
9BF9 A0 FF LDY #&FF
9BFB 20 F4 FF JSR OSBYTE ; OSBYTE &81: read key with time limit
9BFE 86 77 STX &77
9C00 98 TYA
9C01 25 77 AND &77
9C03 C9 FF CMP #&FF
9C05 D0 25 BNE L9C2C
9C07 A9 08 LDA #&08 ; SHIFT-COPY: release records 8-20 ("All Released")
9C09 85 77 STA &77
9C0B A9 14 LDA #&14
9C0D 85 78 STA &78
9C0F 20 52 9D JSR L9D52
9C12 A2 19 LDX #&19
9C14 A0 9C LDY #&9C
9C16 4C 21 AA JMP LAA21
9C19 .db 84 9D 86 41 6C 6C 20 52 65 6C 65 61 73 65 64 20 ; ...All Released
9C29 .db 20 9C 00 ; ..

.L9C2C
9C2C 4C EC A8 JMP LA8EC

.L9C2F
9C2F C9 0D CMP #&0D
9C31 D0 03 BNE L9C36
9C33 4C 28 AB JMP LAB28

.L9C36
9C36 4C 02 99 JMP main_loop
;-----
; Unused keys: issue OSWORD &7A with code &A0 and the key number. ATS doesn't claim codes
; above 27, so this does nothing unless another ROM answers it.
;-----
.L9C39
9C39 98 TYA
9C3A 4A LSR A
9C3B 8D 01 7A STA t_osw+1
9C3E A9 A0 LDA #&A0
9C40 8D 00 7A STA t_osw
9C43 20 38 A3 JSR LA338
9C46 4C 02 99 JMP main_loop
; Sub-page number complete: request page + sub-page.
.L9C49

```

```

9C49 A9 04 LDA #04
9C4B 8D 00 7A STA t_osw
9C4E A9 00 LDA #00
9C50 8D 01 7A STA t_osw+1
9C53 20 38 A3 JSR LA338
9C56 AD 26 7A LDA t_digits
9C59 38 SEC
9C5A E9 30 SBC #30
9C5C 0A ASL A
9C5D 0A ASL A
9C5E 0A ASL A
9C5F 0A ASL A
9C60 8D 04 7A STA t_osw+4
9C63 AD 27 7A LDA t_digits+1
9C66 38 SEC
9C67 E9 30 SBC #30
9C69 0D 04 7A ORA t_osw+4
9C6C 8D 04 7A STA t_osw+4
9C6F AD 28 7A LDA t_digits+2
9C72 38 SEC
9C73 E9 30 SBC #30
9C75 0A ASL A
9C76 0A ASL A
9C77 0A ASL A
9C78 0A ASL A
9C79 8D 05 7A STA t_osw+5
9C7C AD 29 7A LDA t_digits+3
9C7F 38 SEC
9C80 E9 30 SBC #30
9C82 0D 05 7A ORA t_osw+5
9C85 8D 05 7A STA t_osw+5
9C88 4C F9 9A JMP L9AF9
;
; -----
; Show the number being typed at the top left: "P123" or "S0012".
; -----
;
.L9C8B
9C8B AD 2A 7A LDA t_entry_len
9C8E C9 04 CMP #04
9C90 F0 0E BEQ L9CA0
9C92 A9 06 LDA #06
9C94 20 1A BB JSR print_inline_count
9C97 .str '\x1f \x00\x83S '
9C9D 4C B2 9C JMP L9CB2
.L9CA0
9CA0 A9 02 LDA #02
9CA2 20 1A BB JSR print_inline_count
9CA5 .str '\x1e '
9CA7 AD 5E 7B LDA t_hdr_colour
9CAA 20 EE FF JSR OSWRCH
9CAD A9 50 LDA #50
9CAF 20 EE FF JSR OSWRCH ; print 'P'
.L9CB2
9CB2 A2 00 LDX #00
9CB4 AD 25 7A LDA t_ndigits
9CB7 F0 1C BEQ L9CD5
.L9CB9
9CB9 BD 26 7A LDA t_digits,X
9CBC 20 EE FF JSR OSWRCH
9CBF E0 01 CPX #01
9CC1 D0 0C BNE L9CCF
9CC3 AD 2A 7A LDA t_entry_len
9CC6 C9 05 CMP #05
9CC8 D0 05 BNE L9CCF
9CCA A9 3A LDA #3A
9CCC 20 EE FF JSR OSWRCH ; print ':'
.L9CCF
9CCF E8 INX
9CD0 EC 25 7A CPX t_ndigits
9CD3 D0 E4 BNE L9CB9
.L9CD5
9CD5 38 SEC
9CD6 AD 2A 7A LDA t_entry_len
9CD9 ED 25 7A SBC t_ndigits
9CDC 38 SEC
9CDD E9 01 SBC #01
9CDF F0 1A BEQ L9CFB
9CE1 A8 TAY
.L9CE2
9CE2 A9 5F LDA #5F
9CE4 20 EE FF JSR OSWRCH ; print '_'
9CE7 E0 01 CPX #01
9CE9 D0 0C BNE L9CF7
9CEB AD 2A 7A LDA t_entry_len
9CEE C9 05 CMP #05
9CF0 D0 05 BNE L9CF7
9CF2 A9 3A LDA #3A
9CF4 20 EE FF JSR OSWRCH ; print ':'
.L9CF7
9CF7 E8 INX
9CF8 88 DEY

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```

    9CF9 D0 E7    BNE L9CE2
.L9CFB
    9CFB AD 2A 7A  LDA t_entry_len
    9CFE C9 05    CMP #&05
    9D00 F0 05    BEQ L9D07
    9D02 A9 20    LDA #&20
    9D04 20 EE FF  JSR OSWRCH                ; print ' '
.L9D07
    9D07 60      RTS
;-----
; Terminal key table for codes &A0-&B9:
;  &A0 f0 index  &A1-&A4 f1-f4 coloured links  &A5 f5 Telesoftware
;  &A6 f6 reveal &A7 f7 hold  &A8 f8 sub-page  &A9 f9 exit
;  &B0 SHIFT f0 tune  &B1-&B4 SHIFT f1-f4 channels 1-4
;  &B6 SHIFT f6 load page  &B7 SHIFT f7 save page
;-----
.key_table
9D08 .db 27 A5 D0 A4 D9 A4 F3 A4 0D A5 A3 AC 2F 9F 07 9E ; '...../...
9D18 .db 70 9E B6 A2 02 99 02 99 02 99 02 99 02 99 ; p.....
9D28 .db 8E AA 89 9D 8D 9D 91 9D 95 9D 39 9C 96 9F 6B 9F ; .....9...k.
9D38 .db 39 9C 39 9C ; 9.9.
; Clear the link records 1-6.
.L9D3C
9D3C A9 01    LDA #&01
9D3E 85 77    STA &77
9D40 A9 07    LDA #&07
9D42 85 78    STA &78
9D44 A9 01    LDA #&01
9D46 20 52 9D JSR L9D52
9D49 A9 00    LDA #&00
9D4B 8D 54 7A STA &7A54
9D4E 20 45 A1 JSR LA145
9D51 60      RTS
; Clear records &77 to &78-1.
.L9D52
9D52 A5 77    LDA &77
9D54 8D 13 7A STA t_osw2+1
9D57 8D 01 7A STA t_osw+1
9D5A A9 FF    LDA #&FF
9D5C A2 00    LDX #&00
.L9D5E
9D5E 9D 02 7A STA t_osw+2,X
9D61 E8      INX
9D62 E0 0A    CPX #&0A
9D64 D0 F8    BNE L9D5E
9D66 A9 03    LDA #&03
9D68 8D 12 7A STA t_osw2
9D6B A9 00    LDA #&00
9D6D 8D 14 7A STA t_osw2+2
9D70 A9 05    LDA #&05
9D72 8D 00 7A STA t_osw
.L9D75
9D75 20 38 A3 JSR LA338
9D78 20 3F A3 JSR LA33F
9D7B EE 01 7A INC t_osw+1
9D7E EE 13 7A INC t_osw2+1
9D81 AD 01 7A LDA t_osw+1
9D84 C5 78    CMP &78
9D86 D0 ED    BNE L9D75
9D88 60      RTS
;-----
; SHIFT f1-f4: change channel 1-4 and restart (kept pages cleared).
;-----
.L9D89
9D89 A9 00    LDA #&00
9D8B F0 0A    BEQ L9D97
.L9D8D
9D8D A9 01    LDA #&01
9D8F D0 06    BNE L9D97
.L9D91
9D91 A9 02    LDA #&02
9D93 D0 02    BNE L9D97
.L9D95
9D95 A9 03    LDA #&03
.L9D97
9D97 8D 01 7A STA t_osw+1
9D9A A9 01    LDA #&01
9D9C 8D 00 7A STA t_osw
9D9F 20 38 A3 JSR LA338
9DA2 4C 79 98 JMP L9879
; Channel names: BBC1 BBC2 ITV Ch4/S4C.
9DA5 .db 00 42 42 43 31 00 42 42 43 32 00 49 54 56 00 43 ; .BBC1.BBC2.ITV.C
9DB5 .db 68 34 2F 53 34 43 00 ; h4/S4C.
; Print the channel name at the top left.
.L9DBC
9DBC A9 1E    LDA #&1E
9DBE 20 EE FF JSR OSWRCH                ; VDU 30: cursor home
9DC1 A9 20    LDA #&20
9DC3 A2 07    LDX #&07
.L9DC5

```

```

9DC5 20 EE FF JSR OSWRCH          ; print ' '
9DC8 CA      DEX
9DC9 D0 FA    BNE L9DC5
9DCB A9 1E    LDA #&1E
9DCD 20 EE FF JSR OSWRCH          ; VDU 30: cursor home
.L9DD0
9DD0 A5 77    LDA &77
9DD2 48      PHA
9DD3 A5 78    LDA &78
9DD5 48      PHA
9DD6 A9 00    LDA #&00
9DD8 85 77    STA &77
9DDA A2 77    LDX #&77
9DDC A0 00    LDY #&00
9DDE A9 7A    LDA #&7A
9DE0 20 F1 FF JSR OSWORD          ; OSWORD &7A: Teletext system call
9DE3 A5 78    LDA &78
9DE5 AA      TAX
9DE6 E8      INX
9DE7 A0 00    LDY #&00
.L9DE9
9DE9 B9 A5 9D LDA &9DA5,Y
9DEC F0 03    BEQ L9DF1
9DEE C8      INY
9DEF D0 F8    BNE L9DE9
.L9DF1
9DF1 C8      INY
9DF2 CA      DEX
9DF3 D0 F4    BNE L9DE9
.L9DF5
9DF5 B9 A5 9D LDA &9DA5,Y
9DF8 20 EE FF JSR OSWRCH
9DFB C8      INY
9DFC C9 00    CMP #&00
9DFE D0 F5    BNE L9DF5
9E00 68      PLA
9E01 85 78    STA &78
9E03 68      PLA
9E04 85 77    STA &77
9E06 60      RTS
;-----
; f7: HOLD / RELEASE. Hold sets the record's HOLD flag and replaces the clock with
; "HOLD"; release clears it and shows the clock again.
;-----
.L9E07
9E07 A9 02    LDA #&02
9E09 8D 00 7A STA t_osw
9E0C A9 00    LDA #&00
9E0E 8D 01 7A STA t_osw+1
9E11 20 38 A3 JSR LA338
9E14 AD 02 7A LDA t_osw+2
9E17 29 01    AND #&01
9E19 D0 3D    BNE L9E58
9E1B 20 E3 9E JSR L9EE3
9E1E AD 02 7A LDA t_osw+2
9E21 09 01    ORA #&01
9E23 8D 02 7A STA t_osw+2
9E26 A9 03    LDA #&03
9E28 8D 00 7A STA t_osw
9E2B 20 38 A3 JSR LA338
9E2E A9 09    LDA #&09
9E30 8D 00 7A STA t_osw
9E33 A9 FF    LDA #&FF
9E35 8D 01 7A STA t_osw+1
9E38 20 38 A3 JSR LA338
9E3B A9 0B    LDA #&0B
9E3D 20 1A BB JSR print_inline_count
9E40 .str '\x1f \x00 \x82HOLD '
9E4B A9 01    LDA #&01
9E4D 8D 2B 7A STA t_hold
9E50 A9 04    LDA #&04
9E52 8D 2A 7A STA t_entry_len
9E55 4C 02 99 JMP main_loop
.L9E58
9E58 AD 02 7A LDA t_osw+2
9E5B 29 FE    AND #&FE
9E5D 8D 02 7A STA t_osw+2
9E60 A9 03    LDA #&03
9E62 8D 00 7A STA t_osw
9E65 20 38 A3 JSR LA338
9E68 A9 00    LDA #&00
9E6A 8D 2B 7A STA t_hold
9E6D 4C B4 9E JMP L9EB4
; f8: SUB-PAGE ENTRY ("S----") on/off.
.L9E70
9E70 AD 2B 7A LDA t_hold
9E73 F0 03    BEQ L9E78
9E75 4C 02 99 JMP main_loop
.L9E78
9E78 AD 2A 7A LDA t_entry_len

```



```

9E7B C9 04    CMP #&04
9E7D D0 2B    BNE L9EAA
9E7F A9 05    LDA #&05
9E81 8D 2A 7A STA t_entry_len
9E84 A9 04    LDA #&04
9E86 8D 25 7A STA t_ndigits
9E89 A9 30    LDA #&30
9E8B 8D 26 7A STA t_digits
9E8E 8D 27 7A STA t_digits+1
9E91 8D 28 7A STA t_digits+2
9E94 8D 29 7A STA t_digits+3
9E97 A9 09    LDA #&09
9E99 8D 00 7A STA t_osw
9E9C A9 FF    LDA #&FF
9E9E 8D 01 7A STA t_osw+1
9EA1 20 38 A3 JSR LA338
9EA4 20 88 9C JSR L9C8B
9EA7 4C 02 99 JMP main_loop

.L9EAA
9EAA A9 04    LDA #&04
9EAC 8D 2A 7A STA t_entry_len
9EAF A9 00    LDA #&00
9EB1 8D 25 7A STA t_ndigits

.L9EB4
9EB4 A9 09    LDA #&09
9EB6 8D 00 7A STA t_osw
9EB9 A9 01    LDA #&01
9EBB 8D 01 7A STA t_osw+1
9EBE 20 38 A3 JSR LA338
9EC1 A9 04    LDA #&04
9EC3 8D 00 7A STA t_osw
9EC6 A9 00    LDA #&00
9EC8 8D 01 7A STA t_osw+1
9ECB 20 38 A3 JSR LA338
9ECE A9 3F    LDA #&3F
9ED0 8D 04 7A STA t_osw+4
9ED3 A9 7F    LDA #&7F
9ED5 8D 05 7A STA t_osw+5
9ED8 A9 05    LDA #&05
9EDA 8D 00 7A STA t_osw
9EDD 20 38 A3 JSR LA338
9EE0 4C 02 99 JMP main_loop

;-----
; Set the typed page number from the page in the display buffer.
;-----

.L9EE3
9EE3 A9 00    LDA #&00
9EE5 85 82    STA &82
9EE7 A9 4A    LDA #&4A
9EE9 85 83    STA &83
9EEB A2 00    LDY #&00
9EED 20 45 A2 JSR find_row
9EF0 38      SEC
9EF1 A5 84    LDA &84
9EF3 E9 02    SBC #&02
9EF5 85 84    STA &84
9EF7 A5 85    LDA &85
9EF9 E9 00    SBC #&00
9EFB 85 85    STA &85
9EFD A0 00    LDY #&00                ; workspace +&00: grabber records (8 bytes each)
9EFF B1 84    LDA (&84),Y
9F01 18      CLC
9F02 69 30    ADC #&30
9F04 C9 30    CMP #&30                ; '0'
9F06 D0 02    BNE L9F0A
9F08 A9 38    LDA #&38

.L9F0A
9F0A 8D 26 7A STA t_digits
9F0D C8      INY
9F0E C8      INY
9F0F B1 84    LDA (&84),Y
9F11 29 F0    AND #&F0
9F13 4A      LSR A
9F14 4A      LSR A
9F15 4A      LSR A
9F16 4A      LSR A
9F17 18      CLC
9F18 69 30    ADC #&30
9F1A 8D 27 7A STA t_digits+1
9F1D B1 84    LDA (&84),Y
9F1F 29 0F    AND #&0F
9F21 18      CLC
9F22 69 30    ADC #&30
9F24 8D 28 7A STA t_digits+2
9F27 A9 03    LDA #&03
9F29 8D 25 7A STA t_ndigits
9F2C 4C 8B 9C JMP L9C8B

;-----
; f6: REVEAL / CONCEAL by swapping the conceal code &98 and &80 on the screen, then
; HOLD. (Any &80 bytes already on the screen also become &98 when concealing again.)

```

```

;-----
.L9F2F
9F2F A9 00 LDA #&00
9F31 85 82 STA &82
9F33 A9 7C LDA #&7C
9F35 85 83 STA &83
.L9F37
9F37 A0 00 LDY #&00
.L9F39
9F39 B1 82 LDA (&82),Y
9F3B 09 80 ORA #&80
9F3D C9 98 CMP #&98
9F3F D0 07 BNE L9F48
9F41 A9 80 LDA #&80
9F43 91 82 STA (&82),Y
9F45 4C 50 9F JMP L9F50
.L9F48
9F48 C9 80 CMP #&80
9F4A D0 04 BNE L9F50
9F4C A9 98 LDA #&98
9F4E 91 82 STA (&82),Y
.L9F50
9F50 C8 INY
9F51 D0 E6 BNE L9F39
9F53 E6 83 INC &83
9F55 A5 83 LDA &83
9F57 C9 80 CMP #&80
9F59 D0 DC BNE L9F37
9F5B 4C 07 9E JMP L9E07
;-----
; Set the colour of the page number at the top (green searching, white found).
;-----
.L9F5E
9F5E 48 PHA
9F5F A9 03 LDA #&03
9F61 20 1A BB JSR print_inline_count
9F64 .str '\x1f\x07\x00'
9F67 68 PLA
9F68 4C EE FF JMP OSWRCH
;-----
; SHIFT f7: SAVE PAGE: save the MODE 7 screen to a file (OSFILE 0).
;-----
.L9F6B
9F6B 20 E3 9E JSR L9EE3
9F6E 20 45 A1 JSR LA145
9F71 A9 11 LDA #&11
9F73 20 1A BB JSR print_inline_count
9F76 .str '\x1a\x1f\x00\x18 Save Page : '
9F87 20 D6 9F JSR L9FD6
9F8A A9 00 LDA #&00
9F8C A2 00 LDX #&00
9F8E A0 7A LDY #&7A
9F90 20 DD FF JSR OSFILE
9F93 4C BE 9F JMP L9FBE
;-----
; SHIFT f6: LOAD PAGE onto the screen (OSFILE &FF), then "Press SPACE to continue".
;-----
.L9F96
9F96 20 45 A1 JSR LA145
9F99 A9 11 LDA #&11
9F9B 20 1A BB JSR print_inline_count
9F9E .str '\x1a\x1f\x00\x18 Load Page : '
9FAF 20 D6 9F JSR L9FD6
9FB2 A9 FF LDA #&FF
9FB4 A2 00 LDX #&00
9FB6 A0 7A LDY #&7A
9FB8 20 DD FF JSR OSFILE
9FBB 20 3B A0 JSR LA03B
.L9FBE
9FBE A9 09 LDA #&09
9FC0 8D 00 7A STA t_osw
9FC3 A9 01 LDA #&01
9FC5 8D 01 7A STA t_osw+1
9FC8 20 38 A3 JSR LA338
9FCB A9 1A LDA #&1A
9FCD 20 EE FF JSR OSWRCH ; VDU 26: restore default windows
9FD0 20 50 A1 JSR show_page
9FD3 4C 02 99 JMP main_loop
; Read a file name; keys back to normal while typing.
.L9FD6
9FD6 20 43 A8 JSR LA843
9FD9 A9 E5 LDA #&E5
9FDB A2 00 LDX #&00
9FDD 20 B9 A0 JSR LA0B9
9FE0 20 11 A0 JSR LA011
9FE3 20 F9 A0 JSR LA0F9
9FE6 AD 2D 7A LDA t_line
9FE9 C9 0D CMP #&0D
9FEB F0 D1 BEQ L9FBE
9FED A9 06 LDA #&06

```

```

9FEF 20 1A BB JSR print_inline_count
9FF2 .str "\x1c\x01\x18'\x18\x0c"
9FF8 A9 09 LDA #&09
9FFA 8D 00 7A STA t_osw
9FFD A9 00 LDA #&00
9FFF 8D 01 7A STA t_osw+1
A002 20 38 A3 JSR LA338
A005 A0 11 LDY #&11
.LA007
A007 B9 29 A0 LDA &A029,Y
A00A 99 00 7A STA t_osw,Y
A00D 88 DEY
A00E 10 F7 BPL LA007
A010 60 RTS
;
; -----
; Before file operations: *OPT 1,1 (OSBYTE &8B) on tape so that messages show, *OPT 1,0
; on other filing systems.
; -----
.LA011
A011 A9 00 LDA #&00
A013 A8 TAY
A014 20 DA FF JSR OSARGS
A017 C9 04 CMP #&04
A019 B0 09 BCS LA024
A01B A9 8B LDA #&8B
A01D A2 01 LDX #&01
A01F A0 01 LDY #&01
A021 4C F4 FF JMP OSBYTE ; OSBYTE &8B
.LA024
A024 A9 8B LDA #&8B
A026 4C B7 A0 JMP LA0B7
;
; -----
; OSFILE block template: name at &7A2D, the screen at &FFFF7C00.
; -----
A029 .db 2D 7A 00 7C FF FF 00 7C FF FF 00 7C FF FF C0 7F ; -z.|...|...|....
A039 .db FF FF ; ..
;
; -----
; "Press SPACE to continue" on the bottom row; waits for SPACE or "*".
; -----
.LA03B
A03B A0 00 LDY #&00
.LA03D
A03D B9 5E A0 LDA &A05E,Y
A040 99 C0 7F STA &7FC0,Y
A043 C8 INY
A044 C0 28 CPY #&28 ; '('
A046 D0 F5 BNE LA03D
.LA048
A048 20 E0 FF JSR OSRDCH
A04B C9 20 CMP #&20
A04D F0 04 BEQ LA053
A04F C9 2A CMP #&2A ; '*'
A051 D0 F5 BNE LA048
.LA053
A053 48 PHA
A054 A9 1A LDA #&1A
A056 20 EE FF JSR OSWRCH ; VDU 26: restore default windows
A059 20 86 A0 JSR LA086
A05C 68 PLA
A05D 60 RTS
A05E .db 97 9C AC AC AC AC AC AC 87 50 72 65 73 73 85 53 ; .....Press.S
A06E .db 50 41 43 45 87 74 6F 20 63 6F 6E 74 69 6E 75 65 ; PACE.to continue
A07E .db 97 9C AC AC AC AC AC AC ; .....
; Restore the typed number after a temporary change.
.LA086
A086 AD 2A 7A LDA t_entry_len
A089 48 PHA
A08A AD 25 7A LDA t_ndigits
A08D 48 PHA
A08E AD 26 7A LDA t_digits
A091 48 PHA
A092 AD 27 7A LDA t_digits+1
A095 48 PHA
A096 AD 28 7A LDA t_digits+2
A099 48 PHA
A09A A9 04 LDA #&04
A09C 8D 2A 7A STA t_entry_len
A09F 20 E3 9E JSR L9EE3
A0A2 68 PLA
A0A3 8D 28 7A STA t_digits+2
A0A6 68 PLA
A0A7 8D 27 7A STA t_digits+1
A0AA 68 PLA
A0AB 8D 26 7A STA t_digits
A0AE 68 PLA
A0AF 8D 25 7A STA t_ndigits
A0B2 68 PLA
A0B3 8D 2A 7A STA t_entry_len
A0B6 60 RTS
; OSBYTE A with X=1.

```

```

.LA0B7
    A0B7  A2 01    LDX #&01
; OSBYTE A with X, Y=0.
.LA0B9
    A0B9  A0 00    LDY #&00
    A0BB  4C F4 FF  JMP OSBYTE
;-----
; Set up the terminal: MODE 7, function keys &A0+, SHIFT keys &B0+, CTRL keys 1,
; cursor keys as codes, TAB = 9, ESCAPE as a key, flush input. Also sets BRKV.
;-----
.terminal_keys
    A0BE  20 43 A8  JSR LA843
    A0C1  A9 72    LDA #&72
    A0C3  20 B7 A0  JSR LA0B7
    A0C6  A9 02    LDA #&02
    A0C8  20 1A BB  JSR print_inline_count
    A0CB  .str '\x16\x07'
    A0CD  20 2A A8  JSR LA82A
    A0D0  A9 E1    LDA #&E1
    A0D2  A2 A0    LDX #&A0
    A0D4  20 B9 A0  JSR LA0B9
    A0D7  A9 E2    LDA #&E2
    A0D9  A2 B0    LDX #&B0
    A0DB  20 B9 A0  JSR LA0B9
    A0DE  A9 E3    LDA #&E3
    A0E0  20 B7 A0  JSR LA0B7
    A0E3  A9 04    LDA #&04
    A0E5  20 B7 A0  JSR LA0B7
    A0E8  A9 DB    LDA #&DB
    A0EA  A2 09    LDX #&09
    A0EC  20 B9 A0  JSR LA0B9
    A0EF  A9 E5    LDA #&E5
    A0F1  20 B7 A0  JSR LA0B7
    A0F4  A9 0F    LDA #&0F
    A0F6  4C B7 A0  JMP LA0B7
; Read a line (up to 28 characters) into t_line, with DELETE.
.LA0F9
    A0F9  20 26 A8  JSR LA826
    A0FC  A9 00    LDA #&00
    A0FE  8D 2C 7A  STA t_line_len
.LA101
    A101  20 E0 FF  JSR OSRDCH
    A104  C9 7F    CMP #&7F
    A106  D0 13    BNE LA11B
    A108  AD 2C 7A  LDA t_line_len
    A10B  F0 F4    BEQ LA101
    A10D  CE 2C 7A  DEC t_line_len
    A110  A9 03    LDA #&03
    A112  20 1A BB  JSR print_inline_count
    A115  .str '\x08 \x08'
    A118  4C 01 A1  JMP LA101
.LA11B
    A11B  C9 0D    CMP #&0D
    A11D  D0 0A    BNE LA129
    A11F  AC 2C 7A  LDY t_line_len
    A122  99 2D 7A  STA t_line,Y
    A125  20 2A A8  JSR LA82A
    A128  60      RTS
.LA129
    A129  48      PHA
    A12A  AD 2C 7A  LDA t_line_len
    A12D  C9 1C    CMP #&1C
    A12F  D0 04    BNE LA135
    A131  68      PLA
    A132  4C 01 A1  JMP LA101
.LA135
    A135  68      PLA
    A136  AC 2C 7A  LDY t_line_len
    A139  99 2D 7A  STA t_line,Y
    A13C  20 EE FF  JSR OSWRCH
    A13F  EE 2C 7A  INC t_line_len
    A142  4C 01 A1  JMP LA101
; Clear the bottom row of the screen.
.LA145
    A145  A0 27    LDY #&27
    A147  A9 A0    LDA #&A0
.LA149
    A149  99 C0 7F  STA &7FC0,Y
    A14C  88      DEY
    A14D  10 FA    BPL LA149
    A14F  60      RTS
;-----
; DRAW THE PAGE from &A00 onto the screen: page number white if it is the page asked
; for, else green; rows 1-23 found by row number; double height; then the link line.
;-----
.show_page
    A150  A9 04    LDA #&04
    A152  8D 00 7A  STA t_osw
    A155  A9 00    LDA #&00
    A157  8D 01 7A  STA t_osw+1

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A15A 20 38 A3 JSR LA338
A15D AD 02 7A LDA t_osw+2
A160 CD 00 4A CMP page_buf
A163 D0 0C BNE LA171
A165 AD 03 7A LDA t_osw+3
A168 CD 02 4A CMP page_buf+2
A16B D0 04 BNE LA171
A16D A9 87 LDA #87
A16F D0 02 BNE LA173
.LA171
A171 A9 82 LDA #82
.LA173
A173 20 5E 9F JSR L9F5E
A176 A9 00 LDA #00
A178 85 82 STA &82
A17A A9 4A LDA #4A
A17C 85 83 STA &83
A17E 20 C9 A1 JSR LA1C9
A181 A2 18 LDX #18
A183 20 45 A2 JSR find_row
A186 08 PHP
A187 A9 02 LDA #02
A189 8D 00 7A STA t_osw
A18C A9 00 LDA #00
A18E 8D 01 7A STA t_osw+1
A191 20 38 A3 JSR LA338
A194 29 20 AND #20
A196 F0 03 BEQ LA19B
A198 20 48 A3 JSR fetch_links
.LA19B
A19B 28 PLP
A19C 90 1A BCC LA1B8
A19E A9 02 LDA #02
A1A0 8D 00 7A STA t_osw
A1A3 A9 00 LDA #00
A1A5 8D 01 7A STA t_osw+1
A1A8 20 38 A3 JSR LA338
A1AB 29 20 AND #20
A1AD D0 06 BNE LA1B5
A1AF 20 45 A1 JSR LA145
A1B2 4C B8 A1 JMP LA1B8
.LA1B5
A1B5 20 3E A6 JSR link_line
.LA1B8
A1B8 A9 03 LDA #03
A1BA 8D 00 7A STA t_osw
A1BD A9 00 LDA #00
A1BF 8D 01 7A STA t_osw+1
A1C2 8D 02 7A STA t_osw+2
A1C5 20 38 A3 JSR LA338
A1C8 60 RTS
;-----
; Copy rows 1-23 of the page at (&82) to the screen, handling double height and C10
; (inhibit display) from the header.
;-----
.LA1C9
A1C9 A9 00 LDA #00
A1CB 85 86 STA &86
A1CD A9 7C LDA #7C
A1CF 85 87 STA &87
A1D1 A9 08 LDA #08
A1D3 8D 56 7A STA t_dbt_flag
A1D6 A0 00 LDY #00
A1D8 B1 82 LDA (&82),Y
A1DA D0 09 BNE LA1E5
A1DC A0 05 LDY #05
A1DE B1 82 LDA (&82),Y
A1E0 29 08 AND #08
A1E2 8D 56 7A STA t_dbt_flag
.LA1E5
A1E5 A2 01 LDX #01
.LA1E7
A1E7 20 45 A2 JSR find_row
A1EA 90 0F BCC LA1FB
A1EC E0 18 CPX #18
A1EE F0 41 BEQ LA231
A1F0 AD 56 7A LDA t_dbt_flag
A1F3 D0 06 BNE LA1FB
A1F5 20 37 A2 JSR LA237
A1F8 4C 31 A2 JMP LA231
.LA1FB
A1FB A9 00 LDA #00
A1FD 8D 24 7A STA &7A24
A200 20 37 A2 JSR LA237
A203 A0 00 LDY #00 ; workspace +00: grabber records (8 bytes each)
.LA205
A205 B1 84 LDA (&84),Y
A207 09 80 ORA #80
A209 91 86 STA (&86),Y
A20B C9 8D CMP #8D

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A20D D0 03 BNE LA212
A20F EE 24 7A INC &7A24
.LA212
A212 C8 INY
A213 C0 28 CPY #&28 ; '('
A215 D0 EE BNE LA205
A217 AD 24 7A LDA &7A24
A21A F0 15 BEQ LA231
A21C E0 18 CPX #&18
A21E F0 11 BEQ LA231
A220 20 37 A2 JSR LA237
A223 A0 00 LDY #&00 ; workspace +&00: grabber records (8 bytes each)
.LA225
A225 B1 84 LDA (&84),Y
A227 09 80 ORA #&80
A229 91 86 STA (&86),Y
A22B C8 INY
A22C C0 28 CPY #&28 ; '('
A22E D0 F5 BNE LA225
A230 E8 INX
.LA231
A231 E8 INX
A232 E0 19 CPX #&19
A234 90 B1 BCC LA1E7
A236 60 RTS
.LA237
A237 18 CLC
A238 A5 86 LDA &86
A23A 69 28 ADC #&28
A23C 85 86 STA &86
A23E A5 87 LDA &87
A240 69 00 ADC #&00
A242 85 87 STA &87
A244 60 RTS
;
; Find row X in the page at (&82) (42-byte records, up to 30). C clear and (&84) =
; its 40 data bytes if found.
;
.find_row
A245 8A TXA
A246 48 PHA
A247 A5 82 LDA &82
A249 85 84 STA &84
A24B A5 83 LDA &83
A24D 85 85 STA &85
A24F A2 00 LDX #&00
.LA251
A251 A0 01 LDY #&01
A253 68 PLA
A254 48 PHA
A255 D1 84 CMP (&84),Y
A257 F0 1E BEQ LA277
A259 18 CLC
A25A A5 84 LDA &84
A25C 69 2A ADC #&2A
A25E 85 84 STA &84
A260 A5 85 LDA &85
A262 69 00 ADC #&00
A264 85 85 STA &85
A266 E8 INX
A267 E0 1E CPX #&1E
A269 D0 E6 BNE LA251
A26B A9 8C LDA #&8C
A26D 85 84 STA &84
A26F A9 A2 LDA #&A2
A271 85 85 STA &85
A273 68 PLA
A274 AA TAX
A275 38 SEC
A276 60 RTS
.LA277
A277 8A TXA
A278 A8 TAY
A279 18 CLC
A27A A5 84 LDA &84
A27C 69 02 ADC #&02
A27E 85 84 STA &84
A280 A5 85 LDA &85
A282 69 00 ADC #&00
A284 85 85 STA &85
A286 68 PLA
A287 AA TAX
A288 18 CLC
A289 60 RTS
A28A .db 01 00 20 20 20 20 20 20 20 20 20 20 20 20 20 20 ; ..
A29A .db 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 ;
A2AA .db 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 ;
;
; f9: EXIT. Header off, records cleared, teletext interrupts off, restore the caller's
; stack page and BRKV, keys back to normal. With a Tube return to the caller; without,

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; start BASIC.
;-----
.LA2B6
A2B6 A9 09 LDA #&09
A2B8 8D 00 7A STA t_osw
A2BB A9 00 LDA #&00
A2BD 8D 01 7A STA t_osw+1
A2C0 20 38 A3 JSR LA338
A2C3 A9 0E LDA #&0E
A2C5 8D 00 7A STA t_osw
A2C8 20 38 A3 JSR LA338
A2CB A9 0C LDA #&0C
A2CD 8D 00 7A STA t_osw
A2D0 20 38 A3 JSR LA338
A2D3 A2 00 LDX #&00

.LA2D5
A2D5 BD 58 7A LDA t_saved_stack,X
A2D8 9D 00 01 STA stack_page,X
A2DB E8 INX
A2DC D0 F7 BNE LA2D5
A2DE AE 57 7A LDX t_saved_sp
A2E1 9A TXS
A2E2 68 PLA
A2E3 8D 03 02 STA BRKV+1
A2E6 68 PLA
A2E7 8D 02 02 STA BRKV
A2EA A9 16 LDA #&16
A2EC 20 EE FF JSR OSWRCH ; VDU 22: MODE
A2EF A9 07 LDA #&07
A2F1 20 EE FF JSR OSWRCH ; VDU 7: BEL
A2F4 A9 04 LDA #&04
A2F6 A2 00 LDX #&00
A2F8 20 B9 A0 JSR LA0B9
A2FB A9 0C LDA #&0C
A2FD A2 00 LDX #&00
A2FF 20 B9 A0 JSR LA0B9
A302 A9 E1 LDA #&E1
A304 20 B7 A0 JSR LA0B7
A307 A9 E2 LDA #&E2
A309 A2 80 LDX #&80
A30B 20 B9 A0 JSR LA0B9
A30E A9 E3 LDA #&E3
A310 A2 90 LDX #&90
A312 20 B9 A0 JSR LA0B9
A315 A9 E5 LDA #&E5
A317 A2 00 LDX #&00
A319 20 B9 A0 JSR LA0B9
A31C 20 8A 89 JSR L898A
A31F F0 09 BEQ LA32A
A321 68 PLA
A322 AA TAX
A323 68 PLA
A324 A8 TAY
A325 68 PLA
A326 A9 00 LDA #&00
A328 28 PLP
A329 60 RTS

.LA32A
A32A A9 BB LDA #&BB
A32C A2 00 LDX #&00
A32E A0 FF LDY #&FF
A330 20 F4 FF JSR OSBYTE
A333 A9 8E LDA #&8E
A335 4C F4 FF JMP OSBYTE ; OSBYTE &8E: enter language ROM X
; OSWORD &7A with the block at &7A00.
.LA338
A338 A2 00 LDX #&00
A33A A0 7A LDY #&7A
A33C 4C 43 A3 JMP LA343
; OSWORD &7A with the block at &7A12.
.LA33F
A33F A2 12 LDX #&12
A341 A0 7A LDY #&7A
.LA343
A343 A9 7A LDA #&7A
A345 4C F1 FF JMP OSWORD ; OSWORD &7A: Teletext system call
;-----
; FASTEXT: decode the six links of row 27/0 and point records 1-6 at them, with buffers
; &6D00, &6800, &6300, &5E00, &5900, &5400, so that a linked page appears at once.
; If a link names a page already held in another record, its copy is moved over.
;-----
.fetch_links
A348 A9 03 LDA #&03
A34A 8D 12 7A STA t_osw2
A34D A9 01 LDA #&01
A34F 8D 13 7A STA t_osw2+1
A352 A9 00 LDA #&00
A354 8D 14 7A STA t_osw2+2
A357 A9 05 LDA #&05
A359 8D 00 7A STA t_osw

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A35C A9 01 LDA #&01
A35E 8D 01 7A STA t_osw+1
A361 A9 00 LDA #&00
A363 8D 06 7A STA t_osw+6
A366 A9 6D LDA #&6D
A368 8D 07 7A STA t_osw+7
A36B A9 FF LDA #&FF
A36D 8D 08 7A STA t_osw+8
A370 8D 09 7A STA t_osw+9
A373 A9 00 LDA #&00
A375 85 82 STA &82
A377 A9 4A LDA #&4A
A379 85 83 STA &83
A37B 20 90 A4 JSR find_row27
A37E 90 03 BCC LA383
A380 A9 00 LDA #&00
A382 60 RTS
.LA383
A383 A0 01 LDY #&01 ; workspace +&01: grabber records (8 bytes each)
.LA385
A385 B1 84 LDA (&84),Y
A387 AA TAX
A388 BD AF 91 LDA hamming_table,X
A38B 30 43 BMI LA3D0
A38D 29 0F AND #&0F
A38F 99 76 00 STA &0076,Y
A392 C8 INY
A393 C0 07 CPY #&07
A395 D0 EE BNE LA385
A397 20 55 91 JSR L9155
A39A A5 75 LDA &75
A39C 4A LSR A
A39D 4A LSR A
A39E 4A LSR A
A39F 4D 00 4A EOR page_buf
A3A2 F0 2C BEQ LA3D0
A3A4 8D 02 7A STA t_osw+2
A3A7 A5 72 LDA &72
A3A9 C9 FF CMP #&FF
A3AB F0 23 BEQ LA3D0
A3AD A5 72 LDA &72
A3AF 8D 03 7A STA t_osw+3
A3B2 A5 73 LDA &73
A3B4 8D 04 7A STA t_osw+4
A3B7 A5 74 LDA &74
A3B9 8D 05 7A STA t_osw+5
A3BC 05 73 ORA &73
A3BE D0 0A BNE LA3CA
A3C0 A9 3F LDA #&3F
A3C2 8D 04 7A STA t_osw+4
A3C5 A9 7F LDA #&7F
A3C7 8D 05 7A STA t_osw+5
.LA3CA
A3CA 20 38 A3 JSR LA338
A3CD 20 3F A3 JSR LA33F
.LA3D0
A3D0 EE 13 7A INC t_osw+1
A3D3 EE 01 7A INC t_osw+1
A3D6 38 SEC
A3D7 AD 07 7A LDA t_osw+7
A3DA E9 05 SBC #&05
A3DC 8D 07 7A STA t_osw+7
A3DF 18 CLC
A3E0 A5 84 LDA &84
A3E2 69 06 ADC #&06
A3E4 85 84 STA &84
A3E6 A5 85 LDA &85
A3E8 69 00 ADC #&00
A3EA 85 85 STA &85
A3EC AD 01 7A LDA t_osw+1
A3EF C9 07 CMP #&07
A3F1 D0 90 BNE LA383
A3F3 A9 01 LDA #&01
A3F5 8D 01 7A STA t_osw+1
A3F8 A9 00 LDA #&00
A3FA 85 82 STA &82
A3FC A9 6D LDA #&6D
A3FE 85 83 STA &83
.LA400
A400 A9 04 LDA #&04
A402 8D 00 7A STA t_osw
A405 20 38 A3 JSR LA338
A408 A5 82 LDA &82
A40A 85 84 STA &84
A40C A5 83 LDA &83
A40E 85 85 STA &85
.LA410
A410 A0 00 LDY #&00 ; workspace +&00: grabber records (8 bytes each)
A412 B1 84 LDA (&84),Y
A414 CD 02 7A CMP t_osw+2

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A417 D0 52    BNE LA46B
A419 C8       INY
A41A C8       INY
A41B B1 84    LDA (&84),Y
A41D CD 03 7A CMP t_osw+3
A420 D0 49    BNE LA46B
A422 AD 04 7A LDA t_osw+4
A425 C9 3F    CMP #&3F           ; '?'
A427 D0 07    BNE LA430
A429 AD 05 7A LDA t_osw+5
A42C C9 7F    CMP #&7F
A42E F0 10    BEQ LA440
.LA430
A430 C8       INY
A431 B1 84    LDA (&84),Y
A433 CD 04 7A CMP t_osw+4
A436 D0 33    BNE LA46B
A438 C8       INY
A439 B1 84    LDA (&84),Y
A43B CD 05 7A CMP t_osw+5
A43E D0 2B    BNE LA46B
.LA440
A440 A2 00    LDX #&00
A442 A0 00    LDY #&00           ; workspace +&00: grabber records (8 bytes each)
.LA444
A444 B1 84    LDA (&84),Y
A446 91 82    STA (&82),Y
A448 C8       INY
A449 D0 F9    BNE LA444
A44B E6 83    INC &83
A44D E6 85    INC &85
A44F E8       INX
A450 E0 05    CPX #&05
A452 D0 F0    BNE LA444
A454 A9 03    LDA #&03
A456 8D 00 7A STA t_osw
A459 A9 2A    LDA #&2A
A45B 8D 02 7A STA t_osw+2
A45E 20 38 A3 JSR LA338
A461 38       SEC
A462 A5 83    LDA &83
A464 E9 05    SBC #&05
A466 85 83    STA &83
A468 4C 79 A4 JMP LA479
.LA46B
A46B 38       SEC
A46C A5 85    LDA &85
A46E E9 05    SBC #&05
A470 85 85    STA &85
A472 C9 4A    CMP #&4A           ; 'J'
A474 F0 03    BEQ LA479
A476 4C 10 A4 JMP LA410
.LA479
A479 38       SEC
A47A A5 83    LDA &83
A47C E9 05    SBC #&05
A47E 85 83    STA &83
A480 EE 01 7A INC t_osw+1
A483 AD 01 7A LDA t_osw+1
A486 C9 07    CMP #&07
A488 F0 03    BEQ LA48D
A48A 4C 00 A4 JMP LA400
.LA48D
A48D A9 01    LDA #&01
A48F 60       RTS
;-----
; Find row 27/0 in the page at (&82): link control (&7A53) and check word.
;-----
.find_row27
A490 A2 1B    LDX #&1B
A492 20 45 A2 JSR find_row
A495 B0 1C    BCS LA4B3
A497 A0 00    LDY #&00           ; workspace +&00: grabber records (8 bytes each)
A499 B1 84    LDA (&84),Y
A49B AA       TAX
A49C BD AF 91 LDA hamming_table,X
A49F 29 0F    AND #&0F
A4A1 F0 12    BEQ LA4B5
A4A3 18       CLC
A4A4 A5 84    LDA &84
A4A6 69 2A    ADC #&2A
A4A8 85 82    STA &82
A4AA A5 85    LDA &85
A4AC 69 00    ADC #&00
A4AE 85 83    STA &83
A4B0 4C 90 A4 JMP find_row27
.LA4B3
A4B3 38       SEC
A4B4 60       RTS
.LA4B5

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```

A4B5 A0 25 LDY #&25 ; workspace +&25: grabber records (8 bytes each)
A4B7 B1 84 LDA (&84),Y
A4B9 AA TAX
A4BA BD AF 91 LDA hamming_table,X
A4BD 29 0F AND #&0F
A4BF 8D 53 7A STA link_ctrl
A4C2 C8 INY
A4C3 B1 84 LDA (&84),Y
A4C5 8D 4B 7A STA crc_rx_hi
A4C8 C8 INY
A4C9 B1 84 LDA (&84),Y
A4CB 8D 4C 7A STA crc_rx_lo
A4CE 18 CLC
A4CF 60 RTS
;-----
; f1-f4: COLOURED LINK KEYS (records 1-4). The link control bits from row 27 decide
; whether f2-f4 are offered. f0 (INDEX) uses link record 6.
;-----
.LA4D0
A4D0 A9 01 LDA #&01
A4D2 A2 00 LDX #&00
A4D4 A0 6D LDY #&6D
A4D6 4C 30 A5 JMP LA530
.LA4D9
A4D9 AD 53 7A LDA link_ctrl
A4DC 29 08 AND #&08
A4DE F0 0A BEQ LA4EA
A4E0 AD 53 7A LDA link_ctrl
A4E3 29 01 AND #&01
A4E5 D0 03 BNE LA4EA
A4E7 4C 49 A5 JMP LA549
.LA4EA
A4EA A9 02 LDA #&02
A4EC A2 00 LDX #&00
A4EE A0 68 LDY #&68
A4F0 4C 30 A5 JMP LA530
.LA4F3
A4F3 AD 53 7A LDA link_ctrl
A4F6 29 08 AND #&08
A4F8 F0 0A BEQ LA504
A4FA AD 53 7A LDA link_ctrl
A4FD 29 02 AND #&02
A4FF D0 03 BNE LA504
A501 4C 49 A5 JMP LA549
.LA504
A504 A9 03 LDA #&03
A506 A2 00 LDX #&00
A508 A0 63 LDY #&63
A50A 4C 30 A5 JMP LA530
.LA50D
A50D AD 53 7A LDA link_ctrl
A510 29 08 AND #&08
A512 F0 0A BEQ LA51E
A514 AD 53 7A LDA link_ctrl
A517 29 04 AND #&04
A519 D0 03 BNE LA51E
A51B 4C 49 A5 JMP LA549
.LA51E
A51E A9 04 LDA #&04
A520 A2 00 LDX #&00
A522 A0 5E LDY #&5E
A524 4C 30 A5 JMP LA530
.LA527
A527 A9 06 LDA #&06
A529 A2 00 LDX #&00
A52B A0 54 LDY #&54
A52D 4C 30 A5 JMP LA530
;-----
; Show a link record's page at once if it has arrived, else request it.
;-----
.LA530
A530 86 82 STX &82
A532 84 83 STY &83
A534 8D 01 7A STA t_osw+1
A537 8D 13 7A STA t_osw2+1
A53A A9 04 LDA #&04
A53C 8D 00 7A STA t_osw
A53F 20 38 A3 JSR LA338
A542 AD 03 7A LDA t_osw+3
A545 C9 FF CMP #&FF
A547 D0 12 BNE LA55B
.LA549
A549 AD 01 7A LDA t_osw+1
A54C C9 06 CMP #&06
A54E D0 03 BNE LA553
A550 4C 79 98 JMP L9879
.LA553
A553 A9 07 LDA #&07
A555 20 EE FF JSR OSWRCH ; VDU 7: BEL
A558 4C 02 99 JMP main_loop

```

.LA55B

```
A55B A9 87 LDA #&87
A55D 8D 5E 7B STA t_hdr_colour
A560 A9 00 LDA #&00
A562 8D 01 7A STA t_osw+1
A565 A9 00 LDA #&00
A567 8D 06 7A STA t_osw+6
A56A A9 4A LDA #&4A
A56C 8D 07 7A STA t_osw+7
A56F A9 05 LDA #&05
A571 8D 00 7A STA t_osw
A574 20 38 A3 JSR LA338
A577 AD 02 7A LDA t_osw+2
A57A 18 CLC
A57B 69 30 ADC #&30
A57D C9 30 CMP #&30 ; '0'
A57F D0 02 BNE LA583
A581 A9 38 LDA #&38
```

.LA583

```
A583 8D 26 7A STA t_digits
A586 AD 03 7A LDA t_osw+3
A589 29 F0 AND #&F0
A58B 4A LSR A
A58C 4A LSR A
A58D 4A LSR A
A58E 4A LSR A
A58F 18 CLC
A590 69 30 ADC #&30
A592 8D 27 7A STA t_digits+1
A595 AD 03 7A LDA t_osw+3
A598 29 0F AND #&0F
A59A 18 CLC
A59B 69 30 ADC #&30
A59D 8D 28 7A STA t_digits+2
A5A0 A9 03 LDA #&03
A5A2 8D 25 7A STA t_ndigits
A5A5 A9 04 LDA #&04
A5A7 8D 2A 7A STA t_entry_len
A5AA 20 8B 9C JSR L9C8B
A5AD A9 03 LDA #&03
A5AF 8D 00 7A STA t_osw
A5B2 A9 00 LDA #&00
A5B4 8D 02 7A STA t_osw+2
A5B7 20 38 A3 JSR LA338
A5BA A9 00 LDA #&00
A5BC 8D 54 7A STA &7A54
```

.LA5BF

```
A5BF A9 02 LDA #&02
A5C1 8D 12 7A STA t_osw2
A5C4 20 3F A3 JSR LA33F
A5C7 AD 14 7A LDA t_osw2+2
A5CA 29 02 AND #&02
A5CC D0 0F BNE LA5DD
A5CE A9 82 LDA #&82
A5D0 20 5E 9F JSR L9F5E
A5D3 AD 14 7A LDA t_osw2+2
A5D6 29 04 AND #&04
A5D8 F0 61 BEQ LA63B
A5DA 4C BF A5 JMP LA5BF
```

.LA5DD

```
A5DD A9 03 LDA #&03
A5DF 8D 00 7A STA t_osw
A5E2 A9 00 LDA #&00
A5E4 8D 01 7A STA t_osw+1
A5E7 A9 01 LDA #&01
A5E9 8D 02 7A STA t_osw+2
A5EC 20 38 A3 JSR LA338
A5EF A9 00 LDA #&00
A5F1 85 84 STA &84
A5F3 A9 4A LDA #&4A
A5F5 85 85 STA &85
A5F7 A2 00 LDX #&00
```

.LA5F9

```
A5F9 A0 00 LDY #&00
```

.LA5FB

```
A5FB B1 82 LDA (&82),Y
A5FD 91 84 STA (&84),Y
A5FF C8 INY
A600 D0 F9 BNE LA5FB
A602 E6 83 INC &83
A604 E6 85 INC &85
A606 E8 INX
A607 E0 06 CPX #&06
A609 D0 EE BNE LA5F9
A60B A9 03 LDA #&03
A60D 8D 00 7A STA t_osw
A610 A9 00 LDA #&00
A612 8D 01 7A STA t_osw+1
A615 8D 02 7A STA t_osw+2
A618 20 38 A3 JSR LA338
```

```

A61B A9 10 LDA #&10
A61D 8D 00 7A STA t_osw
A620 20 38 A3 JSR LA338
A623 A9 09 LDA #&09
A625 8D 00 7A STA t_osw
A628 A9 01 LDA #&01
A62A 8D 01 7A STA t_osw+1
A62D 20 38 A3 JSR LA338
A630 A9 87 LDA #&87
A632 20 5E 9F JSR L9F5E
A635 20 3C 9D JSR L9D3C
A638 20 50 A1 JSR show_page
.LA63B
A63B 4C 02 99 JMP main_loop
;-----
; Draw the link line on row 24: coloured blocks and page numbers, or keywords.
;-----
.link_line
A63E 20 45 A1 JSR LA145
A641 A9 04 LDA #&04
A643 8D 00 7A STA t_osw
A646 A9 01 LDA #&01
A648 8D 01 7A STA t_osw+1
A64B A9 02 LDA #&02
A64D 8D 12 7A STA t_osw2
.LA650
A650 20 38 A3 JSR LA338
A653 AD 02 7A LDA t_osw+2
A656 C9 FF CMP #&FF
A658 F0 4B BEQ LA6A5
A65A A9 1F LDA #&1F
A65C 20 EE FF JSR OSWRCH ; VDU 31: TAB(x,y)
A65F AD 12 7A LDA t_osw2
A662 20 EE FF JSR OSWRCH
A665 A9 18 LDA #&18
A667 20 EE FF JSR OSWRCH ; VDU 24
A66A AD 01 7A LDA t_osw+1
A66D A8 TAY
A66E B9 BC A6 LDA &A6BC,Y
A671 20 EE FF JSR OSWRCH
A674 A9 03 LDA #&03
A676 20 1A BB JSR print_inline_count
A679 .str '\xff\x87P'
A67C AD 02 7A LDA t_osw+2
A67F 18 CLC
A680 69 30 ADC #&30
A682 C9 30 CMP #&30 ; '0'
A684 D0 02 BNE LA688
A686 A9 38 LDA #&38
.LA688
A688 20 EE FF JSR OSWRCH ; print '8'
A68B AD 03 7A LDA t_osw+3
A68E 29 F0 AND #&F0
A690 4A LSR A
A691 4A LSR A
A692 4A LSR A
A693 4A LSR A
A694 18 CLC
A695 69 30 ADC #&30
A697 20 EE FF JSR OSWRCH
A69A AD 03 7A LDA t_osw+3
A69D 29 0F AND #&0F
A69F 18 CLC
A6A0 69 30 ADC #&30
A6A2 20 EE FF JSR OSWRCH
.LA6A5
A6A5 EE 01 7A INC t_osw+1
A6A8 18 CLC
A6A9 AD 12 7A LDA t_osw2
A6AC 69 09 ADC #&09
A6AE 8D 12 7A STA t_osw2
A6B1 AD 01 7A LDA t_osw+1
A6B4 C9 05 CMP #&05
A6B6 F0 03 BEQ LA6BB
A6B8 4C 50 A6 JMP LA650
.LA6BB
A6BB 60 RTS
; Colour codes for the link line.
A6BC .db 00 81 82 83 86 ; ....
;-----
; Previous page store: keep the previous screen at &7200 when a new page is shown.
;-----
.LA6C1
A6C1 A9 00 LDA #&00
A6C3 85 82 STA &82
A6C5 A9 7C LDA #&7C
A6C7 85 83 STA &83
A6C9 A9 00 LDA #&00
A6CB 85 84 STA &84
A6CD A9 72 LDA #&72

```

```

A6CF 85 85 STA &85
A6D1 A9 00 LDA #&00
A6D3 85 86 STA &86
A6D5 A9 76 LDA #&76
A6D7 85 87 STA &87
A6D9 A0 28 LDY #&28
.LA6DB
A6DB B1 82 LDA (&82),Y
A6DD D1 84 CMP (&84),Y
A6DF D0 04 BNE LA6E5
A6E1 C8 INY
A6E2 D0 F7 BNE LA6DB
A6E4 60 RTS
.LA6E5
A6E5 A2 00 LDX #&00
.LA6E7
A6E7 BD 26 A7 LDA &A726,X
A6EA 85 77 STA &77
A6EC A0 00 LDY #&00 ; workspace +&00: grabber records (8 bytes each)
.LA6EE
A6EE B1 84 LDA (&84),Y
A6F0 91 86 STA (&86),Y
A6F2 B1 82 LDA (&82),Y
A6F4 91 84 STA (&84),Y
A6F6 C8 INY
A6F7 C4 77 CPY &77
A6F9 D0 F3 BNE LA6EE
A6FB E6 83 INC &83
A6FD E6 85 INC &85
A6FF E6 87 INC &87
A701 E8 INX
A702 E0 04 CPX #&04
A704 D0 E1 BNE LA6E7
A706 C6 87 DEC &87
A708 A0 C0 LDY #&C0
A70A A2 00 LDX #&00
.LA70C
A70C BD 2A A7 LDA &A72A,X
A70F 91 86 STA (&86),Y
A711 E8 INX
A712 C8 INY
A713 E0 0E CPX #&0E
A715 D0 F5 BNE LA70C
A717 A9 A0 LDA #&A0
.LA719
A719 91 86 STA (&86),Y
A71B C8 INY
A71C C0 E8 CPY #&E8
A71E D0 F9 BNE LA719
A720 A9 00 LDA #&00
A722 8D 55 7A STA t_prev_state
A725 60 RTS
; Byte counts for screen copies, and " Previous Page" text.
A726 .db 00 00 00 E8 20 50 72 65 76 69 6F 75 73 20 50 61 ; .... Previous Pa
A736 .db 67 65 ; ge
; TAB: swap between the current and previous page.
.LA738
A738 A9 00 LDA #&00
A73A 85 82 STA &82
A73C A9 7C LDA #&7C
A73E 85 83 STA &83
A740 AD 55 7A LDA t_prev_state
A743 30 30 BMI LA775
A745 F0 0B BEQ LA752
A747 A9 00 LDA #&00
A749 85 84 STA &84
A74B A9 72 LDA #&72
A74D 85 85 STA &85
A74F 4C 5A A7 JMP LA75A
.LA752
A752 A9 00 LDA #&00
A754 85 84 STA &84
A756 A9 76 LDA #&76
A758 85 85 STA &85
.LA75A
A75A A2 00 LDX #&00
.LA75C
A75C BD 26 A7 LDA &A726,X
A75F 85 77 STA &77
A761 A0 00 LDY #&00 ; workspace +&00: grabber records (8 bytes each)
.LA763
A763 B1 84 LDA (&84),Y
A765 91 82 STA (&82),Y
A767 C8 INY
A768 C4 77 CPY &77
A76A D0 F7 BNE LA763
A76C E6 83 INC &83
A76E E6 85 INC &85
A770 E8 INX
A771 E0 04 CPX #&04

```

```

A773 D0 E7 BNE LA75C
.LA775
A775 60 RTS
;-----
; " ": OS COMMAND. Read a command, run it with ESCAPE enabled, then "Press SPACE".
;-----
.LA776
A776 20 45 A1 JSR LA145
A779 A9 05 LDA #05
A77B 20 1A BB JSR print_inline_count
A77E .str '\x1a\x1f\x01\x18*'
A783 20 F9 A0 JSR LA0F9
A786 AD 2D 7A LDA t_line
A789 C9 0D CMP #0D
A78B D0 0F BNE LA79C
.LA78D
A78D 20 BE A0 JSR terminal_keys
A790 20 86 A0 JSR LA086
A793 20 8B 9C JSR L9C8B
A796 20 50 A1 JSR show_page
A799 4C 02 99 JMP main_loop
.LA79C
A79C A0 00 LDY #00
.LA79E
A79E B9 FB A7 LDA &A7FB,Y
A7A1 99 C0 7F STA &7FC0,Y
A7A4 C8 INY
A7A5 C0 28 CPY #28 ; '('
A7A7 D0 F5 BNE LA79E
A7A9 A9 15 LDA #15
A7AB 20 1A BB JSR print_inline_count
A7AE .str "\x1a\x1e0S Cmnd\x1c\x00\x17'\x01\x0c\x1c\x00\x16'\x02\x0e"
A7C3 20 43 A8 JSR LA843
A7C6 A9 E5 LDA #E5
A7C8 A2 00 LDX #00
A7CA A0 00 LDY #00
A7CC 20 F4 FF JSR OSBYTE ; OSBYTE &E5: ESCAPE key: produce ASCII 27 / normal
A7CF A2 2D LDX #2D
A7D1 A0 7A LDY #7A
A7D3 20 F7 FF JSR OSCLI
A7D6 A9 E5 LDA #E5
A7D8 A2 01 LDX #01
A7DA A0 00 LDY #00
A7DC 20 F4 FF JSR OSBYTE ; OSBYTE &E5: ESCAPE key: produce ASCII 27 / normal
A7DF A9 0F LDA #0F
A7E1 20 EE FF JSR OSWRCH ; VDU 15
A7E4 20 3B A0 JSR LA03B
A7E7 C9 2A CMP #2A ; '*'
A7E9 D0 08 BNE LA7F3
A7EB A9 1A LDA #1A
A7ED 20 EE FF JSR OSWRCH ; VDU 26: restore default windows
A7F0 4C 76 A7 JMP LA776
.LA7F3
A7F3 A9 1A LDA #1A
A7F5 20 EE FF JSR OSWRCH ; VDU 26: restore default windows
A7F8 4C 8D A7 JMP LA78D
; "Press SHIFT to continue" bar.
A7FB .db 97 AC AC AC AC AC AC AC 87 50 72 65 73 73 85 53 ; .....Press.S
A80B .db 48 49 46 54 87 74 6F 20 63 6F 6E 74 69 6E 75 65 ; HIFT.to continue
A81B .db 97 AC AC AC AC AC AC AC ; .....
; SPACE: just redraw.
.LA823
A823 4C 02 99 JMP main_loop
; Cursor on / off (A82A).
.LA826
A826 A9 01 LDA #01
A828 D0 02 BNE LA82C
.LA82A
A82A A9 00 LDA #00
.LA82C
A82C 48 PHA
A82D A9 02 LDA #02
A82F 20 1A BB JSR print_inline_count
A832 .str '\x17\x01'
A834 68 PLA
A835 20 EE FF JSR OSWRCH
A838 A9 00 LDA #00
A83A A2 08 LDX #08
.LA83C
A83C 20 EE FF JSR OSWRCH ; VDU 0
A83F CA DEX
A840 D0 FA BNE LA83C
A842 60 RTS
;-----
; Point BRKV at the extended vector &FF03 and set that entry to A86B in this ROM, so
; errors during OS commands come back to the terminal.
;-----
.LA843
A843 A9 03 LDA #03
A845 8D 02 02 STA BRKV

```

```

A848 A9 FF LDA #&FF
A84A 8D 03 02 STA BRKV+1
A84D A9 A8 LDA #&A8
A84F A2 00 LDX #&00
A851 A0 FF LDY #&FF
A853 20 F4 FF JSR OSBYTE ; OSBYTE &A8: read address of extended vector table
A856 86 82 STX &82
A858 84 83 STY &83
A85A A0 03 LDY #&03
A85C A9 6B LDA #&6B
A85E 91 82 STA (&82),Y
A860 C8 INY
A861 A9 A8 LDA #&A8
A863 91 82 STA (&82),Y
A865 C8 INY
A866 A5 F4 LDA romsel_copy
A868 91 82 STA (&82),Y
A86A 60 RTS
;-----
; TERMINAL ERROR HANDLER: reset the stack, acknowledge ESCAPE, close files, show
; "Error : <message>" on the bottom row, wait for SPACE, then carry on.
;-----
.terminal_brk
A86B A2 FF LDX #&FF
A86D 9A TXS
A86E A9 E5 LDA #&E5
A870 A2 01 LDX #&01
A872 A0 00 LDY #&00
A874 20 F4 FF JSR OSBYTE ; OSBYTE &E5: ESCAPE key: produce ASCII 27 / normal
A877 A5 FF LDA escape_flag
A879 10 09 BPL LA884
.LA87B
A87B A9 7E LDA #&7E
A87D 20 F4 FF JSR OSBYTE ; OSBYTE &7E: acknowledge ESCAPE
A880 E0 FF CPX #&FF
A882 D0 F7 BNE LA87B
.LA884
A884 A9 00 LDA #&00
A886 A8 TAY
A887 AA TAX
A888 20 CE FF JSR OSFIND
A88B A9 06 LDA #&06
A88D 20 1A BB JSR print_inline_count
A890 .str "\x1c\x00\x18'\x14\x0c"
A896 A9 00 LDA #&00
A898 20 41 BB JSR LBB41
A89B A9 0C LDA #&0C
A89D 20 1A BB JSR print_inline_count
A8A0 .str '\n Error : \x81\x9d\x83'
A8AC A0 01 LDY #&01
.LA8AE
A8AE B1 FD LDA (brk_ptr),Y
A8B0 20 EE FF JSR OSWRCH
A8B3 C8 INY
A8B4 C0 19 CPY #&19
A8B6 F0 04 BEQ LA8BC
A8B8 C9 00 CMP #&00
A8BA D0 F2 BNE LA8AE
.LA8BC
A8BC A9 03 LDA #&03
A8BE 20 1A BB JSR print_inline_count
A8C1 .str ' \x9c'
A8C4 20 3B A0 JSR LA03B
A8C7 A9 1A LDA #&1A
A8C9 20 EE FF JSR OSWRCH ; VDU 26: restore default windows
A8CC A9 0D LDA #&0D
A8CE 8D 00 7A STA t_osw
A8D1 20 38 A3 JSR LA338
A8D4 A9 09 LDA #&09
A8D6 8D 00 7A STA t_osw
A8D9 A9 01 LDA #&01
A8DB 8D 01 7A STA t_osw+1
A8DE 20 38 A3 JSR LA338
A8E1 A9 04 LDA #&04
A8E3 8D 2A 7A STA t_entry_len
A8E6 20 E3 9E JSR L9EE3
A8E9 4C 8D A7 JMP LA78D
;-----
; COPY: KEEP or RELEASE the current page. If a kept record (8 onwards) already holds it,
; release it ("Released"); otherwise take a free record and buffer ("Kept"), or
; "Buffer Full".
;-----
.LA8EC
A8EC 20 45 A1 JSR LA145
A8EF A9 04 LDA #&04
A8F1 8D 12 7A STA t_osw2
A8F4 A9 00 LDA #&00
A8F6 8D 13 7A STA t_osw2+1
A8F9 20 3F A3 JSR LA33F
A8FC A9 08 LDA #&08

```

```

A8FE 8D 01 7A STA t_osw+1
A901 A9 04 LDA #&04
A903 8D 00 7A STA t_osw
.LA906
A906 20 38 A3 JSR LA338
A909 A2 02 LDX #&02
.LA90B
A90B BD 00 7A LDA t_osw,X
A90E DD 12 7A CMP t_osw2,X
A911 D0 2D BNE LA940
A913 E8 INX
A914 E0 06 CPX #&06
A916 D0 F3 BNE LA90B
A918 A9 FF LDA #&FF
A91A A2 02 LDX #&02
.LA91C
A91C 9D 00 7A STA t_osw,X
A91F E8 INX
A920 E0 0A CPX #&0A
A922 D0 F8 BNE LA91C
A924 A9 05 LDA #&05
A926 8D 00 7A STA t_osw
A929 20 38 A3 JSR LA338
A92C A9 03 LDA #&03
A92E 8D 00 7A STA t_osw
A931 A9 00 LDA #&00
A933 8D 02 7A STA t_osw+2
A936 20 38 A3 JSR LA338
A939 A2 62 LDX #&62
A93B A0 AA LDY #&AA
A93D 4C 21 AA JMP LAA21
.LA940
A940 EE 01 7A INC t_osw+1
A943 AD 01 7A LDA t_osw+1
A946 CD 5C 7B CMP t_nrecords
A949 D0 BB BNE LA906
A94B A9 08 LDA #&08
A94D 8D 01 7A STA t_osw+1
.LA950
A950 20 38 A3 JSR LA338
A953 AD 02 7A LDA t_osw+2
A956 C9 FF CMP #&FF
A958 F0 12 BEQ LA96C
A95A EE 01 7A INC t_osw+1
A95D AD 01 7A LDA t_osw+1
A960 CD 5C 7B CMP t_nrecords
A963 D0 EB BNE LA950
A965 A2 71 LDX #&71
A967 A0 AA LDY #&AA
A969 4C 21 AA JMP LAA21
.LA96C
A96C A2 02 LDX #&02
.LA96E
A96E BD 12 7A LDA t_osw2,X
A971 9D 00 7A STA t_osw,X
A974 E8 INX
A975 E0 0A CPX #&0A
A977 D0 F5 BNE LA96E
A979 A9 05 LDA #&05
A97B 8D 00 7A STA t_osw
A97E AD 01 7A LDA t_osw+1
A981 38 SEC
A982 E9 08 SBC #&08
A984 A8 TAY
A985 AD 58 7B LDA t_kept_base
A988 8D 06 7A STA t_osw+6
A98B AD 59 7B LDA t_kept_base+1
A98E 8D 07 7A STA t_osw+7
A991 C0 00 CPY #&00
A993 F0 0C BEQ LA9A1
.LA995
A995 18 CLC
A996 AD 07 7A LDA t_osw+7
A999 69 05 ADC #&05
A99B 8D 07 7A STA t_osw+7
A99E 88 DEY
A99F D0 F4 BNE LA995
.LA9A1
A9A1 20 38 A3 JSR LA338
A9A4 A9 00 LDA #&00
A9A6 8D 02 7A STA t_osw+2
A9A9 A9 03 LDA #&03
A9AB 8D 00 7A STA t_osw
A9AE AD 00 4A LDA page_buf
A9B1 CD 02 7A CMP t_osw+2
A9B4 D0 61 BNE LAA17
A9B6 AD 02 4A LDA page_buf+2
A9B9 CD 03 7A CMP t_osw+3
A9BC D0 59 BNE LAA17
A9BE AD 04 7A LDA t_osw+4

```



```

A9C1 C9 3F    CMP #&3F                ; '?'
A9C3 D0 07    BNE LA9CC
A9C5 AD 05 7A  LDA t_osw+5
A9C8 C9 7F    CMP #&7F
A9CA F0 10    BEQ LA9DC
.LA9CC
A9CC AD 03 4A  LDA page_buf+3
A9CF CD 04 7A  CMP t_osw+4
A9D2 D0 43    BNE LAA17
A9D4 AD 04 4A  LDA page_buf+4
A9D7 CD 05 7A  CMP t_osw+5
A9DA D0 3B    BNE LAA17
.LA9DC
A9DC A2 00    LDX #&00
A9DE A0 00    LDY #&00
A9E0 A9 00    LDA #&00
A9E2 85 82    STA &82
A9E4 A9 4A    LDA #&4A
A9E6 85 83    STA &83
A9E8 AD 06 7A  LDA t_osw+6
A9EB 85 84    STA &84
A9ED AD 07 7A  LDA t_osw+7
A9F0 85 85    STA &85
A9F2 30 23    BMI LAA17
.LA9F4
A9F4 B1 82    LDA (&82),Y
A9F6 91 84    STA (&84),Y
A9F8 C8       INY
A9F9 D0 F9    BNE LA9F4
A9FB E6 83    INC &83
A9FD E6 85    INC &85
A9FF E8       INX
AA00 E0 05    CPX #&05
AA02 D0 F0    BNE LA9F4
AA04 AD 01 7A  LDA t_osw+1
AA07 8D 13 7A  STA t_osw2+1
AA0A A9 03    LDA #&03
AA0C 8D 12 7A  STA t_osw2
AA0F A9 2A    LDA #&2A
AA11 8D 14 7A  STA t_osw2+2
AA14 20 3F A3  JSR LA33F
.LAA17
AA17 20 38 A3  JSR LA338
AA1A A2 83    LDX #&83
AA1C A0 AA    LDY #&AA
AA1E 4C 21 AA  JMP LAA21
;-----
; Show a message (address X/Y) on the bottom row for a few seconds.
;-----
.LAA21
AA21 86 82    STX &82
AA23 84 83    STY &83
AA25 20 45 A1  JSR LA145
AA28 A9 04    LDA #&04
AA2A 20 1A BB  JSR print_inline_count
AA2D .str '\x1f\x00\x18\x07'
AA31 A0 00    LDY #&00
.LAA33
AA33 B1 82    LDA (&82),Y
AA35 20 EE FF  JSR OSWRCH
AA38 C8       INY
AA39 C9 00    CMP #&00
AA3B D0 F6    BNE LAA33
AA3D A2 00    LDX #&00
.LAA3F
AA3F 8A       TXA
AA40 48       PHA
AA41 A9 13    LDA #&13
AA43 20 F4 FF  JSR OSBYTE                ; OSBYTE &13: wait for vertical sync
AA46 A9 80    LDA #&80
AA48 A2 FF    LDX #&FF
AA4A A0 FF    LDY #&FF
AA4C 20 F4 FF  JSR OSBYTE                ; OSBYTE &80
AA4F E0 00    CPX #&00
AA51 D0 08    BNE LAA5B
AA53 68       PLA
AA54 AA       TAX
AA55 E8       INX
AA56 E0 50    CPX #&50                ; 'P'
AA58 D0 E5    BNE LAA3F
AA5A 48       PHA
.LAA5B
AA5B 68       PLA
AA5C 20 50 A1  JSR show_page
AA5F 4C 02 99  JMP main_loop
; "All Released", "Released", "Buffer Full", "Kept".
AA62 .db 84 9D 86 52 65 6C 65 61 73 65 64 20 20 9C 00 81 ; ...Released ...
AA72 .db 9D 83 42 75 66 66 65 72 20 46 75 6C 6C 20 20 9C ; ..Buffer Full .
AA82 .db 00 84 9D 86 4B 65 70 74 20 20 9C 00            ; ....Kept ..
; SHIFT f0: TUNE.

```

```

.LAA8E
AA8E A9 14 LDA #&14
AA90 8D 00 7A STA t_osw
AA93 20 38 A3 JSR LA338
AA96 20 50 A1 JSR show_page
AA99 4C 02 99 JMP main_loop
;-----
; Copy any kept record that has a newer copy of the displayed page into the buffer.
;-----
.LAA9C
AA9C A9 04 LDA #&04
AA9E 8D 00 7A STA t_osw
AAA1 A9 08 LDA #&08
AAA3 8D 01 7A STA t_osw+1
.LAAA6
AAA6 20 38 A3 JSR LA338
AAA9 AD 00 4A LDA page_buf
AAAC CD 02 7A CMP t_osw+2
AAAF D0 69 BNE LAB1A
AAB1 AD 02 4A LDA page_buf+2
AAB4 CD 03 7A CMP t_osw+3
AAB7 D0 61 BNE LAB1A
AAB9 AD 04 7A LDA t_osw+4
AABC C9 3F CMP #&3F ; '?'
AABE D0 07 BNE LAAC7
AAC0 AD 05 7A LDA t_osw+5
AAC3 C9 7F CMP #&7F
AAC5 F0 10 BEQ LAAD7
.LAAC7
AAC7 AD 03 4A LDA page_buf+3
AACA CD 04 7A CMP t_osw+4
AACD D0 4B BNE LAB1A
AACF AD 04 4A LDA page_buf+4
AAD2 CD 05 7A CMP t_osw+5
AAD5 D0 43 BNE LAB1A
.LAAD7
AAD7 A2 00 LDX #&00
AAD9 A0 00 LDY #&00
AADB A9 00 LDA #&00
AADD 85 82 STA &82
AADF A9 4A LDA #&4A
AAE1 85 83 STA &83
AAE3 AD 06 7A LDA t_osw+6
AAE6 85 84 STA &84
AAE8 AD 07 7A LDA t_osw+7
AAEB 85 85 STA &85
AAED 30 2B BMI LAB1A
.LAAEF
AAEF B1 82 LDA (&82),Y
AAF1 91 84 STA (&84),Y
AAF3 C8 INY
AAF4 D0 F9 BNE LAAEF
AAF6 E6 83 INC &83
AAF8 E6 85 INC &85
Aafa E8 INX
AAFB E0 05 CPX #&05
AAFD D0 F0 BNE LAAEF
AAFF A9 02 LDA #&02
AB01 8D 12 7A STA t_osw2
AB04 A9 00 LDA #&00
AB06 8D 13 7A STA t_osw2+1
AB09 20 3F A3 JSR LA33F
AB0C AD 01 7A LDA t_osw+1
AB0F 8D 13 7A STA t_osw2+1
AB12 A9 03 LDA #&03
AB14 8D 12 7A STA t_osw2
AB17 20 3F A3 JSR LA33F
.LAB1A
AB1A EE 01 7A INC t_osw+1
AB1D AD 01 7A LDA t_osw+1
AB20 C9 00 CMP #&00
AB22 F0 03 BEQ LAB27
AB24 4C A6 AA JMP LAAA6
.LAB27
AB27 60 RTS
;-----
; RETURN: TERMINAL STATUS: list the 20 records (link and kept pages).
;-----
.LAB28
AB28 A9 08 LDA #&08
AB2A 20 1A BB JSR print_inline_count
AB2D .str "\x1a\x1c\x00\x18'\x01\x0c\x1a"
AB35 A9 01 LDA #&01
AB37 20 41 BB JSR LBB41
AB3A A9 54 LDA #&54
AB3C 85 82 STA &82
AB3E A9 AC LDA #&AC
AB40 85 83 STA &83
AB42 A9 04 LDA #&04
AB44 8D 00 7A STA t_osw

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AB47 A9 00 LDA #&00
AB49 8D 01 7A STA t_osw+1
.LAB4C
AB4C A9 1F LDA #&1F
AB4E 20 EE FF JSR OSWRCH ; VDU 31: TAB(x,y)
AB51 AD 01 7A LDA t_osw+1
AB54 C9 0A CMP #&0A
AB56 B0 04 BCS LAB5C
AB58 A9 00 LDA #&00
AB5A F0 02 BEQ LAB5E
.LAB5C
AB5C A9 14 LDA #&14
.LAB5E
AB5E 20 EE FF JSR OSWRCH ; VDU 20
AB61 AD 01 7A LDA t_osw+1
AB64 0A ASL A
AB65 C9 14 CMP #&14
AB67 90 03 BCC LAB6C
AB69 38 SEC
AB6A E9 14 SBC #&14
.LAB6C
AB6C 18 CLC
AB6D 69 03 ADC #&03
AB6F 20 EE FF JSR OSWRCH
AB72 AD 01 7A LDA t_osw+1
AB75 C9 08 CMP #&08
AB77 B0 1B BCS LAB94
AB79 A0 00 LDY #&00
.LAB7B
AB7B B1 82 LDA (&82),Y
AB7D 20 EE FF JSR OSWRCH
AB80 C8 INY
AB81 C9 00 CMP #&00
AB83 D0 F6 BNE LAB7B
AB85 18 CLC
AB86 98 TYA
AB87 65 82 ADC &82
AB89 85 82 STA &82
AB8B A5 83 LDA &83
AB8D 69 00 ADC #&00
AB8F 85 83 STA &83
AB91 4C CA AB JMP LABCA
.LAB94
AB94 A2 00 LDX #&00
.LAB96
AB96 BD 9C AC LDA &AC9C,X
AB99 20 EE FF JSR OSWRCH
AB9C E8 INX
AB9D E0 06 CPX #&06
AB9F D0 F5 BNE LAB96
ABA1 AD 01 7A LDA t_osw+1
ABA4 38 SEC
ABA5 E9 07 SBC #&07
ABA7 C9 0A CMP #&0A
ABA9 90 0A BCC LABB5
ABAB 48 PHA
ABAC A9 31 LDA #&31
ABAE 20 EE FF JSR OSWRCH ; print '1'
ABB1 68 PLA
ABB2 38 SEC
ABB3 E9 0A SBC #&0A
.LABB5
ABB5 18 CLC
ABB6 69 30 ADC #&30
ABB8 20 EE FF JSR OSWRCH
ABBB AD 01 7A LDA t_osw+1
ABBE 38 SEC
ABBF E9 07 SBC #&07
ABC1 C9 0A CMP #&0A
ABC3 B0 05 BCS LABCA
ABC5 A9 20 LDA #&20
ABC7 20 EE FF JSR OSWRCH ; print ' '
.LABCA
ABCA A9 02 LDA #&02
ABCC 20 1A BB JSR print_inline_count
ABCF .str ': '
ABD1 20 38 A3 JSR LA338
ABD4 20 EB AB JSR LABEB
ABD7 EE 01 7A INC t_osw+1
ABDA AD 01 7A LDA t_osw+1
ABDD CD 5C 7B CMP t_nrecords
ABE0 B0 03 BCS LABE5
ABE2 4C 4C AB JMP LAB4C
.LABE5
ABE5 20 3B A0 JSR LA03B
ABE8 4C 8D A7 JMP LA78D
.LABEB
ABEB AD 03 7A LDA t_osw+3
ABEE C9 FF CMP #&FF
ABF0 D0 0E BNE LAC00

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ABF2 A2 00 LDX #&00
.LABF4
ABF4 BD 4B AC LDA &AC4B,X
ABF7 20 EE FF JSR OSWRCH
ABFA E8 INX
ABFB E0 09 CPX #&09
ABFD D0 F5 BNE LABF4
ABFF 60 RTS
.LAC00
AC00 A9 50 LDA #&50
AC02 20 EE FF JSR OSWRCH ; print 'P'
AC05 AD 02 7A LDA t_osw+2
AC08 AA TAX
AC09 BD C0 B7 LDA &B7C0,X
AC0C C9 30 CMP #&30 ; '0'
AC0E D0 02 BNE LAC12
AC10 A9 38 LDA #&38
.LAC12
AC12 20 EE FF JSR OSWRCH ; print '8'
AC15 AD 03 7A LDA t_osw+3
AC18 20 48 AC JSR LAC48
AC1B AD 04 7A LDA t_osw+4
AC1E C9 3F CMP #&3F ; '?'
AC20 D0 07 BNE LAC29
AC22 AD 05 7A LDA t_osw+5
AC25 C9 7F CMP #&7F
AC27 F0 14 BEQ LAC3D
.LAC29
AC29 A9 73 LDA #&73
AC2B 20 EE FF JSR OSWRCH ; print 's'
AC2E AD 04 7A LDA t_osw+4
AC31 20 48 AC JSR LAC48
AC34 AD 05 7A LDA t_osw+5
AC37 20 48 AC JSR LAC48
AC3A 4C 47 AC JMP LAC47
.LAC3D
AC3D A9 20 LDA #&20
AC3F A2 05 LDX #&05
.LAC41
AC41 20 EE FF JSR OSWRCH
AC44 CA DEX
AC45 D0 FA BNE LAC41
.LAC47
AC47 60 RTS
; Print A as two hex digits.
.LAC48
AC48 4C D8 B7 JMP LB7D8
; Status and link-line texts.
AC4B .db 5F 5F 5F 5F 20 20 20 20 20 20 50 61 67 65 20 20 ; ---- Page
AC5B .db 20 00 81 4C 69 6E 6B 20 31 87 00 82 4C 69 6E 6B ; ..Link 1...Link
AC6B .db 20 32 87 00 83 4C 69 6E 6B 20 33 87 00 86 4C 69 ; 2...Link 3...Li
AC7B .db 6E 6B 20 34 87 00 20 4C 69 6E 6B 20 35 20 00 20 ; nk 4.. Link 5 .
AC8B .db 49 6E 64 65 78 20 20 00 20 4E 65 78 74 20 20 20 ; Index . Next
AC9B .db 00 20 4B 65 70 74 20 0D ; . Kept .
;-----
; f5: TELESOFTWARE TERMINAL
; "Finding catalogue...": take the initial (index) page from the TSDP, follow its
; catalogue link (row 27 link 5), and download the catalogue file (OSWORD 24 to memory).
; "Finding" is then overprinted with "Loa" to read "Loading catalogue...".
;-----
.LACA3
ACA3 A9 E5 LDA #&E5
ACA5 A2 00 LDX #&00
ACA7 20 B9 A0 JSR LA0B9
ACAA A9 0E LDA #&0E
ACAC 8D 00 7A STA t_osw
ACAF 20 38 A3 JSR LA338
ACB2 A9 19 LDA #&19
ACB4 20 1A BB JSR print_inline_count
ACB7 .str '\x1a\x0c\x1f\x01\x03Finding catalogue...'
ACD0 A9 01 LDA #&01
ACD2 20 41 BB JSR LBB41
.LACD5
ACD5 A9 0F LDA #&0F
ACD7 8D 12 7A STA t_osw2
ACDA 20 3F A3 JSR LA33F
ACDD AD 14 7A LDA t_osw2+2
ACE0 C9 FF CMP #&FF ; no initial page in the TSDP: "No catalogue"
ACE2 D0 0F BNE LACF3
.LACE4
ACE4 00 BRK ; BRK, error &FF "No catalogue"
ACE5 .db FF 4E 6F 20 63 61 74 61 6C 6F 67 75 65 00 ; .No catalogue.
.LACF3
ACF3 AD 13 7A LDA t_osw2+1
ACF6 30 DD BMI LACD5
ACF8 A0 03 LDY #&03
.LACFA
ACFA B9 13 7A LDA t_osw2+1,Y
ACFD 99 02 7A STA t_osw+2,Y
AD00 88 DEY

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AD01 10 F7 BPL LACFA
AD03 A9 AB LDA #&AB
AD05 8D 06 7A STA t_osw+6
AD08 85 82 STA &82
AD0A A9 6B LDA #&6B
AD0C 8D 07 7A STA t_osw+7
AD0F 85 83 STA &83
AD11 A9 05 LDA #&05
AD13 8D 00 7A STA t_osw
AD16 A9 00 LDA #&00
AD18 8D 01 7A STA t_osw+1
AD1B 20 38 A3 JSR LA338
.LAD1E
AD1E A9 03 LDA #&03
AD20 8D 00 7A STA t_osw
AD23 A9 00 LDA #&00
AD25 8D 9B 70 STA dL_to_file
AD28 8D 01 7A STA t_osw+1
AD2B 8D 02 7A STA t_osw+2
AD2E 20 38 A3 JSR LA338
.LAD31
AD31 20 45 B1 JSR LB145
AD34 A9 02 LDA #&02
AD36 8D 00 7A STA t_osw
AD39 20 38 A3 JSR LA338
AD3C AD 02 7A LDA t_osw+2
AD3F 29 02 AND #&02
AD41 F0 EE BEQ LAD31
AD43 A0 00 LDY #&00
AD45 98 TYA
.LAD46
AD46 99 00 7A STA t_osw,Y
AD49 C8 INY
AD4A C0 0F CPY #&0F
AD4C D0 F8 BNE LAD46
AD4E 20 90 A4 JSR find_row27
AD51 A0 19 LDY #&19 ; workspace +&19: grabber records (8 bytes each)
.LAD53
AD53 B1 84 LDA (&84),Y
AD55 AA TAX
AD56 BD AF 91 LDA hamming_table,X
AD59 30 C3 BMI LAD1E
AD5B 29 0F AND #&0F
AD5D 99 5E 00 STA &005E,Y
AD60 C8 INY
AD61 C0 1F CPY #&1F
AD63 D0 EE BNE LAD53
AD65 20 55 91 JSR L9155
AD68 A5 75 LDA &75
AD6A 4A LSR A
AD6B 4A LSR A
AD6C 4A LSR A
AD6D 4D AB 6B EOR &6BAB
AD70 8D 0B 7A STA &7A0B
AD73 A5 72 LDA &72
AD75 C9 FF CMP #&FF
AD77 D0 03 BNE LAD7C
AD79 4C E4 AC JMP LACE4
.LAD7C
AD7C 8D 0C 7A STA &7A0C
AD7F A9 18 LDA #&18 ; OSWORD 24: download the catalogue into memory
AD81 8D 00 7A STA t_osw
AD84 A9 A2 LDA #&A2
AD86 8D 01 7A STA t_osw+1
AD89 A9 AC LDA #&AC
AD8B 8D 02 7A STA t_osw+2
AD8E A9 00 LDA #&00
AD90 8D 03 7A STA t_osw+3
AD93 8D 04 7A STA t_osw+4
AD96 8D 0D 7A STA &7A0D
AD99 8D 0E 7A STA &7A0E
AD9C A9 06 LDA #&06 ; overprint "Fin" with "Loa"
AD9E 20 1A BB JSR print_inline_count
ADA1 .str '\x1f\x01\x03Loa'
ADA7 20 38 A3 JSR LA338
ADAA 4C 60 BB JMP catalogue
;-----
; DOWNLOAD A TELESOFTWARE FILE
; Data is decoded from the Redefinable Telesoftware Format (RTF) a block (one sub-page)
; at a time. With discs or networks each block is written to the file as it completes
; (OPENOUT + OSGBPB); on tape the whole file is kept in memory from OSHWM and saved at
; the end. The block map at &7100 records which blocks have arrived (disordered files).
;-----
.downLoad
ADAD 20 90 B5 JSR LB590
ADB0 20 11 A0 JSR LA011
ADB3 A9 05 LDA #&05
ADB5 20 1A BB JSR print_inline_count
ADB8 .str "\x1c\x00\x18'\x13"
ADB0 A9 00 LDA #&00

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ADBf 8D B3 70 STA dl_ordered
ADc2 8D AB 70 STA dl_blocks_rx
ADc5 8D BC 70 STA dl_handle
ADc8 8D C5 70 STA &70C5
ADCB 8D C6 70 STA &70C6
ADCE 8D C7 70 STA &70C7
ADd1 8D C8 70 STA &70C8
ADd4 8D C3 70 STA &70C3
ADd7 8D C4 70 STA &70C4
ADDA A8      TAY

.LADDB
ADDB 99 00 71 STA dl_block_map,Y      ; clear the block map
ADDE C8      INY
ADDF D0 FA    BNE LADDB
ADE1 A9 FF    LDA #&FF
ADE3 8D BF 70 STA &70BF
ADE6 8D C0 70 STA &70C0
ADE9 A9 01    LDA #&01
ADEB 8D F1 70 STA dl_next_block
ADEE A9 3F    LDA #&3F
ADF0 8D 9F 70 STA dl_sub_hi
ADF3 A9 7F    LDA #&7F
ADF5 8D A0 70 STA dl_sub_lo

.LADF8
ADF8 20 AD B9 JSR LB9AD                ; wait for the first page of the file
ADFB AD 9A 70 LDA dl_blockchar
ADFE C9 91    CMP #&91
AE00 F0 F6    BEQ LADF8
AE02 AD B2 70 LDA dl_keep_tables
AE05 D0 03    BNE LAE0A
AE07 20 12 B0 JSR default_tables      ; default name tables unless "keep tables"

.LAE0A
AE0A 20 EC AF JSR LAFEC
AE0D A9 83    LDA #&83
AE0F 20 F4 FF JSR OSBYTE              ; OSBYTE &83: read OSHWM (PAGE)
AE12 8E BD 70 STX dl_out
AE15 8E AC 70 STX dl_hiwater
AE18 8C BE 70 STY dl_out+1
AE1B 8C AD 70 STY dl_hiwater+1
AE1E A9 01    LDA #&01
AE20 8D EC 70 STA &70EC
AE23 8D EE 70 STA dl_skip

.LAE26
AE26 20 45 B1 JSR LB145                ; decode loop
AE29 20 57 AF JSR next_name           ; next name
AE2C AD FD 70 LDA dl_kind
AE2F C9 03    CMP #&03
AE31 90 03    BCC LAE36
AE33 4C C4 AE JMP LAEC4

.LAE36
AE36 C9 02    CMP #&02                ; kind 2: ignore
AE38 F0 EC    BEQ LAE26
AE3A C9 01    CMP #&01                ; kind 1: data byte
AE3C F0 1D    BEQ LAE5B
AE3E AD FE 70 LDA dl_value
AE41 18      CLC                      ; plus the <ULB>/<URB> adjustment
AE42 6D FA 70 ADC dl_adjust
AE45 20 07 AF JSR put_byte
AE48 AD FA 70 LDA dl_adjust
AE4B C9 58    CMP #&58                ; 'X'
AE4D F0 04    BEQ LAE53
AE4F C9 A8    CMP #&A8
AE51 D0 05    BNE LAE58

.LAE53
AE53 A9 00    LDA #&00
AE55 8D FA 70 STA dl_adjust

.LAE58
AE58 4C 26 AE JMP LAE26

.LAE5B
AE5B AD F9 70 LDA dl_txt_mode          ; escaped name: a command?
AE5E F0 26    BEQ LAE86
AE60 AD FE 70 LDA dl_value
AE63 29 7F    AND #&7F
AE65 8D FE 70 STA dl_value
AE68 AD EE 70 LDA dl_skip
AE6B F0 19    BEQ LAE86
AE6D AD FE 70 LDA dl_value
AE70 C9 61    CMP #&61                ; 'a'
AE72 F0 12    BEQ LAE86
AE74 C9 63    CMP #&63                ; 'c'
AE76 F0 0E    BEQ LAE86
AE78 C9 41    CMP #&41                ; 'A'
AE7A 90 07    BCC LAE83
AE7C C9 5B    CMP #&5B                ; '['
AE7E B0 03    BCS LAE83
AE80 4C 86 AE JMP LAE86

.LAE83
AE83 4C 26 AE JMP LAE26

.LAE86
AE86 A9 50    LDA #&50                ; look it up in the command table at &B150

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```

    AE88 85 86    STA &86
    AE8A A9 B1    LDA #&B1
    AE8C 85 87    STA &87
.LAE8E
    AE8E AD FE 70 LDA dL_value
    AE91 A0 00    LDY #&00
    AE93 D1 86    CMP (&86),Y
    AE95 F0 19    BEQ LAEB0
    AE97 18       CLC
    AE98 A5 86    LDA &86
    AE9A 69 06    ADC #&06
    AE9C 85 86    STA &86
    AE9E A5 87    LDA &87
    AEA0 69 00    ADC #&00
    AEA2 85 87    STA &87
    AEA4 B1 86    LDA (&86),Y
    AEA6 D0 E6    BNE LAE8E
    AEA8 20 92 BA JSR LBA92           ; unknown: "Bad Data..."
    AEAB 48       PHA
    AEAC 48       PHA
    AEAD 4C 50 B4 JMP LB450
.LAEB0
    AEB0 A0 04    LDY #&04
    AEB2 B1 86    LDA (&86),Y
    AEB4 85 77    STA &77
    AEB6 C8       INY
    AEB7 B1 86    LDA (&86),Y
    AEB9 85 78    STA &78
    AEBB 20 C1 AE JSR LAEC1
    AEBE 4C 26 AE JMP LAE26
.LAEC1
    AEC1 6C 77 00 JMP (&0077)
.LAEC4
    AEC4 AD FE 70 LDA dL_value           ; kind 3+: a string
    AEC7 85 86    STA &86
    AEC9 AD FD 70 LDA dL_kind
    AECC 85 87    STA &87
    AECE A0 00    LDY #&00
    AED0 B1 86    LDA (&86),Y
    AED2 8D F8 70 STA &70F8
    AED5 E6 86    INC &86
    AED7 D0 02    BNE LAEDB
    AED9 E6 87    INC &87
.LAEDB
    AEDB A0 00    LDY #&00
    AEDD B1 86    LDA (&86),Y
    AEDF 18       CLC
    AEE0 6D FA 70 ADC dL_adjust
    AEE3 20 07 AF JSR put_byte
    AEE6 AD FA 70 LDA dL_adjust
    AEE9 C9 58    CMP #&58           ; 'X'
    AEEB F0 04    BEQ LAEF1
    AEED C9 A8    CMP #&A8
    AEEF D0 05    BNE LAEF6
.LAEF1
    AEF1 A9 00    LDA #&00
    AEF3 8D FA 70 STA dL_adjust
.LAEF6
    AEF6 C8       INY
    AEF7 CC F8 70 CPY &70F8
    AEFA F0 08    BEQ LAF04
    AEFB B1 86    LDA (&86),Y
    AEFE 20 07 AF JSR put_byte
    AF01 4C F6 AE JMP LAEF6
.LAF04
    AF04 4C 26 AE JMP LAE26
;-----
; Store a decoded byte at the output pointer (unless the block is being skipped) and
; keep the high-water mark.
;-----
.put_byte
    AF07 8D FE 70 STA dL_value
    AF0A 98       TYA
    AF0B 48       PHA
    AF0C AD EE 70 LDA dL_skip
    AF0F D0 25    BNE LAF36
    AF11 A5 86    LDA &86
    AF13 48       PHA
    AF14 A5 87    LDA &87
    AF16 48       PHA
    AF17 AD BD 70 LDA dL_out
    AF1A 85 86    STA &86
    AF1C AD BE 70 LDA dL_out+1
    AF1F 85 87    STA &87
    AF21 A0 00    LDY #&00
    AF23 AD FE 70 LDA dL_value
    AF26 91 86    STA (&86),Y
    AF28 68       PLA
    AF29 85 87    STA &87
    AF2B 68       PLA

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```

AF2C 85 86 STA &86
AF2E EE BD 70 INC dL_out
AF31 D0 03 BNE LAF36
AF33 EE BE 70 INC dL_out+1
.LAF36
AF36 AD BE 70 LDA dL_out+1
AF39 CD AD 70 CMP dL_hiwater+1
AF3C 90 16 BCC LAF54
AF3E D0 08 BNE LAF48
AF40 AD BD 70 LDA dL_out
AF43 CD AC 70 CMP dL_hiwater
AF46 90 0C BCC LAF54
.LAF48
AF48 AD BD 70 LDA dL_out
AF4B 8D AC 70 STA dL_hiwater
AF4E AD BE 70 LDA dL_out+1
AF51 8D AD 70 STA dL_hiwater+1
.LAF54
AF54 68 PLA
AF55 A8 TAY
AF56 60 RTS
;-----
; Next raw byte, looked up in the lone tables; an escape operator switches to the
; escaped tables for the following byte.
;-----
.next_name
AF57 20 A6 BA JSR raw_byte
AF5A 8D FF 70 STA dL_raw
AF5D A8 TAY
AF5E B1 82 LDA (&82),Y
AF60 8D FE 70 STA dL_value
AF63 B1 84 LDA (&84),Y
AF65 8D FD 70 STA dL_kind
AF68 C9 01 CMP #&01
AF6A D0 0D BNE LAF79
AF6C AD FE 70 LDA dL_value
AF6F C9 1B CMP #&1B
AF71 D0 06 BNE LAF79
AF73 20 FF AF JSR LAFFF
AF76 4C 57 AF JMP next_name
.LAF79
AF79 20 EC AF JSR LAFEC
AF7C 60 RTS
;-----
; Next decoded data byte, applying <ULB> (-&58) / <URB> (+&58) prefixes.
;-----
.next_value
AF7D 98 TYA
AF7E 48 PHA
AF7F 8A TXA
AF80 48 PHA
AF81 20 EC AF JSR LAFEC
AF84 20 57 AF JSR next_name
AF87 AD FD 70 LDA dL_kind
AF8A C9 01 CMP #&01
AF8C D0 3B BNE LAFC9
AF8E AD FE 70 LDA dL_value
AF91 C9 60 CMP #&60 ; ''
AF93 D0 04 BNE LAF99
AF95 A9 58 LDA #&58
AF97 D0 06 BNE LAF9F
.LAF99
AF99 C9 7E CMP #&7E ; '~'
AF9B D0 2C BNE LAFC9
AF9D A9 A8 LDA #&A8
.LAF9F
AF9F 8D FA 70 STA dL_adjust
AFA2 20 A6 BA JSR raw_byte
AFA5 8D FE 70 STA dL_value
AFA8 AD F9 70 LDA dL_txt_mode
AFAB F0 08 BEQ LAFB5
AFAD AD FE 70 LDA dL_value
AFB0 29 7F AND #&7F
AFB2 8D FE 70 STA dL_value
.LAFB5
AFB5 AD FE 70 LDA dL_value
AFB8 18 CLC
AFB9 6D FA 70 ADC dL_adjust
AFBC 8D FE 70 STA dL_value
AFBF A9 00 LDA #&00
AFC1 8D FD 70 STA dL_kind
AFC4 8D FA 70 STA dL_adjust
AFC7 F0 1B BEQ LAFE4
.LAFC9
AFC9 AD FF 70 LDA dL_raw
AFCC 8D FE 70 STA dL_value
AFCF A9 00 LDA #&00
AFD1 8D FD 70 STA dL_kind
AFD4 8D FA 70 STA dL_adjust
AFD7 AD F9 70 LDA dL_txt_mode

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```

AFDA F0 08    BEQ LAFE4
AFDC AD FE 70 LDA dL_value
AFDF 29 7F    AND #&7F
AFE1 8D FE 70 STA dL_value
.LAFE4
AFE4 68      PLA
AFE5 AA      TAX
AFE6 68      PLA
AFE7 A8      TAY
AFE8 AD FE 70 LDA dL_value
AFEB 60      RTS
; Use the lone-name tables (&7900 value, &7800 kind).
.LAFEC
AFEC 48      PHA
AFED A9 00    LDA #&00
AFEF 85 82    STA &82
AFF1 A9 79    LDA #&79
AFF3 85 83    STA &83
AFF5 A9 00    LDA #&00
AFF7 85 84    STA &84
AFF9 A9 78    LDA #&78
AFFB 85 85    STA &85
AFFD 68      PLA
AFFE 60      RTS
; Use the escaped-name tables (&7700 value, &7600 kind).
.LAFF
AFFF 48      PHA
B000 A9 00    LDA #&00
B002 85 82    STA &82
B004 A9 77    LDA #&77
B006 85 83    STA &83
B008 A9 00    LDA #&00
B00A 85 84    STA &84
B00C A9 76    LDA #&76
B00E 85 85    STA &85
B010 68      PLA
B011 60      RTS
;
; DEFAULT NAME TABLES: lone names decode to themselves, escaped names are the escape
; operator, lone &1B/&7C/&9B/&FC are escape, escaped letters are commands.
;
.default_tables
B012 A9 00    LDA #&00
B014 8D B0 70 STA &70B0
B017 A9 72    LDA #&72
B019 8D B1 70 STA &70B1
B01C A0 00    LDY #&00
.LB01E
B01E 98      TYA
B01F 99 00 79 STA &7900,Y
B022 A9 00    LDA #&00
B024 99 00 78 STA &7800,Y
B027 A9 1B    LDA #&1B
B029 99 00 77 STA &7700,Y
B02C A9 01    LDA #&01
B02E 99 00 76 STA &7600,Y
B031 C8      INY
B032 D0 EA    BNE LB01E
B034 A9 1B    LDA #&1B
B036 8D 1B 79 STA &791B
B039 8D 7C 79 STA &797C
B03C 8D 9B 79 STA &799B
B03F 8D FC 79 STA &79FC
B042 A9 01    LDA #&01
B044 8D 1B 78 STA &781B
B047 8D 7C 78 STA &787C
B04A 8D 9B 78 STA &789B
B04D 8D FC 78 STA &78FC
B050 A9 00    LDA #&00
B052 8D 1B 76 STA &761B
B055 8D 7C 76 STA &767C
B058 8D 9B 76 STA &769B
B05B 8D FC 76 STA &76FC
B05E A9 1B    LDA #&1B
B060 8D 1B 77 STA &771B
B063 A9 7C    LDA #&7C
B065 8D 7C 77 STA &777C
B068 A9 9B    LDA #&9B
B06A 8D 9B 77 STA &779B
B06D A9 FC    LDA #&FC
B06F 8D FC 77 STA &77FC
B072 A0 61    LDY #&61
.LB074
B074 98      TYA
B075 99 00 77 STA &7700,Y
B078 99 80 77 STA &7780,Y
B07B C8      INY
B07C C0 7B    CPY #&7B
B07E D0 F4    BNE LB074
B080 A0 41    LDY #&41
; '{'

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.LB082
B082 98 TYA
B083 99 00 77 STA &7700,Y
B086 99 80 77 STA &7780,Y
B089 C8 INY
B08A C0 5B CPY #&5B ; '['
B08C D0 F4 BNE LB082
B08E 60 RTS
; &70FC = parity of A.
.parity
B08F 48 PHA
B090 A9 00 LDA #&00
B092 8D FC 70 STA &70FC
B095 68 PLA
B096 48 PHA
B097 A2 08 LDX #&08
.LB099
B099 0A ASL A
B09A 90 03 BCC LB09F
B09C EE FC 70 INC &70FC
.LB09F
B09F CA DEX
B0A0 D0 F7 BNE LB099
B0A2 AD FC 70 LDA &70FC
B0A5 29 01 AND #&01
B0A7 8D FC 70 STA &70FC
B0AA 68 PLA
B0AB 60 RTS
;-----
; <TXT> tables: names with bad parity decode to &7F (error); parity bit removed.
;-----
.LB0AC
B0AC A0 00 LDY #&00
.LB0AE
B0AE 98 TYA
B0AF 20 8F B0 JSR parity
B0B2 AD FC 70 LDA &70FC
B0B5 4D FB 70 EOR &70FB
B0B8 D0 13 BNE LB0CD
B0BA A9 7F LDA #&7F
B0BC 99 00 79 STA &7900,Y
B0BF 99 00 77 STA &7700,Y
B0C2 A9 01 LDA #&01
B0C4 99 00 78 STA &7800,Y
B0C7 99 00 76 STA &7600,Y
B0CA 4C E7 B0 JMP LB0E7
.LB0CD
B0CD B9 00 78 LDA &7800,Y
B0D0 D0 08 BNE LB0DA
B0D2 B9 00 79 LDA &7900,Y
B0D5 29 7F AND #&7F
B0D7 99 00 79 STA &7900,Y
.LB0DA
B0DA B9 00 76 LDA &7600,Y
B0DD D0 08 BNE LB0E7
B0DF B9 00 77 LDA &7700,Y
B0E2 29 7F AND #&7F
B0E4 99 00 77 STA &7700,Y
.LB0E7
B0E7 C8 INY
B0E8 D0 C4 BNE LB0AE
B0EA 60 RTS
;-----
; Teletext format: odd parity, <ULB> = escaped &60, <URB> = escaped &7E.
;-----
.LB0EB
B0EB A9 00 LDA #&00
B0ED 8D FB 70 STA &70FB
B0F0 20 AC B0 JSR LB0AC
B0F3 A9 60 LDA #&60
B0F5 8D E0 77 STA &77E0
B0F8 8D E0 79 STA &79E0
B0FB A9 7E LDA #&7E
B0FD 8D FE 77 STA &77FE
B100 8D FE 79 STA &79FE
B103 A9 00 LDA #&00
B105 8D E0 76 STA &76E0
B108 8D FE 76 STA &76FE
B10B A9 01 LDA #&01
B10D 8D E0 78 STA &78E0
B110 8D FE 78 STA &78FE
B113 60 RTS
; <TX8>: 8-bit format tables.
.LB114
B114 A9 1B LDA #&1B
B116 8D 7C 77 STA &777C
B119 8D 9B 77 STA &779B
B11C 8D FC 77 STA &77FC
B11F A9 01 LDA #&01
B121 8D 7C 76 STA &767C

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B124 8D 9B 76 STA &769B
B127 8D FC 76 STA &76FC
B12A A9 00 LDA #&00
B12C 8D 7C 78 STA &787C
B12F 8D 9B 78 STA &789B
B132 8D FC 78 STA &78FC
B135 A9 7C LDA #&7C
B137 8D 7C 79 STA &797C
B13A A9 9B LDA #&9B
B13C 8D 9B 79 STA &799B
B13F A9 FC LDA #&FC
B141 8D FC 79 STA &79FC
B144 60 RTS
; ESCAPE pressed? Abandon the download.
.LB145
B145 A5 FF LDA escape_flag
B147 30 01 BMI LB14A
B149 60 RTS
.LB14A
B14A 20 55 89 JSR L8955
B14D 4C C4 88 JMP L88C4
;-----
; RTF COMMAND TABLE: escaped name, 3-letter mnemonic, handler. 27 commands.
;-----
.rtf_commands
B150 .db 1B 45 53 43 F3 B1 60 55 4C 42 F3 B1 7E 55 52 42 ; .ESC..`ULB..~URB
B160 .db F9 B1 42 54 58 54 14 B2 43 54 58 38 1C B2 46 54 ; ..BTXT..CTX8..FT
B170 .db 58 4F FF B1 61 44 54 4C 93 B2 62 44 53 42 F8 B2 ; X0..aDTL..bDSB..
B180 .db 63 44 45 42 58 B3 64 44 45 54 80 B4 65 44 53 4C ; cDEBX.dDET..eDSL
B190 .db DF B4 68 44 53 54 D7 B4 69 44 44 54 90 B4 6A 44 ; ..hDST..iDDT..jD
B1A0 .db 43 4F 93 B4 6B 44 49 47 B9 B4 6C 44 4C 41 E3 B4 ; CO..kDIG..LDLA..
B1B0 .db 6D 44 4C 52 24 B5 6E 44 58 41 5F B5 6F 44 58 52 ; mDLR$.nDXA_.oDXR
B1C0 .db 6E B5 70 44 49 52 90 B5 71 44 45 53 AB B5 72 44 ; n.pDIR..qDES..rD
B1D0 .db 45 43 B6 B5 73 44 4C 53 BE B5 74 44 4C 43 C9 B5 ; EC..sDLS..tDLC..
B1E0 .db 78 44 4E 44 FD B6 79 44 45 46 FE B6 7F 55 45 52 ; xDND..yDEF...UER
B1F0 .db F7 B6 00 ; ...
; <ESC>/<ULB>: next byte -&58.
.LB1F3
B1F3 A9 58 LDA #&58
B1F5 8D FA 70 STA dL_adjust
B1F8 60 RTS
; <URB>: next byte +&58.
.LB1F9
B1F9 A9 A8 LDA #&A8
B1FB 8D FA 70 STA dL_adjust
B1FE 60 RTS
; <TX0>: teletext, ordered; then <DSB>.
.LB1FF
B1FF A9 01 LDA #&01
B201 8D B3 70 STA dL_ordered
.LB204
B204 AD B2 70 LDA dL_keep_tables
B207 D0 03 BNE LB20C
B209 20 EB B0 JSR LB0EB
.LB20C
B20C A9 01 LDA #&01
B20E 8D F9 70 STA dL_txt_mode
B211 4C F8 B2 JMP LB2F8
; <TXT>: teletext, disordered.
.LB214
B214 A9 00 LDA #&00
B216 8D B3 70 STA dL_ordered
B219 4C 04 B2 JMP LB204
; <TX8>: 8-bit data.
.LB21C
B21C AD B2 70 LDA dL_keep_tables
B21F D0 03 BNE LB224
B221 20 14 B1 JSR LB114
.LB224
B224 A9 00 LDA #&00
B226 8D F9 70 STA dL_txt_mode
B229 8D B3 70 STA dL_ordered
B22C 4C F8 B2 JMP LB2F8
; Upper-case A.
.LB22F
B22F 29 7F AND #&7F
B231 C9 61 CMP #&61 ; 'a'
B233 90 07 BCC LB23C
B235 C9 7B CMP #&7B ; '{'
B237 B0 03 BCS LB23C
B239 38 SEC
B23A E9 20 SBC #&20
.LB23C
B23C 60 RTS
;-----
; "Different file": the block belongs to another file on the same page; show its name
; and block number and skip it.
;-----
.LB23D
B23D 99 C9 70 STA &70C9,Y

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B240 C8      INY
B241 CA      DEX
B242 F0 06    BEQ LB24A
B244 20 7D AF JSR next_value
B247 4C 3D B2 JMP LB23D
.LB24A
B24A A9 00    LDA #000
B24C 99 C9 70 STA &70C9,Y
B24F A9 12    LDA #12
B251 20 1A BB JSR print_inline_count
B254 .str 'Different file : "'
B266 A0 00    LDY #000
.LB268
B268 B9 C9 70 LDA &70C9,Y
B26B 20 EE FF JSR OSWRCH
B26E C8      INY
B26F C9 00    CMP #000
B271 D0 F5    BNE LB268
B273 A9 03    LDA #003
B275 20 1A BB JSR print_inline_count
B278 .str '" &'
B27B AD F2 70 LDA dL_block
B27E 20 D8 B7 JSR LB7D8
B281 A9 02    LDA #002
B283 20 1A BB JSR print_inline_count
B286 .str '\r\n'
B288 A9 00    LDA #000
B28A 8D F2 70 STA dL_block
B28D A9 FF    LDA #0FF
B28F 8D EE 70 STA dL_skip
B292 60      RTS
;-----
; <DTL> TITLE: compare with the wanted name (case ignored) or adopt it if none given.
;-----
.LB293
B293 20 0A B7 JSR LB70A
B296 8D F8 70 STA &70F8
B299 20 0A B7 JSR LB70A
B29C F0 31    BEQ LB2CF
B29E AA      TAX
B29F A0 00    LDY #000
B2A1 AD DA 70 LDA dL_name
B2A4 C9 0D    CMP #00D
B2A6 F0 15    BEQ LB2BD
.LB2A8
B2A8 20 7D AF JSR next_value
B2AB 99 C9 70 STA &70C9,Y
B2AE 20 2F B2 JSR LB22F
B2B1 D9 DA 70 CMP dL_name,Y
B2B4 D0 87    BNE LB23D
B2B6 C8      INY
B2B7 CA      DEX
B2B8 D0 EE    BNE LB2A8
B2BA 4C CF B2 JMP LB2CF
.LB2BD
B2BD 20 7D AF JSR next_value
B2C0 20 2F B2 JSR LB22F
B2C3 99 DA 70 STA dL_name,Y
B2C6 C8      INY
B2C7 CA      DEX
B2C8 D0 F3    BNE LB2BD
B2CA A9 0D    LDA #00D
B2CC 99 DA 70 STA dL_name,Y
.LB2CF
B2CF CE F8 70 DEC &70F8
B2D2 AD F8 70 LDA &70F8
B2D5 F0 20    BEQ LB2F7
B2D7 20 0A B7 JSR LB70A
B2DA F0 07    BEQ LB2E3
B2DC AA      TAX
.LB2DD
B2DD 20 7D AF JSR next_value
B2E0 CA      DEX
B2E1 D0 FA    BNE LB2DD
.LB2E3
B2E3 CE F8 70 DEC &70F8
B2E6 AD F8 70 LDA &70F8
B2E9 F0 0C    BEQ LB2F7
B2EB 20 0A B7 JSR LB70A
B2EE F0 07    BEQ LB2F7
B2F0 AA      TAX
.LB2F1
B2F1 20 7D AF JSR next_value
B2F4 CA      DEX
B2F5 D0 FA    BNE LB2F1
.LB2F7
B2F7 60      RTS
;-----
; <DSB> START OF BLOCK [block number] [number of blocks]: skip blocks already received or
; (ordered files) out of sequence.

```

```

;-----
.LB2F8
B2F8 A9 00 LDA #&00
B2FA 8D FA 70 STA dL_adjust
B2FD 8D F2 70 STA dL_block
B300 8D EF 70 STA dL_nblocks
B303 20 EC AF JSR LAFEC
B306 20 0A B7 JSR LB70A
B309 8D F8 70 STA &70F8
B30C F0 19 BEQ LB327
B30E 20 40 B7 JSR LB740
B311 AD F4 70 LDA &70F4
B314 8D F2 70 STA dL_block
B317 AD F8 70 LDA &70F8
B31A C9 02 CMP #&02
B31C D0 09 BNE LB327
B31E 20 40 B7 JSR LB740
B321 AD F4 70 LDA &70F4
B324 8D EF 70 STA dL_nblocks

.LB327
B327 AD B3 70 LDA dL_ordered
B32A F0 0C BEQ LB338
B32C AD F2 70 LDA dL_block
B32F CD F1 70 CMP dL_next_block
B332 F0 04 BEQ LB338
B334 A9 09 LDA #&09
B336 D0 0A BNE LB342

.LB338
B338 AC F2 70 LDY dL_block
B33B B9 00 71 LDA dL_block_map,Y
B33E 10 0B BPL LB34B
B340 A9 09 LDA #&09

.LB342
B342 8D 9A 70 STA dL_blockchar
B345 A9 01 LDA #&01
B347 8D EE 70 STA dL_skip
B34A 60 RTS

.LB34B
B34B AD 9A 70 LDA dL_blockchar
B34E C9 92 CMP #&92
B350 D0 05 BNE LB357
B352 A9 00 LDA #&00
B354 8D EE 70 STA dL_skip

.LB357
B357 60 RTS

;-----
; <DEB> END OF BLOCK: on the first block, open the output file; mark the block received,
; write it (disc) and draw the block display. When every block is in, close the file and
; set its load/exec addresses (OSFILE 5/1), or save from memory (tape).
;-----
.LB358
B358 AD EE 70 LDA dL_skip
B35B 10 01 BPL LB35E
B35D 60 RTS

.LB35E
B35E AD EC 70 LDA &70EC
B361 F0 14 BEQ LB377
B363 20 03 B8 JSR LB803
B366 AD 9C 70 LDA dL_save_each
B369 F0 0C BEQ LB377
B36B A2 DA LDX #&DA
B36D A0 70 LDY #&70
B36F A9 80 LDA #&80
B371 20 CE FF JSR OSFIND
B374 8D BC 70 STA dL_handle

.LB377
B377 AD EE 70 LDA dL_skip
B37A D0 03 BNE LB37F
B37C EE AB 70 INC dL_blocks_rx

.LB37F
B37F 20 B7 B8 JSR LB8B7
B382 AD EE 70 LDA dL_skip
B385 F0 01 BEQ LB388
B387 60 RTS

.LB388
B388 AC F2 70 LDY dL_block
B38B A9 FF LDA #&FF
B38D 99 00 71 STA dL_block_map,Y
B390 EE F1 70 INC dL_next_block
B393 A9 01 LDA #&01
B395 8D EE 70 STA dL_skip
B398 AD 9C 70 LDA dL_save_each
B39B F0 52 BEQ LB3EF
B39D AD BC 70 LDA dL_handle
B3A0 F0 4D BEQ LB3EF
B3A2 A9 83 LDA #&83
B3A4 20 F4 FF JSR OSBYTE ; OSBYTE &83: read OSHWM (PAGE)
B3A7 8E BD 70 STX dL_out
B3AA 8C BE 70 STY dL_out+1
B3AD 38 SEC

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B3AE AD AC 70 LDA dL_hiwater
B3B1 ED BD 70 SBC dL_out
B3B4 8D C1 70 STA &70C1
B3B7 AD AD 70 LDA dL_hiwater+1
B3BA ED BE 70 SBC dL_out+1
B3BD 8D C2 70 STA &70C2
B3C0 A0 03 LDY #&03
.LB3C2
B3C2 B9 C5 70 LDA &70C5,Y
B3C5 99 82 00 STA &0082,Y
B3C8 88 DEY
B3C9 10 F7 BPL LB3C2
B3CB AC BC 70 LDY dL_handle
B3CE A2 82 LDX #&82
B3D0 A9 01 LDA #&01
B3D2 20 DA FF JSR OSARGS
B3D5 A2 BC LDX #&8C
B3D7 A0 70 LDY #&70
B3D9 A9 02 LDA #&02
B3DB 20 D1 FF JSR OSGBPB
B3DE A9 83 LDA #&83
B3E0 20 F4 FF JSR OSBYTE ; OSBYTE &83: read OSHWM (PAGE)
B3E3 8E BD 70 STX dL_out
B3E6 8E AC 70 STX dL_hiwater
B3E9 8C BE 70 STY dL_out+1
B3EC 8C AD 70 STY dL_hiwater+1
.LB3EF
B3EF A0 00 LDY #&00
.LB3F1
B3F1 C8 INY
B3F2 B9 00 71 LDA dL_block_map,Y
B3F5 10 58 BPL LB44F
B3F7 CC EF 70 CPY dL_nblocks
B3FA D0 F5 BNE LB3F1
.LB3FC
B3FC A9 0E LDA #&0E
B3FE 8D 00 7A STA t_osw
B401 20 38 A3 JSR LA338
B404 AD 9C 70 LDA dL_save_each
B407 F0 15 BEQ LB41E
B409 A9 00 LDA #&00
B40B AC BC 70 LDY dL_handle
B40E 20 CE FF JSR OSFIND
B411 A9 05 LDA #&05
B413 20 60 B4 JSR LB460
B416 A9 01 LDA #&01
B418 20 60 B4 JSR LB460
B41B 4C 4D B4 JMP LB44D
.LB41E
B41E AD 9B 70 LDA dL_to_file
B421 F0 2A BEQ LB44D
B423 A9 FF LDA #&FF
B425 8D 0C 7A STA &7A0C
B428 8D 0D 7A STA &7A0D
B42B 8D 10 7A STA &7A10
B42E 8D 11 7A STA &7A11
B431 A9 83 LDA #&83
B433 20 F4 FF JSR OSBYTE ; OSBYTE &83: read OSHWM (PAGE)
B436 8E 0A 7A STX &7A0A
B439 8C 0B 7A STY &7A0B
B43C AD AC 70 LDA dL_hiwater
B43F 8D 0E 7A STA &7A0E
B442 AD AD 70 LDA dL_hiwater+1
B445 8D 0F 7A STA &7A0F
B448 A9 00 LDA #&00
B44A 20 60 B4 JSR LB460
.LB44D
B44D 68 PLA
B44E 68 PLA
.LB44F
B44F 60 RTS
; Abandon: stop grabbing and close the file.
.LB450
B450 20 5F 89 JSR LB95F
B453 AC BC 70 LDY dL_handle
B456 F0 F5 BEQ LB44D
B458 A9 00 LDA #&00
B45A 20 CE FF JSR OSFIND
B45D 4C 4D B4 JMP LB44D
; OSFILE A with the download's name, load and exec addresses.
.LB460
B460 48 PHA
B461 A9 DA LDA #&DA
B463 8D 00 7A STA t_osw
B466 A9 70 LDA #&70
B468 8D 01 7A STA t_osw+1
B46B A0 00 LDY #&00
.LB46D
B46D B9 B4 70 LDA dL_load,Y
B470 99 02 7A STA t_osw+2,Y

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B473 C8      INY
B474 C0 08    CPY #&08
B476 D0 F5    BNE LB46D
B478 A2 00    LDX #&00
B47A A0 7A    LDY #&7A
B47C 68      PLA
B47D 4C DD FF JMP OSFILE
; <DET> END OF FILE.
.LB480
B480 A9 01    LDA #&01
B482 8D ED 70 STA &70ED
B485 AD B3 70 LDA dL_ordered
B488 D0 03    BNE LB48D
B48A 4C 58 B3 JMP LB358
.LB48D
B48D 4C FC B3 JMP LB3FC
; <DDT> data type: ignored.
.LB490
B490 4C B9 B4 JMP LB4B9
; <DC0> COMMENT: displayed.
.LB493
B493 20 0A B7 JSR LB70A
B496 8D F8 70 STA &70F8
.LB499
B499 AD F8 70 LDA &70F8
B49C F0 1A    BEQ LB4B8
B49E 20 0A B7 JSR LB70A
B4A1 AA      TAX
B4A2 F0 0E    BEQ LB4B2
.LB4A4
B4A4 20 7D AF JSR next_value
B4A7 09 80    ORA #&80
B4A9 20 EE FF JSR OSHRCH
B4AC CA      DEX
B4AD D0 F5    BNE LB4A4
B4AF 20 E7 FF JSR OSNEWL
.LB4B2
B4B2 CE F8 70 DEC &70F8
B4B5 4C 99 B4 JMP LB499
.LB4B8
B4B8 60      RTS
; <DIG> and others: read and ignore operands.
.LB4B9
B4B9 20 0A B7 JSR LB70A
B4BC 8D F8 70 STA &70F8
.LB4BF
B4BF AD F8 70 LDA &70F8
B4C2 F0 12    BEQ LB4D6
B4C4 20 0A B7 JSR LB70A
B4C7 AA      TAX
B4C8 F0 06    BEQ LB4D0
.LB4CA
B4CA 20 7D AF JSR next_value
B4CD CA      DEX
B4CE D0 FA    BNE LB4CA
.LB4D0
B4D0 CE F8 70 DEC &70F8
B4D3 4C BF B4 JMP LB4BF
.LB4D6
B4D6 60      RTS
.LB4D7
B4D7 A9 01    LDA #&01
B4D9 8D F8 70 STA &70F8
B4DC 4C BF B4 JMP LB4BF
.LB4DF
B4DF 20 40 B7 JSR LB740
B4E2 60      RTS
; -----
; <DLA> LOAD ADDRESS: first block sets the base; others become offsets.
; -----
.LB4E3
B4E3 A9 83    LDA #&83
B4E5 20 F4 FF JSR OSBYTE                ; OSBYTE &83: read OSHWM (PAGE)
B4E8 8E BD 70 STX dL_out
B4EB 8C BE 70 STY dL_out+1
B4EE 20 40 B7 JSR LB740
B4F1 AD F2 70 LDA dL_block
B4F4 C9 01    CMP #&01
B4F6 D0 14    BNE LB50C
B4F8 A0 00    LDY #&00
.LB4FA
B4FA B9 F4 70 LDA &70F4,Y
B4FD 99 B4 70 STA dL_load,Y
B500 A9 00    LDA #&00
B502 99 F4 70 STA &70F4,Y
B505 C8      INY
B506 C0 04    CPY #&04
B508 D0 F0    BNE LB4FA
B50A F0 15    BEQ LB521
.LB50C

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B50C A0 00 LDY #&00
B50E 38 SEC
B50F 08 PHP
.LB510
B510 28 PLP
B511 B9 F4 70 LDA &70F4,Y
B514 F9 B4 70 SBC dL_load,Y
B517 99 F4 70 STA &70F4,Y
B51A 08 PHP
B51B C8 INY
B51C C0 04 CPY #&04
B51E D0 F0 BNE LB510
B520 28 PLP
.LB521
B521 4C 32 B5 JMP LB532
;-----
; <DLR> RELATIVE LOAD ADDRESS: set the output position (file pointer on disc).
;-----
.LB524
B524 A9 83 LDA #&83
B526 20 F4 FF JSR OSBYTE ; OSBYTE &83: read OSHWM (PAGE)
B529 8E BD 70 STX dL_out
B52C 8C BE 70 STY dL_out+1
B52F 20 40 B7 JSR LB740
.LB532
B532 AD 9C 70 LDA dL_save_each
B535 D0 14 BNE LB54B
B537 18 CLC
B538 AD BD 70 LDA dL_out
B53B 6D F4 70 ADC &70F4
B53E 8D BD 70 STA dL_out
B541 AD BE 70 LDA dL_out+1
B544 6D F5 70 ADC &70F5
B547 8D BE 70 STA dL_out+1
B54A 60 RTS
.LB54B
B54B A9 00 LDA #&00
B54D 8D AC 70 STA dL_hiwater
B550 8D AD 70 STA dL_hiwater+1
B553 A2 03 LDX #&03
.LB555
B555 BD F4 70 LDA &70F4,X
B558 9D C5 70 STA &70C5,X
B55B CA DEX
B55C 10 F7 BPL LB555
B55E 60 RTS
; <DXA> EXECUTION ADDRESS.
.LB55F
B55F 20 40 B7 JSR LB740
B562 A0 03 LDY #&03
.LB564
B564 B9 F4 70 LDA &70F4,Y
B567 99 B8 70 STA dL_exec,Y
B56A 88 DEY
B56B 10 F7 BPL LB564
B56D 60 RTS
; <DXR> RELATIVE EXECUTION ADDRESS (relative to OSHWM).
.LB56E
B56E 20 5F B5 JSR LB55F
B571 A9 83 LDA #&83
B573 20 F4 FF JSR OSBYTE ; OSBYTE &83: read OSHWM (PAGE)
B576 8E F4 70 STX &70F4
B579 8C F5 70 STY &70F5
B57C 18 CLC
B57D AD B8 70 LDA dL_exec
B580 6D F4 70 ADC &70F4
B583 8D B8 70 STA dL_exec
B586 AD B9 70 LDA &70B9
B589 6D F5 70 ADC &70F5
B58C 8D B9 70 STA &70B9
B58F 60 RTS
;-----
; Set up: save each block to the file unless the filing system is tape (number < 4).
;-----
.LB590
B590 A9 01 LDA #&01
B592 8D EB 70 STA &70EB
B595 AD 9B 70 LDA dL_to_file
B598 8D 9C 70 STA dL_save_each
B59B A9 00 LDA #&00
B59D A8 TAY
B59E 20 DA FF JSR OSARGS
B5A1 C9 04 CMP #&04
B5A3 B0 05 BCS LB5AA
B5A5 A9 00 LDA #&00
B5A7 8D 9C 70 STA dL_save_each
.LB5AA
B5AA 60 RTS
;-----
; <DES>/<DEC>/<DLS>/<DLC>: redefine an escaped or lone name as a value, string or command.

```



```

;-----
.LB5AB
B5AB 20 DC B5 JSR LB5DC
B5AE A9 01 LDA #&01
B5B0 8D AA 70 STA &70AA
B5B3 4C FF B5 JMP LB5FF

.LB5B6
B5B6 A9 01 LDA #&01
B5B8 8D AA 70 STA &70AA
B5BB 4C 89 B6 JMP LB689

.LB5BE
B5BE 20 DC B5 JSR LB5DC
B5C1 A9 00 LDA #&00
B5C3 8D AA 70 STA &70AA
B5C6 4C FF B5 JMP LB5FF

.LB5C9
B5C9 A9 00 LDA #&00
B5CB 8D AA 70 STA &70AA
B5CE 4C 89 B6 JMP LB689

.LB5D1
B5D1 AD AA 70 LDA &70AA
B5D4 F0 03 BEQ LB5D9
B5D6 4C FF AF JMP LAFFF

.LB5D9
B5D9 4C EC AF JMP LAFEC

.LB5DC
B5DC 20 7D AF JSR next_value
B5DF AD F9 70 LDA dL_txt_mode
B5E2 F0 1A BEQ LB5FE
B5E4 AD FE 70 LDA dL_value
B5E7 29 7F AND #&7F
B5E9 8D FE 70 STA dL_value
B5EC 20 8F B0 JSR parity
B5EF AD FC 70 LDA &70FC
B5F2 29 01 AND #&01
B5F4 D0 08 BNE LB5FE
B5F6 AD FE 70 LDA dL_value
B5F9 09 80 ORA #&80
B5FB 8D FE 70 STA dL_value

.LB5FE
B5FE 60 RTS

.LB5FF
B5FF 20 D1 B5 JSR LB5D1
B602 AC FE 70 LDY dL_value
B605 84 78 STY &78
B607 20 0A B7 JSR LB70A
B60A 8D F8 70 STA &70F8
B60D D0 10 BNE LB61F
B60F 20 D1 B5 JSR LB5D1
B612 A9 02 LDA #&02
B614 A4 78 LDY &78
B616 91 84 STA (&84),Y
B618 A9 00 LDA #&00
B61A 91 82 STA (&82),Y
B61C 4C 86 B6 JMP LB686

.LB61F
B61F C9 01 CMP #&01
B621 D0 14 BNE LB637
B623 20 7D AF JSR next_value
B626 20 D1 B5 JSR LB5D1
B629 A4 78 LDY &78
B62B AD FE 70 LDA dL_value
B62E 91 82 STA (&82),Y
B630 A9 00 LDA #&00
B632 91 84 STA (&84),Y
B634 4C 86 B6 JMP LB686

.LB637
B637 20 D1 B5 JSR LB5D1
B63A A4 78 LDY &78
B63C AD B0 70 LDA &70B0
B63F 91 82 STA (&82),Y
B641 85 86 STA &86
B643 AD B1 70 LDA &70B1
B646 91 84 STA (&84),Y
B648 85 87 STA &87
B64A A0 00 LDY #&00
B64C AD F8 70 LDA &70F8
B64F 91 86 STA (&86),Y
B651 E6 86 INC &86
B653 D0 02 BNE LB657
B655 E6 87 INC &87

.LB657
B657 98 TYA
B658 48 PHA
B659 20 7D AF JSR next_value
B65C 20 D1 B5 JSR LB5D1
B65F 68 PLA
B660 A8 TAY
B661 AD FE 70 LDA dL_value
B664 91 86 STA (&86),Y

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B666 C8      INY
B667 CC F8 70 CPY &70F8
B66A D0 EB    BNE LB657
B66C 18      CLC
B66D AD B0 70 LDA &70B0
B670 6D F8 70 ADC &70F8
B673 8D B0 70 STA &70B0
B676 AD B1 70 LDA &70B1
B679 69 00    ADC #&00
B67B 8D B1 70 STA &70B1
B67E EE B0 70 INC &70B0
B681 D0 03    BNE LB686
B683 EE B1 70 INC &70B1
.LB686
B686 4C EC AF JMP LAFEC
.LB689
B689 20 EC AF JSR LAFEC
B68C 20 DC B5 JSR LB5DC
B68F 20 D1 B5 JSR LB5D1
B692 AC FE 70 LDY dL_value
B695 8C F7 70 STY &70F7
B698 20 7D AF JSR next_value
B69B AD FE 70 LDA dL_value
B69E 29 7F    AND #&7F
B6A0 8D F4 70 STA &70F4
B6A3 20 7D AF JSR next_value
B6A6 AD FE 70 LDA dL_value
B6A9 29 7F    AND #&7F
B6AB 8D F5 70 STA &70F5
B6AE 20 7D AF JSR next_value
B6B1 AD FE 70 LDA dL_value
B6B4 29 7F    AND #&7F
B6B6 8D F6 70 STA &70F6
B6B9 A9 50    LDA #&50
B6BB 85 86    STA &86
B6BD A9 B1    LDA #&B1
B6BF 85 87    STA &87
.LB6C1
B6C1 A0 01    LDY #&01
.LB6C3
B6C3 B1 86    LDA (&86),Y
B6C5 D9 F3 70 CMP &70F3,Y
B6C8 D0 1A    BNE LB6E4
B6CA C8      INY
B6CB C0 04    CPY #&04
B6CD D0 F4    BNE LB6C3
B6CF A0 00    LDY #&00
B6D1 B1 86    LDA (&86),Y
B6D3 48      PHA
B6D4 20 D1 B5 JSR LB5D1
B6D7 AC F7 70 LDY &70F7
B6DA 68      PLA
B6DB 91 82    STA (&82),Y
B6DD A9 01    LDA #&01
B6DF 91 84    STA (&84),Y
B6E1 4C EC AF JMP LAFEC
.LB6E4
B6E4 18      CLC
B6E5 A5 86    LDA &86
B6E7 69 06    ADC #&06
B6E9 85 86    STA &86
B6EB A5 87    LDA &87
B6ED 69 00    ADC #&00
B6EF 85 87    STA &87
B6F1 A0 00    LDY #&00
B6F3 B1 86    LDA (&86),Y
B6F5 D0 CA    BNE LB6C1
; <UER> transmission error: "Bad Data...".
.LB6F7
B6F7 20 92 BA JSR LBA92
B6FA 4C 50 B4 JMP LB450
; <DND>: (nothing to do here)
.LB6FD
B6FD 60      RTS
; <DEF>: default tables.
.LB6FE
B6FE 20 12 B0 JSR default_tables
B701 A9 00    LDA #&00
B703 8D B2 70 STA dL_keep_tables
B706 8D F9 70 STA dL_txt_mode
B709 60      RTS
;-----
; Read a number operand (hex digit or X + two digits in teletext mode).
;-----
.LB70A
B70A AD F9 70 LDA dL_txt_mode
B70D F0 2B    BEQ LB73A
B70F 20 A6 BA JSR raw_byte
B712 20 9D B7 JSR LB79D
B715 C9 11    CMP #&11

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B717 F0 03 BEQ LB71C
B719 C9 00 CMP #&00
B71B 60 RTS
.LB71C
B71C 20 A6 BA JSR raw_byte
B71F 20 9D B7 JSR LB79D
B722 0A ASL A
B723 0A ASL A
B724 0A ASL A
B725 0A ASL A
B726 8D F9 70 STA dL_txt_mode
B729 20 A6 BA JSR raw_byte
B72C 20 9D B7 JSR LB79D
B72F 0D F9 70 ORA dL_txt_mode
B732 48 PHA
B733 A9 01 LDA #&01
B735 8D F9 70 STA dL_txt_mode
B738 68 PLA
B739 60 RTS
.LB73A
B73A 20 A6 BA JSR raw_byte
B73D C9 00 CMP #&00
B73F 60 RTS
; Read a multi-byte number operand into &70F4-&70F7.
.LB740
B740 A9 00 LDA #&00
B742 8D F4 70 STA &70F4
B745 8D F5 70 STA &70F5
B748 8D F6 70 STA &70F6
B74B 8D F7 70 STA &70F7
B74E AD F9 70 LDA dL_txt_mode
B751 F0 2B BEQ LB77E
B753 20 0A B7 JSR LB70A
B756 AA TAX
.LB757
B757 A0 00 LDY #&00
.LB759
B759 0E F4 70 ASL &70F4
B75C 2E F5 70 ROL &70F5
B75F 2E F6 70 ROL &70F6
B762 2E F7 70 ROL &70F7
B765 C8 INY
B766 C0 04 CPY #&04
B768 D0 EF BNE LB759
B76A 8A TXA
B76B 48 PHA
B76C 20 A6 BA JSR raw_byte
B76F 20 9D B7 JSR LB79D
B772 0D F4 70 ORA &70F4
B775 8D F4 70 STA &70F4
B778 68 PLA
B779 AA TAX
B77A CA DEX
B77B D0 DA BNE LB757
B77D 60 RTS
.LB77E
B77E 20 0A B7 JSR LB70A
B781 C9 05 CMP #&05
B783 B0 0C BCS LB791
B785 AA TAX
.LB786
B786 CA DEX
.LB787
B787 20 A6 BA JSR raw_byte
B78A 9D F4 70 STA &70F4,X
B78D CA DEX
B78E 10 F7 BPL LB787
B790 60 RTS
.LB791
B791 AA TAX
.LB792
B792 20 A6 BA JSR raw_byte
B795 CA DEX
B796 E0 05 CPX #&05
B798 B0 F8 BCS LB792
B79A 4C 86 B7 JMP LB786
; Hex character → value (&FF bad, &11 for "X").
.LB79D
B79D A2 00 LDX #&00
B79F 29 7F AND #&7F
.LB7A1
B7A1 DD C0 B7 CMP &B7C0,X
B7A4 F0 08 BEQ LB7AE
B7A6 E8 INX
B7A7 E0 18 CPX #&18
B7A9 D0 F6 BNE LB7A1
B7AB A9 FF LDA #&FF
B7AD 60 RTS
.LB7AE
B7AE E0 16 CPX #&16

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B7B0 B0 0B BCS LB7BD
B7B2 E0 10 CPX #&10
B7B4 B0 02 BCS LB7B8
B7B6 8A TXA
B7B7 60 RTS
.LB7B8
B7B8 8A TXA
B7B9 38 SEC
B7BA E9 06 SBC #&06
B7BC 60 RTS
.LB7BD
B7BD A9 11 LDA #&11
B7BF 60 RTS
; "0123456789ABCDEFabcdefXx"
B7C0 .db 30 31 32 33 34 35 36 37 38 39 41 42 43 44 45 46 ; 0123456789ABCDEF
B7D0 .db 61 62 63 64 65 66 58 78 ; abcdefXx
; Print A in hex.
.LB7D8
B7D8 8D FF 70 STA dL_raw
B7DB 8A TXA
B7DC 48 PHA
B7DD 98 TYA
B7DE 48 PHA
B7DF AD FF 70 LDA dL_raw
B7E2 29 F0 AND #&F0
B7E4 4A LSR A
B7E5 4A LSR A
B7E6 4A LSR A
B7E7 4A LSR A
B7E8 AA TAX
B7E9 BD C0 B7 LDA &B7C0,X
B7EC 20 EE FF JSR OSWRCH
B7EF AD FF 70 LDA dL_raw
B7F2 29 0F AND #&0F
B7F4 AA TAX
B7F5 BD C0 B7 LDA &B7C0,X
B7F8 20 EE FF JSR OSWRCH
B7FB 68 PLA
B7FC A8 TAY
B7FD 68 PLA
B7FE AA TAX
B7FF AD FF 70 LDA dL_raw
B802 60 RTS
;-----
; Block display: up to 80 blocks drawn as white boxes ("Not Received").
;-----
.LB803
B803 AD EF 70 LDA dL_nblocks
B806 F0 04 BEQ LB80C
B808 C9 51 CMP #&51 ; 'Q'
B80A 90 01 BCC LB80D
.LB80C
B80C 60 RTS
.LB80D
B80D A9 86 LDA #&86
B80F 20 F4 FF JSR OSBYTE ; OSBYTE &86: read text cursor position
B812 98 TYA
B813 48 PHA
B814 8A TXA
B815 48 PHA
B816 A9 2B LDA #&2B
B818 20 1A BB JSR print_inline_count
B81B .str '\x1a\x1f\x00\t\x97\xaf\x87Not Received \x92\xaf\x87OK \x91\xaf\x87Error - Retry'
B846 A9 01 LDA #&01
B848 8D B2 70 STA dL_keep_tables
B84B A9 00 LDA #&00
B84D 8D F4 70 STA &70F4
B850 A9 0B LDA #&0B
B852 8D F5 70 STA &70F5
B855 AD EF 70 LDA dL_nblocks
B858 8D F6 70 STA &70F6
B85B F0 41 BEQ LB89E
.LB85D
B85D A9 1F LDA #&1F
B85F 20 EE FF JSR OSWRCH ; VDU 31: TAB(x,y)
B862 AD F4 70 LDA &70F4
B865 8D AE 70 STA &70AE
B868 20 EE FF JSR OSWRCH
B86B AD F5 70 LDA &70F5
B86E 8D AF 70 STA &70AF
B871 20 EE FF JSR OSWRCH
B874 A9 02 LDA #&02
B876 20 1A BB JSR print_inline_count
B879 .str '\x97\xaf'
B87B 18 CLC
B87C AD F4 70 LDA &70F4
B87F 69 02 ADC #&02
B881 8D F4 70 STA &70F4
B884 C9 28 CMP #&28 ; '('
B886 D0 0E BNE LB896

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B888 A9 00 LDA #&00
B88A 8D F4 70 STA &70F4
B88D 18 CLC
B88E AD F5 70 LDA &70F5
B891 69 02 ADC #&02
B893 8D F5 70 STA &70F5
.LB896
B896 CE F6 70 DEC &70F6
B899 AD F6 70 LDA &70F6
B89C D0 BF BNE LB85D
.LB89E
B89E A9 00 LDA #&00
B8A0 8D EC 70 STA &70EC
B8A3 A9 06 LDA #&06
B8A5 20 1A BB JSR print_inline_count
B8A8 .str "\x1c\x00\x18'\x13\x1f"
B8AE 68 PLA
B8AF 20 EE FF JSR OSWRCH
B8B2 68 PLA
B8B3 20 EE FF JSR OSWRCH
B8B6 60 RTS
;-----
; Update the block display: block received (green) or bad (red), "n out of m block(s)
; received."
;-----
.LB8B7
B8B7 AD EC 70 LDA &70EC
B8BA D0 03 BNE LB8BF
B8BC 4C C3 B8 JMP LB8C3
.LB8BF
B8BF 60 RTS
.LB8C0
B8C0 4C 9A B9 JMP LB99A
.LB8C3
B8C3 A9 86 LDA #&86
B8C5 20 F4 FF JSR OSBYTE ; OSBYTE &86: read text cursor position
B8C8 98 TYA
B8C9 48 PHA
B8CA 8A TXA
B8CB 48 PHA
B8CC A9 06 LDA #&06
B8CE 20 1A BB JSR print_inline_count
B8D1 .str '\x1a\x1f\x00\x07\x86&'
B8D7 AD AB 70 LDA d_l_blocks_rx
B8DA 20 D8 B7 JSR LB7D8
B8DD AD EF 70 LDA d_l_nblocks
B8E0 F0 14 BEQ LB8F6
B8E2 A9 09 LDA #&09
B8E4 20 1A BB JSR print_inline_count
B8E7 .str '\x87out of\x86&'
B8F0 AD EF 70 LDA d_l_nblocks
B8F3 20 D8 B7 JSR LB7D8
.LB8F6
B8F6 A9 13 LDA #&13
B8F8 20 1A BB JSR print_inline_count
B8FB .str '\x87block(s) received.'
B90E AD EF 70 LDA d_l_nblocks
B911 C9 51 CMP #&51 ; 'Q'
B913 B0 AB BCS LB8C0
B915 AD F2 70 LDA d_l_block
B918 F0 A6 BEQ LB8C0
B91A C9 51 CMP #&51 ; 'Q'
B91C B0 A2 BCS LB8C0
B91E A9 00 LDA #&00
B920 8D F4 70 STA &70F4
B923 A9 0B LDA #&0B
B925 8D F5 70 STA &70F5
B928 AD EF 70 LDA d_l_nblocks
B92B 8D F6 70 STA &70F6
B92E A9 01 LDA #&01
B930 8D F7 70 STA &70F7
.LB933
B933 AD F7 70 LDA &70F7
B936 CD F2 70 CMP d_l_block
B939 D0 3E BNE LB979
B93B A9 1F LDA #&1F
B93D 20 EE FF JSR OSWRCH ; VDU 31: TAB(x,y)
B940 AD AE 70 LDA &70AE
B943 20 EE FF JSR OSWRCH
B946 AD AF 70 LDA &70AF
B949 20 EE FF JSR OSWRCH
B94C A9 04 LDA #&04
B94E 20 1A BB JSR print_inline_count
B951 .str '\t\n \x1f'
B955 AD F4 70 LDA &70F4
B958 8D AE 70 STA &70AE
B95B 20 EE FF JSR OSWRCH
B95E AD F5 70 LDA &70F5
B961 8D AF 70 STA &70AF
B964 20 EE FF JSR OSWRCH

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```

B967 AD 9A 70 LDA dL_blockchar
B96A 20 EE FF JSR OSWRCH
B96D A9 04 LDA #&04
B96F 20 1A BB JSR print_inline_count
B972 .str '\xaf\x08\n^'
B976 4C 9A B9 JMP LB99A

.LB979
B979 EE F7 70 INC &70F7
B97C 18 CLC
B97D AD F4 70 LDA &70F4
B980 69 02 ADC #&02
B982 8D F4 70 STA &70F4
B985 C9 28 CMP #&28 ; '('
B987 D0 0E BNE LB997
B989 A9 00 LDA #&00
B98B 8D F4 70 STA &70F4
B98E 18 CLC
B98F AD F5 70 LDA &70F5
B992 69 02 ADC #&02
B994 8D F5 70 STA &70F5

.LB997
B997 4C 33 B9 JMP LB933

.LB99A
B99A A9 06 LDA #&06
B99C 20 1A BB JSR print_inline_count
B99F .str "\x1c\x00\x18'\x13\x1f"
B9A5 68 PLA
B9A6 20 EE FF JSR OSWRCH
B9A9 68 PLA
B9AA 4C EE FF JMP OSWRCH

;-----
; Wait for the next page of the file and set the rows to read from it; check its CRC
; (OSWORD 23) and give "Bad Data..." if it fails.
;-----

.LB9AD
B9AD A9 07 LDA #&07
B9AF 8D 00 7A STA t_osw
B9B2 A9 1E LDA #&1E
B9B4 8D 01 7A STA t_osw+1
B9B7 20 38 A3 JSR LA338
B9BA A9 05 LDA #&05
B9BC 8D 00 7A STA t_osw
B9BF A9 00 LDA #&00
B9C1 8D 01 7A STA t_osw+1
B9C4 A2 03 LDX #&03

.LB9C6
B9C6 BD 9D 70 LDA dL_mag,X
B9C9 9D 02 7A STA t_osw+2,X
B9CC CA DEX
B9CD 10 F7 BPL LB9C6
B9CF A9 AB LDA #&AB
B9D1 8D 06 7A STA t_osw+6
B9D4 A9 6B LDA #&6B
B9D6 8D 07 7A STA t_osw+7
B9D9 A9 FF LDA #&FF
B9DB 8D 08 7A STA t_osw+8
B9DE 8D 09 7A STA t_osw+9
B9E1 20 38 A3 JSR LA338
B9E4 A9 03 LDA #&03
B9E6 8D 00 7A STA t_osw
B9E9 A9 00 LDA #&00
B9EB 8D 01 7A STA t_osw+1
B9EE 8D 02 7A STA t_osw+2
B9F1 20 38 A3 JSR LA338

.LB9F4
B9F4 20 45 B1 JSR LB145
B9F7 A9 02 LDA #&02
B9F9 8D 00 7A STA t_osw
B9FC A9 00 LDA #&00
B9FE 8D 01 7A STA t_osw+1
BA01 20 38 A3 JSR LA338
BA04 AD 02 7A LDA t_osw+2
BA07 29 02 AND #&02
BA09 F0 E9 BEQ LB9F4
BA0B A9 3F LDA #&3F
BA0D 8D 9F 70 STA dL_sub_hi
BA10 A9 7F LDA #&7F
BA12 8D A0 70 STA dL_sub_lo
BA15 A9 AB LDA #&AB
BA17 85 82 STA &82
BA19 A9 6B LDA #&6B
BA1B 85 83 STA &83
BA1D 20 90 A4 JSR find_row27
BA20 B0 40 BCS LBA62
BA22 A0 01 LDY #&01 ; workspace +&01: grabber records (8 bytes each)

.LBA24
BA24 B1 84 LDA (&84),Y
BA26 AA TAX
BA27 BD AF 91 LDA hamming_table,X
BA2A 30 36 BMI LBA62

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```

BA2C 29 0F    AND #&0F
BA2E 99 76 00 STA &0076,Y
BA31 C8      INY
BA32 C0 07    CPY #&07
BA34 D0 EE    BNE LBA24
BA36 20 55 91 JSR L9155
BA39 A5 75    LDA &75
BA3B 4A      LSR A
BA3C 4A      LSR A
BA3D 4A      LSR A
BA3E 4D AB 6B EOR &6BAB
BA41 8D 9D 70 STA dL_mag
BA44 A5 72    LDA &72
BA46 8D 9E 70 STA dL_page
BA49 A9 3F    LDA #&3F
BA4B 8D 9F 70 STA dL_sub_hi
BA4E A9 7F    LDA #&7F
BA50 8D A0 70 STA dL_sub_lo
BA53 AD B3 70 LDA dL_ordered
BA56 F0 0A    BEQ LBA62
BA58 A5 73    LDA &73
BA5A 8D 9F 70 STA dL_sub_hi
BA5D A5 74    LDA &74
BA5F 8D A0 70 STA dL_sub_lo
.LBA62
BA62 A9 92    LDA #&92
BA64 8D 9A 70 STA dL_blockchar
BA67 A9 17    LDA #&17
BA69 8D 00 7A STA t_osw
BA6C A9 AB    LDA #&AB
BA6E 8D 01 7A STA t_osw+1
BA71 A9 6B    LDA #&6B
BA73 8D 02 7A STA t_osw+2
BA76 20 38 A3 JSR LA338
BA79 AD 03 7A LDA t_osw+3
BA7C 10 0D    BPL LBA8B
BA7E A9 91    LDA #&91
BA80 8D 9A 70 STA dL_blockchar
BA83 A9 01    LDA #&01
BA85 8D EE 70 STA dL_skip
BA88 20 92 BA JSR LBA92
.LBA8B
BA8B A9 00    LDA #&00
BA8D 85 71    STA &71
BA8F 4C F4 BA JMP LBAF4
; "Bad Data..." with a beep.
.LBA92
BA92 A9 0E    LDA #&0E
BA94 20 1A BB JSR print_inline_count
BA97 .str 'Bad Data...\x07\r\n'
BAA5 60      RTS
;-----
; Next row byte of the page, row by row (rows 1-25 via find_row).
;-----
.raw_byte
BAA6 98      TYA
BAA7 48      PHA
BAA8 8A      TXA
BAA9 48      PHA
BAAA A5 78    LDA &78
BAAC 48      PHA
BAAD A5 82    LDA &82
BAAF 48      PHA
BAB0 A5 83    LDA &83
BAB2 48      PHA
BAB3 A5 84    LDA &84
BAB5 48      PHA
BAB6 A5 85    LDA &85
BAB8 48      PHA
BAB9 AD 99 70 LDA dL_col
BABC C9 28    CMP #&28 ; '('
BABE D0 03    BNE LBAC3
BAC0 20 F4 BA JSR LBAF4
.LBAC3
BAC3 EE 99 70 INC dL_col
BAC6 AD 97 70 LDA dL_rowptr
BAC9 85 77    STA &77
BACB AD 98 70 LDA dL_rowptr+1
BACE 85 78    STA &78
BAD0 A0 00    LDY #&00 ; workspace +&00: grabber records (8 bytes each)
BAD2 B1 77    LDA (&77),Y
BAD4 85 77    STA &77
BAD6 EE 97 70 INC dL_rowptr
BAD9 D0 03    BNE LBADE
BADB EE 98 70 INC dL_rowptr+1
.LBADE
BADE 68      PLA
BADF 85 85    STA &85
BAE1 68      PLA
BAE2 85 84    STA &84

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BAE4 68      PLA
BAE5 85 83    STA &83
BAE7 68      PLA
BAE8 85 82    STA &82
BAEA 68      PLA
BAEB 85 78    STA &78
BAED 68      PLA
BAEE AA      TAX
BAEF 68      PLA
BAF0 A8      TAY
BAF1 A5 77    LDA &77
BAF3 60      RTS
; Next row; after row 25 fetch the next page.
.LBAF4
BAF4 E6 71    INC &71
BAF6 A6 71    LDX &71
BAF8 E0 1A    CPX #&1A
BAFA F0 1B    BEQ LBB17
BAFC A9 AB    LDA #&AB
BAFE 85 82    STA &82
BB00 A9 6B    LDA #&6B
BB02 85 83    STA &83
BB04 20 45 A2 JSR find_row
BB07 A5 84    LDA &84
BB09 8D 97 70 STA dL_rowptr
BB0C A5 85    LDA &85
BB0E 8D 98 70 STA dL_rowptr+1
BB11 A9 00    LDA #&00
BB13 8D 99 70 STA dL_col
BB16 60      RTS
.LBB17
BB17 4C AD B9 JMP LB9AD
;
; PRINT INLINE BYTES: prints the A bytes that follow the JSR and returns after them.
;
.print_inline_count
BB1A 8D 5D 7B STA inline_count
BB1D 68      PLA
BB1E 85 7B    STA &7B
BB20 68      PLA
BB21 85 7C    STA &7C
BB23 A0 00    LDY #&00
.LBB25
BB25 C8      INY
BB26 B1 7B    LDA (&7B),Y
BB28 20 EE FF JSR OSWRCH
BB2B CC 5D 7B CPY inline_count
BB2E D0 F5    BNE LBB25
BB30 A5 7B    LDA &7B
BB32 18      CLC
BB33 6D 5D 7B ADC inline_count
BB36 85 7B    STA &7B
BB38 A5 7C    LDA &7C
BB3A 69 00    ADC #&00
BB3C 48      PHA
BB3D A5 7B    LDA &7B
BB3F 48      PHA
BB40 60      RTS
; Draw a horizontal bar on row A.
.LBB41
BB41 48      PHA
BB42 A9 1F    LDA #&1F
BB44 20 EE FF JSR OSWRCH          ; VDU 31: TAB(x,y)
BB47 A9 00    LDA #&00
BB49 20 EE FF JSR OSWRCH          ; VDU 0
BB4C 68      PLA
BB4D 20 EE FF JSR OSWRCH
BB50 A9 97    LDA #&97
BB52 20 EE FF JSR OSWRCH          ; VDU 151
BB55 A9 AC    LDA #&AC
BB57 A2 27    LDX #&27
.LBB59
BB59 20 EE FF JSR OSWRCH          ; VDU 172
BB5C CA      DEX
BB5D D0 FA    BNE LBB59
BB5F 60      RTS
;
; CATALOGUE BROWSER: the downloaded catalogue file is at OSHWM. Show its two title
; strings, then a scrolling list of the files whose computer type is "BBC" or "ANY",
; with a highlight bar moved by the cursor keys. RETURN downloads the chosen file
; (name, load and exec addresses, magazine and page from the entry); ESCAPE returns to
; the terminal.
;
.catalogue
BB60 A9 E5    LDA #&E5
BB62 20 B7 A0 JSR LA0B7
BB65 A9 04    LDA #&04
BB67 20 B7 A0 JSR LA0B7
BB6A A9 0C    LDA #&0C
BB6C A2 00    LDX #&00

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BB6E 20 B9 A0 JSR LA0B9
BB71 A9 02 LDA #&02
BB73 20 1A BB JSR print_inline_count
BB76 .str '\x1a\x0c'
BB78 A9 83 LDA #&83
BB7A 20 F4 FF JSR OSBYTE ; OSBYTE &83: read OSHWM (PAGE)
BB7D 86 82 STX &82
BB7F 86 84 STX &84
BB81 84 83 STY &83
BB83 84 85 STY &85
BB85 A9 01 LDA #&01
BB87 20 41 BB JSR LBB41
BB8A A0 00 LDY #&00
BB8C 20 73 BF JSR LBF73
BB8F A9 03 LDA #&03
BB91 20 41 BB JSR LBB41
BB94 A9 10 LDA #&10
BB96 20 41 BB JSR LBB41
BB99 20 73 BF JSR LBF73
BB9C A9 12 LDA #&12
BB9E 20 41 BB JSR LBB41
BBA1 A9 00 LDA #&00
BBA3 8D A7 70 STA &70A7
BBA6 8D A8 70 STA &70A8
BBA9 8D A5 70 STA &70A5
BBAC 8D A6 70 STA &70A6
BBAF 8D A3 70 STA &70A3
BBB2 8D A4 70 STA &70A4
BBB5 8D A9 70 STA &70A9
BBB8 8D AA 70 STA &70AA
BBBB 8D A2 70 STA &70A2
BBBE A9 75 LDA #&75
BBC0 20 1A BB JSR print_inline_count
BBC3 .str "\x1f\x01\x14Use the\x85UP\x87&\x85DOWN\x87cursor keys to select\n a file, and press\x85RETURN\x87to download.\r\n\n
Press\x85ESCAPE\x87to exit.\x1c\x00\x0e'\x05\x1f\x01\x00\x9d\x84\x1e"
.LBC38
BC38 20 C2 BE JSR catalogue_entry
BC3B 30 33 BMI LBC70
BC3D D0 1E BNE LBC5D
BC3F AD A9 70 LDA &70A9
BC42 D0 0C BNE LBC50
BC44 AD A7 70 LDA &70A7
BC47 8D A5 70 STA &70A5
BC4A AD A8 70 LDA &70A8
BC4D 8D A6 70 STA &70A6
.LBC50
BC50 EE A9 70 INC &70A9
BC53 AD A9 70 LDA &70A9
BC56 C9 0A CMP #&0A
BC58 F0 16 BEQ LBC70
BC5A 20 E7 FF JSR OSNEWL
.LBC5D
BC5D EE A7 70 INC &70A7
BC60 D0 03 BNE LBC65
BC62 EE A8 70 INC &70A8
.LBC65
BC65 EE A3 70 INC &70A3
BC68 D0 03 BNE LBC6D
BC6A EE A4 70 INC &70A4
.LBC6D
BC6D 4C 38 BC JMP LBC38
.LBC70
BC70 AD A9 70 LDA &70A9
BC73 8D A1 70 STA &70A1
BC76 CE A1 70 DEC &70A1
BC79 A9 00 LDA #&00
BC7B 8D A9 70 STA &70A9
.LBC7E
BC7E 20 E0 FF JSR OSRDCH
BC81 C9 8A CMP #&8A
BC83 F0 03 BEQ LBC88
BC85 4C 10 BD JMP LBD10
.LBC88
BC88 20 B2 BE JSR LBEB2
BC8B A9 20 LDA #&20
BC8D 20 EE FF JSR OSWRCH ; print ' '
BC90 20 EE FF JSR OSWRCH
BC93 AD A9 70 LDA &70A9
BC96 C9 09 CMP #&09
BC98 F0 18 BEQ LBCB2
BC9A CD A1 70 CMP &70A1
BC9D B0 03 BCS LBCA2
BC9F EE A9 70 INC &70A9
.LBCA2
BCA2 20 B2 BE JSR LBEB2
BCA5 A9 9D LDA #&9D
BCA7 20 EE FF JSR OSWRCH ; VDU 157
BCAA A9 84 LDA #&84
BCAC 20 EE FF JSR OSWRCH ; VDU 132
BCAF 4C 7E BC JMP LBC7E

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.LBCB2
BCB2 A9 01 LDA #&01
BCB4 8D A2 70 STA &70A2
BCB7 AD A3 70 LDA &70A3
BCBA 8D A7 70 STA &70A7
BCBD AD A4 70 LDA &70A4
BCC0 8D A8 70 STA &70A8

.LBCC3
BCC3 EE A7 70 INC &70A7
BCC6 D0 03 BNE LBCCB
BCC8 EE A8 70 INC &70A8

.LBCCB
BCCB 20 C2 BE JSR catalogue_entry
BCCE 30 D2 BMI LBCA2
BCD0 D0 F1 BNE LBCC3
BCD2 AD A7 70 LDA &70A7
BCD5 8D A3 70 STA &70A3
BCD8 AD A8 70 LDA &70A8
BCDB 8D A4 70 STA &70A4
BCDE AD A5 70 LDA &70A5
BCE1 8D A7 70 STA &70A7
BCE4 AD A6 70 LDA &70A6
BCE7 8D A8 70 STA &70A8
BCEA A9 15 LDA #&15
BCEC 20 EE FF JSR OSWRCH ; VDU 21

.LBCEF
BCEF EE A7 70 INC &70A7
BCF2 D0 03 BNE LBCEF7
BCF4 EE A8 70 INC &70A8

.LBCF7
BCF7 20 C2 BE JSR catalogue_entry
BCFA D0 F3 BNE LBCEF
BCFC A9 06 LDA #&06
BCFE 20 EE FF JSR OSWRCH ; VDU 6
BD01 AD A7 70 LDA &70A7
BD04 8D A5 70 STA &70A5
BD07 AD A8 70 LDA &70A8
BD0A 8D A6 70 STA &70A6
BD0D 4C A2 BC JMP LBCA2

.LBD10
BD10 C9 8B CMP #&8B
BD12 F0 03 BEQ LBD17
BD14 4C A3 BD JMP LBDA3

.LBD17
BD17 20 B2 BE JSR LBEB2
BD1A A9 20 LDA #&20
BD1C 20 EE FF JSR OSWRCH ; print ' '
BD1F 20 EE FF JSR OSWRCH
BD22 AD A9 70 LDA &70A9
BD25 F0 06 BEQ LBD2D
BD27 CE A9 70 DEC &70A9

.LBD2A
BD2A 4C A2 BC JMP LBCA2

.LBD2D
BD2D A9 02 LDA #&02
BD2F 8D A2 70 STA &70A2
BD32 AD A5 70 LDA &70A5
BD35 8D A7 70 STA &70A7
BD38 AD A6 70 LDA &70A6
BD3B 8D A8 70 STA &70A8

.LBD3E
BD3E AD A7 70 LDA &70A7
BD41 0D A8 70 ORA &70A8
BD44 F0 E4 BEQ LBD2A
BD46 38 SEC
BD47 AD A7 70 LDA &70A7
BD4A E9 01 SBC #&01
BD4C 8D A7 70 STA &70A7
BD4F AD A8 70 LDA &70A8
BD52 E9 00 SBC #&00
BD54 8D A8 70 STA &70A8
BD57 20 C2 BE JSR catalogue_entry
BD5A D0 E2 BNE LBD3E
BD5C AD A7 70 LDA &70A7
BD5F 8D A5 70 STA &70A5
BD62 AD A8 70 LDA &70A8
BD65 8D A6 70 STA &70A6
BD68 AD A3 70 LDA &70A3
BD6B 8D A7 70 STA &70A7
BD6E AD A4 70 LDA &70A4
BD71 8D A8 70 STA &70A8
BD74 A9 15 LDA #&15
BD76 20 EE FF JSR OSWRCH ; VDU 21

.LBD79
BD79 38 SEC
BD7A AD A7 70 LDA &70A7
BD7D E9 01 SBC #&01
BD7F 8D A7 70 STA &70A7
BD82 AD A8 70 LDA &70A8
BD85 E9 00 SBC #&00

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BD87 8D A8 70 STA &70A8
BD8A 20 C2 BE JSR catalogue_entry
BD8D D0 EA BNE LBD79
BD8F A9 06 LDA #&06
BD91 20 EE FF JSR OSWRCH ; VDU 6
BD94 AD A7 70 LDA &70A7
BD97 8D A3 70 STA &70A3
BD9A AD A8 70 LDA &70A8
BD9D 8D A4 70 STA &70A4
BDA0 4C A2 BC JMP LBCA2
.LBDA3
BDA3 C9 0D CMP #&0D
BDA5 F0 07 BEQ LBDAE
BDA7 C9 87 CMP #&87
BDA9 F0 03 BEQ LBDAE
BDAB 4C A3 BE JMP LBEA3
.LBDAE
BDAE A9 15 LDA #&15
BDB0 20 EE FF JSR OSWRCH ; VDU 21
BDB3 AD A5 70 LDA &70A5
BDB6 8D A7 70 STA &70A7
BDB9 AD A6 70 LDA &70A6
BDBC 8D A8 70 STA &70A8
.LBDBF
BDBF AD A9 70 LDA &70A9
BDC2 F0 13 BEQ LBDD7
.LBDC4
BDC4 EE A7 70 INC &70A7
BDC7 D0 03 BNE LBDCC
BDC9 EE A8 70 INC &70A8
.LBDCC
BDCC 20 C2 BE JSR catalogue_entry
BDCE D0 F3 BNE LBDC4
BDD1 CE A9 70 DEC &70A9
BDD4 4C BF BD JMP LBDBF
.LBDD7
BDD7 20 8C BF JSR LBF8C
BDDA A9 03 LDA #&03
BDDC 20 1A BB JSR print_inline_count
BDDF .str '\x06\x1a\x0c'
BDE2 A9 01 LDA #&01
BDE4 20 41 BB JSR LBB41
BDE7 A9 0D LDA #&0D
BDE9 20 1A BB JSR print_inline_count
BDEC .str '\x1f\x01\x03Filename :'
BDF9 20 E2 BF JSR LBFE2
BDFC B1 84 LDA (&84),Y
BDFF 8D F8 70 STA &70F8
BE01 EE F8 70 INC &70F8
BE04 C8 INY
BE05 B1 84 LDA (&84),Y
BE07 09 80 ORA #&80
BE09 20 EE FF JSR OSWRCH
BE0C C8 INY
.LBE0D
BE0D B1 84 LDA (&84),Y
BE0F 20 2F B2 JSR LB22F
BE12 20 EE FF JSR OSWRCH
BE15 99 D8 70 STA &70D8,Y
BE18 C8 INY
BE19 CC F8 70 CPY &70F8
BE1C D0 EF BNE LBE0D
BE1E A9 0D LDA #&0D
BE20 99 D8 70 STA &70D8,Y
BE23 98 TYA
BE24 18 CLC
BE25 65 84 ADC &84
BE27 85 84 STA &84
BE29 A5 85 LDA &85
BE2B 69 00 ADC #&00
BE2D 85 85 STA &85
BE2F 20 E2 BF JSR LBFE2
BE32 20 E2 BF JSR LBFE2
BE35 A9 0B LDA #&0B
BE37 20 1A BB JSR print_inline_count
BE3A .str '\x1f\x01\x05Load : &'
BE45 A0 00 LDY #&00
BE47 A2 03 LDX #&03
.LBE49
BE49 B1 84 LDA (&84),Y
BE4B 20 D8 B7 JSR LB7D8
BE4E 9D B4 70 STA dl_load,X
BE51 C8 INY
BE52 CA DEX
BE53 10 F4 BPL LBE49
BE55 98 TYA
BE56 48 PHA
BE57 A9 0B LDA #&0B
BE59 20 1A BB JSR print_inline_count
BE5C .str '\x1f\x15\x05Exec : &'

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BE67 68 PLA
BE68 A8 TAY
BE69 A2 03 LDX #003
.LBE6B
BE6B B1 84 LDA (&84),Y
BE6D 20 D8 B7 JSR LB7D8
BE70 9D B8 70 STA dL_exec,X
BE73 C8 INY
BE74 CA DEX
BE75 10 F4 BPL LBE6B
BE77 B1 84 LDA (&84),Y
BE79 8D 9D 70 STA dL_mag
BE7C C8 INY
BE7D B1 84 LDA (&84),Y
BE7F 8D 9E 70 STA dL_page
BE82 A9 05 LDA #05
BE84 20 1A BB JSR print_inline_count
BE87 .str "\x1c\x00\x18'\x13"
BE8C A9 00 LDA #00
BE8E 8D B2 70 STA dL_keep_tables
BE91 A9 01 LDA #01
BE93 8D 9B 70 STA dL_to_file
BE96 A9 E5 LDA #E5
BE98 A2 00 LDX #00
BE9A 20 B9 A0 JSR LA0B9
BE9D 20 AD AD JSR downLoad
BEA0 4C A3 AC JMP LACA3
.LBEA3
BEA3 C9 1B CMP #1B
BEA5 D0 08 BNE LBEAF
BEA7 A9 87 LDA #87
BEA9 8D 5E 7B STA t_hdr_colour
BEAC 4C 41 98 JMP L9841
.LBEAF
BEAF 4C 7E BC JMP LBC7E
.LBEB2
BEB2 A9 1F LDA #1F
BEB4 20 EE FF JSR OSWRCH ; VDU 31: TAB(x,y)
BEB7 A9 01 LDA #01
BEB9 20 EE FF JSR OSWRCH ; VDU 1
BEBB AD A9 70 LDA &70A9
BEBF 4C EE FF JMP OSWRCH
;-----
; Is entry n for this computer? The 3-letter computer type is recognised by the sum of
; its characters: &C7 "BBC", &E8 "ANY"; &7E "***" ends the catalogue. Returns 0 to show,
; 1 to skip, &FF at the end.
;-----
.catalogue_entry
BEC2 20 8C BF JSR LBF8C
BEC5 B1 84 LDA (&84),Y
BEC7 C9 03 CMP #03
BEC9 D0 1B BNE LBEE6
BECB 18 CLC
BECC A9 00 LDA #00
BECE C8 INY
BECF 71 84 ADC (&84),Y
BED1 C8 INY
BED2 71 84 ADC (&84),Y
BED4 C8 INY
BED5 71 84 ADC (&84),Y
BED7 C8 INY
BED8 B0 0C BCS LBEE6
BEDA C9 C7 CMP #C7
BEDC F0 0B BEQ LBEE9
BEDE C9 E8 CMP #E8
BEE0 F0 07 BEQ LBEE9
BEE2 C9 7E CMP #7E ; '~'
BEE4 F0 6C BEQ LBF52
.LBEE6
BEE6 4C 4F BF JMP LBF4F
.LBEE9
BEE9 A9 75 LDA #75
BEEB 20 F4 FF JSR OSBYTE ; OSBYTE &75
BEEF 8A TXA
BEF1 30 2A BMI LBF1B
BEF4 AD A2 70 LDA &70A2
BEF6 F0 25 BEQ LBF1B
BEF8 C9 01 CMP #01
BEFA D0 17 BNE LBF11
BEFB A9 1F LDA #1F
BEFC 20 EE FF JSR OSWRCH ; VDU 31: TAB(x,y)
BEFF A9 00 LDA #00
BF01 20 EE FF JSR OSWRCH ; VDU 0
BF04 A9 09 LDA #09
BF06 20 EE FF JSR OSWRCH ; VDU 9
BF09 A9 0A LDA #0A
BF0B 20 EE FF JSR OSWRCH ; VDU 10: line feed
BF0E 4C 1B BF JMP LBF1B
.LBF11
BF11 A9 1E LDA #1E

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BF13 20 EE FF JSR OSWRCH          ; VDU 30: cursor home
BF16 A9 0B LDA #0B
BF18 20 EE FF JSR OSWRCH          ; VDU 11
.LBF1B
BF1B 20 55 BF JSR LBF55
BF1E 8C F8 70 STY &70F8
BF21 A9 86 LDA #86
BF23 20 F4 FF JSR OSBYTE          ; OSBYTE &86: read text cursor position
BF26 A9 1F LDA #1F
BF28 20 EE FF JSR OSWRCH          ; VDU 31: TAB(x,y)
BF2B A9 10 LDA #10
BF2D 20 EE FF JSR OSWRCH          ; VDU 16
BF30 98 TYA
BF31 20 EE FF JSR OSWRCH
BF34 48 PHA
BF35 AC F8 70 LDY &70F8
BF38 20 73 BF JSR LBF73
BF3B A9 1F LDA #1F
BF3D 20 EE FF JSR OSWRCH          ; VDU 31: TAB(x,y)
BF40 A9 1C LDA #1C
BF42 20 EE FF JSR OSWRCH          ; VDU 28: define text window
BF45 68 PLA
BF46 20 EE FF JSR OSWRCH
BF49 20 73 BF JSR LBF73
BF4C A9 00 LDA #00
BF4E 60 RTS
.LBF4F
BF4F A9 01 LDA #01
BF51 60 RTS
.LBF52
BF52 A9 FF LDA #FF
BF54 60 RTS
.LBF55
BF55 B1 84 LDA (&84),Y
BF57 8D F8 70 STA &70F8
BF5A CE F8 70 DEC &70F8
BF5D C8 INY
BF5E B1 84 LDA (&84),Y
BF60 09 80 ORA #80
BF62 20 EE FF JSR OSWRCH
BF65 A9 09 LDA #09
BF67 20 EE FF JSR OSWRCH          ; VDU 9
BF6A 20 EE FF JSR OSWRCH
BF6D AD F8 70 LDA &70F8
BF70 4C 78 BF JMP LBF78
; Print a length-prefixed string with bit 7 set.
.LBF73
BF73 B1 84 LDA (&84),Y
BF75 8D F8 70 STA &70F8
.LBF78
BF78 F0 10 BEQ LBF8A
BF7A A2 00 LDX #00
.LBF7C
BF7C C8 INY
BF7D B1 84 LDA (&84),Y
BF7F 09 80 ORA #80
BF81 20 EE FF JSR OSWRCH
BF84 E8 INX
BF85 EC F8 70 CPX &70F8
BF88 D0 F2 BNE LBF7C
.LBF8A
BF8A C8 INY
BF8B 60 RTS
; (&84) := catalogue entry number &70A7.
.LBF8C
BF8C AD A7 70 LDA &70A7
BF8F 48 PHA
BF90 AD A8 70 LDA &70A8
BF93 48 PHA
BF94 A5 82 LDA &82
BF96 85 84 STA &84
BF98 A5 83 LDA &83
BF9A 85 85 STA &85
BF9C 20 E2 BF JSR LBFE2
BF9F 20 E2 BF JSR LBFE2
.LBFA2
BFA2 AD A7 70 LDA &70A7
BFA5 0D A8 70 ORA &70A8
BFA8 F0 2D BEQ LBF07
BFAA 38 SEC
BFAB AD A7 70 LDA &70A7
BFAE E9 01 SBC #01
BFB0 8D A7 70 STA &70A7
BFB3 AD A8 70 LDA &70A8
BFB6 E9 00 SBC #00
BFB8 8D A8 70 STA &70A8
BFBB 20 E2 BF JSR LBFE2
BFBE 20 E2 BF JSR LBFE2
BFC1 20 E2 BF JSR LBFE2
BFC4 20 E2 BF JSR LBFE2

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BFC7 18      CLC
BFC8 A5 84    LDA &84
BFCA 69 0A    ADC #&0A
BFCC 85 84    STA &84
BFCE A5 85    LDA &85
BFD0 69 00    ADC #&00
BFD2 85 85    STA &85
BFD4 4C A2 BF JMP LBFA2
.LBFD7
BFD7 68      PLA
BFD8 8D A8 70 STA &70A8
BFDB 68      PLA
BFDC 8D A7 70 STA &70A7
BFDf A0 00    LDY #&00
BFE1 60      RTS
; Skip a length-prefixed string.
.LBFE2
BFE2 A0 00    LDY #&00          ; workspace +&00: grabber records (8 bytes each)
BFE4 B1 84    LDA (&84),Y
BFE6 8D F8 70 STA &70F8
BFE9 EE F8 70 INC &70F8
BFEC 18      CLC
BFED A5 84    LDA &84
BFEF 6D F8 70 ADC &70F8
BFF2 85 84    STA &84
BFF4 A5 85    LDA &85
BFF6 69 00    ADC #&00
BFF8 85 85    STA &85
BFFA 60      RTS
BFFB .db 20 47 2E 42 20          ; G.B

```

Files in /home/ste/dev/ttadapter/disassembly: TELEROMU.s and ATS250.s (the listings), teleromu.json / ats250.json (entry points and tables), teleromu.ann / ats250.ann (the commentary), dis.py (the tracing disassembler) and page/ (this page's generator). To regenerate: python3 dis.py teleromu.json > TELEROMU.s (and the same for ats250), then python3 page/build_page.py. Both listings reproduce every byte of their ROM.

Sources: Acorn Teletext System User Guide (1983); schematic 104,000/C redrawn by ZXGuesser (2024) from 100,004 Iss 5 / PCB 0204,000 Iss 10; Advanced Teletext System User Guide (BBC, 1986); SAA5030 data sheet; MOS 1.20 disassembly.